



SAP Cloud Integration

Generated on: 2026-10-06 12:05:13 GMT+0000

SAP Cloud Integration | Cloud

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Original content: https://help.sap.com/docs/CLOUD_INTEGRATION/368c481cd6954bd5d0435479fd4eaf?locale=en-US&state=PRODUCTION&version=Cloud

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Security, Cloud Foundry Environment

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Using a cloud-based integration platform imposes dedicated security measures on the software vendor (SAP) that hosts the platform, as well as on those (the customers) who use the platform.

This section describes the security-related aspects of the integration platform and shows which measures you can take to protect customer data that is passed through the platform during the execution of an integration scenario.

Customers who use SAP Cloud Integration agree that a significant part of their (and their customers') sensitive data is processed by and stored within an infrastructure not owned by themselves.

The core task of an integration platform is to serve as the transit place for messages, which may contain sensitive customer data. First and foremost, these messages must be protected against eavesdropping and unauthorized access.

Therefore, the integration platform must fulfill the following main requirements:

- The integration infrastructure is already designed and built in such a way that it meets the highest security standards.

In particular, it must be guaranteed that the technical system landscape, the communication between the components of the integration platform, and the storage locations of messages are secure.

- The processes related to the usage of Cloud Integration meet the highest security standards.

This relates to the processes at SAP that are related to the development and upgrade of the Cloud Integration software, the processes related to the provisioning and operation of the customers' virtual environment by the infrastructure provider, and the customer onboarding process during which customers set up secure connections between their infrastructure and SAP's integration platform.

- Customers have several options to configure how messages are exchanged within an integration scenario so that the involved data is protected at the highest level.

In particular, when designing integration flows, customers can choose between several options to protect messages by establishing secure communication channels (transport-level security) and by configuring digital encryption and digital signing of messages (message-level security).

This documentation summarizes the measures that are taken by SAP to fulfill these requirements.

Related Information

[Technical Landscape, Cloud Foundry Environment](#)

[Security Aspects of Processes](#)

[Security Aspects of Data, Data Flow](#)

[Identity and Access Management](#)

[Data Storage Security](#)

[Data Protection and Privacy](#)

[Other Security-Related Information](#)

Technical Landscape, Cloud Foundry Environment

The technical infrastructure comprises a set of technical components that can communicate with each other and with remote components in a secure way (using certain protocols such as HTTPS or SFTP, for example).

i Note

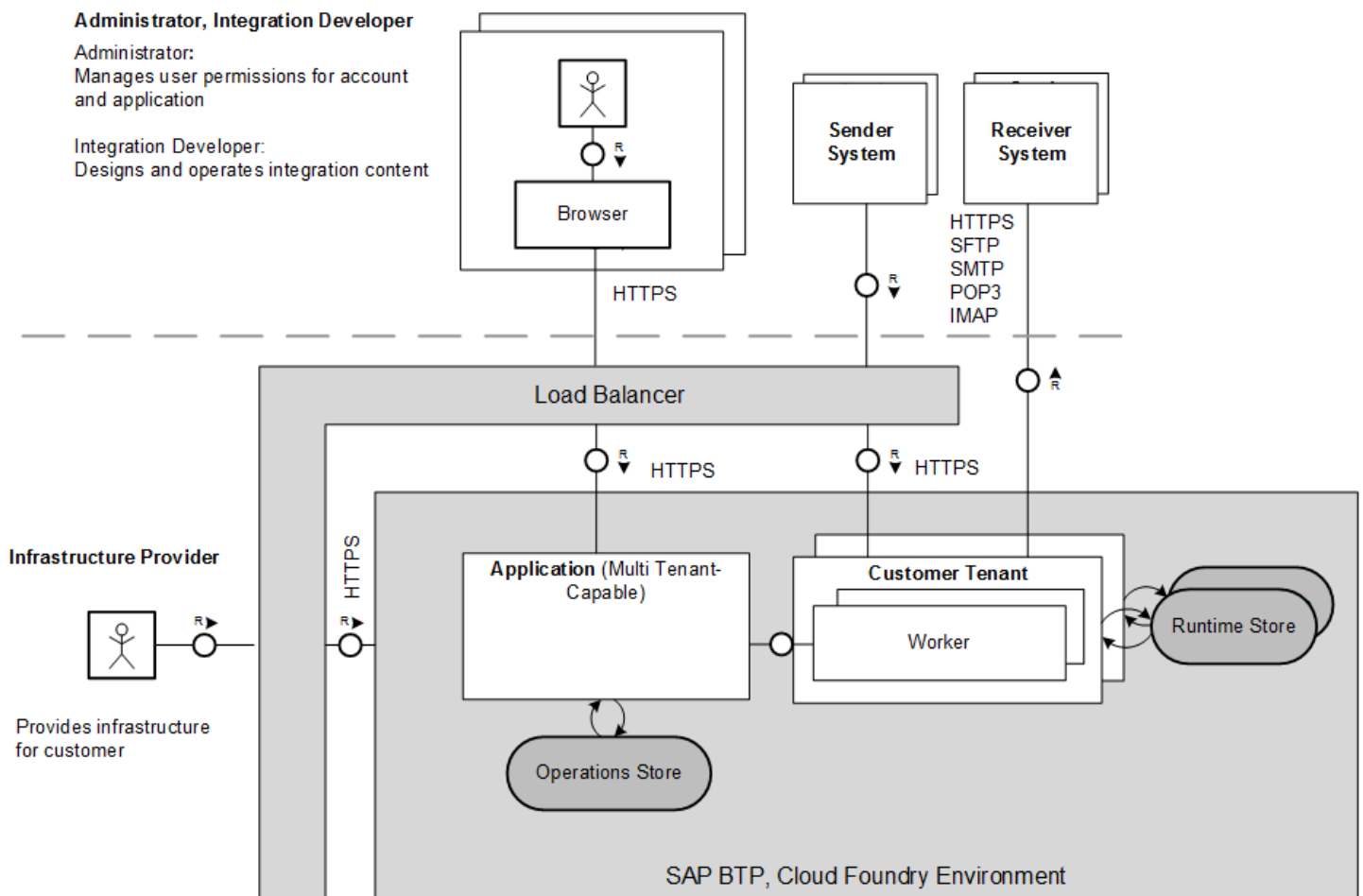
This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Components and Communication Paths

In technical terms, the integration platform is designed as a containerized and clustered integration platform in the cloud. Messages processed by integration flows from different customers are handled on different *parts* of the platform (referred to as tenants).

Tenants processing integration flows from different customers are strictly separated from each other in terms of CPU, data storage, and user access.

The following figure shows a bird's eyes view on the technical architecture.



These are the basic constituents of the virtual platform:

- A multi tenant-capable application comprises a set of microservices (not depicted in the figure) that accomplish tasks related to the management of a tenant and the preparation of monitoring data. It takes requests from the dialog users (for example, when an integration developer deploys an integration flow using the Web user interface).

These microservices run on an application that can be shared across multiple customer tenants.

- A worker (runtime container) processes messages that are exchanged with external systems. Therefore, the worker is connected to the external systems. In other words, workers process customer data that might be confidential and has to be protected.

Workers are operated within customer-specific tenants. These tenants are strictly separated from each other.

As a consequence of this cluster design, the following main communication paths are active during the operation of an integration scenario:

- Communication of tenant cluster and remote components

You can use both cloud systems and on-premise systems (such as on-premise SAP systems) as remote components.

Remote receiver systems are directly connected to a worker through a protocol, which depends on the type of the designed receiver adapter.

If the integration platform communicates with an on-premise receiver system, you can interconnect the SAP Cloud Connector. This component runs as on-premise agent in a secured network and acts as a reverse invoke proxy between the on-premise network and SAP Cloud Integration. Due to its reverse invoke support, you don't need to configure the on-premise firewall to allow external access from the cloud to internal systems.

For inbound communication from a sender targeting Cloud Integration, a load balancer is interconnected between remote sender systems and the involved SAP BTP components. The load balancer terminates incoming Transport Layer Security (TLS) requests and establishes new ones.

Various secure technical protocols can be used for these communication paths. Depending on the adapter type, the following protocols are available:

- Hyper Text Transfer Protocol (HTTP) over Transport Layer Security (TLS), which is referred to as HTTPS
- SSH File Transfer Protocol (SFTP) for the exchange of data with an SFTP server
- Simple Mail Transfer Protocol (SMTP), Post Office Protocol (POP)³, and Internet Message Access Protocol (IMAP) for the exchange of data with mail servers

User Access

Additional components come into play when a dialog user accesses the infrastructure (for example, when an administrator accesses monitoring data or when an integration developer deploys an integration artifact).

People with different roles can access the infrastructure – both on the side of the infrastructure provider and on the customer side. Human access points (for dialog users) are:

- Dedicated experts at the side of the infrastructure provider need access to the infrastructure (for example, to provide a tenant for the customer).
- Experts on the customer side access the infrastructure to design and deploy integration content and to monitor an integration scenario at runtime (integration developers and tenant administrators).

Security Aspects of Processes

Processes that are related to the provisioning, update, and usage of the cloud-based integration platform meet the highest security standards.

Cloud Integration is compliant with various SAP-internal technical policies, procedures, directives, guidelines, and product standards.

For example, SAP software is developed in compliance with the SAP Secure Development Lifecycle (SDLC), which helps to implement measures such as test-driven development and threat modeling.

SAP certifies that the development, maintenance, and operations of Cloud Integration comply with the requirements of various standards:

More information: [SAP Trust Center](#)

Security Aspects of Data, Data Flow

All data in transit, either exchanged with remote components or internal, can be protected by methods such as encryption.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

i Note

Mutual TLS (mTLS) is equivalent to client certificate authentication. While setting up the TLS connection, client and server exchange certificates. With mTLS, not only server certificates, but also client certificates are validated based on the signatures provided by certification authorities. For more information, see [Client Certificate Authentication \(Outbound\)](#) and [Keystore](#).

During a scenario, the connected remote systems exchange data with each other based on the configured transport protocol. These protocols support different options to protect the exchanged data against unauthorized access. In addition to security at the transport level, the content of the exchanged messages can also be protected by means of digital encryption and signature.

Transport-Level Security

Each adapter allows you to set up a specific security level based on the underlying transport protocol.

Transport-Level Security Options

Transport Protocol	Transport-Level Security
HTTP(S) (Hypertext Transfer Protocol Secure)	Hyper Text Transfer Protocol (HTTP) over Transport Layer Security (TLS) (HTTPS) TLS is a protocol for secure communication over a computer network that is widely used on the Internet. As TLS is the enhancement of SSL, these terms are often used synonymously. You can protect communication using Transport Layer Security (TLS). In this case, a symmetric key length of at least 128 bits is used (which is technically enforced). Default length of asymmetric keys provided by SAP is 2048 bits. i Note The following versions are supported: - For inbound communication: TLS 1.2 and 1.3 (in the Cloud Foundry environment) - For inbound communication: TLS 1.2 (in the Neo environment) - For outbound communication: TLS 1.2, and 1.3 i Note The HTTP receiver adapter also allows you to use HTTP URLs. However, we do not recommend using this option when transferring confidential data (including the password for basic authentication). Also, if the network is not entirely trusted, there is no way to verify whether the result of an HTTP request originates from a trustworthy source. Therefore, we do not recommend using this option

Transport Protocol	Transport-Level Security
	for productive scenarios over the Internet. Receiver adapters also support principal propagation via SAP Cloud Connector. Various authentication options (basic authentication using user credentials, client certificates, or OAuth) are supported depending on the selected sender or receiver adapter.  Caution Consider that we do not recommend to use basic authentication in productive scenarios because of the following security aspects: Basic authentication has the risk that authentication credentials, for example, passwords, are sent in clear text. Using TLS (transport-layer security, also referred to as Secure Sockets Layer) as transport-level encryption method (when using HTTPS as protocol) makes sure that this information is nevertheless encrypted on the transport path. However, the authentication credentials might become visible to SAP-internal administrators at points in the network where the TLS connection is terminated, for example, load balancers. If logging is not done properly at such devices, the authentication credentials might become part of log files. Also network monitoring tools used at such devices might expose the authentication information to administrators. Furthermore, the person to whom the authentication credentials belong (in the previous example, the password owner) needs to maintain the password in a secure place. This protocol is supported by all adapter types that allow communication over HTTPS, for example: Ariba, AS2, AS4, ELSTER, Facebook, HTTP, SOAP, IDoc, OData, ODC, OpenConnectors, SAP SuccessFactors, Twitter, XI.
SFTP (Secure Shell File Transfer Protocol)	Secure Shell (SSH) is used to securely transfer files in an open network. SSH is a network protocol that allows you to set up a secure connection to a remote computer, namely an SFTP server. SSH uses a symmetric key length with at least 128 bits to protect FTP communication. Default length of asymmetric keys provided by SAP is 2048 bits. Supported authentication methods: • User name/password authentication (where the SFTP server authenticates the calling component based on the user name and password) • Public key authentication (where the SFTP server authenticates the calling component based on a public key) Secure data transfer with SFTP is based on a combination of symmetric and asymmetric keys. Symmetric (session) keys are used to encrypt and decrypt data within a session. Asymmetric key pairs are used to encrypt and decrypt the session keys. When asymmetric key pairs are used, SFTP also ensures that only authorized public keys are used by the involved participants. Supported versions: • SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture ) • SSH File Transfer Protocol (SFTP) version 3 or higher This protocol is supported by the SFTP sender and receiver adapter.
FTPS (Secure File Transfer Protocol)	This protocol is used to transfer files from and to file servers. This is an extension of the File Transfer Protocol (FTP). This protocol is supported by the FTP sender and receiver adapter.
JMS (Java Message Service)	JMS is an API that supports reliable asynchronous communication based on a JMS Message Broker.

Transport Protocol	Transport-Level Security
	This protocol is supported by the JMS sender and receiver adapter.
JDBC (Java Database Connectivity)	Java Database Connectivity (JDBC) is a protocol or API that defines how a client accesses a database. This protocol is supported by the JDBC receiver adapter.
RFC (Remote Function Call)	Remote Function Call (RFC) is an SAP-proprietary protocol that is used as standard for the communication of SAP systems. This protocol is supported by the RFC receiver adapter.
TCP (Transmission Control Protocol)	TCP is one of the main communication protocols used on the Internet. TLS, previously mentioned, often (but not always) runs on top of TCP. This protocol is supported by the AMQP (TCP) and the LDAP adapter.
WebSocket	WebSocket is a protocol that supports simultaneous communication in both directions (full-duplex). This protocol is used for setting up connections with a messaging systems. This protocol is supported by the AMQP (WebSocket) adapter.
SMTPS (Simple Mail Transfer Protocol Secure)	This protocol enables a computer to exchange emails with a mail server. With SMTPS, SMTP connections can be secured by SSL or TLS. Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol is used by the Mail receiver adapter.
POP3S (Post Office Protocol version 3 over TLS/SSL)	This option enables email clients to retrieve emails from an email server using the Internet Protocol (IP). Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol can be used by the Mail sender adapter.
IMAPS (Internet Message Access Protocol over TLS/SSL)	Internet Message Access Protocol over TLS/SSL (IMAPS). This option enables email clients to retrieve emails from an email server using a TCP/IP connection. Transport encryption is supported via the STARTTLS extended operation.

Transport Protocol	Transport-Level Security
	To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol can be used by the Mail sender adapter.

Message-Level Security

On top of the *transport-level security* options, you can also secure the communication at *message level*, where the content of the exchanged messages can also be protected by means of digital encryption and signatures. Various security standards are available to do this, as summarized in the following table.

To configure message-level security options, you use dedicated integration flow steps (for example, the **Encryptor** and **Signer** step types).

The following standards and algorithms are supported:

Message-Level Security Standards and Algorithms

Standard	Security Feature
PKCS#7/CMS Enveloped Data and Signed Data	Encryption/decryption of message content
	Signing/verification of payload
PKCS#7/CMS Enveloped and Signed Data	Encryption/decryption and signing/verification of payload
Open Pretty Good Privacy (PGP)	Encryption/decryption of message content
	Encryption/decryption and signing/verification of message
XML Signature	Signing/verification of payload
WS-Security	Signing/verification of SOAP body

Related Information

[Connectivity \(Adapters\)](#)

[Authentication and Authorization Options \(Inbound\)](#)

[Authentication Options \(Outbound\)](#)

[Certificate Management](#)

[Message-Level Security](#)

Identity and Access Management

Identity and access management features are used during the lifecycle of an integration scenario.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Access Management

Dialog users who access the platform are authenticated against an identity provider. SAP Identity Service (ID Service) is used by default. SAP ID Service is the central service for the process of managing identities and their lifecycles.

User Management and Authorizations

Access to dedicated functions of the platform is controlled and protected by authorization checks. Permissions are managed through personas — each representing a typical role in an integration project lifecycle and defining a set of tasks the user is authorized to perform.

There are two ways to assign permissions to users:

- **Predefined personas** — Each persona is delivered as a ready-to-use role collection (Cloud Foundry) or authorization group (Neo) covering all the permissions needed for that profile. This is the simpler option and a recommended starting point. For more information, see [Personas](#).
- **Individual roles** — For more fine-grained control, you can define your own role collections (Cloud Foundry) or authorization groups (Neo) by selecting individual roles or role templates that correspond to specific tasks. This approach requires more configuration but lets you tailor permissions precisely to your needs. For more information, see [Tasks and Permissions](#).

i Note

Example:

If the logged-in user has to perform tasks such as designing and deploying integration flows, the user must be assigned the authorization group `AuthGroup.IntegrationDeveloper` (Neo) or the role collection `PI_Integration_Developer` (Cloud Foundry).

Authorization for the Integration Developer

The tasks of persons with integration developer permissions (short: integration developers) constitute a key part of the SAP Cloud Integration lifecycle. Permissions for the integration developer (who is in charge of modeling integration flows) are contained in the authorization group `AuthGroup.IntegrationDeveloper`.

Note that the roles contained in this authorization group give an integration developer full control over message processing during runtime.

During integration flow modeling, the integration developer defines how messages are mapped, which credentials are used, and to which recipients messages are sent. The set of roles provides very powerful permissions and in some cases allows the integration developer to access sensitive data.

i Note

The integration developer can control which credentials are to be used in connections with basic authentication by deploying the associated [User Credentials](#) artifacts on the tenant. These artifacts contain user names and passwords. Note that, however, a password specified in a [User Credentials](#) is never displayed. Furthermore, passwords cannot be downloaded (by either using the user interface or the application programming interface). The integration developer, although having full control over the integration flow, does not have access to credentials of another tenant of the same customer.

Therefore, apply the following measures when designing integration flows for security-sensitive areas:

- Don't give the integration developer access to productive systems.
- Consider applying a four-eyes principle and implement a review process before deploying integration flows to production.
- An integration developer has the option to develop integration flows on a separate development or test tenant. These integration flows can then be transported to the productive tenant by another person.
- Don't share the same secret credentials between tenants with different security levels (for example, between test tenant and productive tenant).
- If you suspect a security violation, check the audit log to find out which user deployed the integration flow in question.
- If read-only access is required to analyze issues in the productive system, use the authorization group `AuthGroup.ReadOnly`.

→ Tip

Instead of using the predefined authorization groups, you can tailor the permissions to your own requirements by applying elementary roles that are defined for individual tasks.

More information: [Tasks and Permissions](#)

Authentication and Authorization Options for Inbound Calls

When a sender system calls the integration platform using HTTPS-based (inbound) requests, there are different ways for the calling sender to authenticate itself against the integration platform. The options are basic authentication, OAuth, and SAML.

i Note

- Authentication

Verifies the identity of the calling entity.

- Authorization

Checks what a user or other entity is authorized to do (for example, as defined by roles assigned to it). In other words, the authorization check evaluates the access rights of a user or other entity.

Related Information

[User Management for SAP BTP, Neo Environment](#)

[Managing Users and Role Assignments, Neo Environment](#)

[Load Balancer Root Certificates Supported by SAP](#)

[Authentication and Authorization Options \(Inbound\)](#)

[Personas](#)

Personas

When you perform user management tasks using SAP BTP cockpit, you find a set of predefined roles that you can assign to users of the account. According to the main tasks associated with integration projects, these roles are associated to certain persona relevant for an integration project.

Personas are predefined sets of permissions that cover the different tasks associated with an integration project. To assign roles that are application-based, see [Configuring User Access to the Application](#). For more fine-grained control over user permissions, see [Tasks and Permissions](#).

i Note

In the different environments, the personas are mapped to different objects.

- In the Neo environment, a persona is realized by an authorization group (beginning with the string **AuthGroup**).
- In the Cloud Foundry environment, a persona is realized by role collection.

Personas			
Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Description
Read-only persona	AuthGroup.ReadOnly	PI_Read_Only	Enables tenant administrator to read messages. This persona provides access to read messages in the Message Store, as well as to integrate with WSDL. This persona authorizes: • Read messages from the Message Store. • View messages in the Message Store. • View messages in the Message Store. • View messages in the Message Store.
Business expert	AuthGroup.BusinessExpert	PI_Business_Expert	Enables read data content payload. This persona authorizes:

Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Descript
			• M f C a • F F a t • \ C • \ e t • E f a F i No This is collect assign other p PI_Inte or PI_F enhanc author busine
Integration developer	AuthGroup.IntegrationDeveloper	PI_Integration_Developer	Enables develope design, a integrati This per: authoriz • \ s F I • C e F a I • I i • \ F • C F

Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Descript
			Integrati deploy u secure p various (However permissi certain s like know keys, or content.
Tenant administrator	AuthGroup.Administrator	PI_Administrator	Enables perform on the te This per: authoriz • H s t c F (s s a • M F U i • M i t i • [c • [c i • [f s • [a
Partner Directory configurator	AuthGroup.TenantPartnerDirectoryConfigurator	There is no predefined role collection for the Partner Directory configurator.	Enables Director;

Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Descript
		You need to assign the role template AuthGroup_TenantPartnerDirectoryConfigurator to the service instance for the related API client.	read and Director

Roles to Define Permissions to Call Integration Flow Endpoints

In order to enable a sender system to process messages on Cloud Integration Using HTTPS/basic authentication (based on a user registered by an identity provider), you need to assign the following entities to the associated user:

- Role template `MessagingSend` (in the Cloud Foundry environment)

Using this role template, you need to define a role collection and assign the role collection to the IdP user.

More information: [Basic Authentication of IdP User for Integration Flow Processing](#)

- Role `ESBmessaging.send` (in the Neo environment)

More information: [Setting Up Inbound HTTP Connections \(with Basic Authentication\), Neo Environment](#)

i Note

Note Related to Role Configuration

Depending on the chosen inbound authorization option, you define permissions for sender systems to call integration flow endpoints in different ways:

Option	Summary of Steps
Basic authentication of a user registered at an identity provider (IdP)	Use SAP BTP cockpit to define a role collection that contains the predefined role template <code>MessagingSend</code> and assign the role collection to the IdP user (under Security > Trust Configuration). The role template <code>MessagingSend</code> is provided by default in your subaccount to define permissions for sender systems to call integration flow endpoints for this use case. See: Basic Authentication of IdP User for Integration Flow Processing
Authentication with an OAuth client (service instance)	Use the Cloud Integration Monitor application and select the User Roles tile under Manage Security. When doing this, you can either use the predefined role <code>ESBmessaging.send</code> or create a custom role. Create service instance and service key using SAP BTP cockpit. During this step, you need the role specified with the User Roles tile. See: • Client Certificate Authentication for Integration Flow Processing • OAuth with Client Credentials Grant for Integration Flow Processing • Basic Authentication with clientId and clientsecret for Integration Flow Processing

Related Information

[Tasks and Permissions](#)

Tasks and Permissions

The following table provides an overview of which roles are required in order to accomplish the various tasks related to Cloud Integration. It's also indicated in how far the tasks and roles are relevant for the main personas defined for Cloud Integration. You can use these individual roles or role templates to define your own custom authorization groups (Neo) or role collections (Cloud Foundry), giving you fine-grained control over user permissions. If you prefer to work with predefined permission sets instead, see [Personas](#).

In the different environments, these persona are mapped to different objects.

- In the Neo environment, a persona is realized by an authorization group (beginning with the string **AuthGroup**).
- In the Cloud Foundry environment, a persona is realized by a role collection.

The mapping of the persona to the authorization groups (Neo) or role collections (Cloud Foundry) is described under [Personas](#).

In the different environments, the permissions to execute certain tasks are given by different objects.

- In the Neo environment, a permission to execute a task is given by a role.
- In the Cloud Foundry environment, a permission to execute a task is given by a role template.

Tasks and Permissions

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Discover	View packages	WebToolingCatalog.OverviewRead	CatalogPackagesRead
Discover	View package artifacts	WebToolingCatalog.OverviewRead WebToolingCatalog.DetailsRead	CatalogPackageArtifactsRead
Discover	Copy package to workspace	WebToolingCatalog.OverviewRead WebToolingWorkspace.Write	CatalogPackagesCopy
Design	View packages and package artifacts	WebToolingWorkspace.Read	WorkspacePackagesRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Design	Create, edit, import, export, delete package with its artifacts	WebToolingWorkspace.Read WebToolingWorkspace.Write	WorkspacePackagesEdit
Design	Update package	WebToolingWorkspace.Read WebToolingWorkspace.Write	WorkspacePackagesEdit
Design	Configure artifacts (integration flows and value mappings)	WebToolingWorkspace.Read WebTooling.IntegrationFlowConfigure WebToolingWorkspace.RuntimeConfigure	WorkspacePackagesConfigure
Design	Deploy/undeploy artifacts	WebToolingWorkspace.Read NodeManager.read GenerationAndBuild.generationandbuildcontent NodeManager.deploycontent WebToolingWorkspace.RuntimeConfigure	WorkspaceArtifactsDeploy
Design	Export Package for transport	WebToolingWorkspace.Read TransportModule.read TransportModule.write	WorkspacePackagesTransport
Design	Import package from transport	WebToolingWorkspace.Read TransportModule.read TransportModule.write	WorkspacePackagesTransport
Design	Update Package from transport	WebToolingWorkspace.Read TransportModule.read TransportModule.write	WorkspacePackagesTransport
Design	Read/write Partner Directory-related content (string/binary parameters, authorized users, alternative partners, and user credentials) Read/write certificate-to-user mappings (only in the Neo environment)	TenantPartnerDirectory.read TenantPartnerDirectory.write NodeManager.deploycredentials NodeManager.readcredentials CertificateUserMappings.Write CertificateUserMappings.Read	AuthGroup_TenantPartnerDirectoryC

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Design	Delete design time artifact locks	WebToolingWorkspace.Read WorkspaceArtifactLocks.Delete	WorkspaceArtifactLocksDelete
Design	Read design time artifact locks	WebToolingWorkspace.Read WorkspaceArtifactLocks.Read	WorkspaceArtifactLocksRead
Monitor	View Monitor Overview	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View message processing logs	IntegrationOperationServer.read	MonitoringDataRead
Monitor	View payload of stored messages from message storage (also trace and message processing log attachments)	esbmessagestorage.read	
Monitor	View tasks	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View tail log	IntegrationOperationServer.read	MonitoringDataRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
		NodeManager.read	
Monitor	View deployed artifact list	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View deployed integration flow in graphical editor	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Download deployed integration flow Download WSDL	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View user roles	Roles.Read	MonitoringDataRead
Monitor	Create, edit, delete user roles	n.a.	UserRolesEdit

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	View deployed security material	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Add credentials	IntegrationOperationServer.read NodeManager.deploycredentials NodeManager.deploycontent	CredentialsEdit
Monitor	Add known host, keystore, PGP keyring artifacts	IntegrationOperationServer.read NodeManager.deploysecuritycontent NodeManager.deploycontent	SecurityMaterialEdit
Monitor	Edit credentials	IntegrationOperationServer.read NodeManager.deploycredentials NodeManager.deploycontent	CredentialsEdit
Monitor	Undeploy credentials	IntegrationOperationServer.read NodeManager.deploycontent NodeManager.deploycredentials	CredentialsEdit
Monitor	Undeploy known host, keystore, PGP keyring artifacts	IntegrationOperationServer.read NodeManager.deploycontent NodeManager.deploysecuritycontent	SecurityMaterialEdit
Monitor	Download keystore, public/private keyring, known host, artifact	IntegrationOperationServer.read NodeManager.read NodeManager.readsecuritycontent	SecurityMaterialDownload
Monitor	View certificate-to-user mappings	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Create/edit/delete certificate-to-user mappings	IntegrationOperationServer.read NodeManager.deploysecuritycontent	SecurityMaterialEdit

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
		NodeManager.read	
Monitor	View keystore entries	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Download public keystore entries	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Add/replace/delete keystore entries	IntegrationOperationServer.read NodeManager.deploysecuritycontent	SecurityMaterialEdit
Monitor	View access policies	IntegrationOperationServer.read AccessPolicies.Read	AccessPoliciesRead
Monitor	Maintain access policies	IntegrationOperationServer.read AccessPolicies.Write	AccessPoliciesEdit
Monitor	View data store entries	IntegrationOperationServer.read ESBDataStore.read	DataStoresAndQueuesRead
Monitor	View data store entries - message payload	IntegrationOperationServer.read ESBDataStore.readPayload	DataStorePayloadsRead
Monitor	View data store entries - variables	IntegrationOperationServer.read ESBDataStore.readPayload	DataStorePayloadsRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	Delete data store entries/variables	IntegrationOperationServer.read ESBDataStore.read ESBDataStore.delete	DataStoresAndQueuesDelete
Monitor	Configure data stores and queues	IntegrationOperationServer.read ESBDataStore.Config	DataStoresAndQueuesConfig
Monitor	View payload of stored messages from message store	esbmessagestorage.read	MessagePayloadsRead
Monitor	View trace configuration	IntegrationOperationServer.read NodeManager.read ConfigurationService.RuntimeBusinessParameterRead	TraceConfigurationRead
Monitor	Edit trace configuration (enable/disable trace)	IntegrationOperationServer.read NodeManager.read ConfigurationService.RuntimeBusinessParameterRead ConfigurationService.RuntimeBusinessParameterWrite	TraceConfigurationEdit
Monitor	Override access policies Get access to data (for example, JMS queues, message processing log attachments) although the data is protected by access policies. Note that when this role is assigned to a user, it overrides all access policies that might prevent the user from accessing data processed by certain artifacts (for example, integration flows). However, this role doesn't grant permission to	AccessPoliciesArtifacts.AccessAll	AccessAllAccessPoliciesArtifacts

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
	access payload data.		
Monitor	Add, edit, or undeploy number ranges, and undeploy integration flows	IntegrationOperationServer.read NodeManager.deploycontent	MonitoringArtifactsDeploy
Monitor	View number ranges	IntegrationOperationServer.read	MonitoringDataRead
Monitor	Activate or deactivate queues	IntegrationOperationServer.read ESBDataStore.Activate	QueuesActivate
Monitor	Retry queues	IntegrationOperationServer.read ESBDataStore.read ESBDataStore.retry	QueuesRetry
Monitor	Delete queues	IntegrationOperationServer.read ESBDataStore.read ESBDataStore.delete	DataStoresAndQueuesDelete
Monitor	View queues	IntegrationOperationServer.read ESBDataStore.read	DataStoresAndQueuesRead
Monitor	View runtime processing locks	IntegrationOperationServer.read MessageProcessingLocks.Read	MessageProcessingLocksRead
Monitor	Access the ID Mapper (using the Message	IntegrationOperationServer.read MessageProcessingLocks.Read	MessageProcessingLocksRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
	Processing Logs OData API).		
Monitor	Delete runtime processing locks	IntegrationOperationServer.read MessageProcessingLocks.Delete	MessageProcessingLocksDelete
Monitor	Test connectivity	IntegrationOperationServer.read NodeManager.readcredentials	MonitoringDataRead (only for tests with authentication) CredentialsRead
Monitor	Activate external logging	IntegrationOperationServer.read ExternalLogging.Activate	ExternalLoggingActivate
Monitor	Read the activation state of external logging	IntegrationOperationServer.read ExternalLoggingActivation.Read	ExternalLoggingActivationRead
Monitor	Change log level	IntegrationOperationServer.read ConfigurationService.RuntimeBusinessParameterWrite NodeManager.read	
Monitor	View audit log entries	IntegrationOperationServer.read AuditLog.Read	
Monitor	Read data metrics for health monitoring API: JMS queue statistics, certificate expiry dates	HealthCheckMonitoringData.Read	HealthCheckMonitoringData.Read
Monitor	Activate archiving Enablement		DataArchivingActivate
Monitor	Read archiving enablement		DataArchivingRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	Change archiving configuration	IntegrationOperationServer.read ConfigurationService.RuntimeBusinessParameterWrite NodeManager.read	TraceConfigurationEdit
Monitor	Change log level		TraceConfigurationEdit
Monitor	Read data metrics for the health check monitoring API. This covers, for example, JMS queue statistics, certificate expiry dates)	ESBDataStore.read NodeManager.read ResourceUsageData.Read	HealthCheckMonitoringDataRead
Operate	Get read-only access to Operations Cockpit. The Operations Cockpit is the central control point for operating edge integration cells and allows the administrator to monitor and adjust system configurations and resources.	n.a.	IntegrationCellOperationCockpitRead
Operate	Get full access to the Operations Cockpit. The Operations Cockpit is the central control point for operating edge integration cells and allows the administrator to monitor and adjust system configurations and resources.	n.a.	IntegrationCellOperationCockpitWrite

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Inspect (only when using SAP Cloud Integration in the Cloud Foundry environment)	Inspect usage of integration resources	n.a.	MonitoringDataRead
Settings	View and change runtime profile	WebToolingSettingsProductProfiles.savetenantconfiguration	
Settings	Set transport system	WebToolingSettingsProductProfiles.savetenantconfiguration	
Settings	Configure Design Guidelines settings	WorkspaceDesignGuidelines.Configure	WorkspaceDesignGuidelinesConfigure
General	Provides permission for the read-only persona (see Personas)	AuthGroup.ReadOnly	AuthGroup_ReadOnly
General	Provides permission for the tenant administrator persona (see Personas)	AuthGroup.Administrator WebToolingWorkspace.RuntimeConfigure	AuthGroup_Administrator
General	Provides permission for the business expert persona (see Personas)	AuthGroup.BusinessExpert	AuthGroup_BusinessExpert
General	Provides permission for the integration developer persona (see Personas)	AuthGroup.IntegrationDeveloper	AuthGroup_IntegrationDeveloper

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
General	This role/role template is only relevant for partners and internal SAP developers developing and publishing content to SAP Business Accelerator Hub. Further details can be found in the partner documentation.	AuthGroup.ContentPublisher	AuthGroup_ContentPublisher

Related Information

[Authorization Entities](#)

Data Storage Security

Customer data can be stored in dedicated steps during message processing.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Customer data stored at rest is strictly separated and isolated for each tenant. Although different tenants might share a common physical infrastructure, each tenant stores its data in a separate schema.

For certain use cases the customer can configure if the data at rest is encrypted.

Message content can be stored encrypted. If this security measure is configured, the encryption key that is generated automatically is unique for each tenant and is renewed periodically.

Data storage encryption uses AES and a key length of 256 bits. The encryption key is not stored in the same location as the encrypted data.

Kinds of Stored Data

The following kinds of data can be stored during the execution of an integration scenario:

- Message content

The runtime node writes message content data to the database in dedicated steps of an integration flow. There is the option to either store message content for a longer time period (the default is 30 days) or temporarily. Temporarily stored message content can be used for subsequent message processing steps. Such steps can then also read message content from the database.

There is the option to configure the retention period of the message content.

- Monitoring data

During message processing, the runtime node also writes monitoring data to the database (which is stored by default for 30 days). Monitoring data comprises the message processing log (MPL), which records the executed processing steps.

Data Protection and Privacy

Various types of customer data are processed by and stored on the integration platform at different times. This data gets the highest level of protection, and SAP takes dedicated measures to guarantee this security level.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

General Information

Governments place legal requirements on industry to protect data and privacy. We provide features and functions to help you meet these requirements.

⚠ Caution

SAP does not provide legal advice in any form. SAP software supports data protection compliance by providing security features and data protection-relevant functions, such as blocking and deletion of personal data. In many cases, compliance with applicable data protection and privacy laws is not covered by a product feature. Furthermore, this information should not be taken as advice or a recommendation regarding additional features that would be required in specific IT environments. Decisions related to data protection must be made on a case-by-case basis, taking into consideration the given system landscape and the applicable legal requirements. Definitions and other terms used in this documentation are not taken from a specific legal source.

⚠ Caution

We assume that you have not maintained any data related to an individual in the tools provided by SAP Cloud Integration (for example, when designing integration content).

We expect that sensitive personal data can only be included in message payloads. This responsibility lies exclusively with you as the operator of an integration scenario using SAP Cloud Integration and remains your responsibility. If you include sensitive personal data within payloads or message attachments, SAP Cloud Integration may store this information on your behalf. This applies also for data maintained in the tools provided by SAP Cloud Integration, however, data within payloads can be protected by enabling encrypted storage.

The knowledge of sensitive personal data lies exclusively with you and remains your responsibility.

The tools of SAP Cloud Integration only use technical users or data without any references to individuals.

User Consent

We assume that software operators, such as SAP customers, collect and store the consent of data subjects, before collecting their personal data. A data privacy specialist can later determine whether data subjects have granted, withdrawn, or denied consent.

Information Report

An information report is a collection of data relating to a data subject. A data privacy specialist may be required to provide such a report or an application may offer a self-service. Cloud Integration assumes that software operators, such as SAP customers, can provide such information.

Erasure of Personal Data

When handling personal data, consider the legislation in the different countries where your organization operates. After the data has passed the end of purpose, regulations may require you to delete the data. However, additional regulations may require you to keep the data longer. During this period you must block access to the data by unauthorized persons until the end of the retention period, when the data is finally deleted.

Data stored on Cloud Integration is only stored for a limited time period (referred to as retention time).

For more information on the retention times for the various kinds of data stored by Cloud Integration, see [Specific Data Assets](#).

Types of Stored Data

Different kinds of data, such as message content or monitoring data, can be stored during the operation of an integration scenario.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Such data needs to be considered as sensitive data as it can contain personal information. The following list provides examples:

- Message content

Messages processed on a runtime node typically contain business data of an integration scenario and therefore can contain sensitive customer data such as addresses, names, or financial information.

When this data is *at-rest*, it can be stored encrypted. Note, however, that in some use cases the customer can configure that the data is not encrypted.

When this data is *in-transit*, several measures can be taken, such as digital message signing or message content encryption.

- Monitoring data

The message processing log records the processing steps of an integration flow. Only users assigned to this tenant and with dedicated permissions can access this data.

- Other data, such as the content of log files

i Note

Personal data processed by and stored on the integration platform is handled according to the Data Processing Agreement, which you can find at <http://www.sap.com/about/agreements.html> under **SAP Cloud Services Customers**.

Due to the tenant isolation concept, data from different customers (stored in different tenants) is strictly isolated. Additionally, SAP has no access to data stored in customer tenants.

The customer can grant people outside its organization permissions to execute specific tasks on its cluster (for example, to SAP employees to execute error analysis tasks in support cases).

Using

Customer data can be stored by dedicated steps during message processing. stores data in the edge environment using persistent volumes. Amazon Web Services (AWS) and Microsoft Azure support data encryption at rest. For more information, check the documentation of the related platform.

Specific Data Assets

Different kinds of data are stored in the SAP Cloud Integration infrastructure during the lifecycle of an integration project.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

The following table lists the different kinds and attributes such like storage location and retention time, for example.

Data Assets

Data	Description	Logical Storage	Classification	Retention Time
Message processing log	Structured information on the processing of a message		Log data	30 days
Message processing log attachments	Data attached to a message processing log during runtime	Message store	Log data Business data	30 days
Integration flow tracing data	Information on the message flow (including the message payload) and on errors that occurred during message processing	Trace store	Log data Business data	60 minutes
Integration content (design time)	Integration flow models and value mappings created or edited by an integration developer	Workspace	Configuration data	Unlimited
Integration content (runtime)	Camel XML representation of integration flows and other design time entities (as deployed on a runtime node)		Configuration data	Unlimited
Data stored by Data Store operations step	Message content stored in dedicated steps in an integration flow (contains information such as message GUID, message processing log GUID, tenant ID, time stamp, and payload). Is used for further message processing in subsequent steps in an integration flow.	Data store	Business data	Can be defined by integration developer (default value: 30 days, maximum possible value is 180 days)
Data stored by Persist step	Message content stored in dedicated steps in an integration flow (contains information such as message GUID,	Message store	Business data	90 days

Data	Description	Logical Storage	Classification	Retention Time
	message processing log GUID, tenant ID, time stamp, and payload) Can be accessed and analyzed after message processing.			
Message content stored by JMS adapter	Message content stored in JMS message queues	JMS queue	Business data	Can be defined by integration developer (default value: 30 days, maximum possible value is 180 days)
Lock entries	Lock entries that are created (in the in-progress repository) to avoid the same message being processed several times in parallel (for example, by different runtime nodes)			

Related Information

[Message Processing Log](#)

[Define Data Store Operations](#)

[Managing Security Material](#)

[Message Locks](#)

Other Security-Related Information

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

User Interface Security

Cloud Integration provides user interfaces for designing and deploying message flows, and monitoring them at runtime.

A Web tool (Web UI) is available to accomplish these tasks. The Web UI is implemented using JavaScript and HTML (UI5).

This user interface is built to prevent vulnerabilities such as cross-site scripting (XSS) and cross-site request forgery (XSRF). The built-in security capabilities of these technologies are used together with secure design and coding principles.

Auditing and Logging Information

Here you can find a list of the security events that are logged by .

i Note

The examples contained in the table are taken from the usage of in the Cloud Foundry environment. Unless otherwise shown, the keywords used to identify an event are identical in both environments.

Security Events Written in Audit Logs

Event grouping	What events are logged	How to identify related log events
Integration design	Import integration package manually	• action: Package_Import_Started/Package_Import_Completed • objectType: Package Example: When importing a package manually, the following events are written: ⌵ Output Code <pre>{ "action": "Package_Import_Started", "objectType": "Package", "objectId": "new package.zip", "attributes": {}, "changedAttributes": {} }" on 2021-06-21T11:02:00.190Z. Security event was related to user "SAP".</pre> ⌵ Output Code <pre>{ "action": "Package_Import_Completed", "objectType": "Package", "objectId": "new package.zip", "attributes": {}, "changedAttributes": {} }" on 2021-06-21T11:02:01.367Z. Security event was related to user "SAP".</pre>
Integration design	Transport integration package	• action: Package_Transport_Started/Package_Transport_Completed • objectType: Package
Integration design	Transport integration artifact	• action: Artifact_Transport_Started/Artifact_Transport_Completed • objectType: <depends on artifact type>

Event grouping	What events are logged	How to identify related log events
Integration design	Create integration artifact	• action: Creation_of_Artifact_Initiated/Creation_of_Artifact_Completed • objectType: <depends on artifact type> Example: When creating an integration flow, the objectType parameter is IntegrationFlow, and the following ever ⌵ Output Code <pre>{ "action": "Creation_of_Artifact_Initiated", "objectType": "IntegrationFlow", "objectId": "My_Flow", "attributes": {}, "changedAttributes": {} }</pre> on 2021-06-21T11:06: Security event was related to user "SAP". ⌵ Output Code <pre>{ "action": "Creation_of_Artifact_Completed", "objectType": "IntegrationFlow", "objectId": "My_Flow", "attributes": {}, "changedAttributes": {} }</pre> on 2021-06-21T11:06: Security event was related to user "SAP".
Integration design	Update integration artifact	• action: Update_of_Artifact_Initiated/Update_of_Artifact_Completed • objectType: <depends on artifact type> Example: When updating a value mapping, the objectType parameter is ValueMapping, and following events are wri ⌵ Output Code <pre>{ "action": "Update_of_Artifact_Initiated", "objectType": "ValueMapping", "objectId": "testvm", "attributes": {}, "changedAttributes": {} }</pre> on 2021-06-25T13:58:2 Security event was related to user "SAP". ⌵ Output Code <pre>{ "action": "Update_of_Artifact_Completed", "objectType": "ValueMapping", "objectId": "testvm", "attributes": {}, "changedAttributes": {} }</pre> on 2021-06-25T13:58:2 Security event was related to user "SAP".

Event grouping	What events are logged	How to identify related log events
Integration design	Update integration artifact as version	• action: Update_of_Version_Initiated/Update_of_Version_Completed • objectType: <depends on artifact type>
Integration design	Update integration flow configuration	• action: Update_of_Configuration_Initiated/Update_of_Configuration_Completed • objectType: <depends on artifact type>
Integration design	Deploy integration artifact	• action: Deploy_of_Artifact_Initiated/Deploy_of_Artifact_Completed • objectType: <depends on artifact type>

Event grouping	What events are logged	How to identify related log events
Integration design	Deploy integration artifact (mass deployment)	
Integration design	Undeploy integration artifact (only logged in Neo)	• action: Undeploy_of_artifact_triggered • objectType: <depends on artifact type>
Integration design	Delete integration artifact	• action: Deletion_of_Artifact_Initiated/Deletion_of_Artifact_Completed • objectType: <depends on artifact type>
Integration design	Download WSDL	

Event grouping	What events are logged	How to identify related log events
Partner Directory	Read Partner Directory entity	• action: Read • objectType: <depends on Partner Directory entity type>
Partner Directory	Create Partner Directory entity	• action: Create • objectType: <depends on Partner Directory entity type> Example: Creation of an alternative partner writes the following log event: ⌵ Output Code <pre>"{"action":"Create","objectType":"PD Alternative Partner","objectId":"Agency\u003c "attributes":{"Pid":"ID1"},"changedAttributes":{}}" on 2021-06-29T15:00:58.948Z. .</pre>
Partner Directory	Update Partner Directory entity	• action: Change • objectType: <depends on Partner Directory entity type>
Partner Directory	Delete Partner Directory entity	• action: Delete • objectType: <depends on Partner Directory entity type> Example: Deletion of an authorized user writes the following log event: ⌵ Output Code <pre>"{"action":"Delete","objectType":"PD Authorized User","objectId":"..."," "attributes":{},"changedAttributes":{}}" on 2021-06-29T15:00:56.218Z. ..."</pre>
Manage security	Create keystore entry	• action: Create • objectType: <depends on keystore entry type> Example: When creating an X.509 key pair, the objectType parameter is X.509 Key-Pair, and the following event is ⌵ Output Code <pre>"{"action":"Create","objectType":"X.509 Key-Pair","objectId":"keypair", "attributes":{"Subject CN":"mydepartment.com","Issuer CN":"mydepartment.com", "Serial Number":"12345678"},"changedAttributes":{}}" on 2021-06-09T10:33:41.421Z. Security event was related to user "SAP".</pre>

Event grouping	What events are logged	How to identify related log events
Manage security	Update keystore entry by uploading a new file	• action: Create • objectType: <depends on keystore entry type> Example: When uploading an X.509 Certificate, the objectType parameter is X.509 Certificate, and the following ⌵ Output Code <pre>{"action": "Create", "objectType": "X.509 Certificate", "objectId": "mycertificate1", "attributes": {"Issuer CN": "OU\u003dSender", "Serial Number": "6818011987146590924"}, "context": "Security event was related to user \"SAP\"."}</pre>
Manage security	Export PGP Keyring	• action: Read • objectType: Public Keyring or Secret Keyring Exporting PGP public keyring example: ⌵ Output Code <pre>{action=Read, objType=Public Keyring Secret Keyring, nodeType="Keymanagement Subsystem", objectId="PGP Public Keyring", objectType="Public Keyring", context="PGP Public Keyring exported successfully"}</pre>
Manage security	Download PGP Key	• action: Read • objectType: Public Keyring or Secret Keyring • objName: 3A7A899C3435D45C Example: Download a specific PGP public key entry output code: ⌵ Output Code <pre>{action=Read, objType=PGP Public Key PGP Secret Key, nodeType="Keymanagement Subsystem", objectId="PGP Public Key", context="PGP Public Key downloaded successfully"}</pre>
Manage security	Import PGP Key(s)	• action: Change • objectType: Public Keyring or Secret Keyring • objName: 3A7A899C3435D45C, 2DDE8A220E3D26B4 Example: Import a PGP Public Keyring output code:

Event grouping	What events are logged	How to identify related log events
		⌵ Output Code <pre>{action=Change, objType=PGP Public Keys PGP Secret Keys, nodeType="Keymanagement</pre>
Manage security	Delete PGP Key	• action: Delete • objectType: Public Keyring or Secret Keyring • objName: 3A7A899C3435D45C Example: Deleting a PGP Public Key entry output code: ⌵ Output Code <pre>{action=Delete, objType=PGP Public Key PGP Secret Key, nodeType="Keymanagement S</pre>
Manage security	Rename alias of keystore entry	• action: Delete/Create • objectType: <depends on keystore entry type> Example: Renaming the alias of a certificate from mycertificate to mycertificate1 provides 2 log entries: ⌵ Output Code <pre>{"action":"Delete","objectType":"X.509 Certificate","objectId":"mycertificate", "attributes":{"Issuer CN":"OU\u003dSender,C\u003dDE","Subject CN":"OU\u003dSender, "Serial Number":"6818011987146590924"},"changedAttributes":{}}" on 2021-06-09T10:4 Security event was related to user "SAP".</pre> ⌵ Output Code <pre>{"action":"Create","objectType":"X.509 Certificate","objectId":"mycertificate1", "attributes":{"Issuer CN":"OU\u003dSender,C\u003dDE","Subject CN":"OU\u003dSender, "Serial Number":"6818011987146590924"},"changedAttributes":{}}" on 2021-06-09T10:4 Security event was related to user "SAP".</pre>
Manage security	Delete keystore entry	• action: Delete • objectType: <depends on keystore entry type>

Event grouping	What events are logged	How to identify related log events
Manage security	Create security material	• action: PasswordStore • objectType: Credential
Manage security	Update security material	• action: PasswordStore • objectType: Credential
Manage security	Delete security material	• action: PasswordDelete • objectType: Credential
Manage security	Upload security material	• action: PasswordStore • objectType: Security Material

Event grouping	What events are logged	How to identify related log events
Manage security	Undeploy security material	• action: PasswordDelete • objectType: Security Material Example: During the undeployment of a PGP secret keyring, the following event is written: ⌵ Output Code <pre>{ "action": "PasswordDelete", "objectType": "Security Material", "objectId": "sekring", "attributes": {}, "changedAttributes": {} } on 2021-06-25T17:50:35.701Z. Security event was related to user "SAP".</pre>
Manage security	Download security material	• action: Read • objectType: Security Material Example: For the download of a known hosts file, the following event is written: ⌵ Output Code <pre>{ "action": "Read", "objectType": "Security Material", "objectId": "known.hosts", "attributes": {}, "changedAttributes": {} } on 2021-06-25T17:53:16.589Z. Security event was related to user "SAP".</pre>
Manage security	Create access policy	• action: Create • objectType: Access Policy Example: For the creation of an access policy, the following event is written: ⌵ Output Code <pre>{ "action": "Create", "objectType": "Access Policy", "objectId": "101:DemoIntegrationFl", "attributes": { "message": "Successfully created access policy!" }, "changedAttributes": {} } Security event was related to user "SAP".</pre>
Manage security	Update access policy	• action: Change • objectType: Access Policy Example: For the update of an access policy, the following event is written: ⌵ Output Code

Event grouping	What events are logged	How to identify related log events
		<pre>{ "action": "Change", "objectType": "Access Policy", "objectId": "101:DemoIntegrationFl", "attributes": {}, "changedAttributes": { "message": { "oldValue": "Successfully updated a", "newValue": "Successfully updated access policy!" } } }</pre> on 2021-06-25T17:58:50.146Z. Security event was related to user "SAP".
Manage security	Delete access policy	• action: Delete • objectType: Access Policy Example: If you delete an access policy that includes artifact references, 2 logs are written, 1 for the artifact references and 1 for the access policy. Output Code <pre>{ "action": "Delete", "objectType": "Artifact Reference", "objectId": "1:my second inte", "attributes": { "ConditionAttribute_1": "Name", "ConditionType_1": "regularExpression", "My First Integration Flow", "Type_1": "INTEGRATION_FLOW", "message": "Successfully de", "changedAttributes": {} } }</pre> on 2021-06-25T18:00:17.523Z. Security event was related to user "SAP". Output Code <pre>{ "action": "Delete", "objectType": "Access Policy", "objectId": "51:my second role", "attributes": { "message": "Successfully deleted access policy!" }, "changedAttributes": {} }</pre> on 2021-06-25T18:00:17.462Z. Security event was related to user "SAP".
Manage security	Create artifact reference (for access policy)	• action: Create • objectType: Artifact Reference Example: For the creation of an artifact reference of an access policy, the following event is written: Output Code <pre>{ "action": "Create", "objectType": "Artifact Reference", "objectId": "51:MyFirstIntegr", "attributes": { "ConditionValue": "MyFirstIntegrationFlow", "ConditionType": "exactStri", "message": "Successfully created artifact references!", "changedAttributes": {} } }</pre> on 2021-06-25T18:00:17.462Z. Security event was related to user "SAP".
Manage security	Update artifact reference (for access policy)	• action: Change • objectType: Artifact Reference Example: For the update of an artifact reference of an access policy, the following event is written: Output Code

Event grouping	What events are logged	How to identify related log events
		<pre> {"action":"Change","objectType":"Artifact Reference","objectId":"51:MyFirstIntegr "attributes":{"Type_51":"INTEGRATION_FLOW"},"changedAttributes":{"ConditionValue_5 "newValue":"MySecondIntegrationFlow"},"ConditionType_51":{"oldValue":"exactString" {"oldValue":"Successfully updated artifact references!","newValue":"Successfully u "ConditionAttribute_51":{"oldValue":"Name","newValue":"Name"}}}" on 2021-06-25T18: Security event was related to user "SAP". </pre>
Manage security	Delete artifact reference (for access policy)	• action: Delete • objectType: Artifact Reference
Manage security	Deploy JDBC material	• action: PasswordStore • objectType: Data Source Example: For the deployment of JDBC material, the following event is written: ⌵ Output Code <pre> {"action":"PasswordStore","objectType":"Data Source","objectId":"jdbc01","attribu "changedAttributes":{}}" on 2021-06-25T18:09:10.084Z. Security event was related t </pre>
Manage security	Undeploy JDBC material	• action: PasswordDelete • objectType: Data Source
Manage security	Create user role (only Cloud Foundry)	(tbd)
Manage security	Update user role (only Cloud Foundry)	(tbd)
Manage security	Delete user role (only Cloud Foundry)	(tbd)

Event grouping	What events are logged	How to identify related log events
Manage security	Create certificate-to-user mapping (only Neo)	• action: Create • objectType: Certificate User Mapping
Manage security	Read certificate-to-user mapping (only Neo)	• action: Read • objectType: Certificate User Mapping
Manage security	Update certificate-to-user mapping (only Neo)	• action: Change • objectType: Certificate User Mapping
Manage security	Save certificate-to-user mapping (only Neo)	• action: Change • objectType: Certificate User Mapping
Manage security	Delete certificate-to-user mapping (only Neo)	• action: Delete • objectType: Certificate User Mapping
Manage stores	Download data store entry	• action: Read • objectType: Message Example: Downloading data store entry HT - 1037 of global data store Products writes the following audit log event: ⌵ Output Code <pre>{ "action": "Read", "objectType": "Message", "objectId": "Products/GLOBAL/HT-1037", "attributes": { "storeName": "Products", "id": "HT-1037" }, "changedAttributes": {} }</pre> on 2021-06-29T15:28:47.695Z.
Manage stores	Delete data store entry	• action: Delete • objectType: Message Example: Deletion of data store entry 126 of global data store CustomerReviews writes the following audit log event: ⌵ Output Code <pre>{ "action": "Delete", "objectType": "Message", "objectId": "CustomerReviews/GLOBAL/126", "attributes": { "ids": [126], "storeName": "CustomerReviews", "id": "126" }, "changedAttr"</pre>

Event grouping	What events are logged	How to identify related log events
		on 2021-06-29T15:22:30.135Z.
Manage stores	Delete data store	• action: Delete • objectType: Message Example: Deletion of global data store CustomerReviews writes the following audit log event: ⌵ Output Code <pre>"{"action":"Delete","objectType":"Message","objectId":"CustomerReviews/GLOBAL/ALL" "attributes":{"storeName":"CustomerReviews"},"changedAttributes":{}}" on 2021-06-2</pre>
Manage stores	Download variable	• action: Read • objectType: Message Example: Download of global variable timestamp writes the following audit log event: ⌵ Output Code <pre>"{"action":"Read","objectType":"Message","objectId":"sap_global_store/GLOBAL/times "attributes":{"storeName":"sap_global_store","id":"timestamp"},"changedAttributes"</pre> Example: Download of local variable ProductId (related to integration flow Write_Product_ID writes the following : ⌵ Output Code <pre>"{"action":"Read","objectType":"Message","objectId":"sap_global_store/Write_Produc "attributes":{"qualifier":"Write_Product_ID","storeName":"sap_global_store","id":" on 2021-06-30T10:04:51.569Z.</pre>
Manage stores	Delete variable	• action: Delete • objectType: Variable Example: Deletion of global variable timestamp writes the following audit log event: ⌵ Output Code <pre>"{"action":"Delete","objectType":"Variable","objectId":"sap_global_store/GLOBAL/ti "attributes":{"ids":["timestamp"],"storeName":"sap_global_store","id":"timestamp"} on 2021-06-30T09:56:48.985Z.</pre>

Event grouping	What events are logged	How to identify related log events
		Example: Deletion of local variable <code>ProductId</code> (related to integration flow <code>Write_Product_ID</code> writes the following at ⌵ Output Code <pre>"{"action":"Delete","objectType":"Variable","objectId":"sap_global_store/Write_Prc "attributes":{"qualifier":"Write_Product_ID","ids":["ProductId"],"storeName":"sap_ "changedAttributes":{}}" on 2021-06-30T10:08:27.385Z.</pre>
Manage stores	Move messages to another queue	• action: Move • objectType: Message Queue
Manage stores	Delete message queue	• action: Delete • objectType: Message Queue Example: Deletion of message queue <code>MyQueue</code> writes the following audit log event: ⌵ Output Code <pre>"{"action":"Delete","objectType":"Message Queue","objectId":"MyQueue", "attributes":{},"changedAttributes":{}}" on 2021-06-30T10:26:08.601Z.</pre>
Manage stores	Delete message from message queue	• action: Delete • objectType: Message Example: Deletion of a message from message queue <code>MyQueue</code> writes the following audit log event: ⌵ Output Code <pre>"{"action":"Delete","objectType":"Message","objectId":"ID:12345-abcde", "attributes":{"Queue":"MyQueue"},"changedAttributes":{}}" on 2021-06-30T10:49:47.9</pre>
Manage stores	Download message from message queue	• action: Read • objectType: Message Example: Downloading a message from message queue <code>MyQueue</code> writes the following audit log event: ⌵ Output Code

Event grouping	What events are logged	How to identify related log events
		<pre>"{"action":"Read","objectType":"Message","objectId":"ID:12345-abcde", "attributes":{"Queue":"MyQueue"},"changedAttributes":{}}" on 2021-06-30T10:49:40.9</pre>
Manage stores	Retry message from message queue	• action: Retry • objectType: Message
Message monitoring	Read / download message payload / header when monitoring integration flow with Trace log level	• action: Read • objectType: Message Payload (Trace) Example: Reading the message payload writes the following audit log event: ⌵ Output Code <pre>"{"action":"Read","objectType":"Message Payload (Trace)", "objectId":"mplId\1234-xyz;traceId\abcd;stepId\u003dEndEvent_2; Odata-Read","attributes":{"message":"Reading Content for Message Payload (Trace) w "changedAttributes":{}}" on 2021-07-05T10:09:35.531Z.</pre>
Message monitoring	Read message processing log attachment	• action: Read • objectType: MPL Attachment

Related Information

[Audit Logging in the Cloud Foundry Environment](#)

[Audit Logging in the Neo Environment](#)

[Auditing and Logging Information for Edge Integration Cell](#)

Rate Limits on SAP Cloud Integration

Rate limiting helps prevent denial-of-service attacks on the entire product or its individual applications.

Reasons for Rate Limits

The most common attacks include:

- A user generates a large number of requests to overload or crash an application.
- A user accidentally writes a script or uses a third-party application that generates so many requests that an application freezes or crashes. This can also result in all resources being allocated to a single tenant or user, making the application unavailable to others.

The requests could come from different sources:

- Too many browser instances (e.g. a large number of users rapidly refresh or script with tools like Selenium)
- Direct access to published APIs (e.g. scripts or third-party applications generate requests without delay)
- Indirect access to internal APIs (e.g. one application calls another application)
- Direct access to private APIs (e.g. someone discovers internal endpoints and call them)

Restrictions

Rate limiting is based on the number of requests per second. It only protects against situations where an unexpectedly high request rate causes issues. It doesn't protect against the following scenarios:

- Memory usage that depends on request parameters
- CPU usage that depends on request parameters
- Load that depends on the amount of data stored or returned
- Load that depends on the number of parallel requests
- Load that depends on the exact number of requests (e.g. two requests per second are acceptable, but three cause a crash)

Rate Limits on Cloud Integration Services

To ensure fair usage across different dimensions, SAP Cloud Integration applies the following rate limitations:

Artifact	Action	Rate per Tenant (requests/second)	Rate per User (requests/second)
Number Range	Deploy	10	5
	List	10	5
Design time Artifacts	All public APIs	15	10
Security Content	All public APIs	12	10
MPL	All public APIs	1000	100
Runtime Artifacts	All public APIs	120	50
Partner Directory- Alternative partners	All public APIs	50	25
Partner Directory- Authorized users	All public APIs	50	25
Partner Directory- Binary Parameters	All public APIs	50	25
Partner Directory- Partners	All public APIs	50	25
Partner Directory- String Parameters	All public APIs	50	25
Metering Connections	All public APIs	20	10

Security, Neo Environment

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Using a cloud-based integration platform imposes dedicated security measures on the software vendor (SAP) that hosts the platform, as well as on those (the customers) who use the platform.

This section describes the security-related aspects of the integration platform and shows which measures you can take to protect customer data that is passed through the platform during the execution of an integration scenario.

Customers who use SAP Cloud Integration agree that a significant part of their (and their customers') sensitive data is processed by and stored within an infrastructure not owned by themselves.

The core task of an integration platform is to serve as the transit place for messages, which may contain sensitive customer data. First and foremost, these messages must be protected against eavesdropping and unauthorized access.

Therefore, the integration platform must fulfill the following main requirements:

- The integration infrastructure is already designed and built in such a way that it meets the highest security standards.

In particular, it must be guaranteed that the technical system landscape, the communication between the components of the integration platform, and the storage locations of messages are secure.

- The processes related to the usage of Cloud Integration meet the highest security standards.

This relates to the processes at SAP that are related to the development and upgrade of the Cloud Integration software, the processes related to the provisioning and operation of the customers' virtual environment by the infrastructure provider, and the customer onboarding process during which customers set up secure connections between their infrastructure and SAP's integration platform.

- Customers have several options to configure how messages are exchanged within an integration scenario so that the involved data is protected at the highest level.

In particular, when designing integration flows, customers can choose between several options to protect messages by establishing secure communication channels (transport-level security) and by configuring digital encryption and digital signing of messages (message-level security).

This documentation summarizes the measures that are taken by SAP to fulfill these requirements.

Related Information

[Technical Landscape, Neo Environment](#)

[Security Aspects of Processes](#)

[Security Aspects of Data, Data Flow](#)

[Identity and Access Management](#)

[Data Storage Security](#)

[Data Protection and Privacy](#)

[Other Security-Related Information](#)

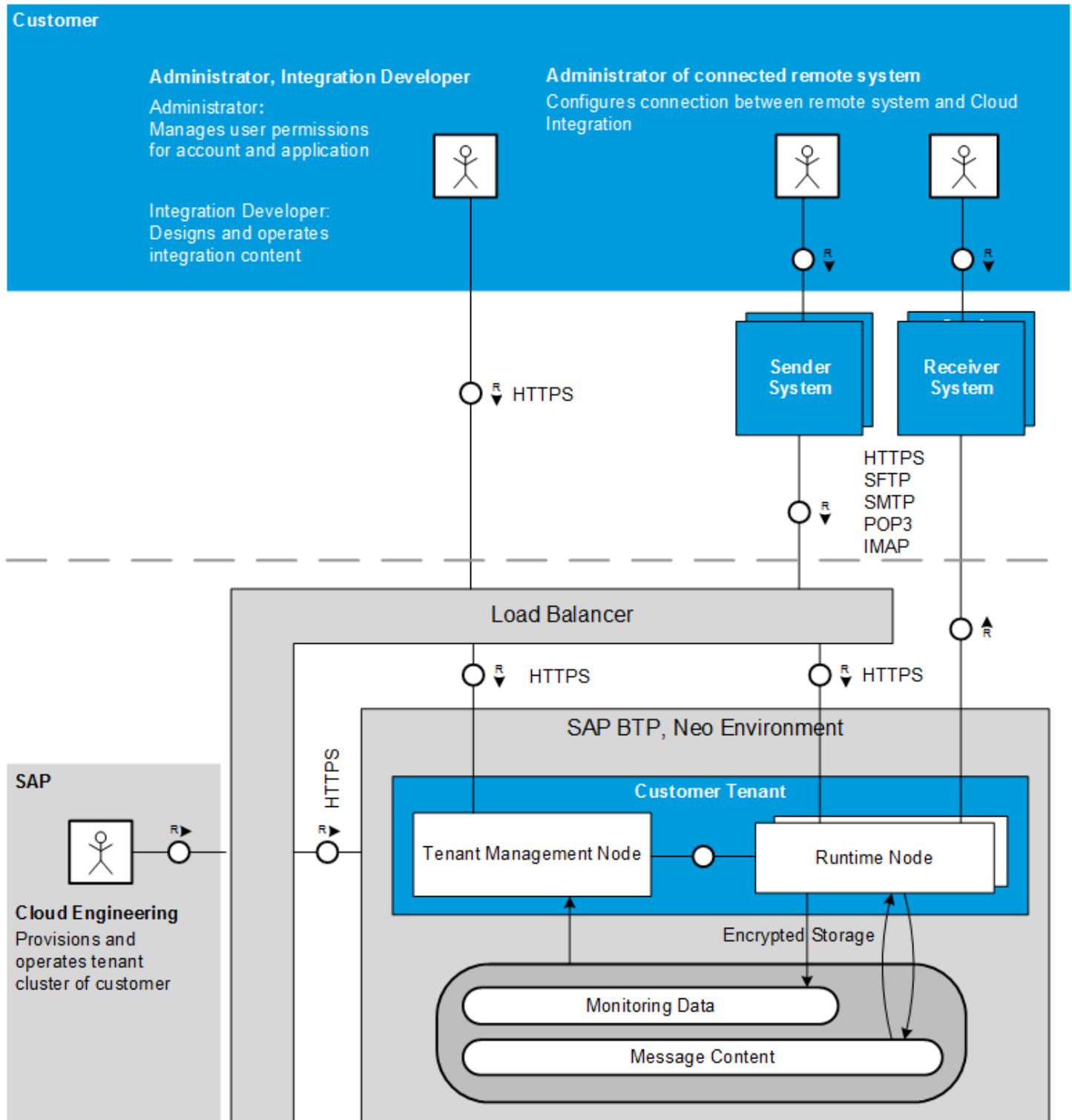
Technical Landscape, Neo Environment

The technical infrastructure comprises a set of technical components that can communicate with each other and with remote components in a secure way based on certain protocols such as HTTPS or SFTP, for example. In addition, user access to the technical infrastructure is designed in such a way that only users with well-defined permissions can access the different segments.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

The following figure shows the high-level technical infrastructure and the main access points and connections. Areas in blue indicate components owned by customers, areas in gray are components that are owned by SAP (in most cases).



Components and Communication Paths

In technical terms, the integration platform is designed as a cluster of virtual machines (nodes) that runs within the SAP cloud. Although all users that connect to the platform through the Internet share the same *physical* infrastructure, each customer gets access to only one or more dedicated *parts* of the platform: accounts or tenants.

On each tenant, a clustered, virtual integration runtime is installed (referred to as a *tenant cluster*). Tenants are strictly isolated from each other with regard to their resources (such as CPU and memory) and to the data that is processed on them. Strictly speaking, each tenant stores data in a separate database schema (this is referred to as *tenant isolation*).

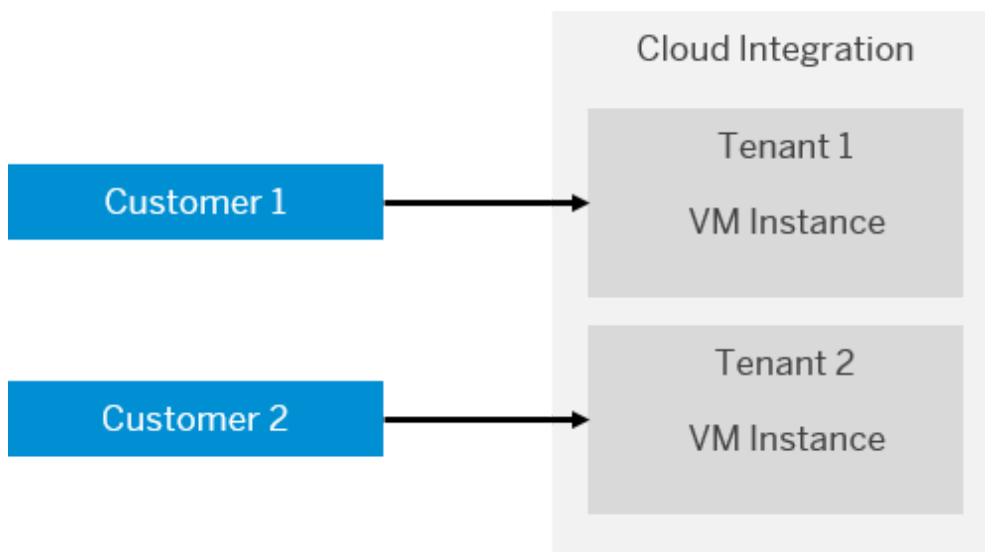
At runtime, the integration platform processes data that is exchanged between the involved participants on a cluster of different virtual machines hosted in the cloud. A cluster associated with a specific tenant is referred to as a **tenant cluster**.

i Note

A virtual machine (VM) is a software implementation of a machine that executes a program like a physical machine.

The integration platform is designed in such a way that the involved virtual machines that process messages at runtime are strictly separated from each other with regard to the related customers. This separation is referred to as *tenant isolation*.

The following figure illustrates this concept for two customers communicating with each other:



The constituents of a tenant cluster are tenant management nodes and runtime nodes.

- A tenant management node accomplishes tasks related to the management of a tenant cluster. It takes requests from the dialog users (for example, when an integration developer deploys an integration flow using the Web user interface). The tenant management node also manages the runtime nodes.
- A runtime node processes messages that are exchanged with external components. Therefore, the runtime node is connected to the external systems. In other words, runtime nodes process customer data that might be confidential and has to be protected.

Typically, one tenant management node has one or more runtime nodes assigned to it. However, in order to support failover scenarios, more than one tenant management node can be operated in a tenant cluster.

As a consequence of this cluster design, the following main communication paths are active during the operation of an integration scenario:

- Communication of tenant cluster and remote components

You can use both cloud systems and on-premise systems (such as on-premise SAP systems) as remote components.

Remote receiver systems are directly connected to the runtime node through a protocol, which depends on the type of the designed receiver adapter. If the integration platform communicates with an on-premise receiver system, you can interconnect the SAP Cloud Connector. This component runs as on premise agent in a secured network and acts as a reverse invoke proxy between the on premise network and SAP Cloud Integration. Due to its reverse invoke support, you don't need to configure the on premise firewall to allow external access from the cloud to internal systems.

For communication in the opposite direction, a load balancer is interconnected between remote sender systems and the involved SAP BTP components. The load balancer terminates incoming Transport Layer Security (TLS) requests and establishes new ones.

- Communication within a tenant cluster: Within a tenant cluster, the tenant management nodes and runtime nodes communicate with each other through a messaging service.

The internal network only allows specific communication (HTTPS) from one virtual machine to another, and this only by taking the loop to the load balancer (not depicted in the figure).

Various secure technical protocols can be used for these communication paths. Depending on the adapter type, the following protocols are available:

- Hyper Text Transfer Protocol (HTTP) over Transport Layer Security (TLS), which is referred to as HTTPS
- SSH File Transfer Protocol (SFTP) for the exchange of data with an SFTP server
- Simple Mail Transfer Protocol (SMTP), Post Office Protocol (POP)³, and Internet Message Access Protocol (IMAP) for the exchange of data with mail servers

Mutual TLS (mTLS) is equivalent to client certificate authentication. While setting up the TLS connection, client and server exchange certificates. With mTLS, not only server certificates, but also client certificates are validated based on the signatures provided by certification authorities. For more information, see [Client Certificate Authentication \(Outbound\)](#) and [Keystore](#).

i Note

Note regarding the timeout behavior for HTTP communication:

In HTTP communication spanning multiple components (for example, from a sender, through the load balancer, to a Cloud Integration tenant, and from there to a receiver), the actual timeout period is influenced by each of the timeout settings of the individual components that are interconnected between sender and receiver (precisely spoken, of those components that can control the Transmission Control Protocol (TCP) session). The component or device with the lowest number set for the idle session timeout will determine the timer that will be used.

- When considering inbound communication (through HTTP-based sender adapters), note that in particular the load balancer has an own timeout setting that has an influence on the overall timeout. For the inbound side, SAP Cloud Integration supports no communication that waits for longer than 10 minutes.
- When considering outbound communication, note that in the involved HTTP-based receiver channel you can configure a dedicated timeout. However, timeout setting has no influence on the TCP timeout when the receiver or any additional component interconnected between the Cloud Integration tenant and the receiver have a lower timeout. For example, consider that you have configured a receiver channel timeout of 10 minutes and there is another component involved with a timeout of 5 minutes. In case, nothing is transferred for a certain time, the connection will be closed after the 5th minute.

User Access

In addition to the above mentioned components that interact with each other when messages are being processed and exchanged between the involved systems, additional components come into play when a dialog user accesses the infrastructure (for example, when an administrator accesses monitoring data or when an integration developer deploys an integration artifact).

People with different roles can access the infrastructure – both on the SAP side (as the provider of the integration infrastructure) and on the customer side. Human access points (for dialog users) are:

- Dedicated experts at SAP access the infrastructure to provide a tenant cluster for the customer.
-

Data Storage

Although the main focus of an integration platform is to receive, process, and forward messages, data can also be stored in dedicated steps during message processing (Experts on the customer side access the infrastructure to design and deploy integration content and to monitor an integration scenario at runtime (integration developers and tenant administrators). *data at-rest*).

The following kinds of data can be stored during the execution of an integration scenario:

- Message content

The runtime node persists message content data in dedicated steps of an integration flow. This data is stored encrypted. The runtime node can also read message content data if the integration flow is designed in such a way that certain steps depend on this data (for example, in content-based routing scenarios).

- Monitoring data

During message processing, the runtime node also persists monitoring data. Monitoring data records the executed processing steps. An administrator (with dedicated permissions) can access this data (for example, using the Web user interface).

Related Information

[Data Storage Security](#)

[Load Balancer Root Certificates Supported by SAP](#)

Security Aspects of Processes

Processes that are related to the provisioning, update, and usage of the cloud-based integration platform meet the highest security standards.

Cloud Integration is compliant with various SAP-internal technical policies, procedures, directives, guidelines, and product standards.

In addition, employees and operators are bound to the SAP code of business conduct and other behavioral security standards such as clean desk and communication.

For example, SAP software is developed in compliance with the SAP Secure Development Lifecycle (SDLC) ,which helps to implement measures such as test-driven development and threat modeling.

SAP certifies that the development, maintenance, and operations of Cloud Integration comply with the requirements of various standards:

More information: [SAP Trust Center](#) 

Security Aspects of Data, Data Flow

All data in transit, either exchanged with remote components or internal, can be protected by methods such as encryption.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

i **Note**

Mutual TLS (mTLS) is equivalent to client certificate authentication. While setting up the TLS connection, client and server exchange certificates. With mTLS, not only server certificates, but also client certificates are validated based on the signatures provided by certification authorities. For more information, see [Client Certificate Authentication \(Outbound\)](#) and [Keystore](#).

Data in transit can be protected at two levels:

During a scenario, the connected remote systems exchange data with each other based on the configured transport protocol. These protocols support different options to protect the exchanged data against unauthorized access. In addition to security at the transport level, the content of the exchanged messages can also be protected by means of digital encryption and signature.

Transport-Level Security

Each adapter allows you to set up a specific security level based on the underlying transport protocol.

Transport-Level Security Options	
Transport Protocol	Transport-Level Security
HTTP(S) (Hypertext Transfer Protocol Secure)	Hyper Text Transfer Protocol (HTTP) over Transport Layer Security (TLS) (HTTPS) TLS is a protocol for secure communication over a computer network that is widely used on the Internet. As TLS is the enhancement of SSL, these terms are often used synonymously. You can protect communication using Transport Layer Security (TLS). In this case, a symmetric key length of at least 128 bits is used (which is technically enforced). Default length of asymmetric keys provided by SAP is 2048 bits. i Note The following versions are supported: • For inbound communication: TLS 1.2 and 1.3 (in the Cloud Foundry environment) • For inbound communication: TLS 1.2 (in the Neo environment) • For outbound communication: TLS 1.2, and 1.3 i Note The HTTP receiver adapter also allows you to use HTTP URLs. However, we do not recommend using this option when transferring confidential data (including the password for basic authentication). Also, if the network is not entirely trusted, there is no way to verify whether the result of an HTTP request originates from a trustworthy source. Therefore, we do not recommend using this option for productive scenarios over the Internet. Receiver adapters also support principal propagation via SAP Cloud Connector. Various authentication options (basic authentication using user credentials, client certificates, or OAuth) are supported depending on the selected sender or receiver adapter. ⚠ Caution Consider that we do not recommend to use basic authentication in productive scenarios because of the following security aspects: Basic authentication has the risk that authentication credentials, for example, passwords, are sent in clear text. Using TLS (transport-layer security, also referred to as Secure Sockets Layer)

Transport Protocol	Transport-Level Security
	as transport-level encryption method (when using HTTPS as protocol) makes sure that this information is nevertheless encrypted on the transport path. However, the authentication credentials might become visible to SAP-internal administrators at points in the network where the TLS connection is terminated, for example, load balancers. If logging is not done properly at such devices, the authentication credentials might become part of log files. Also network monitoring tools used at such devices might expose the authentication information to administrators. Furthermore, the person to whom the authentication credentials belong (in the previous example, the password owner) needs to maintain the password in a secure place. This protocol is supported by all adapter types that allow communication over HTTPS, for example: Ariba, AS2, AS4, ELSTER, Facebook, HTTP, SOAP, IDoc, OData, ODC, OpenConnectors, SAP SuccessFactors, Twitter, XI.
SFTP (Secure Shell File Transfer Protocol)	Secure Shell (SSH) is used to securely transfer files in an open network. SSH is a network protocol that allows you to set up a secure connection to a remote computer, namely an SFTP server. SSH uses a symmetric key length with at least 128 bits to protect FTP communication. Default length of asymmetric keys provided by SAP is 2048 bits. Supported authentication methods: • User name/password authentication (where the SFTP server authenticates the calling component based on the user name and password) • Public key authentication (where the SFTP server authenticates the calling component based on a public key) Secure data transfer with SFTP is based on a combination of symmetric and asymmetric keys. Symmetric (session) keys are used to encrypt and decrypt data within a session. Asymmetric key pairs are used to encrypt and decrypt the session keys. When asymmetric key pairs are used, SFTP also ensures that only authorized public keys are used by the involved participants. Supported versions: • SSH version 2 (as specified at The Secure Shell (SSH) Protocol Architecture 📄) • SSH File Transfer Protocol (SFTP) version 3 or higher This protocol is supported by the SFTP sender and receiver adapter.
FTPS (Secure File Transfer Protocol)	This protocol is used to transfer files from and to file servers. This is an extension of the File Transfer Protocol (FTP). This protocol is supported by the FTP sender and receiver adapter.
JMS (Java Message Service)	JMS is an API that supports reliable asynchronous communication based on a JMS Message Broker. This protocol is supported by the JMS sender and receiver adapter.
JDBC (Java Database Connectivity)	Java Database Connectivity (JDBC) is a protocol or API that defines how a client accesses a database. This protocol is supported by the JDBC receiver adapter.
RFC (Remote Function Call)	Remote Function Call (RFC) is an SAP-proprietary protocol that is used as standard for the communication of SAP systems. This protocol is supported by the RFC receiver adapter.
TCP (Transmission Control Protocol)	TCP is one of the main communication protocols used on the Internet. TLS, previously mentioned, often (but not always) runs on top of TCP.

Transport Protocol	Transport-Level Security
	This protocol is supported by the AMQP (TCP) and the LDAP adapter.
WebSocket	WebSocket is a protocol that supports simultaneous communication in both directions (full-duplex). This protocol is used for setting up connections with a messaging systems. This protocol is supported by the AMQP (WebSocket) adapter.
SMTPS (Simple Mail Transfer Protocol Secure)	This protocol enables a computer to exchange emails with a mail server. With SMTPS, SMTP connections can be secured by SSL or TLS. Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol is used by the Mail receiver adapter.
POP3S (Post Office Protocol version 3 over TLS/SSL)	This option enables email clients to retrieve emails from an email server using the Internet Protocol (IP). Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol can be used by the Mail sender adapter.
IMAPS (Internet Message Access Protocol over TLS/SSL)	Internet Message Access Protocol over TLS/SSL (IMAPS). This option enables email clients to retrieve emails from an email server using a TCP/IP connection. Transport encryption is supported via the STARTTLS extended operation. To authenticate against the e-mail server, you can send user name and password in plain text or encrypted (the latter only in case the e-mail server supports this option). i Note The (optional) password-based authentication only applies to communication between the Cloud Integration system and the mail server. Communication between mail servers is usually not authenticated. Therefore, you must not assume that data received by mail comes from a trustworthy source, unless other security measures (such as digital signatures at message level) are applied. This protocol can be used by the Mail sender adapter.

Message-Level Security

On top of the *transport-level security* options, you can also secure the communication at *message level*, where the content of the exchanged messages can also be protected by means of digital encryption and signatures. Various security standards are available to do this, as summarized in the following table.

To configure message-level security options, you use dedicated integration flow steps (for example, the **Encryptor** and **Signer** step types).

The following standards and algorithms are supported:

Message-Level Security Standards and Algorithms

Standard	Security Feature
PKCS#7/CMS Enveloped Data and Signed Data	Encryption/decryption of message content
	Signing/verification of payload
PKCS#7/CMS Enveloped and Signed Data	Encryption/decryption and signing/verification of payload
Open Pretty Good Privacy (PGP)	Encryption/decryption of message content
	Encryption/decryption and signing/verification of message
XML Signature	Signing/verification of payload
WS-Security	Signing/verification of SOAP body

Related Information

[Connectivity \(Adapters\)](#)

[Authentication and Authorization Options \(Inbound\)](#)

[Authentication Options \(Outbound\)](#)

[Certificate Management](#)

[Message-Level Security](#)

Identity and Access Management

Identity and access management features are used during the lifecycle of an integration scenario.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Access Management

Dialog users who access the platform are authenticated against an identity provider. SAP Identity Service (ID Service) is used by default. SAP ID Service is the central service for the process of managing identities and their lifecycles.

The authentication of inbound calls to the platform depends on the chosen authentication mode. If the client sends a client certificate, the authentication is done by the load balancer. The load balancer terminates the TLS connection; therefore, it checks the client certificate of the calling component against a list of trusted certification authorities (CAs). This certificate is mapped to a user. If basic authentication is configured, the calling entity is checked by the connected identity provider. Besides client certificate authentication, the platform supports basic authentication, OAuth, and Security Assertion Markup Language (SAML).

User Management and Authorizations

Access to dedicated functions of the platform is controlled and protected by authorization checks. A number of authorization groups are available to manage the authorizations of dialog users. An authorization group is based on a persona and defines a set of dedicated permissions relating to the tasks that come into play during the lifecycle of an integration project.

i Note

Example:

If the logged-in user has to perform tasks such as designing and deploying integration flows, the user must be assigned the authorization group `AuthGroup.IntegrationDeveloper`.

Authorization for the Integration Developer

The tasks of persons with integration developer permissions (short: integration developers) constitute a key part of the SAP Cloud Integration lifecycle. Permissions for the integration developer (who is in charge of modeling integration flows) are contained in the authorization group `AuthGroup.IntegrationDeveloper`.

Note that the roles contained in this authorization group give an integration developer full control over message processing during runtime.

More information: [Personas](#)

During integration flow modeling, the integration developer defines how messages are mapped, which credentials are used, and to which recipients messages are sent. The set of roles provides very powerful permissions and in some cases allows the integration developer to access sensitive data.

i Note

The integration developer can control which credentials are to be used in connections with basic authentication by deploying the associated **User Credentials** artifacts on the tenant. These artifacts contain user names and passwords. Note that, however, a password specified in a **User Credentials** is never displayed. Furthermore, passwords cannot be downloaded (by either using the user interface or the application programming interface). The integration developer, although having full control over the integration flow, does not have access to credentials of another tenant of the same customer.

Therefore, apply the following measures when designing integration flows for security-sensitive areas:

- Don't give the integration developer access to productive systems.
- Consider applying a four-eyes principle and implement a review process before deploying integration flows to production.
- An integration developer has the option to develop integration flows on a separate development or test tenant. These integration flows can then be transported to the productive tenant by another person.
- Consider using client certificate-based authentication rather than basic authentication. When using client certificates, the private key remains in the sender system and no secret password is transmitted through the network.
- Don't share the same secret credentials between tenants with different security levels (for example, between test tenant and productive tenant).
- If you suspect a security violation, check the audit log to find out which user deployed the integration flow in question.
- If read-only access is required to analyze issues in the productive system, use the authorization group `AuthGroup.ReadOnly`.

→ Tip

Hint: Instead of using the predefined authorization groups, you can tailor the permissions to your own requirements by applying elementary roles that are defined for individual tasks.

Authentication and Authorization Options for Inbound Calls

When a sender system calls the integration platform using HTTPS-based (inbound) requests, there are different ways for the calling sender to authenticate itself against the integration platform. The options are client certificate authentication, basic authentication, OAuth, and SAML.

Detailed access to the resources of the integration platform is controlled by certain authorization options. In sender channels that are based on transport protocol HTTP, the authorization options are [User Role](#) and [Client Certificate](#),

i Note

[Client Certificate](#) is now deprecated. Instead of this, a certificate-to-user mapping should be maintained. When using this option, within user management a role has to be assigned to the user to grant access to the resource.

i Note

- Authentication

Verifies the identity of the calling entity.

- Authorization

Checks what a user or other entity is authorized to do (for example, as defined by roles assigned to it). In other words, the authorization check evaluates the access rights of a user or other entity.

Combinations of Authentication and Authorization (Inbound)

For inbound communication based on HTTPS, the authentication and authorization options can be combined in a specific way.

Combination of Authentication/Authorization Options

Authentication Option ...	Can Be Used with the Following Authorization Option ...
Basic authentication The sender (client) authenticates itself against the server based on user credentials (user name and password). The HTTP header of the inbound message (from the sender) contains the user name and password.	Role-based authorization For this user, the authorizations are checked based on user-to-role assignments defined on the tenant. i Note When you use Cloud Integration in the Cloud Foundry environment, as user credentials you can also use clientid and clientsecret from a Process Integration service instance with plan integration-flow and client_credentials grant type.
Client-certificate authentication and certificate-to-user mapping (only in the Neo environment) The sender (client) authenticates itself against the server based on a digital client certificate. Furthermore, this certificate is mapped to	Role-based authorization For the user derived from the certificate-to-user mapping, the authorizations are checked based on user-to-role assignments defined on the tenant.

Authentication Option ...	Can Be Used with the Following Authorization Option ...
a user (based on the information contained in a Certificate-to-User Mapping artifact deployed on the tenant). i Note You can map multiple certificates to the same user (n:1 certificate-to-user mappings possible).	
Client-certificate authentication (without certificate-to-user mapping) The sender (client) authenticates itself against the server based on a digital client certificate.	Subject/Issuer DN authorization check of a certificate In a subsequent authorization check, the permissions of the sender are checked on the tenant by evaluating the distinguished name (DN) of the client certificate of the sender.
OAuth Grants access to resources of SAP Cloud Integration without the need to share passwords with the client. i Note This option is supported for the following sender adapter types: SOAP (SOAP 1.x), SOAP (SAP RM), HTTPS, and OData.	Role-based authorization

Related Information

- [User Management for SAP BTP, Neo Environment](#)
- [Managing Users and Role Assignments, Neo Environment](#)
- [Load Balancer Root Certificates Supported by SAP](#)
- [Authentication and Authorization Options \(Inbound\)](#)
- [Personas](#)

Personas

When you perform user management tasks using SAP BTP cockpit, you find a set of predefined roles that you can assign to users of the account. According to the main tasks associated with integration projects, these roles are associated to certain persona relevant for an integration project.

Personas are predefined sets of permissions that cover the different tasks associated with an integration project. To assign roles that are application-based, see [Configuring User Access to the Application](#). For more fine-grained control over user permissions, see [Tasks and Permissions](#).

i Note

In the different environments, the personas are mapped to different objects.

- In the Neo environment, a persona is realized by an authorization group (beginning with the string [AuthGroup](#)).
- In the Cloud Foundry environment, a persona is realized by role collection.

Personas			
Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Descript

Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Descript
Read-only persona	AuthGroup.ReadOnly	PI_Read_Only	Enables tenant a message This per: access (in the Me well as to integrati WSDL. This per: authoriz • M f c a • \ s t • \ t [• \ e f f a c l
Business expert	AuthGroup.BusinessExpert	PI_Business_Expert	Enables read dat contain: content payload: This per: authoriz • M f c a • F f a t • \ c • \ e t

Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Descript
			• E - f - æ - F i No This is collect assign other p PI_Inte or PI_F enhance author busine
Integration developer	AuthGroup.IntegrationDeveloper	PI_Integration_Developer	Enables develop design, æ integrati This pers authoriz • \ - è - F I • (- è - F - æ I • [- i • \ - F • (- F - F - t - æ - F Integrati deploy u secure p various (However permissi certain s like know keys, or content.

Persona	Authorization Group in Neo	Role Collection in Cloud Foundry	Description
Tenant administrator	AuthGroup.Administrator	PI_Administrator	Enables perform on the te This per: authoriz • F • s • t • c • F • (• s • s • a • M • F • U • i • M • i • t • i • [• c • [• c • i • [• f • s • [• a
Partner Directory configurator	AuthGroup.TenantPartnerDirectoryConfigurator	There is no predefined role collection for the Partner Directory configurator. You need to assign the role template AuthGroup_TenantPartnerDirectoryConfigurator to the service instance for the related API client.	Enables Director; read and Director;

Roles to Define Permissions to Call Integration Flow Endpoints

In order to enable a sender system to process messages on Cloud Integration Using HTTPS/basic authentication (based on a user registered by an identity provider), you need to assign the following entities to the associated user:

- Role template **MessagingSend** (in the Cloud Foundry environment)

Using this role template, you need to define a role collection and assign the role collection to the IdP user.

More information: [Basic Authentication of IdP User for Integration Flow Processing](#)

- Role `ESBmessaging.send` (in the Neo environment)

More information: [Setting Up Inbound HTTP Connections \(with Basic Authentication\), Neo Environment](#)

i Note

Note Related to Role Configuration

Depending on the chosen inbound authorization option, you define permissions for sender systems to call integration flow endpoints in different ways:

Option	Summary of Steps
Basic authentication of a user registered at an identity provider (IdP)	Use SAP BTP cockpit to define a role collection that contains the predefined role template <code>MessagingSend</code> and assign the role collection to the IdP user (under Security > Trust Configuration). The role template <code>MessagingSend</code> is provided by default in your subaccount to define permissions for sender systems to call integration flow endpoints for this use case. See: Basic Authentication of IdP User for Integration Flow Processing
Authentication with an OAuth client (service instance)	Use the Cloud Integration Monitor application and select the User Roles tile under Manage Security. When doing this, you can either use the predefined role <code>ESBMessaging.send</code> or create a custom role. Create service instance and service key using SAP BTP cockpit. During this step, you need the role specified with the User Roles tile. See: • Client Certificate Authentication for Integration Flow Processing • OAuth with Client Credentials Grant for Integration Flow Processing • Basic Authentication with clientId and clientsecret for Integration Flow Processing

Related Information

[Tasks and Permissions](#)

Tasks and Permissions

The following table provides an overview of which roles are required in order to accomplish the various tasks related to Cloud Integration. It's also indicated in how far the tasks and roles are relevant for the main personas defined for Cloud Integration. You can use these individual roles or role templates to define your own custom authorization groups (Neo) or role collections (Cloud Foundry), giving you fine-grained control over user permissions. If you prefer to work with predefined permission sets instead, see [Personas](#).

In the different environments, these persona are mapped to different objects.

- In the Neo environment, a persona is realized by an authorization group (beginning with the string `AuthGroup`).
- In the Cloud Foundry environment, a persona is realized by a role collection.

The mapping of the persona to the authorization groups (Neo) or role collections (Cloud Foundry) is described under [Personas](#).

In the different environments, the permissions to execute certain tasks are given by different objects.

- In the Neo environment, a permission to execute a task is given by a role.
- In the Cloud Foundry environment, a permission to execute a task is given by a role template.

Tasks and Permissions

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Discover	View packages	WebToolingCatalog.OverviewRead	CatalogPackagesRead
Discover	View package artifacts	WebToolingCatalog.OverviewRead WebToolingCatalog.DetailsRead	CatalogPackageArtifactsRead
Discover	Copy package to workspace	WebToolingCatalog.OverviewRead WebToolingWorkspace.Write	CatalogPackagesCopy
Design	View packages and package artifacts	WebToolingWorkspace.Read	WorkspacePackagesRead
Design	Create, edit, import, export, delete package with its artifacts	WebToolingWorkspace.Read WebToolingWorkspace.Write	WorkspacePackagesEdit
Design	Update package	WebToolingWorkspace.Read WebToolingWorkspace.Write	WorkspacePackagesEdit
Design	Configure artifacts (integration flows and value mappings)	WebToolingWorkspace.Read WebTooling.IntegrationFlowConfigure WebToolingWorkspace.RuntimeConfigure	WorkspacePackagesConfigure
Design	Deploy/undeploy artifacts	WebToolingWorkspace.Read NodeManager.read	WorkspaceArtifactsDeploy

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
		GenerationAndBuild.generationandbuildcontent NodeManager.deploycontent WebToolingWorkspace.RuntimeConfigure	
Design	Export Package for transport	WebToolingWorkspace.Read TransportModule.read TransportModule.write	WorkspacePackagesTransport
Design	Import package from transport	WebToolingWorkspace.Read TransportModule.read TransportModule.write	WorkspacePackagesTransport
Design	Update Package from transport	WebToolingWorkspace.Read TransportModule.read TransportModule.write	WorkspacePackagesTransport
Design	Read/write Partner Directory-related content (string/binary parameters, authorized users, alternative partners, and user credentials) Read/write certificate-to-user mappings (only in the Neo environment)	TenantPartnerDirectory.read TenantPartnerDirectory.write NodeManager.deploycredentials NodeManager.readcredentials CertificateUserMappings.Write CertificateUserMappings.Read	AuthGroup_TenantPartnerDirectoryCo
Design	Delete design time artifact locks	WebToolingWorkspace.Read WorkspaceArtifactLocks.Delete	WorkspaceArtifactLocksDelete
Design	Read design time artifact locks	WebToolingWorkspace.Read WorkspaceArtifactLocks.Read	WorkspaceArtifactLocksRead
Monitor	View Monitor Overview	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	View message processing logs	IntegrationOperationServer.read	MonitoringDataRead
Monitor	View payload of stored messages from message storage (also trace and message processing log attachments)	esbmessagestorage.read	
Monitor	View tasks	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View tail log	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View deployed artifact list	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	View deployed integration flow in graphical editor	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Download deployed integration flow Download WSDL	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	View user roles	Roles.Read	MonitoringDataRead
Monitor	Create, edit, delete user roles	n.a.	UserRolesEdit
Monitor	View deployed security material	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Add credentials	IntegrationOperationServer.read NodeManager.deploycredentials NodeManager.deploycontent	CredentialsEdit
Monitor	Add known host, keystore, PGP	IntegrationOperationServer.read NodeManager.deploysecuritycontent	SecurityMaterialEdit

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
	keyring artifacts	NodeManager.deploycontent	
Monitor	Edit credentials	IntegrationOperationServer.read NodeManager.deploycredentials NodeManager.deploycontent	CredentialsEdit
Monitor	Undeploy credentials	IntegrationOperationServer.read NodeManager.deploycontent NodeManager.deploycredentials	CredentialsEdit
Monitor	Undeploy known host, keystore, PGP keyring artifacts	IntegrationOperationServer.read NodeManager.deploycontent NodeManager.deploysecuritycontent	SecurityMaterialEdit
Monitor	Download keystore, public/private keyring, known host, artifact	IntegrationOperationServer.read NodeManager.read NodeManager.readsecuritycontent	SecurityMaterialDownload
Monitor	View certificate-to-user mappings	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Create/edit/delete certificate-to-user mappings	IntegrationOperationServer.read NodeManager.deploysecuritycontent NodeManager.read	SecurityMaterialEdit
Monitor	View keystore entries	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead
Monitor	Download public keystore entries	IntegrationOperationServer.read NodeManager.read	MonitoringDataRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	Add/replace/delete keystore entries	IntegrationOperationServer.read NodeManager.deploysecuritycontent	SecurityMaterialEdit
Monitor	View access policies	IntegrationOperationServer.read AccessPolicies.Read	AccessPoliciesRead
Monitor	Maintain access policies	IntegrationOperationServer.read AccessPolicies.Write	AccessPoliciesEdit
Monitor	View data store entries	IntegrationOperationServer.read ESBDataStore.read	DataStoresAndQueuesRead
Monitor	View data store entries - message payload	IntegrationOperationServer.read ESBDataStore.readPayload	DataStorePayloadsRead
Monitor	View data store entries - variables	IntegrationOperationServer.read ESBDataStore.readPayload	DataStorePayloadsRead
Monitor	Delete data store entries/variables	IntegrationOperationServer.read ESBDataStore.read ESBDataStore.delete	DataStoresAndQueuesDelete
Monitor	Configure data stores and queues	IntegrationOperationServer.read ESBDataStore.Config	DataStoresAndQueuesConfig
Monitor	View payload of stored messages from message store	esbmessagestorage.read	MessagePayloadsRead
Monitor	View trace configuration	IntegrationOperationServer.read NodeManager.read ConfigurationService.RuntimeBusinessParameterRead	TraceConfigurationRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	Edit trace configuration (enable/disable trace)	IntegrationOperationServer.read NodeManager.read ConfigurationService.RuntimeBusinessParameterRead ConfigurationService.RuntimeBusinessParameterWrite	TraceConfigurationEdit
Monitor	Override access policies Get access to data (for example, JMS queues, message processing log attachments) although the data is protected by access policies. Note that when this role is assigned to a user, it overrides all access policies that might prevent the user from accessing data processed by certain artifacts (for example, integration flows). However, this role doesn't grant permission to access payload data.	AccessPoliciesArtifacts.AccessAll	AccessAllAccessPoliciesArtifacts
Monitor	Add, edit, or undeploy number ranges, and undeploy integration flows	IntegrationOperationServer.read NodeManager.deploycontent	MonitoringArtifactsDeploy
Monitor	View number ranges	IntegrationOperationServer.read	MonitoringDataRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	Activate or deactivate queues	IntegrationOperationServer.read ESBDataStore.Activate	QueuesActivate
Monitor	Retry queues	IntegrationOperationServer.read ESBDataStore.read ESBDataStore.retry	QueuesRetry
Monitor	Delete queues	IntegrationOperationServer.read ESBDataStore.read ESBDataStore.delete	DataStoresAndQueuesDelete
Monitor	View queues	IntegrationOperationServer.read ESBDataStore.read	DataStoresAndQueuesRead
Monitor	View runtime processing locks	IntegrationOperationServer.read MessageProcessingLocks.Read	MessageProcessingLocksRead
Monitor	Access the ID Mapper (using the Message Processing Logs OData API).	IntegrationOperationServer.read MessageProcessingLocks.Read	MessageProcessingLocksRead
Monitor	Delete runtime processing locks	IntegrationOperationServer.read MessageProcessingLocks.Delete	MessageProcessingLocksDelete
Monitor	Test connectivity	IntegrationOperationServer.read NodeManager.readcredentials	MonitoringDataRead (only for tests with authentication) CredentialsRead
Monitor	Activate external logging	IntegrationOperationServer.read ExternalLogging.Activate	ExternalLoggingActivate
Monitor	Read the activation state of external logging	IntegrationOperationServer.read ExternalLoggingActivation.Read	ExternalLoggingActivationRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
Monitor	Change log level	IntegrationOperationServer.read ConfigurationService.RuntimeBusinessParameterWrite NodeManager.read	
Monitor	View audit log entries	IntegrationOperationServer.read AuditLog.Read	
Monitor	Read data metrics for health monitoring API: JMS queue statistics, certificate expiry dates	HealthCheckMonitoringData.Read	HealthCheckMonitoringData.Read
Monitor	Activate archiving Enablement		DataArchivingActivate
Monitor	Read archiving enablement		DataArchivingRead
Monitor	Change archiving configuration	IntegrationOperationServer.read ConfigurationService.RuntimeBusinessParameterWrite NodeManager.read	TraceConfigurationEdit
Monitor	Change log level		TraceConfigurationEdit
Monitor	Read data metrics for the health check monitoring API. This covers, for example, JMS	ESBDataStore.read NodeManager.read ResourceUsageData.Read	HealthCheckMonitoringDataRead

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
	queue statistics, certificate expiry dates)		
Operate	Get read-only access to Operations Cockpit. The Operations Cockpit is the central control point for operating edge integration cells and allows the administrator to monitor and adjust system configurations and resources.	n.a.	IntegrationCellOperationCockpitRead
Operate	Get full access to the Operations Cockpit. The Operations Cockpit is the central control point for operating edge integration cells and allows the administrator to monitor and adjust system configurations and resources.	n.a.	IntegrationCellOperationCockpitWrite
Inspect (only when using SAP Cloud Integration in the Cloud Foundry environment)	Inspect usage of integration resources	n.a.	MonitoringDataRead
Settings	View and change runtime profile	WebToolingSettingsProductProfiles.savetenantconfiguration	
Settings	Set transport system	WebToolingSettingsProductProfiles.savetenantconfiguration	
Settings	Configure Design Guidelines settings	WorkspaceDesignGuidelines.Configure	WorkspaceDesignGuidelinesConfigure

Area	Task	Role (Neo)	Role-Templates (Cloud Foundry)
General	Provides permission for the read-only persona (see Personas)	AuthGroup.ReadOnly	AuthGroup_ReadOnly
General	Provides permission for the tenant administrator persona (see Personas)	AuthGroup.Administrator WebToolingWorkspace.RuntimeConfigure	AuthGroup_Administrator
General	Provides permission for the business expert persona (see Personas)	AuthGroup.BusinessExpert	AuthGroup_BusinessExpert
General	Provides permission for the integration developer persona (see Personas)	AuthGroup.IntegrationDeveloper	AuthGroup_IntegrationDeveloper
General	This role/role template is only relevant for partners and internal SAP developers developing and publishing content to SAP Business Accelerator Hub. Further details can be found in the partner documentation.	AuthGroup.ContentPublisher	AuthGroup_ContentPublisher

Related Information

[Authorization Entities](#)

Data Storage Security

Customer data can be stored in dedicated steps during message processing.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Customer data stored at rest is strictly separated and isolated for each tenant. Although different tenants might share a common physical infrastructure, each tenant stores its data in a separate schema.

For certain use cases the customer can configure if the data at rest is encrypted.

Message content can be stored encrypted. If this security measure is configured, the encryption key that is generated automatically is unique for each tenant and is renewed periodically.

Data storage encryption uses AES and a key length of 256 bits. The encryption key is not stored in the same location as the encrypted data.

Kinds of Stored Data

The following kinds of data can be stored during the execution of an integration scenario:

- Message content

The runtime node writes message content data to the database in dedicated steps of an integration flow. There is the option to either store message content for a longer time period (the default is 30 days) or temporarily. Temporarily stored message content can be used for subsequent message processing steps. Such steps can then also read message content from the database.

There is the option to configure the retention period of the message content.

- Monitoring data

During message processing, the runtime node also writes monitoring data to the database (which is stored by default for 30 days). Monitoring data comprises the message processing log (MPL), which records the executed processing steps.

Physical Site

Customer data is stored in various regions worldwide. Here, highest security standards are met. To mention a few examples, redundant power supplies are used and physical access is restricted by means such as biometric access control mechanisms. All of these measures are regularly checked and audited.

More information: <http://www.sapdatacenter.com> 

All license editions of SAP Cloud Integration include the embedded use of the SAP ASE Platform Edition with a limited disk space.

Data Protection and Privacy

Various types of customer data are processed by and stored on the integration platform at different times. This data gets the highest level of protection, and SAP takes dedicated measures to guarantee this security level.

General Information

Governments place legal requirements on industry to protect data and privacy. We provide features and functions to help you meet these requirements.

Caution

SAP does not provide legal advice in any form. SAP software supports data protection compliance by providing security features and data protection-relevant functions, such as blocking and deletion of personal data. In many cases, compliance with applicable data protection and privacy laws is not covered by a product feature. Furthermore, this information should not be taken as advice or a recommendation regarding additional features that would be required in specific IT environments. Decisions related to data protection must be made on a case-by-case basis, taking into consideration the given system

landscape and the applicable legal requirements. Definitions and other terms used in this documentation are not taken from a specific legal source.

Caution

We assume that you have not maintained any data related to an individual in the tools provided by SAP Cloud Integration (for example, when designing integration content).

We expect that sensitive personal data can only be included in message payloads. This responsibility lies exclusively with you as the operator of an integration scenario using SAP Cloud Integration and remains your responsibility. If you include sensitive personal data within payloads or message attachments, SAP Cloud Integration may store this information on your behalf. This applies also for data maintained in the tools provided by SAP Cloud Integration, however, data within payloads can be protected by enabling encrypted storage.

The knowledge of sensitive personal data lies exclusively with you and remains your responsibility.

The tools of SAP Cloud Integration only use technical users or data without any references to individuals.

User Consent

We assume that software operators, such as SAP customers, collect and store the consent of data subjects, before collecting their personal data. A data privacy specialist can later determine whether data subjects have granted, withdrawn, or denied consent.

Read Access Logging

Read Access Logging is used to monitor and log read access to sensitive data. Data may be categorized as sensitive by law, by external company policy, or by internal company policy. Read Access Logging enables you to answer questions about who accessed particular data within a specified time frame. Such questions could be:

- Who accessed the data of a given business entity, for example, a bank account?
- Who accessed personal information, such as health data?
- Who accessed personal data of accounts or business partners?

A tenant administrator can display audit logs for a tenant using the Monitoring application of the Web UI (under [Manage Security](#) in the [Audit Log](#) tile).

Additionally, within SAP, audit logs can be displayed for teams in charge of maintaining the virtual cloud environment and to analyze and resolve error situations. Audit logs related to different customers are separated from each other (according to the tenant isolation feature).

More information: [Security-Relevant Logging and Tracing](#)

Information Report

An information report is a collection of data relating to a data subject. A data privacy specialist may be required to provide such a report or an application may offer a self-service. SAP Cloud Integration assumes that software operators, such as SAP customers, can provide such information.

Erasure of Personal Data

When handling personal data, consider the legislation in the different countries where your organization operates. After the data has passed the end of purpose, regulations may require you to delete the data. However, additional regulations may require you to keep the data longer. During this period you must block access to the data by unauthorized persons until the end of the retention period, when the data is finally deleted.

Data stored on the SAP Cloud Integration tenant is only stored for a limited time period (referred to as retention time).

For more information on the retention times for the various kinds of data stored by SAP Cloud Integration, see [Specific Data Assets](#).

Related Information

[Types of Stored Data](#)

[Specific Data Assets](#)

[Security-Relevant Logging and Tracing](#)

[Data Storage Security](#)

Types of Stored Data

Different kinds of data, such as message content or monitoring data, can be stored during the operation of an integration scenario.

Such data needs to be considered as sensitive data as it can contain personal information. The following list provides examples:

- Message content

Messages processed on a runtime node typically contain business data of an integration scenario and therefore can contain sensitive customer data such as addresses, names, or financial information.

When this data is *at-rest*, it can be stored encrypted. Note, however, that in some use cases the customer can configure that the data is not encrypted.

When this data is *in-transit*, several measures can be taken, such as digital message signing or message content encryption.

- Monitoring data

The message processing log records the processing steps of an integration flow. Only users assigned to this tenant and with dedicated permissions can access this data.

- Other data, such as the content of log files

i Note

Personal data processed by and stored on the integration platform is handled according to the Data Processing Agreement, which you can find at <http://www.sap.com/about/agreements.html> under **SAP Cloud Services Customers**.

Due to the tenant isolation concept, data from different customers (stored in different tenants) is strictly isolated. Additionally, SAP has no access to data stored in customer tenants.

The customer can grant people outside its organization permissions to execute specific tasks on its cluster (for example, to SAP employees to execute error analysis tasks in support cases).

Related Information

[Specific Data Assets](#)

Specific Data Assets

Different kinds of data are stored in the SAP Cloud Integration infrastructure during the lifecycle of an integration project.

The following table lists the different kinds and attributes such like storage location and retention time, for example.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Data Assets

Data	Description	Logical Storage	Classification	Retention Time	Backup Available
Message processing log	Structured information on the processing of a message		Log data	30 days	14 days
Message processing log attachments	Data attached to a message processing log during runtime	Message store	Log data Business data	30 days	14 days
Audit log	Information on events such as data read accesses or system configuration changes		Log data	30 days	14 days
System log	Information about errors that occurred during HTTP inbound processing		Log data	7 days	
Integration flow tracing data	Information on the message flow (including the message payload) and on errors that occurred during message processing	Trace store	Log data Business data	60 minutes	14 days
Integration content (design time)	Integration flow models and value mappings created or edited by an integration developer	Workspace	Configuration data	Unlimited	14 days
Integration content (runtime)	Camel XML representation of integration flows and other design time entities (as deployed on a runtime node)		Configuration data	Unlimited	14 days
Data stored by Data Store operations	Message content stored in dedicated	Data store	Business data	Can be defined by integration	14 days

Data	Description	Logical Storage	Classification	Retention Time	Backup Available
step	steps in an integration flow (contains information such as message GUID, message processing log GUID, tenant ID, time stamp, and payload). Is used for further message processing in subsequent steps in an integration flow.			developer (default value: 30 days)	
Data stored by Persist step	Message content stored in dedicated steps in an integration flow (contains information such as message GUID, message processing log GUID, tenant ID, time stamp, and payload) Can be accessed and analyzed after message processing.	Message store	Business data	90 days	14 days
Lock entries	Lock entries that are created (in the in-progress repository) to avoid the same message being processed several times in parallel (for example, by different runtime nodes)				14 days

Limited retention times make sure that data is stored in the system only for a limited time period.

Related Information

[Define Data Store Operations](#)

[Managing Security Material](#)

[Message Locks](#)

Security-Relevant Logging and Tracing

Audit logs allow administrators at SAP or the tenant administrator to monitor events such as data read accesses or system configuration changes. This enables administrators to take adequate measures to prevent malicious usage of the system.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

A tenant administrator can display system access logs for his or her tenant using the **Monitoring** application of the Web UI (under **Access Logs** in the **Audit Log** and **System Log Files** tile).

The **Audit Log** section allows you to monitor changes to the configuration of the tenant cluster such as (examples):

- Access to the system (for example, read access to message content)
- Deletion of data from the message store
- Change in the content of a keystore

An audit log records the following type of information for each logged event:

- Type of the event
- Date and time of the event
- Initiator of the event (user)
- Source of the event (IP address of the source that issued the action)

To view audit logs, the tenant administrator needs to have assigned the roles `IntegrationOperationServer.read` and `AuditLog.Read`.

The audit log data retention time in the database is 30 days.

The **System Log Files** section provides information related to errors that occurred during HTTP inbound processing and contains 2 different log types: the **HTTP Access Logs** and the **Worker Nodes Default Traces**.

- For both log types, the access is granted through personas or composite roles such as (`IntegrationDeveloper`, `Administrator`, `BusinessExpert`, `ReadOnly`, `SystemDeveloper`). They all can open the log files as the composite roles always contain the single role `IntegrationOperationServer.read`. You need this role to download logs.
- Retention time for both logs is 7 days. After that period, the log files are deleted.

Stored information

The information stored depends on the log type.

For the **HTTP Access Logs**, the system stores information about the incoming HTTP request on the worker node, such as:

- The incoming time of the call
- The IP address of the sender client
- The User Id, if included in the request
- The HTTP method (HEAD, POST, PUT, GET, DELETE)
- The call path such as e.g. `/cxf/SOD/incoming`
- The HTTP status code (e.g. 200, 401, 403, 500)

- The call size in bytes
- The duration of the call.

This information is stored to be able to identify authentication or authorization problems arising in an early message processing phase, since no Message Processing Log Entries are logged yet. It is used by integration developers or administrators for trouble-shooting and/or configuration purposes.

The **Default Trace** contains general information about system activities according to the log level you have set.

In case of the default settings, errors and stack traces are logged. The default trace format is :

- Log entry time stamp
- Log level (*ERROR, INFO, WARNING, DEBUG,...*)
- Class name of the logger
- Tenant
- Log text

This information is stored for trouble-shooting purposes, for example if an component monitor throws an error message during deployment of the corresponding integration flow and this error message does contain enough details. In this case the developer or the administrator might have a look into the *Worker Nodes Default Traces*.

i Note

Audit logs managed by SAP

Audit logs are also generated SAP-internally and displayed for people with dedicated permissions. Such audit logs are generated for all virtual machines (nodes) of the customer's tenant clusters as well as for the load balancer. These logs can cover events such as:

- Access to the system (for example, read access to message content)
- Change to the setup of a tenant cluster (for example, if VM has been started or stopped)
- Change to the configuration (for example, an integration flow has been deployed or undeployed)
- Change of the tenant configuration

Audit logs related to different customers are separated from each other (according to the tenant isolation feature).

Other Security-Related Information

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

User Interface Security

Cloud Integration provides user interfaces for designing and deploying message flows, and monitoring them at runtime.

A Web tool (Web UI) is available to accomplish these tasks. The Web UI is implemented using JavaScript and HTML (UI5).

This user interface is built to prevent vulnerabilities such as cross-site scripting (XSS) and cross-site request forgery (XSRF). The built-in security capabilities of these technologies are used together with secure design and coding principles.

Security Measures for Remote API

You can access certain functions of Cloud Integration through application programming interfaces (APIs).

The OData API is protected by basic authentication.

In order to protect the API against CSRF (cross-site request forgery) attacks, modifying operations (for example, POST, DELETE) should be used in conjunction with session-based authentication and client-side CSRF handling.

Auditing and Logging Information

More information: [Auditing and Logging Information](#)

Integration Settings

1. Choose **Settings** from the left navigation pane.
2. Configure the settings. Navigate to every tab to configure different settings.
3. Save your settings.

[Runtime Profiles](#)

Cloud Integration allows you to design integration content for different target integration platforms. Accordingly, different runtime profiles are available to adapt the user interface of the integration content designer to the specifications and capabilities of the target integration platform.

[Importing Content from SAP Process Orchestration System](#)

Import content from an SAP Process Orchestration system and reuse in the cloud world.

[Creating Custom Tags](#)

You can maintain custom attributes in your editable integration packages.

[Malware Scanner](#)

Scan your design time artifacts for malware before the users upload them to the tenant.

[Design Guidelines](#)

As the integration lead for your organization, help your integration developers design enterprise-grade artifacts.

Runtime Profiles

Cloud Integration allows you to design integration content for different target integration platforms. Accordingly, different runtime profiles are available to adapt the user interface of the integration content designer to the specifications and capabilities of the target integration platform.

A **runtime profile** defines a set of capabilities for content design supported by a specific target integration platform. In particular, a specific runtime profile supports the configuration of a specific set of integration flow steps, APIs, and adapter types.

The following runtime profiles are available:

Runtime Profiles

Runtime Profile	Description	Supported Artifacts	Important Links
Cloud Integration	Cloud-based integration runtime of Cloud Integration	See	
SAP Process Orchestration (for a specific Process Orchestration release, for example: SAP Process Orchestration 7.5, SP05)	On-premise integration runtime of SAP Process Integration		

→ Tip

To display and manage runtime profiles, choose **Settings > Integrations**. When you select a runtime profile, the features enabled and supported by that profile are displayed. That way, you can find out which component versions (for example, which version of a dedicated adapter) are supported by the selected runtime profile.

If you apply a runtime profile to an integration flow, the integration flow editor restricts you to configuring only the supported features.

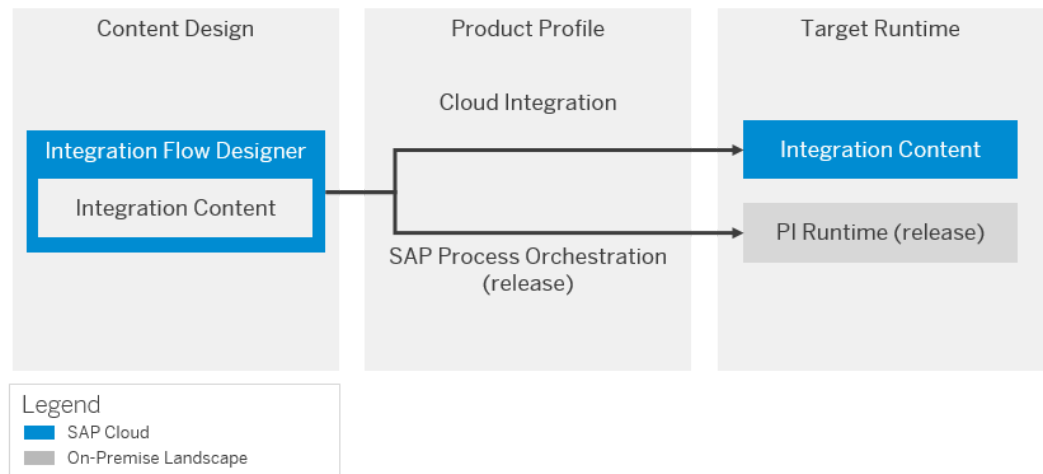
Before start working with and designing integration content, you must know on which target runtime you want to deploy the integration content.

All the profiles appear in the **Runtime Profiles** tab. You can enable or disable a runtime profile based on your requirements. Only the enabled runtime profiles appear in the **Runtime Configuration** tab of the integration flow editor.

i Note

- At any point of time, there's only one Cloud Integration runtime profile and you can't disable it. The option to disable this profile is grayed out.
- The runtime profiles can't be disabled. The **Current Version** column displays the version on which the runtime profile is currently running.
- The runtime profiles are updated at cloud speed, which means you must upgrade them monthly. If your runtime is outdated, content deployment and undeployment get blocked. For more information, see .
- Tenant administrator can decide to enable or disable the **SAP Process Orchestration** supported packages (SPs). They can take the decision based on the availability of SPs in the on-premise integration platform.
- If you make an **SAP Process Orchestration** runtime as the default profile and disable it, Cloud Integration automatically becomes the default profile.
- If you disable a runtime profile that is already used in an integration flow, the integration flow editor sends an alert message indicating that the runtime profile isn't available. The default runtime profile is automatically used for validations.

The following figure illustrates a use case for the runtime profiles SAP Cloud Integration and SAP Process Orchestration 7.5 SPO.



When you've decided on the runtime profile in question, follow the steps mentioned in [Set Default Runtime Profile](#).

You have the option to configure a runtime profile also for an individual integration flow (under [Runtime Configuration](#)). See: [Configure Runtime Profile for an Integration Flow](#)

Set Default Runtime Profile

The tenant administrator can view and configure a runtime profile, to mark one of them as default for the tenant.

Prerequisites

- You have connected to SAP Cloud Integration.
- You have assigned `WebToolingSettingsProductProfiles.savetenantconfiguration` or the tenant administrator role.

Context

Runtime profile is a collection of capabilities such as success factor adapter, splitter or datastore elements, available for a particular product. You can consume these capabilities at the time of designing integration flows.

The tool enables you to design for multiple runtimes at the same time. You should select specific runtime profile to develop content for the respective runtime. You can execute the following steps to configure a runtime profile for a tenant.

Procedure

1. Choose [Settings](#).

i Note

You can click on the runtime profile name under [Name](#) field to see the available capabilities for the relevant profile.

2. Choose [Edit](#) in the [Runtime Profiles](#) tab page.
3. Select the radio button for the runtime profile you want to mark as default.

4. Choose **Save** or **Cancel** as appropriate.

i Note

You can also configure runtime profile at integration flow level. For more details please refer to [Configure Runtime Profile for an Integration Flow](#).

Related Information

[Runtime Profiles](#)

Configure Runtime Profile for an Integration Flow

You can use runtime profile in an integration flow, to develop content for a particular runtime.

Prerequisites

- Your user is assigned with integration developer role.
- You have opened an integration flow in the editor.

Context

Runtime profile is a collection of capabilities such as OData adapter, splitter or datastore elements, available for a particular product. You can consume these capabilities at the time of designing integration flows.

The tool enables you to design for multiple runtimes at the same time. You should select specific runtime profile to develop content for the respective runtime.

i Note

If a runtime profile does not support a particular capability then the checks report errors for unsupported components in the integration flow.

Procedure

1. Choose the **Runtime Configuration** tab page.
2. Select relevant runtime profile from the **Runtime Profile** dropdown list.

i Note

- If you want to publish content in the catalog, then it is a recommendation to select a specific runtime profile.
- If you switch runtime profiles, then the system reloads the palette and displays relevant capabilities.

3. Choose **Save** or **Deploy** as appropriate.

Related Information

[Runtime Profiles](#)

Importing Content from SAP Process Orchestration System

Import content from an SAP Process Orchestration system and reuse in the cloud world.

What Is System Settings

This setting enables you to connect to an SAP Process Orchestration system so that you reuse the integration content that you already have and avoids the overhead of creating that content again in Cloud Integration.

To enable this setting, you've to configure the connectivity to Integration Directory and Enterprise Services Repository. Since these systems are on-premise systems, you must connect to them via Cloud Connector. For more information on [Cloud Connector](#), see [SAP Cloud Connector documentation](#).

What Are the Steps Involved

API endpoints and subpaths are used to extract data from your SAP Process Orchestration system. Make sure that SAP Destination service can access the following endpoints. See [Destination Service](#).

- Directory Content
 - /CommunicationChannelInService
 - /IntegratedConfigurationInService
 - /SenderAgreementInService
 - /AlertRuleInService
 - /IntegratedConfiguration750InService
 - /ValueMappingInService
 - /ConfigurationScenarioInService
 - /BPMFacadeBeanImplService
 - /ReceiverAgreementInService
 - /InterfaceDeterminationInService
 - /ReceiverDeterminationInService
- Enterprise Services Repository (ESR) Content
 - /rep/read/ext
 - /dir/read/ext
 - /rep/support/SimpleQuery
 - /rep/query/ext
- For Message mapping: /rep/query/
- For Value mapping: /dir/query/

The endpoints are constructed according in the pattern <protocol>://<hostname>:<port><endpoint>. For example, <https://po75-systema.sap:443/AlertRuleInService>. You must use Cloud Connector to securely expose these endpoints:

- Ensure that Cloud Connector is connected to your subaccount. See [Establish Connections to SAP BTP](#).
- Create a new HTTP destination (cloud to on-premise) with back-end type **SAP Process Integration**. See [Configure Access Control \(HTTP\)](#).
- Limit access to the previously mentioned endpoints and subpaths by changing **Access Policy** to **Path And All Sub-Paths**.

How Does This Setting Benefit You

By configuring the connectivity using Cloud Integration **System** setting, you avoid creating destinations in your SAP BTP subaccount. This setting in Cloud Integration, based on the inputs you provide about your SAP PO system, automatically creates the destination for you in your SAP BTP subaccount. With the connectivity to an SAP PO system in place, you can import content like message mappings, function libraries, and message types.

After configuring connectivity to an SAP PO system, you can test the connectivity to the system using the **Test Connection** option.

You can configure connectivity to more than one SAP PO system so that you import content from all of them in one Cloud Integration tenant.

Changes to Already Configured ES Repository

Until the 2304 release of Cloud Integration, this tab was named **ES Repository** where you could configure connectivity to one ES Repository (ESR) only. If you had configured an ESR in the past, the same would now appear in the renamed tab. Upon first time editing of the previously configured ESR, you're prompted to enter the user ID and password of the technical user from your SAP PO system.

For Neo customers, user ID and password aren't required. The connectivity configuration continues to exist with the deployed credential name.

Configuring Connectivity to an SAP Process Orchestration System

Learn how to connect an SAP Process Orchestration system.

Recommended Approach: Using **System** Settings

Prerequisites

- You must have tenant administrator role to configure connectivity to an SAP Process Orchestration system.
- You have completed the tasks mentioned in [Importing Content from SAP Process Orchestration System](#).

Procedure

1. Choose **Settings**.
2. Choose the **System** tab.

If there are SAP Process Orchestration systems already added, you see them. Select an existing system and choose **Edit** to update the configurations.

3. Choose **+ Add** to connect an SAP PO system.
4. Provide a **System Name** for the PO system that you want to connect to.

The name that you provide here appears at all places in the tenant when integration developers try to fetch content from a PO system. Make sure that you provide a meaningful name.

5. Select the **Version** of your PO system.
6. Optional: Provide a description that explains the PO system.
7. Enter values in fields based on description in the following table:

Integration Directory

Field	Value
Address	Provide the URL from Cloud Connector that connects to your PO system. i Note If the test connection fails, try removing the character "_" (underscore) in the address field in your Cloud Connector. Then, provide the updated URL. An underscore is a likely candidate for such connection failures.
Credential Name Credential name is applicable only for Neo environment. For CF, use User ID and Password .	Enter the name of the deployed user credentials.
User ID	Add the user ID (technical user) from your SAP Process Orchestration system. i Note The user must be assigned to the following roles in your SAP Process Orchestration system: ◦ SAP_XI_API_DISPLAY_J2EE ◦ SAP_XI_API_DEVELOP_J2EE ◦ SAP_XI_MONITOR_J2EE
Password	Add the user's password.
Confirm Password	Enter the password again.
Location ID	Add the location ID.

8. **Optional:** If you're using a central Enterprise Services Repository (ESR), uncheck the **Same as Integration Directory** check box. Enter values in fields based on description in the following table:

ES Repository

Field	Value
Address	Provide the URL from Cloud Connector that connects to your PO system. i Note If the test connection fails, try removing the character "_" (underscore) in the address field in your Cloud Connector. Then, provide the updated URL. An underscore is a likely candidate for such connection failures.

Field	Value
User ID	Add the user ID (technical user) from your SAP Process Orchestration system. i Note The user must be assigned to the following roles in your SAP Process Orchestration system: ◦ SAP_XI_API_DISPLAY_J2EE ◦ SAP_XI_API_DEVELOP_J2EE ◦ SAP_XI_MONITOR_J2EE
Password	Add the user's password.
Confirm Password	Enter the password again.
Location ID	Add the location ID.

9. Choose **Save**.

Alternative Approach: Creating Destinations in SAP BTP Subaccount

Prerequisites

- You must have tenant administrator role to configure connectivity to an SAP Process Orchestration system.
- You have completed the tasks mentioned in [Importing Content from SAP Process Orchestration System](#).

Context

Execute the following steps in your SAP BTP subaccount. After you create a destination, an entry for that SAP Process Orchestration system is automatically created in the **System** settings tab and it is ready to use.

Though creating destinations directly in the SAP BTP subaccount is a valid approach, we recommend you to use the **System** settings tab.

Procedure

1. Log on to SAP BTP cockpit and navigate to your subaccount.

In SAP BTP, Destination service is used to store your connectivity configuration.

2. Navigate to **Connectivity > Destinations** and choose **New Destination**.

For general information on creating destinations in your subaccount, see [Create a Destination](#).

3. Configure a new destination with the following details specific to your system:

Integration Directory Destination

Field	Content
Name	P0_<systemname>_DIR

Field	Content
Type	HTTP
Description	Add a meaningful description
URL	Add your system URL
Proxy Type	OnPremise
Authentication	Basic Authentication
Location ID	Add your location ID
User	Add the user ID (technical user) from your SAP Process Orchestration system i Note The user must be assigned to the following roles in your SAP Process Orchestration system: ◦ SAP_XI_API_DISPLAY_J2EE ◦ SAP_XI_API_DEVELOP_J2EE ◦ SAP_XI_MONITOR_J2EE
Password	Add the user's password
Additional Properties:	
systemname	Add the name of the system you created in Migration Assessment
esrdestination	Optional: If you use a central ESR and need to create an ESR destination (see next step), add the destination's name here

4. **Optional:** If you're using a central Enterprise Services Repository (ESR), create a second destination with the details listed in the following table.

If you're not using a central ESR, you only need the Integration Directory destination.

ESR Destination

Field	Content
Name	P0_<systemname>_ESR
Type	HTTP
Description	Add a meaningful description
URL	Add your system URL
Proxy Type	OnPremise
Authentication	Basic Authentication
Location ID	Add your location ID
User	Add the user ID (technical user) from your SAP Process Orchestration system i Note The user must be assigned to the following roles in your SAP Process Orchestration system:

Field	Content
	◦ SAP_XI_API_DISPLAY_J2EE ◦ SAP_XI_API_DEVELOP_J2EE ◦ SAP_XI_MONITOR_J2EE
Password	Add the user's password
Additional Properties:	
systemname	Add the name of the system you created in Migration Assessment

5. Choose **Save**.

Importing Mapping Content from ES Repository

Prerequisites

- You've configured a connection to ES Repository under **Settings** (⚙️) tab. For more information, see [Configuring Connectivity to an SAP Process Orchestration System](#).
- You've opened the integration flow in which you want to add the integration content and you are editing it.

Context

After you've configured the connection to ES Repository, you can import content from it in the **References** tab of the integration flow editor. Currently, you can import:

- Mapping
 - [Message Mapping](#)
 - [Operation Mapping](#)
 - Value Mapping
- Schema
 - WSDL

! Restriction

- Import of operation mapping fails if:
 - It contains XSLT references.
 - The interfaces are asynchronous in type.
- The import process fails if the message mapping contains:
 - Imported archives
 - Parameters
- Importing, viewing, and editing of Java UDF is supported.

- Message Mapping containing reference to Value Mapping isn't supported at runtime.
- Import of a message mapping object fails if it contains a XSD in the source or target message.

Here's how you can do it:

Procedure

1. Choose **References** > **Local** tab in the integration flow properties sheet.

If you can't find the **References** tab, choose the empty space in the integration flow editor. You will find the **References** tab at the end of the integration flow editor.

2. Choose **Add** and select a supported resource. See the supported sources here: [Manage References](#).
3. Choose **ES Repository** as the source.
4. Select a ES Repository from the **Name** list. Choose **Connect**.

A connection is established to the ES Repository. A table with the list of available content is displayed.

5. Choose the resource that you want to import and choose **Select**.

A summary of all the artifacts that are available for import is displayed.

6. Choose **Add**.

You see a confirmation about all the artifacts that are successfully imported.

Creating Custom Tags

You can maintain custom attributes in your editable integration packages.

You can tag your integration package using custom attributes that can be created under the **Settings** tab. The tags can help you identify and track your packages based on attributes, for example, such as Author, Line of Business etc.

→ Remember

You must have the tenant administrator role assigned to access the **Custom Tags** settings. See [Tasks and Permissions](#).

1. Navigate to the **Settings** (⚙️) in the web UI and choose **Custom Tags**.
2. Choose **Edit** and select **Add** to add custom attributes.
3. Enter a name for the tag under **Tag Name**.
4. By default, the **Mandatory** check box is enabled to enforce integration developers to add specific values before saving the package. Uncheck the check box if you don't want to set your tag mandatory.
5. In the **Permitted Values** field, type in the possible custom values from which you want the integration developers to choose. For example, with a custom tag name **Department**, you can provide permitted values like **HR**, **Finance**, **Sales**, and **Marketing**. Leave the field blank, if you want the integration developers to type their own values.
6. Choose **Save**. You can't rename the tags after saving them. The tags become read-only.

i Note

- After adding the necessary custom tags, you can inform the integration developers to maintain the custom tags in the existing packages.

- For transport scenarios, the custom tags can be exported from source tenants and imported into the target tenant before the package transport, to reflect the custom tag values maintained at source.

The integration developers can now start maintaining the attribute values under the **Tags** tab in the newly created packages. For existing packages without the attributes, the values can be added by editing the package or using OData API.

To know how to work with custom tags using OData API, see [Working with Custom Tags](#) and [Create New Custom Tags Configuration](#).

Saving of an edited package or a newly created package fails if the mandatory attributes aren't maintained.

i Note

Deleting a custom tag in the **Settings** that is already maintained in a package only hides the tag from the **Tags** section of the package. But the value of the tag along with the tag name is persisted in the package for easy retrieval. Read this [blog](#) to know more about custom tags and their behaviours.

Malware Scanner

Scan your design time artifacts for malware before the users upload them to the tenant.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Users of the tenant upload various design time files like integration artifacts and their resources to the tenant. Such files are prone to malware attacks, which in turn, compromises the security of the tenant.

SAP Cloud Integration provides a malware scanner that scans such files when a user attempts to upload them. By default, the scanner is disabled. You can enable the option under **Settings** (selecting the **Malware Scanner** tab). With malware scan enabled, files of larger size can take a little longer time to complete the upload than when it's disabled. The files that are supported by the malware scanner are:

- Documents that you upload within a package.
- Integration artifacts like integration flows, value mappings, script collection artifacts, and API-based integration flow and their resources.
- Integration packages
- Keystore files like – X.509 certificate, key pair, and keystore.
- BinaryParameters resources when created for the Partner Directory using the OData API.

i Note

Currently, JDBC drivers aren't under the purview of malware scanner.

Though not supported by the malware scanner, security materials like Known Host files, PGP public keyrings, and PGP secret keyrings are validated for format during upload. This format check prevents the upload of malicious content.

To enable the malware scanner:

1. Go to **Settings**.

2. Move to the **Malware Scanner** tab and choose **Edit**.

3. Enable the **Malware Scan** feature and choose **Save**.

Design Guidelines

As the integration lead for your organization, help your integration developers design enterprise-grade artifacts.

What are Design Guidelines

Design guidelines for artifacts enable your integration developers to design artifacts in a robust fashion to safeguard your business processes.

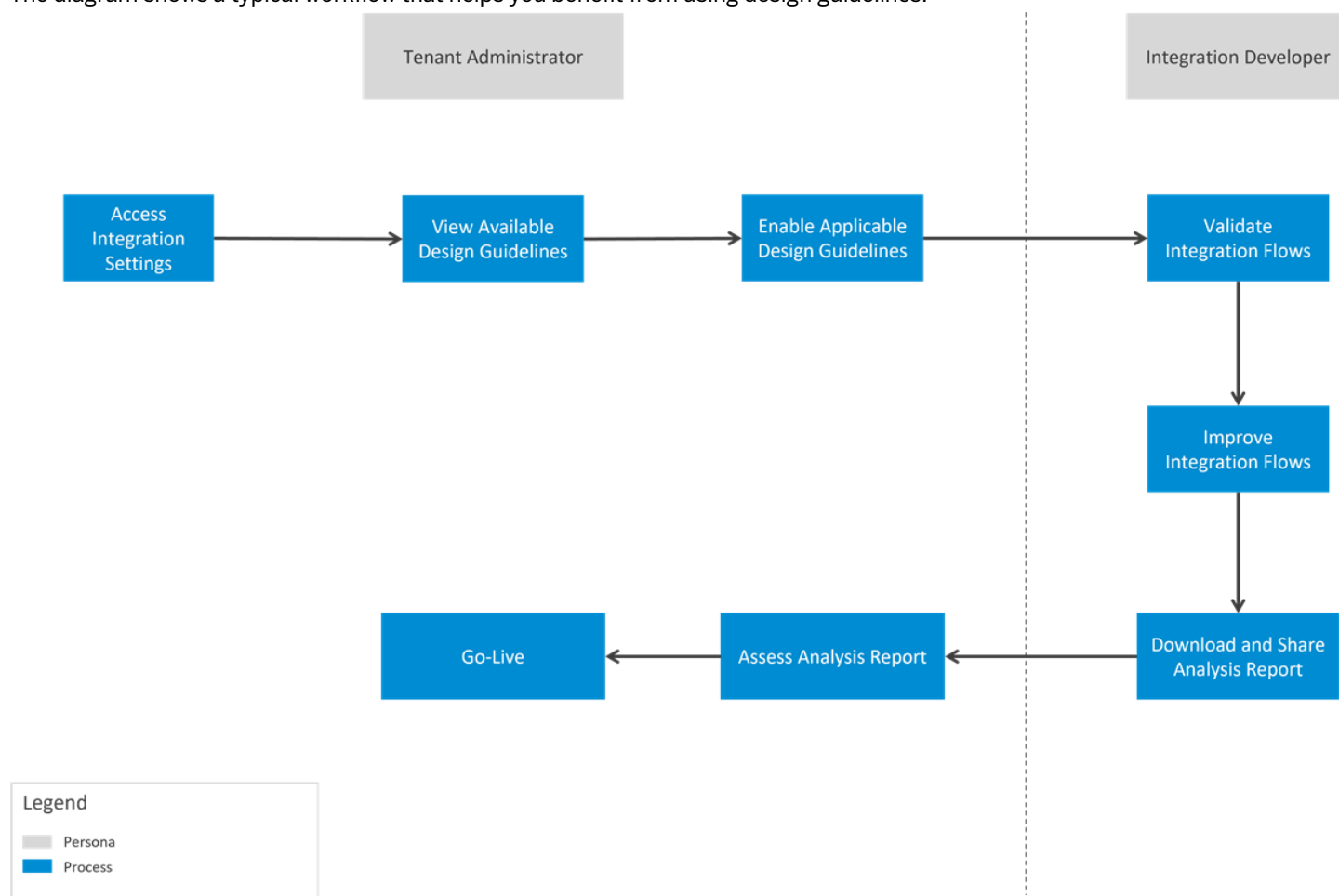
In the past, these design guidelines were recommendations that were published as product help and prepackaged content on SAP Business Accelerator Hub. You can find the guidelines in the [Integration Flow Design Guidelines](#) section. Now, the same guidelines are embedded within the software as rules that help integration developers.

Why Design Guidelines

Each design guideline acts as a rule that helps integration developers design an improved artifact. For example, keeping your artifact readable, easy to understand, avoid memory leaks and performance issues, handling attachments in the right way, applying the appropriate security standards, or implement a specific pattern. Enable all or a subset of available design guidelines that you think are applicable for your organization's business needs.

The design guidelines are logically grouped. For example, all transaction handling related guidelines are grouped under a single category. Some of the logical groupings are: **Handle errors gracefully** and **Run an integration flow under well-defined boundary conditions**.

The diagram shows a typical workflow that helps you benefit from using design guidelines:



Enable Design Guidelines

Before you enable, understand every design guideline and its implications. Consume the in-app help available for each design guideline to learn more.

Follow these simple steps to enable or disable all or a subset of available design guidelines:

1. Go to **Settings > Integrations > Design Guidelines**.
2. Choose **Edit**.
3. Use to toggle in the **Activate** column to enable the design guidelines that you think are appropriate for your organization's requirements. Similarly, disable the ones that you think aren't appropriate anymore.

i Note

Consider the severity of each guideline to take this decision. Severity indicates how strongly it should be followed.

4. Choose **Save**.

You can enable or disable a design guideline anytime.

→ Tip

As a tenant administrator, you can assign a dedicated Integration Lead to enable the applicable design guidelines. You can [define a custom role collection](#) in the SAP BTP cockpit and add the role `WorkspaceDesignGuidelinesConfigure` to it. Later, assign this role collection to a user.

Harmonize All Tenants

It's critical that the various development tenants that you've follow the same design guidelines so that the experience of artifact validation is the same. To achieve this experience, you can export the design guidelines with the enabled/disabled status from one tenant and import the same in another tenant. This way, you don't have to manually enable the same design guidelines in multiple tenants.

Follow these simple steps to export and import the design guidelines with their status:

1. In your source tenant, go to **Settings > Integrations > Design Guidelines**.
2. Choose **Export** and save the file to your local drive.
3. In your target tenant, go to **Settings > Integrations > Design Guidelines**.
4. Choose **Import** and select the exported file.

The design guidelines with their status from the source tenant are maintained in your target tenant now.

Review and Productize Artifacts

After you enable design guidelines, all artifacts designed in your tenant must abide by the enabled design guidelines. As the integration lead, you can review the design guidelines report for each artifact, and then make an informed decision to go-live with every new artifact.

If you follow a practice of creating an artifact in a development tenant and then try out in a production tenant, you can transport the validated artifact or package from one tenant to other.

→ Remember

The design guidelines don't disrupt the execution of the existing and new artifact.

For more information on how integration developers can validate their artifacts, see [Design Guidelines View](#).

→ Remember

You can use design guidelines to validate not only your custom integration content but also the prepackaged integration content that is shipped by SAP.

Related Information

[Integration Flow Design Guidelines](#)

Understanding the Basic Concepts

This section provides an overview of the concepts and terms.

Runtime in Detail

For different customers, separate resources (in terms of: memory, CPU, file system) of the cloud-based integration platform are allocated – although all customers share the same hardware. This concept is also referred to as tenant isolation.

At runtime, data is processed on a tenant allocated for a connected customer.

i Note

A tenant represents the resources of the cloud-based integration platform allocated for a customer. Typically one tenant is defined for each customer connected to the platform.

It is always made sure that each tenant uses a separate database schema, which guarantees that data of different customers is strictly separated. The architecture guarantees that different tenants are unable to interfere and that physical resources of the cloud platform are partitioned per tenant.

For more information on the architecture, see:

- [Technical Landscape, Cloud Foundry Environment](#) (for the Cloud Foundry environment)
- [Technical Landscape, Neo Environment](#) (for the Neo environment)

Virtual System Landscapes

There are different virtual system landscapes.

Landscapes, Virtual Servers and IP Addresses

For the list of available landscapes and respective IP addresses, see: [Regions and Hosts Available for the Neo Environment](#).

Ports for Outbound Communication

When configuring an **outbound** channel (receiver adapter), make sure to use the following standard ports:

- Outbound **HTTP/HTTPS** connections: By default, **port 443 and all ports > 1024** are opened in the SAP BTP firewall.

In case of any issues, open a ticket.
- For **SFTP** connections, make sure that the SSH data channel between SAP Cloud Integration and the SFTP server is opened.
Use **port 22** at SAP BTP side.
- For **SMTP** connections, use **port 25**.

IP Addresses for Trial (Cloud Foundry Environment)

When using SAP Cloud Integration with an SAP BTP trial account, the following landscapes are available with the following IP addresses:

Region / SAP Cloud Integration Domain	IPs for outbound requests from SAP Cloud Integration	Load balancer IPs for inbound requests to SAP Cloud Integration
*eu10-001.hana.ondemand.com	3.123.188.210	3.124.250.14
	3.124.136.11	18.185.50.121
	3.124.47.101	3.124.200.210
	3.67.214.188	
	3.122.185.150	
	3.66.178.54	
	3.66.41.248	
	3.67.212.190	
	3.65.22.214	
	3.67.244.246	
	52.57.199.163	
	3.121.5.26	
	3.67.214.217	
	35.157.77.220	
	18.157.166.207	
	3.67.217.171	
	3.67.50.130	
	3.66.207.74	
	3.66.114.211	
	35.158.167.12	
	3.65.56.47	
	3.67.164.208	
	3.66.111.58	
	3.124.140.47	

Region / SAP Cloud Integration Domain	IPs for outbound requests from SAP Cloud Integration	Load balancer IPs for inbound requests to SAP Cloud Integration
	3.67.203.162 35.157.51.108 18.195.153.225	
*.eu10-002.hana.ondemand.com	18.198.196.89 18.193.21.232 3.65.9.91 18.197.134.65 52.29.190.137 3.67.255.232 3.67.182.154 3.68.44.236 3.66.249.150	3.64.227.236 3.126.229.22 18.193.180.19
*.us10.hana.ondemand.com	52.200.16.71 52.23.123.125 52.202.170.155 18.211.235.11 54.156.172.106 34.234.191.59 34.192.134.47 18.204.173.15 3.213.197.54 184.73.43.82 18.210.47.160 3.216.16.207 34.225.190.250 52.2.110.230 54.234.93.200 35.153.88.132 52.204.111.138 3.88.250.160 52.20.242.182 52.71.83.110 52.200.165.163 54.208.119.130	52.23.189.23 52.4.101.240 52.23.1.211

Region / SAP Cloud Integration Domain	IPs for outbound requests from SAP Cloud Integration	Load balancer IPs for inbound requests to SAP Cloud Integration
	34.202.136.35 34.192.100.96 54.85.65.82 54.205.71.200 54.221.30.9	
*.us10-001.hana.ondemand.com	18.213.153.162 52.0.214.195 54.227.144.195 34.233.151.91 3.225.73.158 3.222.22.16 23.23.172.117 34.194.239.31 34.238.1.234	
*.ap21.hana.ondemand.com	40.90.179.153 20.198.169.214 20.198.168.45 20.198.169.5 40.90.170.226 20.198.225.78 20.198.225.102 20.198.225.27 40.90.162.117 20.191.154.174 20.191.154.191 20.191.154.193	20.184.61.122

i Note

The **IPs for outbound requests from SAP Cloud Integration** addresses are related to **outbound communication**, that means: for calls from SAP Cloud Integration to a receiver system. They are required by the customers to configure the firewall settings (*IP allowlisting*) to enable their system to connect to the cloud-based integration platform.

Related Information

[Tool Access](#)

Stable URL

You can use this URL to access the tenant.

Management URL

Tenant	SSL Host	URL
Productive	HCI	https://<Account Short Name>-tmn.hci.<Landscape Host> For example, https://I0001-tmn.hci.eu1.hana.ondemand.com
Test	HCI	https://<Account Short Name>-tmn.hci.<Landscape Host>

Runtime URL

Tenant	SSL Host	URL
Productive	HCISBP	https://<Account Short Name>-iflmap.hcisbp.<Landscape Host> For example, https://I0001-iflmap.hcisbp.eu1.hana.ondemand.com
Test	HCISBT	https://<Account Short Name>-iflmap.hcisbt.<Landscape Host>

i Note

- In the URL, account short name refers to account detail mentioned in SAP email. You receive this email after onboarding the tenant. Also, you can access the following link to see the list of available landscapes and respective IP addresses: [Landscape Hosts](#).
- In SAP email, account short names begin with a letter followed by 4-5 digits. For example, **A0001**.

Configuring Custom Domains

Allows you to customize the default tenant URL or domain as per your needs and access the tenant using your own the domain.

There are use cases where you do not wish to expose the default domain provided by SAP Cloud Integration. In such a scenario, you can construct custom domain names and TLS settings for specific Cloud Integration instance.

Use the procedure here to create and add custom domains.

i Note

When a custom domain is used, the domain name and the server certificate of the domain are owned by the customer.

1. To use a custom domain for your application, you must fulfill a number of preliminary steps. For more information, see [Prerequisites](#).
2. [Create an SSL Host](#) - the host holds the mapping between your chosen custom domain and the application on SAP BTP as well as the SSL configuration for secure communication through this custom domain.

3. [Upload a Certificate](#) - it will be used as a server certificate on the SSL host.
4. [Bind the Certificate to the SSL Host](#).
5. [Add the Custom Domain](#) - this maps the custom domain to the Cloud Integration URL.

i Note

The format of the default application URL is `https://<application_name><provider_subaccount>-<consumer_subaccount>.<domain>`.

You can find the required components for the above URL in the SAP BTP Cockpit:

URL Elements	Where to find it?
<application_name>	In Application Subscriptions find the subscribed application details.
<provider_subaccount>	In Application Subscriptions and then choose the relevant Application . In the Subscribed Application - <account ID> Overview page, find the Provider Subaccount details. For example, Provider Subaccount: myacct (wc077ad).
<consumer_subaccount>	Under the Subaccount Information section, use the subaccount Name .

i Note

After creating an custom domain, you can use both the default URL as well as the custom domain to access the application.

6. [Configure DNS](#)- you can create a CNAME mapping.
7. [Configure Single Sign-On](#) - if you have a custom trust configuration in your subaccount, you need to enable single logout.

The configuration of custom domains has different setups related to the subscriptions of your subaccount. For more information about custom domains for applications that are part of a subscription, see [Custom Domains for Multitenant Applications](#).

Enabling TLS Protocol Versions

Configure and enable TLS protocol of your choice for inbound communication to SAP Cloud Integration. For more information, see [How to replace SSL with TLS](#).

i Note

Use `--supported-protocols` parameter to enable the specified TLS protocol.

Setting-Up Custom Domains in CF

Allows you to customize the default tenant URL.

There are use cases where you don't wish to expose the default domain provided by SAP Cloud Integration. In such a scenario, you can construct custom domain names and TLS settings for a specific Cloud Integration instance. In the following you can configure a custom domain for accessing the runtime worker, for example `runtime.subdomain.mydomain.com`.

i Note

When you've set up a custom domain, there are currently the following limitations:

- To access Cloud Integration through your browser (to perform tasks such like integration flow design, monitoring, for example), you need to use the URL that contains the original SAP domain provided when subscribing to Cloud Integration service. Currently, it's not possible to access the Cloud Integration user interface through an address that contains your custom domain.
- You can use runtime URLs containing the custom domain to call integration flows with HTTP-based sender adapters. However, when checking the endpoint URL of your integration flows in the **Monitor** application (under **Manage Integration Content**), the original SAP domain is shown in the base address of the endpoint URL, not the custom domain. If you develop a workaround to relate both endpoint URLs with each other, note that you know the mapping between both: You anyway need to execute this mapping when setting up the custom domain.

1. Initial Steps:

To use the Custom Domain Service you have to complete some initial steps, described here: [Initial Setup](#).

2. Prerequisites

Before configuring custom domains, you have to fulfill a number of prerequisites. See: [Prerequisites](#).

3. Configuring Custom Domains

- To get a better overview of the creation process of custom domains with TLS/SSL server authentication, see: [Creating Custom Domains with TLS/SSL Server Authentication](#).
 - To create custom domains, see: [Create Custom Domains](#).
 - To create a certificate signing request, see: [Create a Certificate Signing Request](#). We recommend to create a wildcard certificate for both custom domains.
 - To upload and activate the signed certificate, see: [Upload and Activate the Signed Certificate](#).
 - To configure the DNS for a custom domain, see: [Configure the DNS for a Custom Domain](#).
- To map the Cloud Integration instance to a custom domain, see: [Map a SaaS Application to a Custom Domain](#).

Mapping example for the runtime:

```
cf custom-domain-map-route <tenantname>.landscape-runtimeid.cfapps.datacentername.hana.on
```

- To test your custom domain, and make sure it's reachable and secured by a certificate, send a *postman* request to the runtime URL.

Remember that certain steps during the setup process of your custom domain can take up to 24 hours and the DNS configuration can take several hours depending on your provider.

- To add trusted certificates for client authentication to a custom domain, see: [Adding Trusted Certificates for Client Authentication to a Custom Domain](#).

4. Managing Custom Domains

To manage custom domains, check the steps described in: [Managing Custom Domains](#).

5. Removing the Custom Domain Service Instance

To remove the Custom Domain Service Instance, see [Remove the Custom Domain Service Instance](#).

If you encounter issues with your custom domains, we recommend that you follow the procedure described in [Monitoring and Troubleshooting](#).

Application URLs, Neo Environment

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Application URLs provide various options to connect to the SAP Cloud Integration virtual environment.

Application URLs are defined both for the tenant management node application and for the runtime node application.

The following table provides a summary of the available application URLs:

Application URLs

Node Type	Application URL	Description
Tenant management node	<code>http://<tenant management node address>/Operations</code>	Operations URL to be used when connecting to the tenant using Eclipse (for example, when using the Adapter Development Kit)
Tenant management node	<code>http://<tenant management node address>/api</code>	Base URL to access the SAP Cloud Integration application programming interface (API) Note that the SAP Cloud Integration OData API is published on the SAP Business Accelerator Hub at: https://api.sap.com/shell/discover/contentpackage/CloudIntegrationAPI
Tenant management node	<code>http://<tenant management node address>/itspaces</code>	Web UI URL for the tenant (for dialog users to design integration flows, monitor message processing, and execute additional tasks within an integration project)
Tenant management node	<code>http://<tenant management node address>/cxf</code>	Endpoint address under which SAP Solution Manager can retrieve alert notifications (if configured accordingly)
Runtime node	<code>http://<runtime node address>/as2</code>	AS2 sender adapter endpoint base URL
Runtime node	<code>http://<runtime node address>/cxf</code>	SOAP sender adapter endpoint base URL
Runtime node	<code>http://<runtime node address>/gw/odata</code>	OData endpoint URL
Runtime node	<code>http://<runtime node address>/http</code>	HTTPS sender adapter endpoint base URL For example, assume that you have configured an integration flow with an HTTPS sender adapter and as Address you have specified the following value: <code>/myEndpoint</code> . Then an HTTP client can call this integration flow through the following endpoint address: <code>http://<tenant address>/http/myEndpoint</code>

i Note

To display the available application URLs, open the subaccount, select ► **Applications** ► **Subscriptions** ►, and then the required application (either the tenant management node application ending with **tmn** or the runtime node application ending with **iflmap**).

Overview

Destinations

Roles

Home [Europe (Rot)] / [redacted] / [redacted] / [redacted]

Subscribed Application: [redacted] - Overview

Subscribed Java Application

Provider Subaccount: [redacted]

Subscribed Application: [redacted]

Application State

✓ Started [redacted]

Application URLs

https://[redacted].com/

https://[redacted].com/as2

https://[redacted].com/as4

https://[redacted].com/cxf

https://[redacted].com/gw

https://[redacted].com/http

User Management for SAP BTP, Neo Environment

Users management tasks are required for different activities associated with the setup and operation of the cluster.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

- To create users, you register at blogs.sap.com (👤).
- Managing user-to-role assignments (using SAP BTP cockpit).

This task includes initial activities by the tenant administrator to define authorizations for people who are supposed to work on the tenant cluster.

Related Information

[Creating a User for Cloud Integration, Neo Environment](#)

[Managing Users and Role Assignments, Neo Environment](#)

Creating a User for Cloud Integration, Neo Environment

For several tasks associated with SAP Cloud Integration, you need a user account on blogs.sap.com.

Context

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

There are two types of users who can access SAP Cloud Integration:

1. **S-User:** If you have an S-User ID, you are automatically a member of blogs.sap.com.
2. **P-User:** If you do not have an S-User ID, you can create a public user (P-User) account directly by referring to this [blog](#) .

After you have created a new user account, you can find your user ID in your account settings.

Managing Users and Role Assignments, Neo Environment

You specify the members of the account and assign roles to them.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

If this function is not available for your account, ask your SAP contact to activate it.

This task includes initial activities by the tenant administrator to assign roles to the users associated with all people who are supposed to work on the tenant cluster.

The task is sub divided into the following main activities:

- Adding members to the account:

With this step, you specify all users who should have assigned the same role like the tenant administrator

In addition to that, you can also define all users who should have a *restricted* tenant administrator role (on the account). This specific role allows you to assign application-specific roles to other users of the account (like, for example, the integration developer role).

- Defining authorizations for individual integration team members:

With this step, you assign the actual application-specific roles to the integration team members (like, for example, the integration developer role)

To perform these steps, open the SAP BTP cockpit using the S-user ID provided to you by SAP (in the mail that contains also the information on the account).

The URL of the SAP BTP cockpit depends on the data center.

SAP BCP Cockpit URLs

Region	URL
Europe (Rot)	https://account.hana.ondemand.com/cockpit
US East (Ashburn)	https://account.us1.hana.ondemand.com/cockpit
Australia (Sydney)	https://account.ap1.hana.ondemand.com/cockpit

Related Information

[Adding Members to an Account](#)

Adding Members to an Account

You specify the members of the account to define who is involved in an integration project.

Context

Perform the following steps.

Procedure

1. To specify all users who should get assigned the tenant administrator role, open the SAP BTP cockpit using the S-user ID provided to you by SAP (in the mail that contains also the information on the account).
2. Choose **Members** ►► **Add Members** ►.
3. Enter the user ID.
4. Select the role which should be assigned to the user.

Add Members

Enter up to 100 SAP user IDs. To separate them, you can use commas, spaces, semicolons.

User IDs:

myUserID

Comment (optional, visible to all members):

Assign roles:

☒ Administrator

☐ Developer

☐ Support User

☐ Application User Admin

☐ Cloud Connector Admin

Add Members **Cancel**

Assign role **Administrator** to the user who is supposed to have the full permissions of an administrator.

Assign role **Application User Admin** to the user who is supposed to have restricted administrator permissions.

i Note

These roles are **not** specific to any tasks related to integration projects.

When you define more specific authorizations for integration team members, you assign other roles (as described in the corresponding section).

5. Choose **Add Members**.

Related Information

[Defining Authorizations for Integration Team Members](#)

[Personas](#)

[Support Tasks](#)

Defining Authorizations for Integration Team Members

To authorize selected people to work on the account as part of the integration team in the context of SAP Cloud Integration (for example, as integration developers), you assign roles to the associated users.

Context

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

Authorizations can also be given to users that are not associated with any integration team member (for example a user associated with a system that is to be authorized to send a message to Cloud Integration).

Perform the following steps for all users for which to assign authorizations.

i Note

You can also define **Role Collections** and assign roles to role collections. That way, you can assign roles at once for all users who should to get identical permissions on the account.

i Note

SAP has provided you with the link to access SAP BTP cockpit.

Procedure

1. In SAP BTP cockpit, select your subaccount and then choose **Security > Authorizations** . Enter your user ID, and choose **Assign**.
2. In the navigation pane, choose **Security > Authorizations** .
3. If you like to first create a user group, on page **Authorization Management** choose the tab **Groups**.
Enter a group name and add the relevant users to it (as shown in the figure).
4. On page **Authorization Management** enter the user group or user for which you like to assign authorization groups.

Authorization Management

Users Groups Token


Assign Web Roles and Groups to Individual Users

User: *

Roles	Assign	Unassign All	Groups	Assign	Unassign All
Account	Application	Role	Group		

5. Choose **Assign**.

6. On page **Assign Roles to User <User Name>** choose the authorization group to be assigned.

Assign roles to user DI <User Name> ✕

Subaccount:

Application:

Role:

Note: Changes will affect new sessions only.

You can assign groups of roles (also referred to as **authorization groups** and starting with **AuthGroup**) or individual roles. Authorization groups cover the different tasks associated with an integration project.

As **Application**, choose the one which ends with **tmn** (for tenant management node).

7. Choose **Save**.

Related Information

[Personas](#)

Defining Permissions for Senders to Process Messages on a Runtime Node

Context

→ Remember

There are currently certain limitations when working in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#).

i Note

SAP has provided you with the link to access SAP BTP Cockpit.

i Note

In order to authorize a sender system to call a tenant (using HTTPS/basic authentication) and to get messages processed on the tenant, you need to assign to the associated technical user the specific role **ESBmessaging.send**.

Procedure

1. In SAP BTP cockpit, select your subaccount and then choose **Security > Authorizations**. Enter your user ID, and choose **Assign**.
2. In the navigation pane, choose **Security > Authorizations**.
3. On page **Authorization Management**, enter the user for which you like to define the permission.

Authorization Management

Users Groups Token

Assign Web Roles and Groups to Individual Users

User: *

Roles	Assign	Unassign All
Account	Application	Role

Groups	Assign	Unassign All
Group		

4. Choose **Assign**.
5. As **Application**, choose the one, which ends with **iflmap** (corresponding to a runtime node of the cluster that actually is in charge of processing the message).
6. Select the role **ESBmessaging.send**.

Assign roles to user myUser ✕

Account: ▼

Application: ▼

Role: ▼

Note: Changes will affect new sessions only.

Save
Cancel

Next Steps

i Note

When configuring inbound authorization settings in the corresponding sender adapter, you can keep the role name **ESBmessaging.send** pre-entered by default in the **User Role** field (when you have for **Authorization** selected the option **User Role**).

Alternatively, you can define a custom role for the runtime node by performing the following steps:

1. Select your subaccount.
2. Choose **Subscriptions**.
3. Under **Application**, select the one ending with **iflmap** (for your runtime node).
4. Choose **Roles**.
5. Choose **New Role** and enter the name of your custom role.

i Note

When creating a new user role, be aware of the fact that for role names in the Cloud Foundry environment there are certain restrictions. In the Cloud Foundry environment, only the following characters are allowed in user role names:

- a-z
- A-Z
- 0-9
- _
(underscore)
- .
(period)

If you don't consider these restrictions when creating a new role in the Neo environment (as described here), you run into problems when migrating your integration content to the Cloud Foundry environment.

i Note

When configuring inbound authorization settings in the corresponding sender adapter, you have to enter this role name in the **User Role** field (when you have for **Authorization** selected the option **User Role**).

Activating Enterprise Messaging

Activate SAP Event Mesh.

Prerequisites

i Note

This procedure is only relevant when you use SAP Cloud Integration in the Neo and Cloud Foundry environment. JMS queues are included in SAP Integration Suite without the need for an additional license, except in Basic Edition (where the feature is not supported).

- You must have purchased a minimum of 10 SKUs of SAP Event Mesh, in which five SKUs are consumed by Message Monitor and remaining five SKUs are used to activate JMS capabilities.
- Contact your Account Executive for more details and for purchasing additional SKUs for SAP Event Mesh.

Context

Use procedure to activate SAP Event Mesh for consuming JMS capabilities.

Procedure

1. The following steps are executed by a global account administrator:
 - a. Log into the accounts cockpit (SAP BTP cockpit).
 - b. Choose a global account.
 - c. Navigate to the **Subaccounts** tab.
 - d. Choose a subaccount.
 - e. Navigate to the **Entitlements** page.
 - f. Choose **Edit** to assign the required quota to a relevant subaccount in the **Event Mesh** section.

i Note

The quota must be in 1:5 ratio, where one messaging queue on Cloud Integration corresponds to five quotas.

- g. Saves the changes before modifying the quota for other subaccounts.

2. Execute following steps to assign tenant administrator role:
 - a. Navigate to the **Subscription** tab.
 - b. Choose **Provision** application for **hcisvc** provider account.
 - c. Navigate to **Roles** tab.
 - d. Assign **AuthGroup.Administrator** authgroup to activate the application.

3. The following steps are executed by Tenant Administrator to activate SAP Event Mesh and to modify the messaging queues:

- a. Navigate to the account cockpit.
- b. Navigate to **Services** tab.
- c. Choose the **Process Integration** tile.
- d. Choose the **Configure Process Integration** link under **Take Action**.
- e. Choose the **Event Mesh** tab, and then choose **Activate** to start the service.

i Note

- **Google Chrome, Firefox and Safari** web browsers support provisioning application.
- Choose **Deactivate** to deactivate the messaging service.

Perform the steps here to increase the SAP Event Mesh queues for a specific subaccount.

- a. Communicate with the Global Account Administrator to increase the quota to the relevant subaccount.
- b. In the **Cloud Integration** application, select **Event Mesh** tab.
- c. Choose **Edit** to increase the message queues.

Perform the steps here to decrease the SAP Event Mesh queues for a specific subaccount.

- a. Decrease the message queues by selecting **Event Mesh** and choose **Edit** to decrease the message queues.

i Note

You must remove unused JMS queues from your tenant before decreasing the message queues. For more information, see [Managing Message Queues](#).

- b. Communicate with the Global Account Administrator to decrease the quota to the relevant subaccount in the **Entitlements** page of the account cockpit.

For more information, read the blog on [Activating and Managing Enterprise Messaging Capabilities](#) .

Monitor

Check the status of messages and integration content artifacts for a tenant cluster.

The start page is subdivided into the following sections, each covering a specific task area.

Sections on Monitor Start Page

Container	Allows you to ...
Monitor Status Overview	Monitor the status overview of your messages according to your custom filters.
Monitor Message Processing	Monitor message processing on the tenant. Tiles in this section show the number and status of processed messages within a specified time window.
Manage Integration Content	Manage integration content for the tenant. Tiles in this section show the number and status of integration content artifacts (such as integration flows). In the Cloud Foundry environment, you can also specifically monitor the Integration Adapter .

Container	Allows you to ...
Manage Security	Manage security artifacts for the tenant. Tiles in this section allow you to manage certain tasks related to the setup of secure connections between your tenant and remote systems. The Security Material tile provides access to and allows you to deploy security-related artifacts such as user credentials artifacts. The Keystore tile provides access to the content of the tenant keystore and allows you to manage its content and also the lifecycle of keys and certificates. The PGP Keys tile allows you to check the information of the PGP keys. The Access Policies tile provides an overview of the existing access policies and allows you to maintain them. The Certificate-to-User Mappings tile allows you to manage certificate-to-user mappings (relevant for the setup of inbound connections). i Note Certificate-to-User Mappings are only available in the Neo environment. The JDBC Material tile provides an overview of the used JDBC data sources (artifact connections interacting with a database) as well as of the JDBC drivers. The User Roles tile provides an overview of the existing user roles and allows you to maintain them. i Note User Roles are only available in the Cloud Foundry environment. The Connectivity Tests tile allows you to test the connectivity to a receiver system.
Manage Stores	Manage temporary data storages on the tenant. The Data Stores tile provides an overview of storages on the tenant, which are temporarily used to persist data of different kind during message processing (when using the Data Store Operations step type). The Variables tile allows you to monitor variables used in integration flows. The Message Queues tile provides an overview of the active queues of a tenant and allows you to manage them i Note You can only monitor message queues, if Enterprise Messaging is activated and a JMS Message broker is provisioned. (See: Activating Enterprise Messaging)

Container	Allows you to ...
	The Number Ranges tile provides an overview of number ranges that are used in business-to-business scenarios.
Access Logs	Access the System Log Files and to analyze errors that occurred during inbound HTTP processing (and documented in system log files). i Note In the Cloud Foundry environment, you can access the system logs only if the Cloud Logging Service is available for your Cloud Integration application. This feature isn't yet enabled on all data centers. To access audit logs (resulting from system changes) i Note Audit logs are only available in the Neo environment.
Manage Locks	Manage message lock as well as the design artifact lock entries. The Message Lock tile provides an overview of the lock entries that are created (in the in-progress repository) to avoid the same message being processed several times in parallel, and allows you to manage them. The Designtime Artifact Locks tile provides an overview of the locked design time artifacts and allows you to manage them.

Each section contains tiles, which show filter settings and the number of messages/artifacts that correspond to the filter settings.

Selecting a tile opens a page with more information.

The start page is automatically refreshed every 5 seconds. The autorefresh ends after 5 minutes (indicated by a message).

i Note

All times displayed on the Monitoring pages are local times (with regard to the client of the Web application).

Managing Tiles

You can customize the start page to fit your personal needs by adding new tiles (by selecting an empty tile containing a + symbol) or deleting existing ones. You can also rearrange the start page by dragging and dropping tiles to another location. For example, you can customize the start page so that it shows only data related to integration flows that are edited by you.

i Note

You can rearrange tiles only within the same section.

You can add, delete, or edit tiles in the following section types:

- **Monitor Message Processing**
- **Manage Integration Content**

i Note

The personal settings are persisted in the client of the user. So, you can log out, close the browser, and log on again to the monitoring, and the previously specified user settings are kept.

To edit a tile, right-click the tile and choose **Edit**.

When you add or edit a tile, you can specify the following filter categories:

- For **Monitor Message Processing** tiles you can specify **Status**, **Time**, and the related **Integration Artifact** or **Package** for the messages to be displayed.

In the dropdown list for the **Artifact**, the display name and type are shown. The tooltip for an entry additionally shows the technical name and the package (if available) of the corresponding integration artifact.

- For **Manage Integration Content** tiles, you can specify the **Status**, and the **Type** of the integration artifact to be displayed.

Related Information

[Monitor Message Processing](#)

[Manage Integration Content](#)

[Manage Security](#)

[Managing Security Material](#)

[Managing Keystore Entries](#)

[Managing PGP Keys](#)

[Managing Certificate-to-User Mappings, Neo Environment](#)

[Creating Custom Roles for Access Policies, Neo Environment](#)

[Creating Custom Roles for Access Policies, Cloud Foundry Environment](#)

[Managing User Roles, Cloud Foundry Environment](#)

[Managing JDBC Material](#)

[Performing Connectivity Tests](#)

[Manage Stores](#)

[Managing Data Stores](#)

[Managing Variables](#)

[Managing Number Ranges](#)

[Access Logs, Neo Environment](#)

[Monitoring Audit Log, Neo Environment](#)

[Access Logs, Cloud Foundry Environment](#)

[Manage Locks](#)

[Message Locks](#)

[Design-time Artifact Locks](#)

Monitor with AI

Optimize the monitoring of your artifacts with artificial intelligence.

i Note

Availability of this feature depends upon the SAP Integration Suite service plan that you use. For more information about different service plans and their supported feature set, see SAP Note [2903776](#).

The following AI features are available in the **Monitor** section:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

Feature	Description	More Information
AI-assisted Error Resolution	Analyze and resolve message processing errors using AI.	AI-assisted Error Resolution

AI-assisted Error Resolution

Use artificial intelligence to analyze and resolve message processing errors efficiently.

Prerequisites

- Enable the **AI-assisted Error Resolution** feature in your SAP Integration Suite tenant. For more information, see [Artificial Intelligence](#).
- Ensure that you have the role `PI_Integration_Developer` or `PI_Administrator` assigned.

Context

You can use AI assistance to analyze failed message processing logs. The AI assistant identifies the main error cause and proposes a solution.

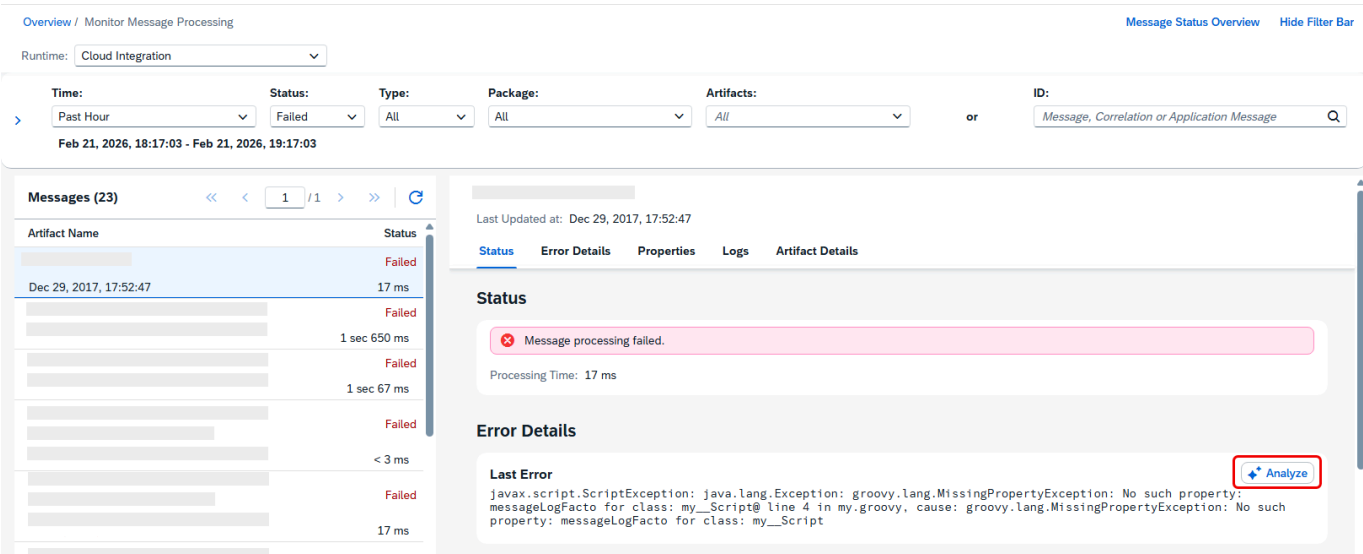
→ Remember

This feature is only available for **Failed** MPLs.

To access the message monitor, select a tile in the **Monitor Message Processing** area. The system displays messages based on the tile's filter settings.

Procedure

1. Select a failed message from the list to view more information.
2. In the **Error Details** section, choose the **Analyze** button to start the error analysis.



i Note

AI analysis may not run if users reach the usage quota. A fair usage limit is available each month. Every AI analysis request uses quota, and this quota resets monthly.

3. The AI agent analyzes the error and displays the following results:

- **Root Cause:** The primary reason for the error, based on the analysis of the failing step.
- **First Failed Step:** The earliest step where the processing stopped. Use this as the starting point for troubleshooting. This field is specifically helpful in case of parallel processing flow steps, such as parallel multicast, parallel splitting, and aggregation.
- **Suggested Solution:** An actionable recommendation to resolve the issue.

❖ Example

The screenshot displays the SAP Message Processing Monitor interface. At the top, there's a navigation bar with 'Overview / Monitor Message Processing' and links for 'Message Status Overview' and 'Hide Filter Bar'. Below this is a filter bar with 'Runtime: Cloud Integration' and a dropdown menu. The main area is divided into two sections. The left section, titled 'Messages (23)', shows a table of messages with columns for 'Artifact Name', 'Status', and 'Time'. The right section, titled 'Error Details', shows the analysis of a specific message. It includes a 'Last Error' section with a stack trace, a 'To prevent losing your analysis results, save them before opening a new message.' warning, and an 'AI-Generated Analysis' section with 'Root Cause', 'First Failed Step', and 'Suggested Solution' details.

Artifact Name	Status	Time
Dec 29, 2017, 17:52:47	Failed	17 ms
	Failed	1 sec 650 ms
	Failed	1 sec 67 ms
	Failed	< 3 ms
	Failed	17 ms
	Failed	< 3 ms
	Failed	270 ms

Error Details

Last Error

```
javax.script.ScriptException: java.lang.Exception: groovy.lang.MissingPropertyException: No such property: messageLogFacto for class: my__Script@ line 4 in my.groovy, cause: groovy.lang.MissingPropertyException: No such property: messageLogFacto for class: my__Script
```

AI-Generated Analysis

Root Cause

The error is caused by a Groovy script in the integration flow referencing a non-existent property 'messageLogFacto'. This is a clear scripting mistake (typo) in the Groovy script content.

First Failed Step

Groovy Script 1

Suggested Solution

Correct the Groovy script by fixing the typographical error in the variable name. The script refers to 'messageLogFacto', which should likely be 'messageLogFactory'. Update the variable name to the correct one.

4. **Copy** the AI-generated analysis into a secure location, or select **Save As** to download the results to your local file system.

→ Tip

The system doesn't save the analysis. To prevent data loss and avoid the need to regenerate error analyses, save your analysis before opening a new message.

There are two options under **Save As**:

- **New File.** The default, editable file name is based on the integration flow name.
- **Existing File.** You can use an existing file to collect related MPL failure analyses in one sheet. This enables correlation across multiple analyses.

→ Tip

Save your analysis in a shared folder and reuse saved files when possible. This helps prevent early quota exhaustion, especially when multiple integration developers use the feature. Audit log trails identify users leveraging the AI feature for collaboration purposes.

5. Apply the suggested solution to your integration artifact to solve the error.

Troubleshooting

If the following warning or error messages pop up when analyzing errors, perform the corresponding actions:

Message	Action
The [integration flow] and its dependencies have been updated, which may have resolved the issue. Do you still want to analyze the error?	Some dependencies have changed since the MPL was created. This update might affect the accuracy of the analysis. You can either confirm to start the analysis or cancel the process.
[integration flow] references some artifacts that are currently undeployed, and the analysis may be inaccurate. Do you still want to analyze the error?	During dependency checking, the system found related artifacts (e.g. correlated integration flows, or script collections) that aren't deployed. This can affect the accuracy of the analysis. You can either confirm to start the analysis or cancel the process.
AI analysis is temporarily unavailable because you reached the monthly usage limit. Try again after the limit resets.	AI analysis can't run because you reached the fair usage limit. Try again after the quota resets. i Note The quota resets each month.
The error could not be analyzed because [integration flow] is currently undeployed.	AI analysis cannot run because the main integration flow for the specified MPL ID is not deployed. Deploy your artifact and try again.
The error could not be analyzed because no message processing log exists with this ID.	The system can't find any MPLs with the provided ID.

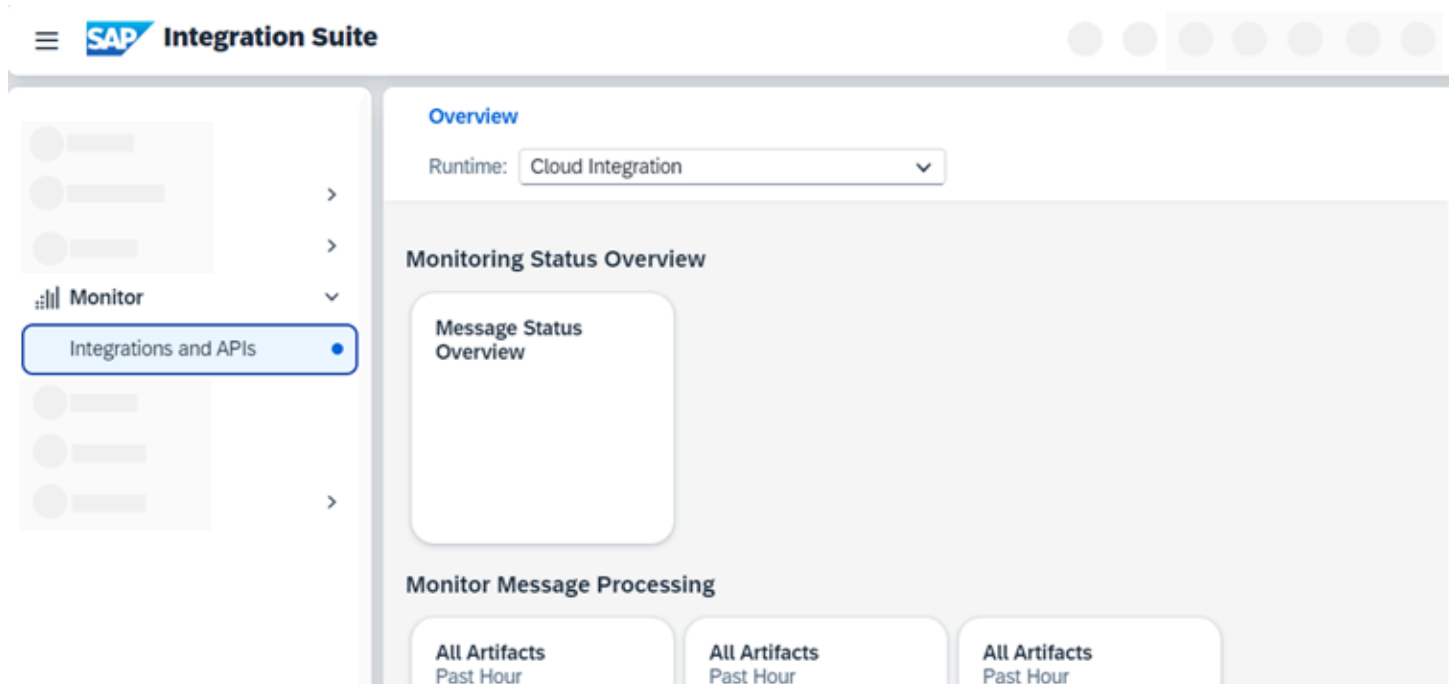
Monitor Message Status Overview

The [Message Status Overview](#) page provides a customizable overview of the messages processed and their respective statuses in a selected time frame as well as quick navigation to and from the Message Monitor.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

You access the Message Status Overview table by clicking the tile Message Status Overview in the section Monitoring Status Overview.



Message Status Overview only works with edge locations that are running runtime version 8.20x or later, and Cloud Integration version 6.58.x. If you are using an earlier version, you will receive the error `No handler registered for command 'com.sap.it.op.tmn.commands.dashboard.webui.MplStatusOverviewCommand'`. To resolve this issue, update your runtime version to 8.21.x or newer.

Message Status Overview Table

The table displays the collected data of the selected runtime, which you can change in the dropdown menu Runtime.

The table shows the individual list names, and the status filters you selected. For certain options, such as **Artifacts** or **Integration Flows**, it also shows the version. If the **Version** column is empty, this can be because it is still in **Draft**, the data can't be retrieved or it is a row that contains combined data, such as the first row **All Artifacts** or **All Integration Flows**.

Overview / Message Status Overview

Hide Filter Bar

Runtime: Cloud Integration

Time: Past Hour

Overview By: Artifacts

Line Items (23)

Filter by Artifact Name

Artifact Name	Version	Completed	Failed	Retry	Processing	Escalated	Total
All Artifacts		121	61	0	0	0	206
	1.0.0	60	0	0	0	0	62
		0	60	0	0	0	62
	1.0.9	60	0	0	0	0	62
	1.0.4	0	0	0	0	0	1
	1.0.3	0	0	0	0	0	1
	2.36.0	0	1	0	0	0	1

By default, the inline items are displayed in table format and sorted by the total number of messages in descending order. You can change this by selecting [Settings](#) [Sort](#).

Overview / Message Status Overview

Hide Filter Bar

Runtime:

Cloud Integration

Time:

Past Hour

Overview By:

Artifacts

Line Items (23)

Filter by Artifact Name

🔍

↺

⚙️

📄

🗃️

🕒

📊

You can also choose to display your data as a pie chart or column chart by selecting **Pie Chart** or **Column Chart** respectively.

Runtime: Cloud Integration

Time:

Overview By:

Past Hour

Artifacts

Line Items (23)

Filter by Artifact Name



Once you switch to the pie or column chart, you can choose for example for which artifact, sender, or receiver (according to your previously selected data under [Overview By](#)) you want to see the data. To do so, use the dropdown menu above the chart. By default, the chart displays the combined data of the first row of the table.

The statuses shown in the charts correspond to the statuses in the table. You can change the selection and order the same way you would for the table version. For more information, see [Status](#) below.

Navigation to and from Message Monitor

You can navigate to the [Message Monitor](#) page by clicking any non-zero value in the table. The settings for the filters [Time](#), [Status](#) and [Overview By](#) are then used on the [Message Monitor](#) page.

If you display your data in pie or column form, you can click on the respective parts or bars, which will take you to the [Message Monitor](#) page in the same way.

If you navigate back to [Message Status Overview](#) from the [Message Monitor](#) with the button [Message Status Overview](#), the filters [Time](#) and [Overview By](#) you previously selected on the [Message Status Overview](#) page remain in place.

Filter Options

You can customize your overview with various filter options. The table also allows you to filter for specific artifacts by entering the artifact name in the search field. This is case insensitive.

Overview By

You can filter your message statistics according to different criteria. You can get an overview for all artifacts or integration flows for example, or you can get more specific statistics for artifacts with senders or receivers.

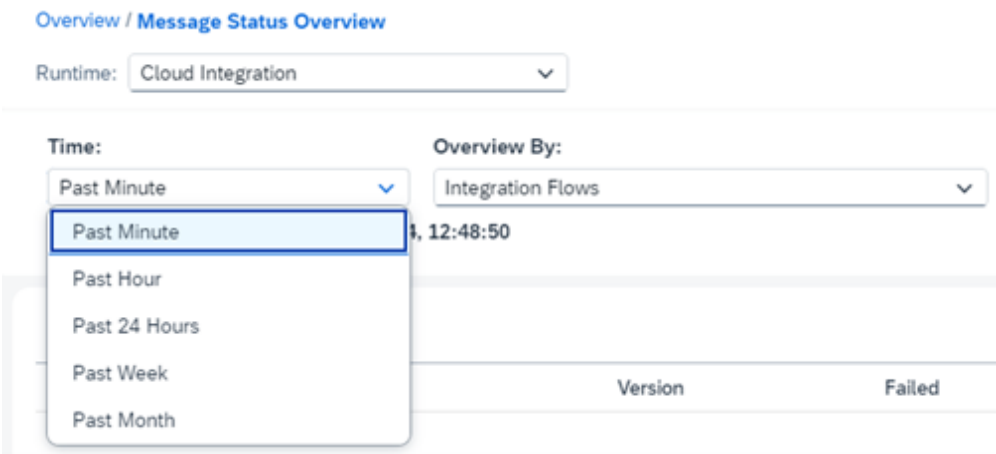
Filter which data you want to show by selecting one of the following options from the dropdown list. By default, [Artifacts](#) is selected.

- Artifacts
- Integration Flows
- APIs (Edge runtime only)
- OData APIs
- SOAP APIs
- REST APIs
- Application Message Type
- Sender

- Receiver

Time

Specify your selection by showing only messages processed in a specific time span, such as the last hour.



! Restriction

Typically, all time periods are available regardless of database usage. However, for technical reasons, the collected data can be slightly incomplete and might not cover the entire period specified by the user. For example, when there's been a recent update to the software for collecting data.

Status

Filter your messages according to their status. You can select multiple statuses which will be shown as columns in the table.

By default, five of the eight statuses are selected. You can adjust these under ►Table Settings►Columns► and choose additional statuses to display.



To change the order of the status columns, simply drag the status up or down in the list. If you select a status, you can also use the respective buttons next to the status name.

Table Settings

Reset

Columns

Sort

Search

Show Selected

Field (5/8)

☒ Failed

☒ Retry

☒ Completed

☒ Processing

☒ Escalated

☐ Cancelled

☐ Discarded

☐ Abandoned

OK

Cancel

For more information on the different message statuses, see [Message Status](#).

Export Your Data

You can export your data as an Excel .xlsx file. This will take into account your current filter settings and order of columns. To do so, simply select **Export** and you will be asked whether you want to download the file.

You can choose a custom name for the file. The default name is determined by your **OverviewBy** filter settings.

Overview / Message Status Overview

Hide Filter Bar

Runtime: Cloud Integration

Time: Past Hour

Overview By: Artifacts

Line Items (23)

Filter by Artifact Name

Note that this option is only available if you display your data in table form.

Monitor Message Processing

The message monitor provides an overview of the messages processed on a tenant and allows you to display the details for individual messages.

This is custom documentation. For more information, please visit [SAP Help Portal](#).

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To access the message monitor, choose a tile in the **Monitor Message Processing** area. The system displays a table with all processed messages based on the selected filter settings.

i Note

The system separates message counting from message retrieval. It loads the message list (up to 50 messages at once) first and computes the message count in the background. During high load scenarios, the system displays three dots instead of the message count to indicate that counting is in progress. This way, you can see all retrieved messages even if counting fails. This reduces the risk of timeouts when the number of messages in the selected time interval is too high.

Filter Settings

You can control which messages appear by changing the filter. You can filter messages by the following categories:

Filter Attributes

Attribute	Description
Time	Specify the time interval freely with the option Last X Minutes/Hours, or select from the following predefined time intervals: • Past Minute • Past Hour • Past 24 Hours • Past Week • Past Month • Custom Date Range i Note When you select the Custom Date Range option, you can display messages based on when their status last changed within a specific time period. Define this period with the From and To filters. Select Last Updated to load the message list more quickly and avoid timeouts. → Tip If you encounter timeouts, reduce the time period to hours or minutes. The specified time interval appears above the message list. When you browse different pages in the message monitor, the time interval remains the same. To change the time interval, adjust the filter settings or refresh the message monitor. This applies to all time intervals except for Custom.
Status	Filter messages according to their status: • All • Failed • Retry • Completed • Processing • Escalated • Cancelled • Discarded

Attribute	Description
	• Abandoned See: Message Status
Type	Filter all artifacts by a specific type: • All • Integration Flow • API • MCP • OData API • SOAP API • REST API
Package	Display all messages that belong to a specific package . If you select a package, you can also filter for the different types or artifacts that belong to that package. If you change the package or type in the filter, the system clears the artifact selection as well.
Artifact	Display messages related to specific artifacts. You can select multiple artifacts to display their messages. By default, the system shows all artifacts. The list includes all artifacts that have message processing logs. When you place the cursor on an artifact in the list, a tooltip displays the artifact ID. You can filter artifacts by entering a specific sequence of characters in their name or ID. The search does not consider case sensitivity.
ID	Filter messages using different IDs. See: Using IDs to Filter Messages

→ Remember

After selecting the appropriate filters, choose [Go](#) to see the list of messages.

[Overview](#) / Monitor Message ProcessingMessage Status Overview

Monitor Message Processing

Go

Hide Filter Bar

Filters

Time: *
Past Hour

Status: *
All

Type: *
All

Package: *
All

Artifacts: *
All

Log Level:
All

Messages (12)

Aug 07, 2026, 15:58:08 - Aug 07, 2026, 16:58:08

Artifact Name	Status	Last Updated At	Artifact Type	Processing Time	Log Level	
TestJsonAttachmentNewest	Completed	Aug 07, 2026, 16:55:06	Integration Flow	6 sec 285 ms	Info	

Sender: Receiver:

<< < 1 / 1 > >>

Choose [Filters](#) to display additional filter criteria. For more information, see [Using Additional Filter Criteria](#).

Messages Table

After you select the filters, a list of all relevant messages appears under [Messages](#).

i Note

If the number of filtered messages exceeds 50, the list is divided across multiple pages. You can browse the different pages by selecting the navigation options. You can also enter a page number to go directly to a specific page in the list.

You can adjust the table view by choosing the settings icon () and selecting the information you want to see about each message. The following attributes are available for each message:

Message Attributes in Message Overview	
Attribute	Description
Artifact Name	The name of the artifact. For example, this is the name of the integration flow that specifies message processing. i Note The tooltip shows the technical name, the artifact, and the package name.
Status	The status of end-to-end message processing. If messages have a critical status, you can open them to view error details.
Last Updated at	The time at which the message processing log was last updated.
Processing Time	The total message processing time.
Artifact Type	The type of integration artifact.

Attribute	Description
Log Level	The granularity of information that the message processing log collects
Sender	The system from which the message originates.
Receiver	The system to which the message is sent.
Application Message Type	The identifier set by a dedicated header (SAP_MessageType)..
Application Message ID	The identifier of the application message.
Archiving Status	Indicates whether the message processing log has been archived
Artifact ID	The identifier used to correlate different messages with each other that are processed jointly in the context of an integration scenario.
Correlation ID	The identifier used to correlate different messages with each other that are processed jointly in the context of an integration scenario.
Custom Status	A custom processing status that can be set by the user.
Message ID	The unique identifier that is associated with the message processing log.
Predecessor ID	The identifier of the previous message processing log in a chain of interconnected integration flows.

Message Details

Select a message from the table to view more information about it. When you choose a message, the table slides to the left and a message details pane opens on the right. You can expand the **Message Details** screen by choosing the full screen icon (). To close the details and return to the message table, choose **X**.

In the new pane that opens, select one of the following tabs to view more details about the selected message:

- **Status.** This section contains status information about the message, such as the **Processing Time**. If the message does not have the status COMPLETED, the system displays the last error message if one is available.
- **Properties:**
 - **Message ID**
 - **Correlation ID**
 - **Application Message ID**
 - **Predecessor ID**

For more information, see: [Using IDs to Filter Messages](#).

 - **Sender**
 - **Receiver**
 - **Application Message Type**
 - **Custom Status**
 - **Retention Periods** for completed and uncompleted messages.

i Note

Sender, **Receiver**, **Application Message ID**, **Application Message Type**, and **Custom Status** are optional and can be set via the Content Modifier. They're only shown if they aren't empty.

The **Custom Status** is only visible if it differs from the overall processing status.

- **Custom Header Properties.** You can set these properties using the script step. If set, the system displays them in this section.

For more information, see: [Use Custom Header Properties to Search for Message Processing Logs](#)

- **Logs.** This section displays the **Log Level** and the **Runtime Node**. Select the log level link to view the message processing log in a visual format. For more information, see [Message Processing Log](#).

To access the text version, select **Open Text View**, see [Message Processing Log – Text View](#).

If the processing of an integration flow fails and the retry runs are performed, these runs appear in a table in the **Logs** section. The table displays up to 50 retries.

If you activate **Data Archiving** for the logs of an integration artifact, the system displays the archiving status in this section. The possible statuses are:

- **Not Archived:**

- **Data Archiving** is not activated for logs of this integration artifact.
- **Data Archiving** does not consider logs of discarded message processing.
- **Data Archiving** is configured to consider logs of successfully completed messages only.

- **Archiving Pending:** **Data Archiving** is activated for logs of this integration artifact.

- **Archived:** **Data Archiving** is completed for this log.

- **Attachments.** This section appears only when the message processing log includes attachments. The system displays a list of all attachments that can be displayed as text. You can browse the different pages to view the complete list of attachments. Select the relevant navigation options to do so.

The following attachment formats are supported:

- Plain text
- XML. XML attachments are displayed as formatted plain text (including tags).
- Text files with comma-separated values
- Text files with tab-separated values. In these files, each entry appears as a single line in the text file.
- HTML. HTML attachments are displayed as plain text (including tags).

You can open the attachments on a new screen and download them to your local file system.

- **Artifact Details.** This section displays the artifact **Name**, **ID**, **Type**, and **Package**, if available.

You can also use any of the following navigation links to continue working with your artifacts:

Hyperlink	Description
Manage Integration Content	Navigate to the integration content monitor. When you navigate to the integration content monitor, the system opens the active artifact that is currently deployed.

Hyperlink	Description
	See: Manage Integration Content
View deployed Artifact	Navigate to the deployed artifact. If the artifact is an integration flow, the system displays the model for the deployed integration flow. You can't edit this model.
Navigate to Artifact Editor	Navigate to the artifact editor which allows you to edit the artifact. If the artifact is an integration flow, you can edit the integration flow model.

Message Status Overview

You can quickly navigate to the [Message Status Overview](#) page by choosing the [Message Status Overview](#) button.

This view offers a detailed summary of the statuses of messages for various artifacts over a specified time period. For more information, see [Monitor Message Status Overview](#).

Related Information

[Message Processing Log – Text View](#)

[Message Status](#)

[Configuring Archiving Settings](#)

Using IDs to Filter Messages

You can filter message processing logs using different IDs.

The following IDs can be used:

- **Message ID**

The unique identifier that is associated with the message processing log.

- **Correlation ID**

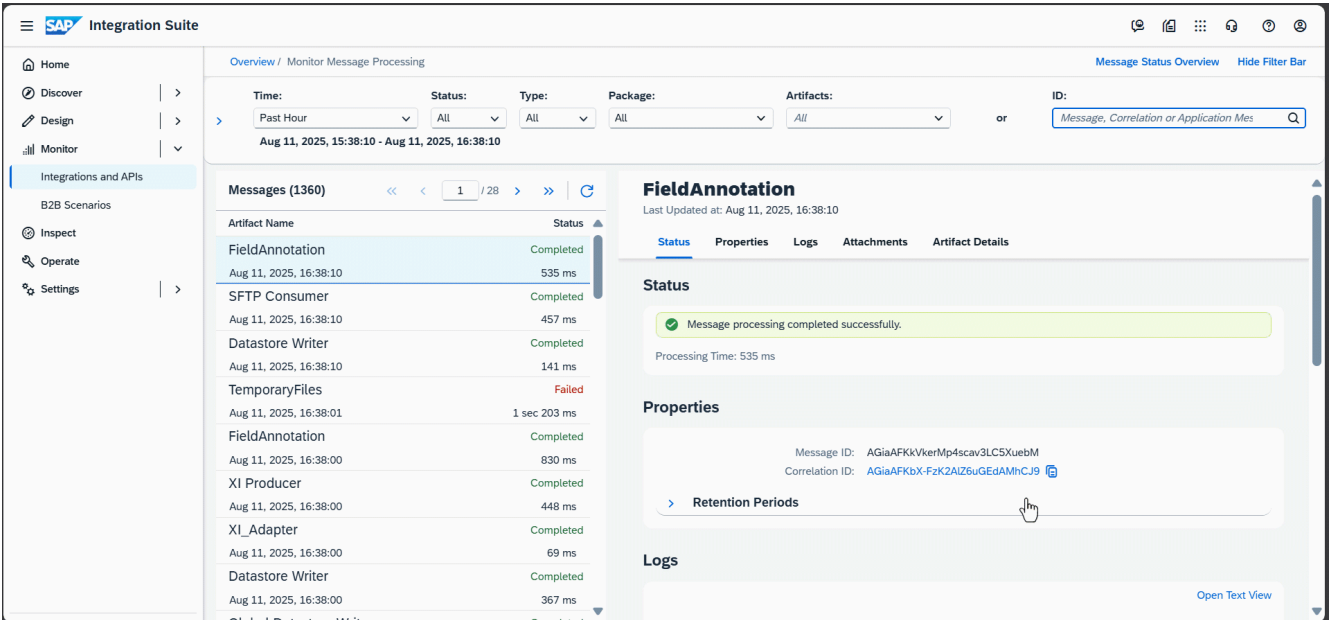
The identifier used to correlate different messages with each other that are processed jointly in the context of an integration scenario.

❖ Example

For example, consider a scenario where one integration flows (*integration flow 1*) calls another one (*integration flow 2*) using the **ProcessDirect** adapter. In this case, for one message processing run two message processing logs are written: one integration flow 1 and another one for integration flow 2. Both message processing logs have different message IDs values but the same correlation ID.

➔ Tip

To filter all the message processing logs with the same **Correlation ID**, choose the **Correlation ID** link from the **Properties** section of a message.



i Note

To correlate the HTTP inbound and HTTP outbound message processing logs, pass the correlation ID via the content modifier step or HTTP header property.

• Application Message ID

The identifier set by a dedicated header (SAP_ApplicationID). You can set this ID using, for example, a **Content Modifier** step.

• Predecessor ID

Use this ID to navigate to the previous message processing log in a chain of interconnected integration flows. You can copy and search this predecessor ID to build a linked list of message or integration flow dependencies. This helps detect faulty integration flows in a chain of interconnected flows, reduces application downtime, and helps identify areas that need stabilization.

! Restriction

- The **Predecessor ID** doesn't work if an aggregator flow step is used in the integration flow, as this leads to multiple predecessors.
- You can't obtain a list of all predecessor IDs at once.
- This feature is available for adapters that send the SAP_MessageProcessingLogId header to the called integration flow. The sender adapter in the receiving integration flow then retains this header.

! Example

When you use an HTTP sender adapter with an HTTP receiver adapter in the sending integration flow, the sender adapter retains the `SAP_MessageProcessingLogId` header to assign the MPL ID as the predecessor ID.

- Currently, this feature doesn't apply to AS2 and AS4 adapter variants.

The ID search process checks for message processing logs that have the specified ID as:

- Message ID:** If not found, the system checks whether there's a message with the specified **Correlation ID**.
- Correlation ID:** If not found, the system searches for messages with the given **Application ID**.
- Application ID:** This search attribute can't find messages where the **Correlation ID** is the same as an existing **Message ID**, or the **Application ID** equals a **Correlation ID** or **Message ID** (by coincidence). Searching for messages by using this attribute is helpful for support use cases (for root cause analysis, for example).
- Predecessor ID:** If not found, check the restrictions mentioned above.

Using Additional Filter Criteria

You can filter message processing logs using additional filter criteria.

Use the **Filters** option to control which message processing logs are displayed .

The dialog that opens provides two views: **List** and **Group**. The **List** view displays all active filters individually; the **Group** view organizes them by category. The **Edit** button allows you to modify filter configurations, and the **Sort** button changes the display order of the filters.

You can use the **Search for Filters** field to locate a specific filter by name and use the **Add Filter** to add new filters to the list.

The screenshot shows the SAP Integration Suite interface for monitoring message processing. The left sidebar contains navigation links: Home, Discover, Design, Modernize, Monitor, Integrations and APIs, B2B Scenarios, and Inspect. The main content area is titled 'Monitor Message Processing' and includes a search bar for 'Message, Correlation or Application Message ID'. Below the search bar are filter options for Time (Past Hour), Status (All), Type (All), Package (All), Artifacts (All), and Log Level (All). A 'Go' button is present next to the filters. The interface also shows a 'Message Status Overview' link and a 'Filters' button. The main table displays 12 messages, with the first message being 'TestJsonAttachmentNewest' in a 'Completed' status, last updated on Aug 10, 2026, at 14:55:06, with an artifact type of 'Integration Flow', a processing time of 6 sec 477 ms, and a log level of 'Info'. The table also includes columns for Artifact Name, Status, Last Updated At, Artifact Type, Processing Time, and Log Level. A pagination bar at the bottom shows '1 / 1'.

The following additional filter criteria are available (depending on the integration flow design):

- Sender**

Identifier set by a dedicated header (SAP_Sender).

- Receiver

Identifier set by a dedicated header (SAP_Receiver).

- Custom Status

Identifier set by a dedicated exchange property (SAP_MessageProcessingLogCustomStatus).

- Application Message Type

Identifier set by a dedicated header (SAP_MessageType).

- Custom Header

Filter for a dedicated value of a custom header by entering an expression like:

<custom header name> = <custom header value>

For more information on how to set up a custom header, check out [Use Custom Header Properties to Search for Message Processing Logs](#).

You can also get custom header properties using an OData V2 application programming interface (API). See: [Get Custom Header Properties](#).

→ Remember

After selecting the appropriate filters, choose **Go** to see the list of messages.

[Overview](#) / [Monitor Message Processing](#)Message Status Overview

Monitor Message Processing

Q

Go

Hide Filter Bar

Filters

Time: *

Past Hour

🗓️

Status: *

All

▼

Type: *

All

▼

Package: *

All

📁

Artifacts: *

All

📁

Log Level:

All

▼

Messages (12)

🔄 ⚙️

Aug 07, 2026, 15:58:08 - Aug 07, 2026, 16:58:08

Artifact Name	Status	Last Updated At	Artifact Type	Processing Time	Log Level	
TestJsonAttachmentNewest	Completed	Aug 07, 2026, 16:55:06	Integration Flow	6 sec 285 ms	Info	>

Sender: Receiver:

<< < 1 / 1 > >>

i Note

A value help is available for the **Sender**, **Receiver**, **Custom Status**, and **Application Message Type** fields. The list of available values depends on your integration design.

To filter for a dedicated value of a custom header, in the **Custom Header** field, enter an expression like: <custom header name> = <custom header value>

Alternatively, you can search for IDs in the **ID** field, and display messages associated with various IDs such as:

Message Status

The message processing status indicates how a messages has been processed at runtime.

Message Status

Status	Description
COMPLETED	Message has been delivered to receiver successfully.
PROCESSING	Message is currently being processed.
RETRY	Status retry is set if an error occurred during message processing, and a retry was automatically started. i Note If an external broker delivers the message and then the message is removed, the MPL remains in this status since the system doesn't attempt to deliver it again.
ESCALATED	During message processing an error occurred, and no retry has been triggered. For synchronous messages, an error message is sent to the sender.
FAILED	Message processing failed, message hasn't been delivered to receiver, and no retries are possible. In other words: FAILED is a final status, and message processing ultimately has failed.
CANCELLED	Manual cancellation of entries in the JMS queue - MPL is set to status cancelled. i Note In general, the last final status is accepted, but if the message processing is still ongoing, and the status CANCELLED is set, it's overwritten by another "final status" such as COMPLETED, after completion of the message processing.
DISCARDED	For scheduler triggered integration flows, the MPL is shown on the worker node where the message processing started first. For all subsequent message processing starts, the message status is set to DISCARDED. For example, assume that an integration flow is initiated by a Timer event (scheduler) and a worker node goes into an out of memory state. The system starts the worker node again and synchronizes the integration flow as soon as the node is active. The message is restarted in that case, and a new message ID generated. The message with the original ID goes into status DISCARDED.
ABANDONED	Message processing was interrupted or log wasn't updated for an uncommonly long time. The status can change in case processing is resumed. i Note When the message processing is interrupted because of a re- or undeployment of an integration flow, or a controlled worker node shutdown, the MPL is immediately set to ABANDONED. This status isn't final and the processing can be resumed if retries are

Status	Description
	configured. If there are hard worker node crashes, the status is changed to ABANDONED by an asynchronous system job, after some time of inactivity

An aggregated message processing log (MPL) can have the following status values:

Status of Aggregated Messages

Status	Description
PROCESSING	Is set as soon as the aggregation process is started and remains as long as the aggregate is still open for further messages. Note that this status can in many situations be shown for quite some time (even days) – depending on the applied aggregation use case.
FAILED	Is set when the aggregated message (after aggregation process has successfully been finished) fails.
RETRY	Is set when the aggregated message (after aggregation process has successfully been finished) runs into an error and sending it's retried.

Related Information

[Define End Message Event](#)

Message Processing Log

The message processing log displays structured information of a message.

Select a message from the list. In the view section, the header shows the integration flow name, the processing date and the time of its last update.

→ Remember

The dates and times displayed are converted to the local time zone set for the browser.

Next, you can select and display **Status**, **Properties** and **Logs**. If the processing of a integration flow failed, the retry runs performed are displayed in the **Logs** section.

Messages (581)

Artifact Name

Relax Dependencies - Apply Retry Pattern - JMS Outbound

Retry

326 ms

Relax Dependencies - Apply Retry Pattern - JMS Outbound

Last Updated at:

Status

Properties

Logs

Attachments

Artifact Details

2

18 ms

Info

Retry

1

326 ms

Info

Retry

Attachments

Name	Type	Modified At	Size	Actions
ODataV2_Adapter_Request_Headers	text/plain		1 KB	↓
ODataV2_Adapter_Response_Body	text/plain		1 KB	↓
ODataV2_Adapter_Response_Headers	text/plain		1 KB	↓

Artifact Details

Name:

Relax Dependencies - Apply Retry Pattern - JMS Outbound

ID:

RelaxDependencies_ApplyRetryPattern_JMSOutbound

Type:

Integration Flow

Package:

Integration Flow Design Guidelines - Relax Dependencies to External Components

The logs section shows the number of runs as well as the following information:

Runs

Property	Description
#	Displays the sequential number of the run.
Started At	Displays the date and time of the last run.
Duration	Displays the duration of the processing run in <i>ms</i> .
Log Level	Displays the log level.
Process ID	Displays the ID of the worker node on which the run was executed.
Status	Displays the intermediate status of the message processing log after this run.

You open the message processing log by selecting the **log level** of one of the runs.

The detailed view shows the list of the **Run Steps**.

If an error occurred in one of the run steps, it is shown by a red exclamation mark in the run step list. To display the error message, follow the exclamation mark.

If a **Split** or **Multicast Step** is used, the steps belonging to one sub-message get the same segment number. They can therefore be grouped together.

To view the **Log Content**, select one run step from the list and choose **Log Content**. The log content displays the **Properties** and the **Activities** of the step. If more than one step is related to a flow element, all of them are selected within the list and you can scroll from one step to the next by using the arrows in the element info bar and view the properties and activities accordingly.

You can only review the **Message Content** section, if the log level has been set to **trace**.

i Note

The names of the flow elements can only be shown in the run step list if the integration flow model is loaded. If the model cannot be loaded, the system only shows the run steps IDs.

The **Integration Flow Model** is displayed on the right hand side, if it is deployed.

You can view the path taken by the message in the integration flow. If you select a run step for the list, it is highlighted in the integration flow model.

You can also select an element of the **Integration Flow Model** and the related run step will be highlighted in the run step list on the left hand side.

The properties view of the selected integration flow element is displayed at the end of the page.

The availability of the trace data is also visualized in the integration flow model as envelope icons.

Envelope Icons

Icons	Description
	Successfully processed step with trace data available.
	Error in processed step with trace data available.
	Successfully processed step with no trace data available.
	Error in processed step with no trace data available.

In order to have a visual distinction between steps with available trace data and steps with processing information only, inverted icons are displayed if there is no trace data available.  for a successfully processed step and if the step processing resulted in an error.

For more information on the **Trace** log level check out [Tracing the Execution of an Integration Flow](#).

i Note

If the log level is set to *Info* and the processing does not show any error, the link to the detailed view is not active. If the processing shows at least one error, the link is active and you can access the detailed view.

i Note

This component stores data on your tenant. Note that the tenant space is limited and shared with other tenant data like message processing logs.

Related Information

[Message Processing Log – Text View](#)

[Monitor Message Processing](#)

[Message Processing Log - Adapter Tracing](#)

[Setting Log Levels](#)

[Message Status](#)

Message Processing Log – Text View

The message processing log displays structured information on the processing of a message.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

The message processing log comprises a log header and log steps. In the log header you find more general information . The log steps show specific processing events and steps as well as further information, depending on the messages logged. This could be an `EscalationEvent` for example, which will trigger an alert showing up in the message processing log.

The log header shows the following properties:

MPL Header

Property	Description
StartTime	Start of message processing.
StopTime	End of message processing.
OverallStatus	Specifies the end-to-end status of message processing and corresponds to the Status attribute in the Message Monitoring editor. For more information on statuses see Message Status in the Related Information section. .
ChildCount	Indicates the serial number of the current processing step.
ChildrenCounter	Specifies the total number of message processing steps.
ContextName	Integration flow name.
CorrelationId	ID that identifies correlated messages. Messages can be correlated, for example, when different integration flows on the same tenant communicate with each other. A correlation ID is a base64-encoded ID that is generated in this case by the first integration flow and stored in the message header. As part of the message header, the CorrelationId is then propagated across all related integration flows (that are in charge of processing the correlated messages).
CustomHeaderProperties	Can only be added by defining a script in the script API and the properties are displayed in the message processing log header. For more information see Define Scripts in the Related Information section.
ExecutedMapping	Indicates the name of the message mapping artifact that was executed. Useful when you use the dynamic message mapping feature to understand which message mapping artifact was executed by the integration flow.
Id	Is displayed in the MPL header if the ID has been previously defined as header property. For more information see Headers and Exchange Properties in the Related Information section.
IntermediateError	True if an error occurred (even temporarily) during message processing, or message processing took more than 1 minute. The header then contains an additional property: LastError showing the error type.
MessageGuid	Key that identifies the message uniquely in the database.
MessageType	Is displayed in the MPL header if the message type has been previously defined as header property. For more information see Headers and Exchange Properties in the Related Information section.

Property	Description
Node	Host name of the node that processed the message.
ReceiverId	Is displayed in the MPL header if the receiver ID has been previously defined as header property. For more information see Headers and Exchange Properties in the Related Information section.
SenderId	Is displayed in the MPL header if the sender ID has been previously defined as header property. For more information see Headers and Exchange Properties in the Related Information section.
LocalSuccessor Id	Is set during aggregation and incoming message process processing logs get the aggregated MPL as a successor.
LocalPredecessor Id	Is set during creation of a new MPL, if the message header SAP_MessageProcessingLogID is present or the message header SapPredecessorMessageProcessingLogId is set.
TransactionId	This is the unique identifier for the processing transaction. It correlates fully to the trace ID from the <code>transparent</code> header when received, helping correlate log entries and trace message activity across systems. If no <code>transparent</code> header is present, it correlates to the SAP password. If neither exists, the transaction ID is generated at runtime.

The log steps show the following properties:

Log

Message Processing Log:

```

StartTime      = Tue Nov 28 01:10:21.407 UTC 2017
StopTime       = Tue Nov 28 01:10:21.433 UTC 2017
OverallStatus  = COMPLETED
ChildCount     = 0
ChildrenCounter = 4
ContextName    = com.sap.it.op.test.example.jms.submitter.for.adk.api.test
CorrelationId  = AFoct305vzfzZ2TzPvnHOsR7VRvy
IntermediateError = false
MessageGuid    = AFoct32wL7_0U3D8IlqnoR3i1MdR
Node           = vsa3290317

```

Branch:

```

Entering Camel route route13:
  StartTime      = Tue Nov 28 01:10:21.411 UTC 2017
  ChildCount     = 1
  ModelStepId    = TimerEventDefinition_1

  Exchange ID-vsa3290317-40179-1511831409839-33-2 created in Endpoint
[quartz2://com.sap.it.op.test.example.jms.submitter.for.adk.api.testTimerEventDefinition1?
fireNow=true&trigger.repeatCount=0&trigger.repeatInterval=0]:
  StartTime      = Tue Nov 28 01:10:21.412 UTC 2017
  Status         = PROCESSING
  ChildCount     = 2
  ModelStepId    = TimerEventDefinition_1

  Exchange ID-vsa3290317-40179-1511831409839-33-2 completed:
  StartTime      = Tue Nov 28 01:10:21.433 UTC 2017
  Status         = COMPLETED
  ChildCount     = 3
  ModelStepId    = TimerEventDefinition_1

Exiting Camel route route13:
  StartTime      = Tue Nov 28 01:10:21.433 UTC 2017
  ChildCount     = 4

```

MPL Steps

Property	Description
Branch	Indicates that the following steps have been processed in the same branch. If a Split or Multicast Step is used, the steps belonging to one sub-message are grouped together within one Branch .
ModelStepId	This ID is used to specify relation between modeled step (in integration flow) and MPL entry. The integration flow model steps are fragmented in the Camel runtime environment into several processing steps. For more information see Error Classification in the Related Information section.
StepId	ID of the related integration flow step. It is assigned by the Camel framework.

i Note

- The content of the message processing log varies according to the log level you have set.
- The step properties displayed may vary from step to step, according to the message processing.

- The payload is displayed with header and content. You can also download it by clicking on the **Download** button on top right corner. This provides a zip file containing one header and one payload file.

→ Remember

- All string values are limited to 1000 characters. Any string value longer than 1000 characters is cut off after 1000 characters.
- The MPL property names are limited to 100 characters. Any property name longer than 100 characters is cut off after 100 characters.

You can create attachments to the message processing log by using the **Script API**.

i Note

The amount of MPL attachments which can be written is limited to 1 GB per 24 hours. If the limit is reached, MPL attachments will no longer be stored until the amount of MPL attachments written in the last 24 hours is again below 1 GB.

Related Information

[Setting Log Levels](#)

[Monitor Message Processing](#)

[Define a Local Script Step](#)

[Headers and Exchange Properties Provided by the Integration Framework](#)

[Error Classification](#)

[Message Status](#)

Message Processing Log - Adapter Tracing

The adapter tracing is part of the regular tracing feature and the payloads are recorded if you have set the log level to **Trace**.

The adapter tracing is only possible for adapters that transform the message either before sending or upon reception, such as *AS2*, *Ariba*, *LDAP*, *Mail*, *SuccessFactors* and *HTTP/HTTPS*. You also get tracing information for integration flows containing *CFX* based adapters such as *IDOC*, *SOAP*, *SAPRM*, but in this case you have to redeploy the integration flow to get the tracing data recorded in the regular tracing feature and displayed.

Adapter tracing is not possible for *SMS*, *SFTP*, *Facebook*, *Twitter* and *Process direct*, as they do not modify the payload.

→ Remember

For log level **Trace**, detailed information is recorded for all steps and in addition, the message content is tracked. The trace function expires after a certain time (default value: 10 minutes). After expiry the log level switches back to the log level set before. The recorded message content is also retained for a certain time (default value: 1 hour).

To view trace information, select an integration flow from the overview list. Go to the **Log** section and open the log level link. In the integration flow model, the system displays trace entries as envelope icons.

Select a step from the run steps panel and switch to the **Message Content** tab. There you can find headers, exchange properties, and payloads. You can also download all the details by choosing **Download** in the top right corner. A zip file containing one header file and one payload file is saved to your local file system.

To find the information you need, check the description field. This field contains contextual information such as, **Request as sent over communication protocol by Receiver Adapter**, **Response as received over communication protocol by Receiver Adapter**, or **Message before Step**.

Overview / Monitor Message Processing / Message Processing Run

Run Steps (4)

SOAP1 / 1

EndSegment 159 ms

SOAPSegment 11 sec 399 ms

HTTPSSegment 161 ms

HTTPSSegment 1< 3 ms

Integration Flow ModelLog ContentMessage Content

Request as sent over communication protocol by Receiver AdapterDownload<<< 2 / 3>>>

HeaderExchange PropertiesPayload

Encoding: UTF-8Download Payload

<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"><soap:Body><soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"><soap:Header><Action xmlns="http://www.w3.org/2005/08/addressing" http://cxfr.apache.org/jaxws/dispatch/DummyImpl/InvokeRequest</Action><MessageID xmlns="http://www.w3.org/2005/08/addressing" urn:uuid:82ec57a0-e079-4201-8fa5-d4a52fe2b5c7</MessageID><To xmlns="http://www.w3.org/2005/08/addressing" href="https://mbt01-iflmap.hc1sb.int.sap.hana.ondemand.com/cxf/S00/incoming"><ReplyTo xmlns="http://www.w3.org/2005/08/addressing" Address="http://www.w3.org/2005/08/addressing/anonymous"><Address></ReplyTo></soap:Header><soap:Body><n0:TestSodOutboundMessage xmlns:n0="http://sapcd.com/sodTestTypes"><field1>1</field1><field2>2</field2><field3>3</field3></n0:TestSodOutboundMessage></soap:Body></soap:Envelope></soap:Body></soap:Envelope>

Header

The **Header** tab displays the HTTP headers of the message as received by the adapter. These headers are set by the sender and carry metadata such as content type, authorization, and tracing identifiers.

To observe and correlate a message end-to-end across all participating systems, every message must carry a globally unique identifier in its transport headers.

The following identifier headers can be found on your message:

Header	Description
sap-passport	SAP's internal correlation identifier. When two SAP systems communicate, the message carries a SAP Passport that encodes a uniquely generated trace identifier.
traceparent	adopts the W3C Trace Context standard as the universal trace identifier for cross-system tracing. W3C Trace Context is an open web standard defined by the World Wide Web Consortium (W3C). SAPCloud Integration includes the W3C Trace ID inside the SAP Passport so that the correlation chain is preserved when a message crosses from an OpenTelemetry-instrumented system into the SAP ecosystem. If the incoming request includes a W3C trace context header (traceparent), captures the trace parent in the message processing log. This allows you to correlate the MPL entry with the corresponding record in an external monitoring tool. i Note For SAP-to-SAP communication, the SAP Passport remains the primary carrier of trace information.

W3C Trace Context Propagation Scenarios

evaluates inbound requests for the presence of an SAP Passport and a W3C Trace Context header (traceparent) and applies the following rules to determine how trace context is propagated. Regardless of the inbound scenario, both the sap-passport and traceparent headers are included in outbound requests by default.

→ Remember

The outbound `traceparent` is derived from the SAP Passport, with the W3C Trace ID corresponding to the SAP Passport Transaction ID.

The following table describes the behavior for each combination of inbound headers:

Inbound Request	Inbound Behavior	Outbound Behavior
<code>sap-passport</code> only	If an SAP Passport is already present in the integration flow prior to the inbound request, the SAP Passport from the request is not used.	Both <code>sap-passport</code> and <code>traceparent</code> are included. The <code>traceparent</code> is generated from the SAP Passport.
<code>traceparent</code> only	If no SAP Passport is present in the integration flow, an SAP Passport is generated from the inbound <code>traceparent</code> header. The W3C Trace ID maps to the SAP Passport Transaction ID.	Both <code>sap-passport</code> and <code>traceparent</code> are included. The outbound <code>traceparent</code> carries the same Trace ID as the inbound <code>traceparent</code> , but with a different parent ID.
Both <code>sap-passport</code> and <code>traceparent</code>	The SAP Passport takes priority. The inbound <code>traceparent</code> is ignored. If no SAP Passport is already present in the integration flow, the SAP Passport from the request is used.	Both <code>sap-passport</code> and <code>traceparent</code> are included. Because the inbound <code>traceparent</code> was ignored, the outbound <code>traceparent</code> may differ entirely from the inbound one, as it is generated from the SAP Passport.
Neither <code>sap-passport</code> nor <code>traceparent</code>	If no SAP Passport is already present in the integration flow, an SAP Passport is generated. If the <code>X-SAP-PASSPORT-TRANSACTION-ID</code> header is present in the request, its value is used as the Transaction ID. Otherwise, a new Transaction ID is generated.	Both <code>sap-passport</code> and <code>traceparent</code> are included. The <code>traceparent</code> is generated from the SAP Passport.

i Note

You can prevent the propagation of the `traceparent` header in outbound requests by setting the `SAP_DisableExternalW3CTraceContextTransmission` exchange property to `true` in the **Content Modifier** step of the integration flow. This property controls only the automatic derivation of `traceparent` from the SAP Passport. A `traceparent` header explicitly added in the **Content Modifier** step is not affected by this property and still propagates to the outbound request. See: [Define Content Modifier](#).

Exchange Properties

The **Exchange Properties** tab displays runtime properties set on the message exchange during processing. These are key-value pairs written by integration flow steps and adapters: for example, routing decisions, message IDs, or adapter-specific context values. Exchange properties are internal to the integration flow and aren't transmitted to the receiver.

Payload

Adapter tracing captures detailed payload information at several stages of message processing. Initially, each step records its payload before execution begins. However, adapters that perform message transformation can capture additional payloads: the outbound message as it's sent over the communication protocol and the inbound response as it's received. This results in multiple payload entries.

When multiple payloads exist for a single step, a paging area appears in the interface. You need to navigate between pages to access all available payload data.

i Note

It is possible that the adapter logs different payloads at different times. This can result in multiple entries for the step on the left-hand side panel.

Manage Integration Content

The Manage Integration Content section provides an overview of integration content artifacts, such as integration flows, that have been deployed on the tenant.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

In the **Manage Integration Content** area, a set of tiles is displayed that show the number of deployed integration content artifacts for a particular type (for example, Integration Flow) and with a particular status (for example, Started). You can configure each tile to filter by a different artifact type or status, or both. To do this, position the cursor on a tile and choose **Edit** from the context menu.

To open the integration content monitor for a particular set of artifacts (as defined by the tile filter), click a tile.

When you click a tile, a list of integration artifacts is displayed (for the filter settings as defined for the tile). On the left side of the screen, the list of artifacts is shown in a table with the following attributes for each artifact:

- **Name**

Below the name, the artifact type (for example, Value Mapping) is shown.

- **Status**

For more information on the available statuses, see [Runtime Status](#).

To customize the list of displayed artifacts, you can either search for specific artifacts by artifact name or ID or you can filter the table content by attributes such as Status or Type.

To sort and filter the content of the table, choose **Table Settings** (). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

i Note

The following paragraph is relevant only when you use SAP Cloud Integration in the Neo environment.

Deployment of Integration Adapter

To deploy an integration adapter, choose **+**. In the pop up, select **Browse** to choose an integration adapter from the files listed, (the standard file extension for integration adapters is `.esa`). **Name**, **ID**, and **Version** are filled in automatically.

On the right side of the screen, more details about the artifact selected on the left side are shown. Depending on the artifact type, other functions can be accessed as well.

Integration Content Details

The header area provides the following information about the selected artifact:

Integration Content Details	
Information	Description
Deployed on	Time when the artifact was deployed.
Deployed by	User who deployed the artifact.
Version	Version of the artifact, for example, the integration flow version.
ID	Artifact ID
Package	Name of the package the artifact was deployed from.
Mode (if applicable)	Indicates whether it is configure-only content.
Repository (if applicable)	Name of the GitHub repository to which the integration flow is linked to.
Branch (if applicable)	Name of the GitHub repository branch to which the integration flow is linked to.

The header area also provides certain functions (the available functions depend on the type of the selected artifact):

- Restart

Allows you to restart an artifact (for example, an integration flow that has been stopped on the node).

- Undeploy

Allows you to undeploy an active integration flow.

 Caution

If you choose to undeploy your integration flow while message processing is still in progress, the undeployment is delayed for up to 1 minute to allow time to finish the message processing. If the message can't be processed within 1 minute, an error is thrown and the message processing cannot be completed.

- Download

Allows you to download the artifact to your computer.

i Note

The functions **Restart**, **Undeploy**, and **Download** are not available for artifacts with status **Stopping** (see [Runtime Status](#)).

Below the header, the following sections contain detailed information about the selected artifact:

- Endpoints

Shows the URLs of the services exposed by the artifact for a sending application. You can copy the URL by choosing the

Copy button 

For example, if the artifact is an integration flow with a SOAP adapter configured as the sender channel, the endpoint URL specifies how the service exposed by the integration flow can be reached by a SOAP client.

- **Status Details**

Shows the status of the artifact with regard to its usage at runtime (for example, if it is starting).

Also shows the **Polling Information** for all message polling adapters (such as SFTP or mail adapter) in use. The view provides the respective information on:

- the adapter information,
- the precise date and time when the latest poll was executed,
- whether further polls are scheduled or not (No further polls scheduled/Further polls scheduled),
- the status of the poll (Successful/Failed),
- (if available) a detailed error message.

If an error occurs during the lifecycle of an artifact, detailed information about the error is displayed. You can copy this information to the ticket you open to get the error solved.

i Note

Errors can occur in various steps during the lifecycle of an artifact. If you think of the artifact as an integration flow, steps are executed on the tenant management node (which the user is connected to when deployment of the integration flow is triggered through the Web UI) as well as on the runtime node where the integration flow is processed at runtime.

Let's look at an example of an error that is related to the runtime node: If deployment of an integration flow is triggered, which requires a runtime node with a specific node profile, but no runtime node with this profile has been started in the cluster, the artifact cannot be distributed from the tenant management node to the runtime node. In this case, an error is displayed together with the information that no suitable runtime node has been started for this use case.

- **Artifact Details**

The type of the selected artifact determines which kind of information is displayed.

Artifact Details

Artifact Type	You can:
Integration flows	access the Monitor Message Processing View deployed artifact Navigate to Artifact Editor
Value mapping	View deployed artifact Navigate to Artifact Editor
SOAP API	access the Monitor Message Processing View deployed artifact Navigate to Artifact Editor
REST API	access the Monitor Message Processing View deployed artifact Navigate to Artifact Editor

Artifact Type	You can:
OData API	access the Monitor Message Processing Navigate to Artifact Editor
Integration Adapter	access the Vendor
Script Collection	View deployed artifact Navigate to Artifact Editor
Message Mapping	View deployed artifact Navigate to Artifact Editor

Caution

Availability of APIs depends upon the SAP Integration Suite service plan that you use. For more information about different service plans and their supported feature set, see [2903776](#) .

- **Log Configuration**

If you deploy an artifact for the first time, the log level is set to `info`. If you want to change the log level, see: [Setting Log Levels](#)

- If you have set up Data Archiving, [Configure](#) allows you to select the data to be archived.

Related Information

[Runtime Status](#)

[Setting Log Levels](#)

[Configuring Archiving Settings](#)

[Archiving Data, Cloud Foundry Environment](#)

Setting Log Levels

The log level for the message processing log specifies the granularity of information collected by the message processing log

→ **Remember**

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

You can set the log level for an integration flow scenario in the [Manage Integration Content](#) page of the [Web-based Monitor](#).

Select your integration flow in the [Manage Integration Content](#) and set the log level for this integration flow in the [Log Configuration](#) section.

You can choose the following log levels:

Log Level	Description
None	No data is recorded during message processing and no data is shown in data monitoring.

Log Level	Description
	i Note In case data archiving is available on your tenant and configured for your integration flow, no data is archived for this specific integration flow.
Info	Basic information is recorded during message processing. The header is always displayed. If there are failed messages, the message processing log retains and displays additional detailed information about the last 50 message steps.
Error	The message processing log records basic information for failed message executions only. The message processing log retains and displays detailed information about the last 50 message steps. Use this log level, if you've no interest in recording completed messages. No message processing log attachments are recorded. i Note In case data archiving is available on your tenant and configured for your integration flow, no data is archived for this specific integration flow. Specifics for log level Error: If you've defined a retry-scenario in your integration flow and set log level Error, a final successful retry isn't recorded. In particular, the overall Message Processing Log Status stays in status Ret ry or Fail ed although the message is finally successfully delivered.
Debug	⚠ Caution As the log level Debug is a high resource consuming feature, we recommend setting log level Debug only in a development or test environment! The message processing log records detailed information for all steps during message processing. While all steps (with headers and additional information) are accessible in the graphical UI, only the last 100 are shown in the text view of the message processing log. The log level debug expires after a certain period of time. After expiry, the log level switches back to the log level set before. i Note in the Neo environment, the expiry time can be set in the tenant configuration. To request an extension of the expiry time, open a ticket on the component L0D-HCI-PI-OPS. In the Cloud Foundry environment, the expiry time is fixed and set to 24 hours.
Trace	⚠ Caution

Log Level	Description
	As in the NEO environment , the log level Trace is a high resource consuming feature, we recommend setting log level Trace only in a development or test environment! SAP can disable this function for a specific tenant or adjust expiry and retention time. The message processing log records detailed information for all steps during message processing and additionally tracks the message content. The trace function expires after a certain period of time (default value: 10 minutes). After expiry, the log level switches back to the log level set before. The recorded message content is also only retained for a certain time (default value: 1 hour) Specifics for log level Trace: • If the downloaded trace data, such as header value size and exchange properties exceed 10000 characters, they're truncated.

i Note

If you update an integration flow, that is, deploy an integration flow without deleting the predecessor version, the log level configuration is kept stable. If you undeploy an integration flow and then deploy a newer version, it's treated as a new integration flow, which is logged with log level **Info**.

Changes to the log level don't affect messages already in process and only apply to newly created messages after the time of modification.

i Note

In order to conserve resources, you can use a Groovy script (in a **Script** integration flow step) that retrieves additional information associated with particular log levels. You can model the scenario in such a way that the information is only determined when it is really needed.

For instance, using the following script, you can limit the inclusion of the payload as an attachment by specifying that the attachment should only be saved if the log level is set to a specific value. For more information about scripts, see [Add Information to the Message Processing Log](#).

≡ Sample Code

```
import com.sap.it.api.msglog.MessageLogFactory;
import com.sap.gateway.ip.core.customdev.util.Message;
import java.util.HashMap;

def Message processData(Message message){
    def logLevel = message.getProperty("SAP_MPL_LogLevel_Overall");
    def logLevelInternal = message.getProperty("SAP_MPL_LogLevel_Internal");
    def logLevelExternal = message.getProperty("SAP_MPL_LogLevel_External");

    def messageLog = messageLogFactory.getMessageLog(message);
    messageLog.addCustomHeaderProperty("Log Level", logLevel);
    messageLog.addCustomHeaderProperty("Internal Log Level", logLevelInternal);
    messageLog.addCustomHeaderProperty("External Log Level", logLevelExternal);
}
```

```
    if(logLevel.equals("DEBUG") || logLevel.equals("TRACE")) {
        def body = message.getBody();
        messageLog.addAttachmentAsString("Payload", body, "text/plain");
        messageLog.addAttachmentAsString("Debugging information", "data", "text/plain");
    }
    return message;
}
```

The log level is represented in three different ways:

- *Internal* log level (from property `SAP_MPL_LogLevel_Internal`) refers to the setting that is applicable to the monitoring data accessible through the SAP Cloud Integration user interface and the public OData API.
- *External* log level (from property `SAP_MPL_LogLevel_External`) refers to the log level selected for sending logging information to an external logging system. This is only relevant if you have activated the corresponding feature.
- *Overall* log level (from property `SAP_MPL_LogLevel_Overall`) represents the highest log level between *Internal* and *External* log levels, following the order of `None`, `Error`, `Info`, `Debug`, `Trace`. If you are using both logging options, the *Overall* log level allows you to consider only one value for comparison. However, if you are not using external logging, then considering the *Internal* log level alone is sufficient.

The *Overall* log level serves as a convenience level if you only want to focus on one value that is achieved by at least one of the two log levels.

Examples:

- Internal = ERROR, External = INFO => Overall = INFO
- Internal = DEBUG, External = INFO => Overall = DEBUG
- Internal = INFO, External = NONE => Overall = INFO

Related Information

- [Message Processing Log – Text View](#)
- [Monitor Message Processing](#)
- [Message Processing Log - Adapter Tracing](#)

Runtime Status

Indicates if a deployed artifact is ready to operate.

The following status values are possible.

Statuses

Status	Description
Started	The artifact is ready to be used. An artifact with this status is in a final state.
Error	The artifact requires attention by the user (administrator). An artifact with this status is in a final state.

Status	Description
Starting	The artifact is in the process of being deployed on the worker node(s) and, if deployed, is being started. An artifact with this status is in an intermediate state.
Stopping	The artifact is in the process of being stopped on the worker node(s) and, if stopped, is being undeployed. An artifact with this status is in an intermediate state. During the process of undeployment, the artifact has status Stopping. An artifact with this status cannot be restarted, undeployed or downloaded.

→ Tip

You can also use the Cloud Integration OData API to get the runtime status of a deployed integration flow. For more information, see [Get Runtime Status of Deployed Integration Flow](#).

Related Information

[Deployment Status View](#)

Endpoint Visualization

You can visualize the adapter endpoints in the [Manage Integration Content](#) section.

With the new release of integration service, the endpoint definitions of some adapters are extracted.

Go to [Manage Integration Content](#) and select an integration flow from the list and go to [Endpoints](#) in the detailed view.

You can visualize the endpoint URL and the definition as specified in the following table, copy the URL to the clipboard or download the definition:

Adapter	Show Endpoint (access) URL	Show Definition
SOAP	X	X
IDOC	X	
HTTP	X	
AS2	X	

Download Options

Depending on the sender adapter type contained in the integration flow, you have different download options.

Adapter	Download Options
SOAP sender adapter	Download  Caution

Adapter	Download Options
	In case you want to download the WSDL in Cloud Foundry via a technical client, you need a service instance of the plan <code>api</code> assigned with the <code>MonitoringDataRead</code> role. It is not possible to use a service instance of plan <code>integration</code> flow. See: Creating Service Instance and Service Key for Inbound Authentication and Tasks and Permissions. • WSDL Policies There are two options to download the WSDL: with or without policies. The WSDL with policies includes WS-policy assertions defining special requirements the sender system must adhere to. If the sender system doesn't understand WSDLs with policies, the WSDL without policies can be used instead.
OData sender adapter	Download You can download the EDMX files. → Remember • If such integration flows use User Role as authorization, download of EDMX files work only if you disable Single Sign-On (SSO) authentication in your browser. • For an OData API Project, download of EDMX files work only if you disable Single Sign-On (SSO) authentication.

Manage Security

The Manage Security section allows you to manage various kinds of security material (for example, user credentials, keystore entries), and to perform outbound connectivity tests.

Related Information

[Managing Security Material](#)

[Managing Keystore Entries](#)

[Managing PGP Keys](#)

[Managing Certificate-to-User Mappings, Neo Environment](#)

[Performing Connectivity Tests](#)

Managing Security Material

The Manage Security Material area provides an overview of security-related artifacts.

You open the **Manage Security Material** area with the following actions:

Select the tile **Security Material**.

Security Material Overview

A list of security material is displayed in a table. For each artifact, the following attributes are displayed:

Attributes of Security-Related Artifacts	
Attribute	Description
Name	Display name of the artifact
Type	Possible values: - User Credentials For <i>User Credentials</i> artifacts (a tooltip indicates the kind of the artifact). - OAuth2 Credentials - OAuth2 Authorization Code - Secure Parameter For <i>Secure Parameter</i> artifacts (a tooltip indicates the kind of the artifact). - Known Hosts (SSH)
Status	Displays the state with regard to the artifact deployment. This status indicates whether an artifact has been deployed successfully on a tenant or whether it still needs to be authorized. Possible values: - Stored i Note Refresh the list to see the current status. - Deployed - Error - Unauthorized (for OAuth2 Authorization Code)
Deployed By	User who deployed the artifact
Deployed On	Time when the artifact was deployed

i Note

To add PGP keyrings, see [Managing PGP Keys](#). To maintain keystore entires, see [Managing Keystore Entries](#)

To sort and filter the content of the table, choose **Table Settings** (⚙️). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

Actions

Actions

Create	To create/deploy a new artifact, choose Create and select the artifact type (possible for the following artifact types: User Credentials , OAuth2 Client Credentials , OAuth2 Password Credentials , OAuth2 SAML Bearer Assertion , OAuth2 Authorization Code and Secure Parameter).
Upload	To add a Known Hosts (SSH) file, select Upload .
Authorize	To authorize an existing artifact, select the artifact in the table and choose Authorize (only supported for OAuth2 Authorization Code).
Edit	To edit and redeploy an existing artifact, select the artifact in the table and choose Edit (only supported for Credentials).
Download	To download an artifact, select the artifact in the table and choose Download (only supported for Known Hosts).
Delete	You can also delete an artifact (supported for all artifact types except Keystore).

Related Information

[Deploying a User Credentials Artifact](#)

[Deploying an SSH Known Hosts Artifact](#)

[Deploying a Secure Parameter Artifact](#)

[Deploying an OAuth2 Client Credentials Artifact](#)

[Deploying a PGP Secret Keyring](#)

[Deploying a PGP Public Keyring](#)

Deploying a User Credentials Artifact

To set up a connection using basic authentication or username token authentication, you have to specify the required attributes (for example, user name and password).

Context

User Credentials (security artifact) on the corresponding tenant. You can specify basic authentication either for a connected SuccessFactors system (through the SuccessFactors adapter) or for general SOAP connectivity or an OpenConnectors service (through the OpenConnector adapter). To enable basic authentication for a runtime node, you specify the user credentials and deploy the attributes.

i Note

You can also edit and redeploy an existing artifact. To do that, select the artifact in the table under **Manage Security Material** and choose **Edit**.

Procedure

1. Choose **Monitor**.
2. Select the **Security Material** tile in the **Manage Security** section.

3. Choose **Add**.

4. As **Type**, select **User Credentials**.

5. Specify the following properties:

Option	Description
Name	A name for the artifact. The artifact name is used as an alias for the confidential data assigned by this parameter.
Description	A description of the artifact (optional).
Type	This allows you to configure a specific system to enable a connection with your integration flow artifact. If you select: ◦ SuccessFactors: Specify the Company ID (which indicates the client instance used to connect to the SuccessFactors system). ◦ OpenConnectors: Specify the Organization and Element (value obtained from the authorization header that helps to authenticate with the specific instance). See: OpenConnectors Receiver Adapter
User	The user that calls the receiver system.
Password	Password against which the user has to be authenticated. This parameter has a maximum length of 255 characters.

6. Choose **Deploy**.

Results

When you refresh the **Manage Security Material** page, the new artifact is displayed (with Type **Credentials**) in the artifact table.

Next Steps

i Note

When you have a User Credentials artifact, you also have to re-deploy and restart all integration flows that use this artifact.

Related Information

[Managing Security Material](#)

Deploying an OAuth2 Client Credentials Artifact

Many web servers use OAuth 2.0 for authorization purposes. If you want to connect to a system that uses OAuth 2.0 authentication, you need to deploy an OAuth2 Credentials artifact using the following procedure.

Context

You can edit and deploy an OAuth2 Client Credentials artifact.

i Note

Set the token expiry to at least 10 minutes. Otherwise, since the token is refreshed 5 minutes before its expiry, it can create a continuous refresh loop and trigger rate limits from the token server.

i Note

- The adapter using the OAuth 2 Client Credentials accesses the OAuth token and caches it against the name of the credential. The adapter uses the same token until it expires. Also, if you use the same credential in another adapter simultaneously, the token is reused.
- Every time you edit an OAuth2 Client Credentials artifact, you must re-enter the Client Secret.

i Note

Read [SAP Cloud Integration – Principal Propagation with SuccessFactors OData V2](#), to design and deploy an integration flow that talks to SuccessFactors OData V2 endpoint with OAuth2 authentication.

More information on OAuth: <https://tools.ietf.org/html/rfc6749>

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Add**.
4. As **Type**, select **OAuth2 Client Credentials**.
5. Specify the following attributes:

Attribute	Description
Name	Name for the credentials. This name is also called as "alias" when using in an adapter.
Description	Description for the credentials.
Token Service URL	URL of the OAuth2 authorization server that issues the access token.
Grant Type	Allows you to use grant type as one of the following. ◦ Send as part of URL: This includes grant type as part of the URL and is set by default for credentials that are already deployed or existing. ◦ Send as part of body: This includes grant type in the request body sent to the token authentication server.
Client ID	ID of the client that you're connecting to.
Client Secret	Secret key of the client that you're connecting to.

Attribute	Description
	OAuth2 uses a multiple step authentication pattern: Client credentials (Client ID and Client Secret , as specified in the artifact) are used by the client application to initially request an access token. The access token is then used to authorize the client (for as long as the token is valid) to access the server's resources (for example, the resources that are used in the associated integration flow). In many OAuth2 scenarios, the access token is issued (or generated) by an authorization server.
Client Authentication	Allows you to access an application using Client ID and Client Secret. By default the Send as Body Parameter is selected, this option sends the Client ID and Client Secret as a JSON content to the authentication server in the request body. If you select the Send as Request Header option, then the Client ID and Client Secret are encoded, and send to the server as an Authorization header.
Scope	Provide the OAuth2 scope information to be included in the request body.
Content Type	Request content type to indicate the media type.
Resource	Enter the identifier of the application or service that shares the same client secret. The identifier varies depending on the service that you want to connect with.
Audience	Enter the identifier of the application or service that shares the same client secret. The identifier varies depending on the service that you want to connect with.

Custom Parameter

Add or delete additional parameters in the custom parameter table, which includes three columns: **Key**, **Value**, and **Send as Part of**.

Each custom parameter can be sent in the request body, request header, or request URL, depending on specific requirements.

i Note

- Duplicate combinations of **Key**, **Value**, and **Send as Part of** fields are not allowed.
- The **Key** cannot be "client_id," "client_secret," or "grant_type".
- Fields cannot be empty or consist only of whitespace.
- The maximum length of the fields can be 1024 characters.
- You can add a maximum of 20 key-value pairs.

6. Choose **Deploy**.

Related Information

Deploying an OAuth2 Password Credentials

Deploy an OAuth2 Password Credentials to securely connect to systems that use OAuth 2.0 password grant authentication. The deployed credentials can then be referenced by integration adapters to obtain and reuse access tokens.

Context

You can edit and deploy an OAuth2 Password Credentials artifact.

i Note

Set the token expiry to at least 10 minutes. Otherwise, since the token is refreshed 5 minutes before its expiry, it can create a continuous refresh loop and trigger rate limits from the token server.

i Note

- The adapter using the OAuth 2 Password Credentials accesses the OAuth token and caches it against the name of the credential. The adapter uses the same token until it expires. Also, if you use the same credential in another adapter simultaneously, the token is reused.
- Every time you edit an OAuth2 Password Credentials artifact, you must re-enter the **Client Secret**. This option is available when **Enable Client Authentication** checkbox is selected.

i Note

Read [SAP Cloud Integration – Principal Propagation with SuccessFactors OData V2](#) , to design and deploy an integration flow that talks to SuccessFactors OData V2 endpoint with OAuth2 authentication.

For more information about OAuth, see <https://tools.ietf.org/html/rfc6749> .

How OAuth2 Authentication Works

When an integration flow runs, sends the Client ID and Client Secret to the authorization server of the target system. Once validated, an access token is issued from this server. then uses this token to access the target system's resources for the duration of the token's validity.

i Note

The Client ID and Secret is issued by the target system's authorization server when is registered as a client application.

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Create** > **OAuth2 Password Credentials** > .
4. Specify the following attributes:

Attribute	Description
Name	Name for the credentials. This name is also called as "alias" when using in an adapter.
Description	Description for the credentials.
Token Service URL	URL of the OAuth2 authorization server that issues the access token.
Grant Type	Allows you to use grant type as one of the following. ◦ Send as part of URL: This includes grant type as part of the URL and is set by default for credentials that are already deployed or existing. ◦ Send as part of body: This includes grant type in the request body sent to the token authentication server.
Username	Enter the username required to authenticate with the authorization server.
Password	Enter the password required to authenticate with the authorization server.
Enable Client Authentication	Select this checkbox to enable client authentication. When selected, you can specify the client ID, client secret and client authentication method.
Client ID (Available only if Enable Client Authentication checkbox is selected)	Unique identifier assigned to the application () by the target system's authorization server. It identifies as the registered client application and must always be used together with the Client Secret for authentication.
Client Secret (Available only if Enable Client Authentication checkbox is selected)	Confidential credential assigned to your application (Cloud Integration) by the target system's authorization server. It is used together with the Client ID to authenticate Cloud Integration before an access token is issued from the server.
Client Authentication (Available only if Enable Client Authentication checkbox is selected)	Allows you to access an application using Client ID and Client Secret. By default the Send as Body Parameter is selected, this option sends the Client ID and Client Secret as a JSON content to the authentication server in the request body. If you select the Send as Request Header option, then the Client ID and Client Secret are encoded, and send to the server as an Authorization header.
Scope	Provide the OAuth2 scope information to be included in the request body.
Content Type	Request content type to indicate the media type.
Resource	Enter the identifier of the application or service that shares the same client secret. The identifier varies depending on the service that you want to connect with.
Audience	Enter the identifier of the application or service that shares the same client secret. The identifier varies depending on the

Attribute	Description
	service that you want to connect with.

Custom Parameter

Add or delete additional parameters in the custom parameter table, which includes three columns: **Key**, **Value**, and **Send as Part of**.

Each custom parameter can be sent in the request body, request header, or request URL, depending on specific requirements.

i Note

- Duplicate combinations of **Key**, **Value**, and **Send as Part of** fields are not allowed.
- The **Key** cannot be "client_id," "client_secret," or "grant_type".
- Fields cannot be empty or consist only of whitespace.
- The maximum length of the fields can be 1024 characters.
- You can add a maximum of 20 key-value pairs.

5. Choose **Deploy**.

Deploying an OAuth2 SAML Bearer Assertion

Many web servers use OAuth 2.0 for authorization purposes. If you want to connect to a system that uses OAuth 2.0 authentication, you need to deploy an OAuth2 Credentials artifact using the following procedure.

Prerequisites

For technical user propagation workflow, you've created a key pair. See [Creating a Key Pair/SSH Key Pair](#).

Context

You can design your OAuth2 credential either via principal propagation or technical user propagation.

- Using principal propagation, you forward the identity of the logged-in user to the target OAuth2-supported system. This method works well during the runtime (execution) of integration artifacts.
- Using technical user propagation, you propagate a dedicated technical user profile mentioned in the key-pair common name as the principal to your target system. Currently, this method works with SuccessFactors OData V2 Receiver, SuccessFactors SOAP Sender, and SuccessFactors SOAP Receiver Adapters. Especially, in the query modeler, use technical user propagation. To know more about how to implement technical user propagation, read the [blog](#) .

i Note

Read [SAP Cloud Integration – Principal Propagation with SuccessFactors OData V2](#) , to understand how to design, and deploy an integration flow that communicates to SuccessFactors OData V2 endpoint with OAuth2 authentication.

More information on OAuth: <https://tools.ietf.org/html/rfc6749> .

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Add**.
4. As **Type**, select **OAuth2 SAML Bearer Assertion**.
5. Specify the following attributes:

Attribute	Description
Name	Name for the artifact that you want to deploy on the tenant.
Description	Description of the artifact name you're deploying on the tenant.
Audience	Provide the host name of the target system, to which you want to establish the connection.  Example www.successfactors.com
Client Key	A unique identifier created by the target system to identify the client. For fetching a client key for SuccessFactors, see Registering Your OAuth2 Client Application. i Note Every time you edit an OAuth2 credentials artifact, you must re-enter the client key.
Token Service URL	Provide the URL that generates OAuth2 token for the registered OAuth2 client.
Target System Type	Specify the relevant host system for authenticating the user against the system. Select one of the following systems : - SuccessFactors - SAP BTP (Neo) - SAP BTP (CF)
Company ID (only if you select SuccessFactors for target system type)	Specify the company ID of your SuccessFactors instance.
User ID (only if you select SuccessFactors for target system type)	Choose one of the following: - Principal Propagation - Key Pair Common Name (CN)
Additional Properties (only if you select SuccessFactors for target system type and Principal Propagation for user ID)	Maintain the Value of the key field userIdSource for each environment as follows: If your tenant is on Neo environment, set the value as mail.

Attribute	Description
	If your tenant is on Cloud Foundry environment, set the value as email.
Key Pair Alias (only if you select SuccessFactors for target system type and Key Pair Common Name (CN) for user ID)	Provide the alias name that you defined in the prerequisites.
Token Service User (only if you select SAP BTP for target system type)	Username required to access the Token Service URL .
Token Service Password (only if you select SAP BTP for target system type)	Password required to access the Token Service URL .

6. Choose **Deploy**.

Related Information

[Managing Security Material](#)

Deploying an OAuth2 Authorization Code

Many web servers can use OAuth 2.0 for authorization purposes. If you use an OAuth2 Authorization Code, the authentication is done via a token service as an intermediary step. The client can exchange the OAuth2 Authorization Code for an access token. Your user credentials are never shared with the client.

Context

You can edit and deploy an OAuth2 Authorization Code. See also: [OAuth 2.0 Authorization Code Grant](#) and for more information on OAuth: <https://tools.ietf.org/html/rfc6749> .

- [Deploying an OAuth2 Client Authorization Code with Generic Provider](#)
- [Deploying an OAuth2 Client Authorization Code with Microsoft 365 as Provider](#)

Related Information

[OAuth 2.0 Authorization Code Grant](#)

[OAuth 2.0](#)

Deploying an OAuth2 Client Authorization Code with Generic Provider

Create an OAuth2 Client Authorization Code with the Generic Provider.

Context

Cloud Integration supports OAuth Clients with various update processes for their refresh tokens:

- **Refresh Tokens that expire and are invalidated:** There are some OAuth clients that create a new refresh token that automatically and immediately invalidates the old refresh token. This means, the old refresh token can no longer be used as soon the new refresh token is available.

Example: OAuth Client of GoToWebinar.

- **Refresh Tokens that expire but aren't invalidated:** There are some OAuth Clients that create a new refresh token, however, the old refresh token is still valid until its expiry date. In this case, a refresh token is valid during the whole expiry period, that is from creation time to creation time + expiry period.

Example: NEO OAuth2 Client;

- **Refresh Tokens that don't expire:** There are OAuth clients whose refresh token don't expire. In this case the refresh token doesn't need to be updated at all.

i Note

The following update processes aren't supported:

- OAuth Clients with refresh tokens that do expire but that can't be updated (rotated) aren't supported, since the connection doesn't work anymore after the refresh token has expired. This applies, for instance, to the SAP XSUAA OAuth client.
- OAuth Clients which require proprietary parameters for updating the refresh token in the token request, for instance Microsoft 365, which will only return the refresh token in the token request, if you specify the scope "offline_access". See: [Deploying an OAuth2 Client Authorization Code with Microsoft 365 as Provider](#).

There are OAuth clients which require additional query parameters in the token or authorization URL.

Cloud Integration supports these kinds of OAuth clients. You can just specify these parameters in the URL, for example:

- the **Google OAuth client** requires the additional query parameters "prompt=consent" and "access_type=offline" in the authorization URL: `https://accounts.google.com/o/oauth2/v2/auth?prompt=consent&access_type=offline;`
- the **NEO OAuth client** requires the additional query parameter "token_format=JWT" to support offline token verification in the token URL: `https://oathasservices-<application ID>.<data center>.hana.ondemand.com/oauth2/api/v1/token?token_format=JWT;`
- the **Dropbox client** requires the additional query parameter "token_access_type=offline" in the authorization URL: `https://www.dropbox.com/oauth2/authorize?client_id=<app_id>&token_access_type=offline&response_type=code&redirect_uri=http%3A%2F%2Flocalhost%2Fcode&state=123;`

⚠ Caution

- The Refresh Token expiry period must be at least 3 days.
- The update of the Refresh Token must occur at least 2 days before the Refresh Token expires.
- Refresh Tokens that are invalidated after their first usage are not supported.
- A refresh token must not be invalidated by a succeeding refresh token before 20 per cent of the expiry period is reached.
- The Access Token expiry period must be at least 30 min. We recommend an expiry period of 1 hour.
- The maximum number of OAuth2 Authorization Code credentials is 500 in Cloud Foundry and 60 in Neo (including the Microsoft 365 credentials).

- You can't use the same OAuth Client that supports only one valid Refresh Token in one point in time in more than one OAuth2 Authorization Code credential with the same user.

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Add**.
4. As **Type**, select **OAuth2 Authorization Code**.
5. Specify the following attributes:

Attribute	Description
Name	Name of the artifact that you want to deploy on the tenant.
Description	Description of the artifact you're deploying on the tenant.
Provider	Enter the name of the provider of the platform on which you created the OAuth2 client.
Authorization URL	Provide the Authorization URL for authorizing the OAuth client to access resources of a user. You can add also additional query parameters, as described at the beginning of this chapter. Be aware that the following query parameters are added automatically: <code>client_id=<client_id>&response_type=code&redirect_uri=<url encoded redirect uri>&state=<state guid>&scope=<scope></code>. The scope query part is only added if the scope attribute is not empty, see below.
Token Service URL	Provide the URL that generates the OAuth2 access and refresh token for the registered OAuth2 client and the provided user. You can add also additional query parameters, as described at the beginning of this chapter. The scope attribute (see below) is added as query parameter automatically in case of the "authorization_code" grant type request; the scope attribute is not added in case of the "refresh_token" grant type request.
Refresh Token Expiry	Provide an amount of time (in days) after which the refresh token expires. The minimum value is 3 days. You can also leave this attribute blank (don't type in anything), in case your refresh token doesn't expire (see: Supported OAuth Client Types in this chapter). The default value is blank.
Redirect URL	Displays the URL you need, when creating the OAuth Clients/App in OAuth Authorization Server/Token Server. For more information, see OAuth2.0 Authorization Code Grant
Client ID	ID of the client you want to connect to.
Client Secret	Secret key of the client that you're connecting to. See: OAuth 2.0.
Send As	◦ Body Parameter: Send the Client ID and Client Secret in the request body when calling the Authorization URL or Token Service URL. ◦ Basic Auth Header: Send the Client ID and Client Secret via Basic Authorization when calling the Authorization URL or Token Service URL.
User Name	Name of the user whose resources the OAuth2 client gets access to.
Scope	OAuth2 scopes protecting the access to the resources.

Attribute	Description
	The scope value is added to the Authorization URL (see above) as query parameter and is also added to the Token Service URL, in case of the "authorization_code" grant type request; but is not added to the Token Service URL in case of the "refresh_token" grant type request (see above). i Note If you add more than 1 scope, you need to separate your scopes by a blank space.

i Note

Test your new setup regularly in the beginning, to waterproof the functionality of your setup. Especially test the replacement of refresh tokens, whether they expire or the new refresh token invalidates the old refresh token.

6. Choose **Deploy**.

Related Information

[Managing Security Material](#)

[OAuth 2.0](#)

[Deploying an OAuth2 Client Authorization Code with Microsoft 365 as Provider](#)

Deploying an OAuth2 Client Authorization Code with Microsoft 365 as Provider

Context

You can edit and deploy an OAuth2 Authorization Code. See also: [OAuth 2.0](#) and for more information on OAuth: <https://tools.ietf.org/html/rfc6749> .

i Note

Refer to [Deploying an OAuth2 Client Authorization Code with Generic Provider](#), if you want to use other providers than "Microsoft 365".

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Add**.
4. As **Type**, select **OAuth2 Authorization Code**.
5. Specify the following attributes:

Attribute	Description
Name	Name of the artifact that you want to deploy on the tenant.
Description	Description of the artifact you are deploying on the tenant.

Attribute	Description
Provider	Enter the name of the provider of the platform on which you created the OAuth2 client.
Authorization URL	Provide the Authorization URL for authorizing the OAuth client to access resources of a user.
Token Service URL	Provide the URL that generates the OAuth2 access and refresh token for the registered OAuth2 client and the provided user.
Refresh Token Expiry	Provide an amount of time after which the refresh token expires. The maximum is set to 90 days and the minimum is set to 3 days. The SAP Cloud Integration system will update the refresh token before the expiration.
Redirect URL	Displays the URL you need, when creating the OAuth Clients/App in OAuth Authorization Server/Token Server. For more information, see OAuth 2.0 Authorization Code Grant .
Client ID	ID of the client you want to connect to.
Client Secret	Secret key of the client that you are connecting to. See: OAuth 2.0 .
Send As	- Body Parameter: Send the Client ID and Client Secret in the request body when calling the Authorization URL or Token Service URL. - Basic Auth Header: Send the Client ID and Client Secret via Basic Authorization when calling the Authorization URL or Token Service URL.
User Name	Name of the user whose resources the OAuth2 client gets access to.
Scope	OAuth2 scopes protecting the access to the resources. i Note Configure Mail Sender and Receiver Adapter: - The scope https://outlook.office.com/SMTP.Send is necessary for sending e-mails from Outlook 365. - The scope https://outlook.office.com/IMAP.AccessAsUser.All is necessary for reading e-mails from Outlook 365. - You can also request an offline_access scope . When a user approves the offline_access scope, your app can receive refresh tokens from the Microsoft identity platform token endpoint. If the offline_access scope is not provided, is adding this scope automatically to the credential since Cloud Integration needs the refresh token to create access tokens. - If you add more than one scope, you need to separate your scopes by a blank space.

6. Choose **Deploy**.

Related Information

[Managing Security Material](#)

[Deploying an OAuth2 Client Authorization Code with Generic Provider](#)

[OAuth 2.0](#)

Deploying a Secure Parameter Artifact

Use the secure parameter artifact to deploy confidential data, for example, for custom adapters.

Context

You can store secure data like passwords in a secure store and use an alias name to access this data in an integration flow. The secure parameter artifact contains this alias name and confidential data.

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Add**.
4. As **Type**, select **Secure Parameter**.
5. Specify the following attributes.

Option	Description
Name	Provide a name for the artifact. The artifact name is used as an alias for the confidential data assigned by this parameter.
Description	Provide a more detailed description of the artifact.
Secure Parameter	Enter the confidential value of the attribute. i Note The possible length of the secure parameter varies: - o for Cloud Foundry: 4096 characters max. (including spaces) - o for Neo: 255 characters max. (including spaces)
Repeat Secure Parameter	Repeat the confidential value of the attribute.

6. Choose **Deploy**.

Results

When you refresh the **Manage Security Material** page, the new artifact is displayed in the table.

Related Information

[Managing Security Material](#)

Deploying an SSH Known Hosts Artifact

This artifact type specifies the known hosts file used when configuring secure connectivity based on SSH File Transfer Protocol (SFTP).

Context

An SSH known hosts artifact contains the public keys and addresses of trusted SFTP servers.

Procedure

1. Choose **Monitor**.
2. Click the **Security Material** tile in the **Manage Security** section.
3. Choose **Create** and select **Known Hosts (SSH)**.
4. In the **Create SSH Known Hosts** dialog, you can import entries from a file using **Browse**, add entries manually using **Add**, or edit entries directly in the table.

Caution

Using **Browse** replaces all existing known host entries in the table with the entries from the imported file.

5. Choose **Deploy**.

i Note

For detailed instructions on creating, editing, and maintaining known host entries, see [Maintaining SSH Known Hosts for SFTP Connectivity](#).

Results

When you refresh the **Manage Security Material** page, the new artifact is displayed in the table.

Related Information

[Managing Security Material](#)

Managing Keystore Entries

The Keystore Monitor allows a tenant administrator to manage the tenant keystore and its entries (X.509 certificates and key pairs).

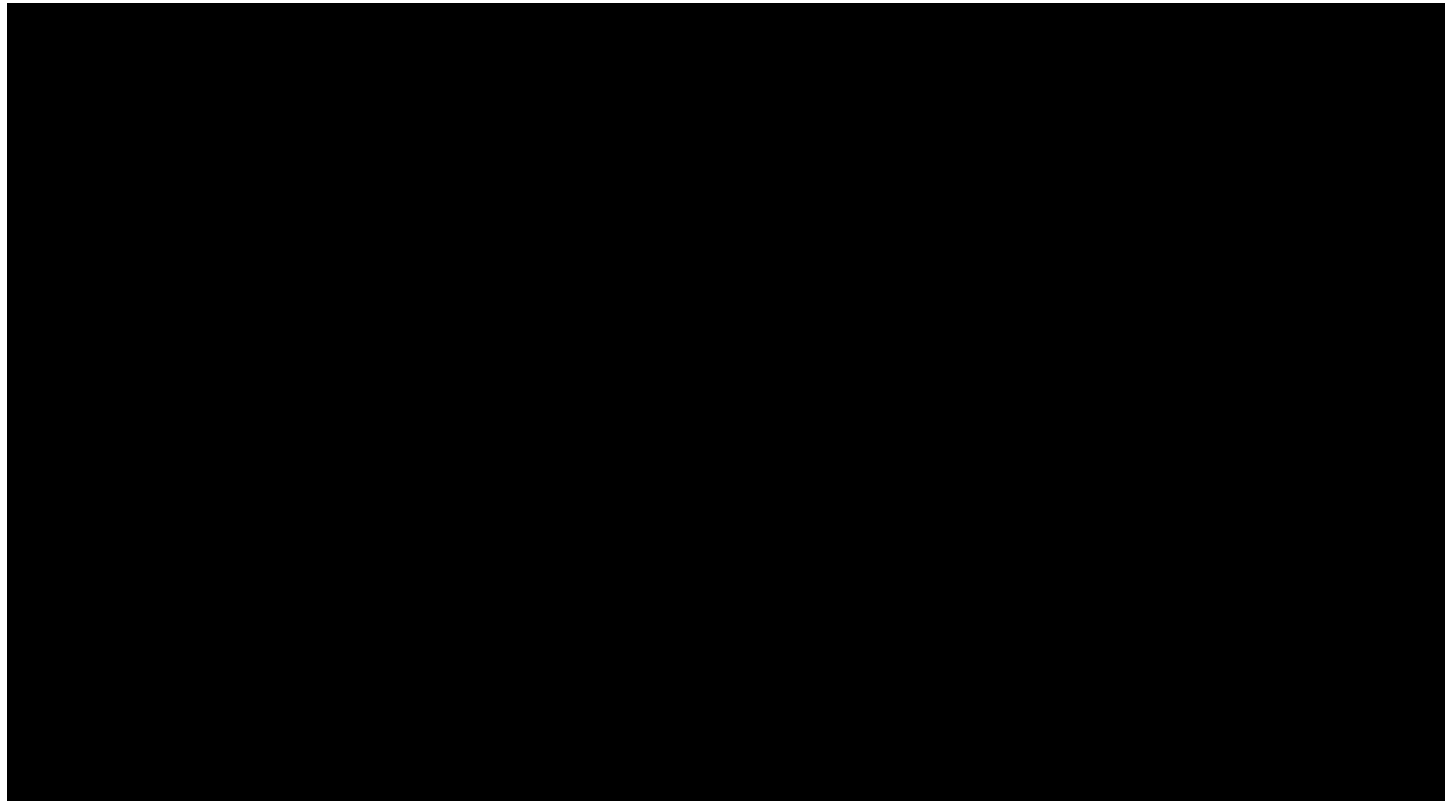
Context

Connections between a Cloud Integration tenant and a remote system can be secured using client certificate authentication (with X.509 certificates). The key pairs and certificates required to implement this authentication option are stored in a keystore.

A keystore typically contains entries that belong to the tenant administrator (customer) and entries that are owned by SAP. Each entry is uniquely identified by an alias.

The operations you can perform on a keystore entry depend on whether it's owned by SAP or the tenant administrator. You (as tenant administrator) can, for example, only delete or back up entries that are owned by the tenant administrator. Vice versa, SAP can't access or download any keystore entries owned by the tenant administrator.

The following GIF gives you a brief overview of the **Keystore** tile and its functions:



Keystore entries are identified by an alias.

There are the following entry types:

- **Key Pair** entry

Consists of a private key and its X.509 certificate chain.

All private keys of a keystore are encrypted with the same password. This password is also used as the keystore password (for checking the integrity of the keystore). The keystore is never stored in the same database as the encrypted/signed application data. The password is stored in a separate database.

The certificate chain typically consists of the public key certificate and the intermediate certification authority (CA) certificate with which the signature of the public key certificate can be verified.

- **Certificate** entry

In many cases this is an X.509 root certificate.

More information: [Keystore](#)

Caution

Note the following restrictions when managing keystore entries:

The maximum size of a keystore is 2 MB (when using the Neo environment).

- The 2 MB limit corresponds to around 2000 X.509 certificates.
- A key pair with a chain of three X.509 certificates consumes about 3 KB, so if the keystore only contains key pairs of this type, then you can store around 600 key pairs in the keystore.

The maximum size of a keystore is 6 MB (when using the Cloud Foundry environment).

- The 6 MB limit corresponds to around 6000 X.509 certificates.

- A key pair with a chain of three X.509 certificates consumes about 3 KB, so if the keystore only contains key pairs of this type, then you can store around 1800 key pairs in the keystore.

If you upload a whole keystore (.jks file) to the tenant, the maximum keystore size is limited to 2 MB.

A locking mechanism prevents different users from changing or deleting entries of the keystore at the same time. If a user is changing the content of a keystore, the keystore is locked for other users during that time.

Private keys can't be downloaded from a keystore.

Procedure

1. Choose **Monitor**.
2. Choose the **Keystore** tile in the **Manage Security** section.
3. The **Current** tab shows the keystore entries available for the tenant.

For each keystore, the single entries are displayed (by alias).

The Keystore Monitor shows all keystore entries in a table. For each entry, the following attributes are displayed:

Attribute	Description
Alias	Uniquely identifies the keystore entry. A keystore entry can be either be owned by the tenant administrator or by SAP. In case a keystore entry it is owned by SAP, the lock icon (🔒 <i>Maintained by SAP</i>) is shown next to the alias. SAP-owned keystore entries provide a reduced set of functions. For example, you cannot update or delete an SAP-owned keystore entry.
Type	Indicates whether the entry is a Certificate (X.509 certificate) or a Key Pair (with public and private key and an X.509 certificate chain). ◦ Key Pair Consists of a private key and its X.509 certificate chain. ◦ Certificate In many cases this is an X.509 root certificate.
Valid Until	Indicates the expiration date. If keys and certificates have expired, the date is highlighted in red.
Last Modified At	Indicates the date and time the entry was last modified.
Subject DN	Indicates the Subject distinguished name (DN) which entails information about the owner of the certificate such as the common name of the certificate owner (CN), the organization (O), or the country or region (C).
Issuer DN	Indicates the Issuer distinguished name (DN) which entails information about the issuer of the certificate such as common name of the certificate issuer (CN), the organization (O), or the country or region (C).
Actions	Choose the  (<i>Actions</i>) button for a key pair or certificate to perform an action. The available actions depend on the type of the keystore entry and whether it is owned by SAP. ◦ Rename

Attribute	Description
	◦ Update ◦ Download ◦ Delete ◦ Download Certificate ◦ Download Public OpenSSH Key ◦ Update Signing Response ◦ Download Certificate Chain ◦ Download Root Certificate ◦ Download Signing Request

i Note

SAP-owned entries are indicated by a lock icon. You can't change, delete, or back them up.

Next Steps

The Keystore Monitor also gives you the following options:

Actions

Create	To create a Key Pair or an SSH Key Pair, select Create and specify the artifact. Fill in the necessary details. See also Creating a Key Pair/SSH Key Pair.
Add	To upload an entry to the keystore or to add a keystore, select Add. You can upload individual entries to an existing keystore. In the latter case, you can overwrite existing entries or keep them. SAP-owned keystore entries are indicated by a lock icon. You can't change or delete them. There are the following options: • Add > Certificate See: Uploading a Certificate • Add > Key Pair See: Uploading a Key Pair • Add > SSH Key See: Uploading an SSH Key • Add > RSA Key See: Uploading an RSA Key • Add > Keystore See: Uploading a Keystore

Download	To download the public content of a keystore or a single keystore entry, select Download. See also: • Downloading a Keystore • Downloading Single Keystore Entries
Back Up	To back up key store entries, see: Backing Up Keystore Entries
Download Back Up	To download backed-up keystore entries, select Back Up . See also: Downloading Backed-Up Keystore Entries .
Delete	To delete an artifact, go to ► Monitor ► Managing Security Material ► and choose Delete after the selecting the artifact in the table. See also: • Deleting Keystore Entries • Managing Security Material
Restore	To restore backed-up keystore entries, see: Restoring Backed-Up Keystore Entries
Update	To update an artifact, see: • Updating a Key Pair • Updating a Key Pair with a Signing Response • Updating an RSA Key • Updating a Certificate • Changing the Alias of a Keystore Entry

Related Information

[Managing Security Material](#)

[Displaying Properties of a Keystore Entry](#)

[Certificate Management](#)

[Client Certificate Authentication and Certificate-to-User Mapping \(Inbound\), Neo Environment](#)

[Client Certificate Authentication \(Inbound\), Neo Environment](#)

[Client Certificate Authentication \(Outbound\)](#)

[Managing the Lifecycle of Keys](#)

[Creating a Key Pair/SSH Key Pair](#)

[Cloud Integration – Keystore Monitor now available for Tenant Administrator](#) ➡

Displaying Properties of a Keystore Entry

Display properties of a selected keystore entry.

Context

There are the following entry types:

- **Key Pair** entry

Consists of a private key and its X.509 certificate chain.

All private keys of a keystore are encrypted with the same password. This password is also used as the keystore password (for checking the integrity of the keystore). The keystore is never stored in the same database as the encrypted/signed application data. The password is stored in a separate database.

The certificate chain typically consists of the public key certificate and the intermediate certification authority (CA) certificate with which the signature of the public key certificate can be verified.

- **Certificate** entry

In many cases this is an X.509 root certificate.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. Click the alias of a keystore entry to show the details.
3. The following attributes are shown for the selected keystore entry.

In case, the keystore entry is part of a certificate chain, the certificate chain is displayed as a hierarchy tree at left.

The following details are displayed for the selected keystore entry. When you click a certain element in the certificate chain, the details are adapted to show the details of this element.

- The **Keystore** attribute indicates if the displayed entry is part of the currently active keystore (**Current**) or of the backed-up keystore (**Backup**).
- The **Alias** is used to refer to a specific public key from a keystore.
- The **Owner** can either be **SAP** or **Tenant Administrator**.

Administration Attributes

Attribute	Description
Created By	User who created the certificate
Created On	Time when certificate has been created
Last Modified By	User that modified the keystore entry at the latest
Last Modified On	Time when the keystore entry has been modified at the latest

i Note

The user shown in **Last Modified By** is anonymized (the string **SAP** is shown) in case the keystore is owned by SAP.

General Attributes

Attribute	Description
-----------	-------------

Attribute	Description
Type	- Key Pair Consists of a private key and its X.509 certificate chain. - Certificate In many cases this is an X.509 root certificate.
Version	Certificate version
Serial Number	Serial number of certificate as provided by the issuer
Subject DN	Distinguished name of the entity associated with the public key of the certificate
Issuer DN	Distinguished name of the issuer (the certification authority that issued and signed the certificate)
Key Type	Cryptographic standard the keystore entry is using (for example, RSA)
Key Size	Number of bits (also referred to as <i>Key Length</i>)
Signature Algorithm	Algorithm used to create the signature of the key entry (for example, SHA - 512)
Valid From	Start of validity period
Valid Until	End of validity period

The section **Fingerprints** shows different sequences defined to identify the public key. A fingerprint is generated out of the public key applying a hash function on the public key. For each fingerprint, the hash algorithm is also displayed (for example, SHA - 512).

Next Steps

The following functions are available:

- You can delete the keystore entry in case it is owned by the Tenant Administrator.
- You can download the keystore entry.
- You can change the alias of the keystore entry in case it is owned by the Tenant Administrator.

Related Information

[Changing the Alias of a Keystore Entry](#)

Creating a Key Pair/SSH Key Pair

Create a key pair or a Secure Shell (SSH) key pair.

Context

You create a key pair to use it for SSL, decryption, signature, and client certificate authentication.

You create an SSH key pair to connect to the SFTP server.

Procedure

1. In the Monitor view, choose ► **Manage Security** ► **Keystore** ►.
2. In the **Current** tab, choose ► **Create** ► **Key Pair** ► or ► **Create** ► **SSH Key** ► depending upon your requirement.
3. In the next screen, enter the required information.

Attribute	Description
Alias	Alias for the artifact that you want to create. The alias must be unique. You can't create an alias that is already existing in the tenant for another key pair or SSH key.
Key Type	The available options are: ◦ RSA ◦ EC ◦ DSA (only for key pair)
Key Size	Pick a key-bit value of your choice from the list.
Signature Algorithm	Pick a hash algorithm of your choice from the list.
Common Name (CN)	Provide a common name for the technical user. You use this common name while deploying an OAuth2 credential.
Organizational Unit (OU)	(Optional): Enter the department name within your organization.
Organization (O)	(Optional): Enter the name of your organization.
Location (L)	(Optional): Enter the name of the city or town.
State or Province (ST)	(Optional): Enter the name of your state or province.
Country/Region (C)	Enter the two-letter ISO code for the country.
E-Mail (E)	(Optional): Enter the email address associated to the user.
Valid From	Enter the date from which you want the key pair to be active.
Valid Until	Enter the date until which you want the key pair to be active.

For more information on certificates, see [X.509 Certificates](#).

4. Choose **Create**.

Related Information

[Creating SFTP Keys](#)

[Uploading a Key Pair](#)

[Updating a Key Pair](#)

Uploading a Certificate

Upload a single certificate to the tenant keystore.

Context

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, above the table, choose **Add**, then **Certificate**.
3. Enter the alias (**Alias** field).

i Note

You cannot import a certificate with an alias starting with **sap_** (as this namespace is reserved for SAP-owned keystore entries).

Make sure you don't enter an alias that is already used by an existing keystore entry.

4. Choose **Browse** and select the certificate on your local disk.
5. Choose **Add**.

i Note

If there is already an entry with this alias in the keystore, an error message is shown.

Currently, the size limit for uploading certificates to the keystore is at 10240 bytes.

If the system cannot verify the trustworthiness of the certificate (for example, it cannot verify its signature), a warning message is displayed, which also contains the fingerprints.

A fingerprint is generated out of the public key applying a hash function to it. You can check the trustworthiness of the public key by comparing the fingerprint with a fingerprint for the public key provided by your communication partner via a trustworthy channel (phone, signed e-mail).

Uploading a Key Pair

Upload a private/public key pair to the tenant keystore.

Context

i Note

You cannot import a key pair for a keystore entry with an alias starting with **sap_** or **hccertificate** (as these entries are owned by SAP).

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, above the table, choose **Add > Key Pair >**.
3. Enter the alias (**Alias** field).

i Note

You cannot import a key pair with an alias starting with (as this namespace is reserved for SAP-owned keystore entries). You cannot import a key pair with an alias starting with `sap_` (as this namespace is reserved for SAP-owned keystore entries). You cannot import a key pair with an alias starting with

Make sure you don't enter an alias that is already used by an existing keystore entry.

4. Choose **Browse** and select the key pair on your local disk.

You can also upload files with the extension `.pfx` and `.p12`.

⚠ Caution

Note that a `.pfx` or `.p12` file can contain one or more keys. In case the file contains only one key, you can upload it as single key pair with this feature. If the file contains more than one key, it is recommended to upload it as keystore (via **Add > Keystore**). The reason is: If you try to upload a `.pfx` or `.p12` file that contains more than one key via **Add > Key Pair**, the system will only upload one key and ignore the others.

5. Enter the password associated with the private key.
6. Choose **Add**.

i Note

If there is already an entry with this alias in the keystore, an error message is shown.

Currently, the size limit for uploading key pairs to the keystore is at 30720 bytes.

If the system cannot verify the trustworthiness of the key pair (for example, if it is self-signed), a warning message is displayed, which also contains the fingerprints.

A fingerprint is generated out of the public key applying a hash function to it. You can check the trustworthiness of the public key by comparing the fingerprint with a fingerprint for the public key provided by your communication partner via a trustworthy channel (phone, signed e-mail).

Uploading an RSA Key

Upload an RSA key to the tenant keystore.

Context

You can add an RSA key to the keystore monitor instead of creating one.

Using this action, you create a new keystore entry based on an uploaded RSA key file.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, above the table, choose **Add > RSA Key**.
3. Choose **Browse** and select the RSA key on your local disk.

You can only upload PEM-encoded RSA keys in PKCS#1 format.

The alias is generated based on the file name.

4. Select the signature algorithm.

By default, **SHA-256/RSA** is set.

5. Enter the required information for the following attributes.

These attributes represent the coordinates of the organization associated with the certificate (such like **Organization (O)**, **Organizational Unit (OU)**, **Country/Region (C)**, and so forth).

Note that for **Common Name (CN)** you need to specify an address (for example, myhost . com). Which address to specify, depends on the usage of the key pair. For example, if you use the key pair to sign a message, the address of the signing component is to be specified.

For more information on certificates, see: [X.509 Certificates](#).

6. Check the settings vor the validity (**Valid From** and **Valid Until** fields) and, if desired, change them (by default, the validity is set to 10 years).

7. Choose **Add** to create the key entry.

Uploading a Keystore

Upload a new keystore. You can replace an existing keystore completely or update an existing one by adding new entries. In the latter case, you can overwrite existing entries or keep them.

Context

i Note

You cannot upload keystore entries with aliases starting with sap_ or hcicertificate. Theses aliases are reserved for SAP-owned keystore entries.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, above the table, choose ► **Add** ► **Keystore** ►.
3. Choose **Browse** and select the keystore on your local disk.

Upload of keystores supports the following formats:

- JKS format (file extension . j ks) for Java keystores
- PKCS#12format (file extension . pfx or . p12)

A . pfx or . p12 file can contain one or more keys. In case the file contains only one key, you can also upload it as single key pair (via ► **Add** ► **Key Pair** ►).

After the import of a . pfx or . p12 file, the root certificates of all keys contained in the keystore will be displayed separately in the keystore overview (in Web UI **Monitoring** under **Manage Security** in the **Keystore** tab).

4. Enter the keystore password.
5. In the **Action** field, select one of the following options:

Option	Description
Add	Adds the entries from the uploaded keystore so that they are merged with the ones in the existing keystore.

Option	Description
Replace	Replaces the whole keystore with the uploaded one. You need to confirm the replacement of the existing keystore. i Note Exception: SAP-owned entries are preserved as they cannot be changed or deleted by the tenant administrator.

- If you have selected **Add** (in the **Action** field), you can specify whether existing entries are to be overwritten by uploaded entries with the same alias.
- Choose **Deploy**.

Next Steps

A dialog is displayed showing a summary of the added, preserved, and overwritten entries. If any entries could not be uploaded, information is provided to explain why (for example, you tried to upload a keystore entry with an alias starting with `sap`).

Uploading an SSH Key

Upload an SSH key or a putty key to the tenant keystore.

Context

It is possible to add an SSH or putty key to the keystore monitor instead of creating one. If you want to learn more about how to setup connections to the sftp servers, go and visit this blog: [Cloud Integration - How to Setup Secure Connection to sftp Server](#) 🐼

Procedure

- Choose the **Keystore** tile in the **Manage Security** section.
- On the **Current** tab, above the table, choose **Add > SSH Key**.
- Enter an alias for the keystore entry (**Alias** field).
- Choose **Browse** and select the SSH key or putty key on your local disk.
- Specify the password for the key and define key specific values.
- Enter the required information for the certificate attributes.

These attributes represent the coordinates of the organization associated with the certificate (such like **Organization (O)**, **Organizational Unit (OU)**, **Country/Region (C)**, and so forth).

Note that for **Common Name (CN)** you need to specify an address (for example, `myhost.com`). Which address to specify, depends on the usage of the key pair. For example, if you use the key pair to sign a message, the address of the signing component is to be specified.

For more information on certificates, see: [X.509 Certificates](#).

- Check the settings vor the validity (**Valid From** and **Valid Until** fields) and, if desired, change them (by default, the validity is set to 10 years).
- Choose **Add** to create the key.

Changing the Alias of a Keystore Entry

Context

An alias is a reference to a single keystore entry. You can use an alias to refer to and select a specific public key from a keystore.

i Note

You can only change the alias of a keystore entry owned by the Tenant Administrator.

The maximal length of a keystore alias is 250 characters.

You cannot use aliases that are reserved for SAP (for example, starting with `sap`).

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. Click the alias of a keystore entry to show the details.
3. Choose **Rename**.
4. Enter the new alias and choose **Save**.

Next Steps

Alternatively, you can change the alias of a keystore entry in the following way: Choose the **Keystore** tile in the **Manage Security** section and for a keystore entry click the  (Actions) icon and choose **Rename**.

Updating a Key Pair

Update a key pair keeping the alias of the keystore entry unchanged.

Context

This option is not available for keystore entries owned by SAP.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, select a key pair, choose the  (Actions) icon, then select **Update**. Alternatively, you can click the key pair alias to open the key pair details, and then choose  **Update** .
3. Browse to the key pair file on your computer and enter the password (as the key pair also contains a private key that is protected by a password).

i Note

You cannot change the alias.

4. Choose **Update**.

The key pair is updated in the keystore.

Updating a Key Pair with a Signing Response

Upload a signing response from a certification authority and use it to update the key pair in your keystore, keeping the alias of the keystore entry unchanged.

Prerequisites

You have created a certificate signing request (CSR) for the key pair and used this to request a signed certificate from a certification authority (CA). The certification authority has provided a signing response.

Context

This option is not available for keystore entries owned by SAP.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, select the key pair that the signing response was created for.
3. Choose the  (Actions) icon, then select **Update Signing Response**. Alternatively, you can click the key pair alias to open the key pair details, and then choose  **Update** .
4. Browse to the signing response file that you want to upload.

i Note

The following signing response formats are supported:

- PKCS#7 in binary or PEM or base64 textual encoding (*.p7c)
- Chain of DER-PEM encoded X.509 certificates in one file (*.*)
- X.509 PKI path (sequence of certificates with top CA certificate at index 0) (*.pkipath)
- Software Publisher Certificate (SPC) (*.spc)

5. Choose **Update**.

i Note

Currently, the upload size limit for signing responses is at 30720 bytes.

Results

The key pair is updated in the keystore. The key pair can now be used to set up secure connections to external receiver back ends.

Related Information

[Cloud Integration – How to Setup Secure Outbound HTTP Connection using Keystore Monitor](#) 

Updating an RSA Key

Update an RSA key.

Context

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, select the RSA key.
3. Choose the  (Actions) icon, then select **Update**. Alternatively, you can click the alias to open the RSA key details, and then choose .
4. Browse for the RSA key file on your computer.

You can only upload PEM-encoded RSA keys in PKCS#1 format.

The alias is changed based on the file name.

5. Choose **Update**.

Updating a Certificate

Update a certificate keeping the alias of the keystore entry unchanged.

Context

This option is not available for keystore entries owned by SAP.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, select the certificate.
3. Choose the  (Actions) icon, then select **Update**. Alternatively, you can click the certificate alias to open the certificate details, and then choose .
4. Browse for the certificate file on your computer.

You cannot change the alias.

5. Choose **Update**.

The new certificate is added to the keystore entry.

Downloading a Keystore

Download a keystore to your local disk.

Context

When you've downloaded the keystore file, you can, for example, open it with an external tool.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, choose **Download** (above the table).

Next Steps

The public part of the keystore is downloaded (in a file with the name `PublicKeyKeystore.jks`).

The file is not password protected. You can open it with a third-party keystore editor (for example, KeyStore Explorer).

For private key entries, the certificate chains are resolved into single entries. The alias of the public key certificate is the same as the alias of the original key-pair entry. The intermediate certificate gets an additional suffix `_1` in the alias name. The root certificate gets an additional suffix `_2`.

Downloading Single Keystore Entries

Download a single keystore entry to your local disk.

Context

You can download a key, a single certificate, a complete certificate chain, a root certificate, or a signing request (depending on the keystore entry type).

i Note

You can download only public keystore content (no private keys).

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, select a keystore entry.
3. Choose the  (Actions) icon, then select one of the options listed in table below. Alternatively, you can click the keystore entry alias to open the details, and then choose the option under **Download**.

The options available depend on the type of keystore entry you have selected.

Option	Description
Key Pair entry	You can download the public key (X.509 format). Download Certificate A file with the name <code><alias>.cer</code> is downloaded. Download Certificate Chain A file with the name <code><alias>.p7b</code> is downloaded. The file contains the whole certificate chain. Download Public OpenSSH Key Downloads the public key in OpenSSH format. You can store the downloaded <code><alias>.pub</code> file on the connected SFTP server to enable public key authentication on the server. i Note To configure public key authentication on certain SFTP servers, public keys need to be available in OpenSSH format. In such cases, use this download option.

Option	Description
	→ Tip Want to know how to get an SSH2 public key for the id_rsa/id_dsa key-pair of the tenant keystore? You can use ssh-keygen for this purpose (to be installed on Windows via Cygwin, for example). In ssh-keygen, perform the following command: <pre>\$ ssh-keygen -e -f id_rsa.pub -m RFC4716 > id_rsa.pub_ssh2</pre> Example result file: <pre> ---- BEGIN SSH2 PUBLIC KEY ---- Comment: "1024-bit RSA, converted by d023101@WDFN33785618A from OpenSS" AAAAB3NzaC1yc2EAAAADAQABAAQgQCRB+U0PmnPF9W4cqn6h1z3V5izPFA0TCXSF4cfnw ZivjhV8ZAcKq6QwV/Sy0XkXWp/w0bjKgTXtiHngdJ0ky0Q+66E1eq/yh04NDJ0QM3Vzv15 IhL+eLYBtynk1ddF5l6kDLgSz1evA9F988wWKiz/vpI8DVjbY8HJjlQbE8w0SQ== ---- END SSH2 PUBLIC KEY ---- </pre> ○ Download Root Certificate A file with the name <alias>_rootCA.cer is downloaded. i Note If the key pair is self-signed, this action performs the same function as Download Certificate. ○ Download Signing Request A file with the name <alias>.csr is downloaded.
Certificate entry	Download A file with the name <alias>.cer is downloaded. It contains the base64-encoded certificate.

For keystore entries owned by SAP, a restricted set of options is available. For an SAP key you can, for example, not download any signing request.

Next Steps

You can import the downloaded files into other keystores and open them using third-party tools (for example, KeyStore Explorer).

Deleting Keystore Entries

Context

This option is not available for keystore entries owned by SAP.

Procedure

1. Choose the [Keystore](#) tile in the [Manage Security](#) section.

2. On the **Current** tab, select a keystore entry.
3. Choose the  (Actions) icon, then select **Delete**. Alternatively, you can click the keystore entry alias to open the details, and then choose **Delete**.

Caution

You cannot delete SAP-owned keystore entries.

Backing Up Keystore Entries

Back up keystore entries owned by the tenant administrator.

Context

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Current** tab, choose **Back Up** (above the table).

Next Steps

You can view the backed-up entries on the **Backup** tab.

Related Information

[Restoring Backed-Up Keystore Entries](#)

<https://blogs.sap.com/2017/08/14/cloud-integration-backuprestore-using-keystore-monitor/> 

Restoring Backed-Up Keystore Entries

Restore backed-up entries of the keystore (owned by the tenant administrator).

Context

You can restore keystore entries that you have backed up. For example, you can restore entries if you delete them from the active tenant keystore by mistake.

i Note

Restored keystore entries from the keystore backup replace all existing entries of the tenant administrator. Therefore, be careful using this feature as you can delete by mistake existing keystore entries.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. On the **Backup** tab, the backed-up keystore entries are displayed in a table.

The following attributes are displayed for each entry:

Attribute	Description
Alias	Uniquely identifies the keystore entry. A keystore entry can be either be owned by the tenant administrator or by SAP. In case a keystore entry it is owned by SAP, the lock icon ( (<i>Maintained by SAP</i>)) is shown next to the alias. SAP-owned keystore entries provide a reduced set of functions. For example, you cannot update or delete an SAP-owned keystore entry.
Type	Indicates whether the entry is a Certificate (X.509 certificate) or a Key Pair (with public and private key and an X.509 certificate chain). - Key Pair Consists of a private key and its X.509 certificate chain. - Certificate In many cases this is an X.509 root certificate.
Valid Until	Indicates the expiration date. If keys and certificates have expired, the date is highlighted in red.
Last Modified At	Indicates the date and time the entry was last modified.
Subject DN	Indicates the Subject distinguished name (DN) which entails information about the owner of the certificate such as the common name of the certificate owner (CN), the organization (O), or the country or region (C).
Issuer DN	Indicates the Issuer distinguished name (DN) which entails information about the issuer of the certificate such as common name of the certificate issuer (CN), the organization (O), or the country or region (C).
Actions	Choose the  (<i>Actions</i>) button for a key pair or certificate to perform an action. The available actions depend on the type of the keystore entry and whether it is owned by SAP. Rename Update Download Delete Download Certificate Download Public OpenSSH Key Update Signing Response Download Certificate Chain Download Root Certificate Download Signing Request

3. Choose **Restore** (above the table).

The selected entries are restored in the deployed tenant keystore.

Related Information

Downloading Backed-Up Keystore Entries

Download the public content of the backup keystore to your local disk.

Context

You can download public keystore entries that you have backed up.

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. Go to the **Backup** tab, where the backed-up keystore entries are displayed in a table.
3. Choose **Download** (above the table).

The public part of the backup keystore is downloaded (in a file with the name `PublicContentBackupKeystore.jks`).

Managing the Lifecycle of Keys

To enable secure communication between the tenant and connected remote systems, the system keystore that is deployed on the tenant must contain the up-to-date keys owned by the tenant administrator and SAP.

Context

SAP Cloud Integration comes with a set of features that facilitate the tenant administrator's task of renewing keys provided by SAP on the tenant.

The following different keystore types enable the tenant administrator to renew keys in an efficient way:

- System keystore

This keystore is located in the `system.jks` file provided with the tenant and contains all keys that are actively used by the deployed integration flows.

- New SAP Keys keystore

This keystore contains keys already prepared by SAP before a key pair expires.

As the key expiry date approaches, the tenant administrator can do the following:

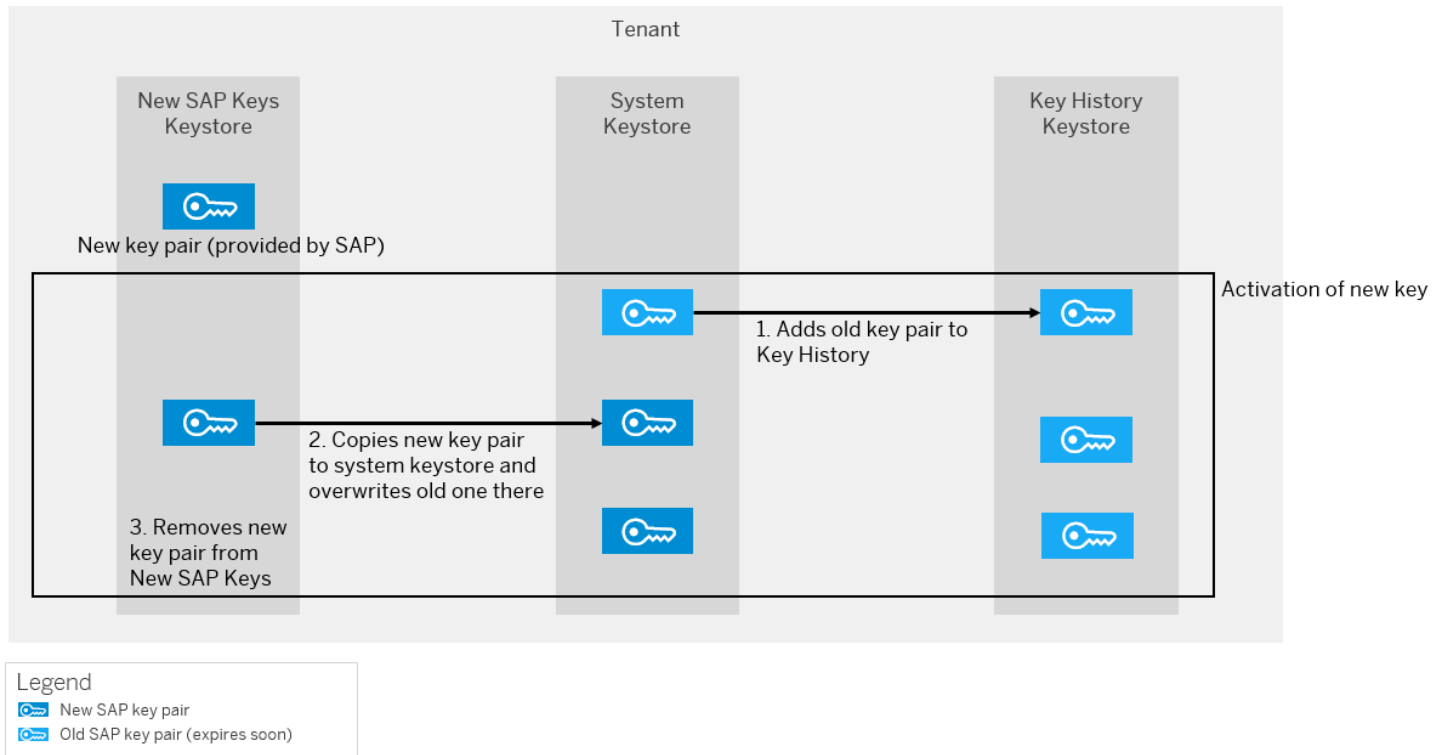
- Download the new SAP keys from the KeyRenewal keystore to share them with the administrators of the connected back ends
- Activate the keys relevant for the tenant

- SAP Key History keystore

This keystore contains old (expired) keys.

If required, the tenant administrator can restore a key from the Key History keystore and use it to replace a key in the system keystore.

During the activation of an SAP key, the system performs three steps, as shown in the following figure.



1. The old key pair (which expires soon) is added by the system to the Key History keystore.
2. The old key pair (in the system.jks keystore) is overwritten by the new key pair from the New SAP Keys keystore.
3. In the New SAP Keys keystore, the new key pair is removed.

i Note

To restore an old key pair, the tenant administrator can copy the key pair from the SAP Key History keystore to the New Keys keystore and activate it there.

Related Information

[Activating a New Key Pair Provided by SAP](#)

[Restoring a Key Pair from the Key History](#)

[Renewal of Keys Provided by SAP](#)

Activating a New Key Pair Provided by SAP

Activate a new key pair provided by SAP in order to replace an old key pair which is supposed to expire soon.

Context

⚠ Caution

Make sure that you activate a new SAP key pair as part of a dedicated key renewal process that you have agreed on with the administrators of all connected sender and receiver systems.

For more information on the recommended processes, see the separate topics.

i Note

Note regarding the case if you don't renew an SAP key prior to its expiration date:

A system job makes sure that in this case the new key is automatically activated (one day before the expiration date).

This job checks daily if there are new entries in the New SAP Keys keystore and if an SAP key pair entry in the system keystore (system.jks) is due to expire within the next 24 hours. If this is the case, the job checks whether the New SAP Keys keystore contains an entry with the same alias. If this is the case, the key pair entry is activated (through the same sequence of steps as shown in the figure).

Procedure

- 1. Go to the **Monitor** section and select the **Keystore** tile (under **Manage Security**).
- 2. Choose the **New SAP Keys** tab.
- 3. Select the entry and choose **Activate**.

Related Information

[Renewal of Keys Provided by SAP](#)
[Activating a New SAP Key Pair on the Tenant](#)

Downloading a New Key Pair Provided by SAP

Download a single Key Pair to your local disk.

Context

You can download a key, a single certificate, a complete certificate chain or a root certificate from the **New SAP Keys** tab.

i Note

You can download only public keystore content (no private keys).

Procedure

- 1. Choose the **Keystore** tile in the **Manage Security** section.
- 2. On the **New SAP Keys** tab, select a keystore entry.
- 3. Choose one of the following options for the selected keystore entry.

Download only supports Key Pair entries in the **New SAP Keys** tab.

Option	Description
Key Pair entry	You can download the public key (X.509 format). ◦ Download Certificate A file with the name <alias>.cer is downloaded. ◦ Download Certificate Chain A file with the name <alias>.p7b is downloaded.

Option	Description
	The file contains the whole certificate chain. ◦ Download Root Certificate A file with the name <alias>_rootCA.cer is downloaded.

Next Steps

You can import the downloaded files into other keystores and open them using third-party tools (for example, KeyStore Explorer).

Restoring a Key Pair from the Key History

Restore a key pair from the SAP Key History.

Context

You use this process to restore an entry from the SAP Key History in the New SAP Keys keystore. In a subsequent step, you can then reactivate the key to make it available in the tenant's system keystore.

Caution

Make sure that you activate a new SAP key pair according to the dedicated key renewal process that you have agreed with the administrators of all connected sender and receiver systems.

For more information about the recommended processes, see the separate topics.

Procedure

1. Choose the [SAP Key History](#) tile in the [Manage Security Material](#) section.
2. Select the entry and choose [Add to New SAP Keys](#).

In the subsequent dialog, the old SAP alias name is proposed, but you have the option to overwrite it with a new SAP alias.

→ Tip

You can specify a new SAP alias if you want to avoid overwriting already active key pairs with this step.

Related Information

[Renewal of Keys Provided by SAP](#)

[Activating a New SAP Key Pair on the Tenant](#)

Downloading a Key Pair from the Key History

Download a key pair from the [SAP Key History](#) tab.

Context

You can download a key, a complete certificate, a certificate chain, or a root certificate from the [SAP Key History](#) tab.

i Note

You can download only public keystore content (no private keys).

Procedure

- 1. Choose the [Keystore](#) tile in the [Manage Security](#) section.
- 2. On the [SAP Key History](#) tab, select a keystore entry.
- 3. Choose one of the following options for the selected keystore entry.

Option	Description
Key Pair entry	You can download the public key (X.509 format). ◦ Download Certificate A file with the name <alias>.cer is downloaded. ◦ Download Certificate Chain A file with the name <alias>.p7b is downloaded. The file contains the whole certificate chain. ◦ Download Root Certificate A file with the name <alias>_rootCA.cer is downloaded.

Next Steps

You can import the downloaded files into other keystores and open them using third-party tools (for example, KeyStore Explorer).

Related Information

[Restoring a Key Pair from the Key History](#)

Managing PGP Keys

The PGP Keys monitor allows a tenant administrator to manage the public and private PGP keys.

Overview

Select [Monitor](#).

Under [Manage Security](#), select [PGP Keys](#).

A list of public and secret PGP keys is displayed in a table. For each artifact, the following attributes are displayed:

Attributes of PGP Keyrings

Attribute	Description
User ID	States the User ID of this PGP key.
Type	Indicates whether the entry is a public PGP Key or a secret PGP key.
Key ID	States the key ID.
Validity State	Indicates the validity state: - Valid: The PGP key is valid. - Critical: The PGP key will expire within the next 14 days. - Expired: The PGP key is no longer valid.
Valid Until	Indicates the expiration date.
Modified On	Indicates the date and time the entry was last modified.

To sort and filter the content of the table, choose **Table Settings** (⚙️). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

Actions

Actions

Display key properties.	To display properties of a dedicated key, click the key. See: Displaying Properties of a PGP Key
Add a keyring.	To add a public keyring, select ►Add ► Public Keys ►. To add a secret keyring, select ►Add ► Secret Keys ►. When adding a secret keyring, you need to specify the key passphrase. See: Deploying a PGP Public Keyring Deploying a PGP Secret Keyring
Download a keyring.	To download public keys, select ►Download ► Public Keys ►. To download secret keys, select ►Download ► Secret Keys ►. When downloading secret keys, you need to specify the key passphrase twice. i Note It is not recommended to download secret keys. If, for certain reasons, you nevertheless need to download secret keys, make sure to use of a sufficiently secure storage location for the downloaded PGP secret keyring file, for example, a password manager. Why do you need to apply a passphrase again when downloading secret keys? During upload, all keys are decrypted with the passphrase specified by you. In a next step, behind the scenes,

	the system re-encrypts the keys with a system-generated passphrase that is not disclosed to the user. This step further increases the protection level of your keys. Due to these additional steps, it is necessary that you apply a passphrase also when downloading secret keys. This passphrase is used to encrypt all keys in the downloaded secret keyring file.
Download a single key as file.	To download a dedicated key as a file, click the download button (↓) at the end of the row of the key. When downloading a secret key, you need to specify the key passphrase twice.
Delete	To delete a dedicated key, click the delete button (🗑) at the end of the row of the key.
Search	To search for a key, start typing the user id or key id of a dedicated key in the search field above the table.

Related Information

- [Manage Security](#)
- [Define PGP Encryptor](#)
- [Define PGP Decryptor](#)

Deploying a PGP Public Keyring

This artifact contains the public key that enables the tenant to encrypt or verify messages using the Pretty Good Privacy (PGP) standard.

Prerequisites

You have created a PGP keyring file and stored it on your computer.

Procedure

1. Select the tile **PGP Keys**.
2. Go to **Add** and select the option **Public Keys** to upload an artifact.
3. Browse for the public keyring file on your computer.

i Note

The selected keyring file can contain 1 or several PGP secret keys in one of the following formats:

- Binary format (typical file extension: **.pgp**)
- ASCII armored format (typical file extension: **.asc**)

4. For **Action**, select one of the following options:

Option	Description
Add	Adds the entries from the uploaded keyring. These are merged with the existing keys.

Option	Description
	When you select the option Overwrite Existing Keys, existing entries are overwritten by uploaded entries with the same KeyId value. i Note A PGP key is considered <i>identical</i> to another one if the hexadecimal KeyId value of both keys matches. Note that there can be several distinct PGP keys with the same UserId.
Replace	Replaces existing keyring including all keys with the uploaded one. Replaces the whole keyring with the uploaded one (replaces only the keys that are already available). You need to confirm the replacement of the existing keys.

5. Confirm to replace existing keys (only if the option **Overwrite Existing Keys** is selected in the previous step).

6. Select **Deploy** to deploy your artifact.

i Note

After a successful upload, a dialog is displayed showing the summary of the added, removed, changed, and not imported keys, if there are any. If any keys were not uploaded, information is provided to explain why (for example, the key with the same key id already exists). The keys in the table refresh automatically. You can also manually refresh the list by clicking the refresh button (↻).

Results

If deployment isn't successful, the following errors are raised:

- The key type you try to upload is invalid.

Is displayed if you try to upload secret keys instead of public keys.

- File upload impossible due to invalid file format. Select one of the supported file formats.

Is displayed if you try to upload a file with a wrong format (not .asc and not .pgp).

Related Information

[Managing PGP Keys](#)

Deploying a PGP Secret Keyring

This artifact contains the PGP secret keys for the usage of Open Pretty Good Privacy (PGP). The private key enables the tenant to decrypt or sign messages.

Prerequisites

You've created a PGP secret keyring and stored it on your computer.

More information: [Creating a Key Pair](#)

Procedure

1. Select the tile **PGP Keys**.
2. Go to **Add** and select the option **Secret Keys** to upload an artifact.
3. Browse for the secret keyring file on your computer and enter the secret key passphrase.

i Note

The selected keyring file can contain 1 or several PGP secret keys in one of the following formats:

- Binary format (typical file extension: **.pgp**)
- ASCII armored format (typical file extension: **.asc**)

To guarantee an adequate security level, SAP Cloud Integration doesn't allow you to upload unencrypted secret keys. Therefore, when uploading a PGP secret key, you need to provide a passphrase. Make sure to specify a secure passphrase that meets the common safety standards.

If you upload a file that contains multiple PGP keys, all of them have to be encrypted with the same passphrase. Import of the keyring file fails if this condition isn't met. This situation can happen, for example, when you import secret keys from different sources.

i Note

It is not recommended to download secret keys. If, for certain reasons, you nevertheless need to download secret keys, make sure to use of a sufficiently secure storage location for the downloaded PGP secret keyring file, for example, a password manager.

Why do you need to apply a passphrase again when downloading secret keys? During upload, all keys are decrypted with the passphrase specified by you. In a next step, behind the scenes, the system re-encrypts the keys with a system-generated passphrase that is not disclosed to the user. This step further increases the protection level of your keys. Due to these additional steps, it is necessary that you apply a passphrase also when downloading secret keys. This passphrase is used to encrypt all keys in the downloaded secret keyring file.

4. For **Action**, select one of the following options:

Option	Description
Add	Adds the entries from the uploaded keyring. These are merged with the existing keys. When you select the option Overwrite Existing Keys, existing entries are overwritten by uploaded entries with the same KeyId value. i Note A PGP key is considered <i>identical</i> to another one if the hexadecimal KeyId value of both keys matches. Note that there can be several distinct PGP keys with the same UserId.
Replace	Replaces existing keyring including all keys with the uploaded one.

Option	Description
	Replaces the whole keyring with the uploaded one (replaces only the keys that are already available).
	You need to confirm the replacement of the existing keys.

5. Confirm to replace existing keys (only if the option **Overwrite Existing Keys** is selected in the previous step).

6. Click **Deploy** to deploy your artifact.

i Note

After a successful upload, a dialog is displayed showing the summary of the added, removed, changed, and not imported keys, if there are any. If any keys were not uploaded, information is provided to explain why (for example, the key with the same key id already exists). The keys in the table refresh automatically. You can also manually refresh the list by clicking the refresh button (↻).

Results

If deployment isn't successful, the following errors are raised:

- The key type you try to upload is invalid.

Is displayed if you try to upload public keys instead of secret keys.

- Invalid passphrase.

Is displayed if you upload secret key with a wrong passphrase.

- File upload impossible due to invalid file format. Select one of the supported file formats.

Is displayed if you try to upload a file with a wrong format (not .asc and not .pgp).

Related Information

[Managing PGP Keys](#)

Displaying Properties of a PGP Key

Display properties of a selected PGP key.

Context

There are the following key types:

- **Public**

Public PGP key

- **Secret**

Public/secret key pair

See: [How OpenPGP Works](#)

Procedure

1. Choose the **Keystore** tile in the **Manage Security** section.
2. Click a key to show the details.
3. The **Key Details** tab shows the following attributes for the selected key.

Attribute	Description
Fingerprint	Character sequence that identifies the public key. A fingerprint is generated out of the public key applying a hash function on the public key.
User IDs	User IDs associated with the key. In the context of PGP, a user ID indicates the entity that uses the key to perform a dedicated action on the message content. The user ID can be a name, an email address, or a combination of both. Examples: ◦ A user ID associated with a public PGP key indicates the entity that receives the message encrypted with the public key. ◦ A user ID associated with a secret PGP key indicates the entity that sends the encrypted message that is to be decrypted using the secret key.
PGP Keys	This section shows a set of attributes for the key and (if defined) its subkeys. The following attributes are displayed: ◦ ID Key ID that uniquely identifies the key or sub key. ◦ Type Asymmetric key algorithm. Supported values: RSA, DSA, Elgamal In particular, elyptic curve algorithms are not supported. ◦ Strength The length of the key in bits. ◦ Usage (Key Flags) Shows for which activity the key is used, for example, for message encryption. ◦ Valid From ◦ Valid Until Indicates the expiration date. ◦ Modified On

Attribute	Description
	Indicates the date and time the key was last modified.

Managing Certificate-to-User Mappings, Neo Environment

The **Manage Security** area provides an overview of security-related artifacts. It also provides access to all certificate-to-user mappings defined for the tenant.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

You can display all certificate-to-user mappings deployed on the tenant by clicking the **Certificate-to-User Mapping** tile under **Manage Security Material**.

Certificate-to-User Mapping Overview

A list of certificate-to-user mappings is displayed in a table. For each artifact, the following attributes are displayed:

Attributes of Certificate-to-User Mapping Artifacts

Attribute	Description
User Name	Name of the user to which the certificate is to be mapped
Subject DN	Indicates the subject distinguished name (DN) which entails information about the owner of the certificate such as the common name of the certificate owner (CN), the organization (O), or the country or region (C).
Issuer DN	Indicates the Issuer distinguished name (DN) which entails information about the issuer of the certificate such as common name of the certificate issuer (CN), the organization (O), or the country or region (C).
Serial Number	Unique serial number of the certificate
Valid Until	Time when the certificate expires
Last Modified by	Name of user who has modified the certificate-to-user mapping the last time
Last Modified on	Date and time when certificate-to-user mapping has been modified the last time

i Note

Limit for certificate-to-user mapping (when using the Neo environment): 2 MB (corresponds to about 2000 X.509 certificates)

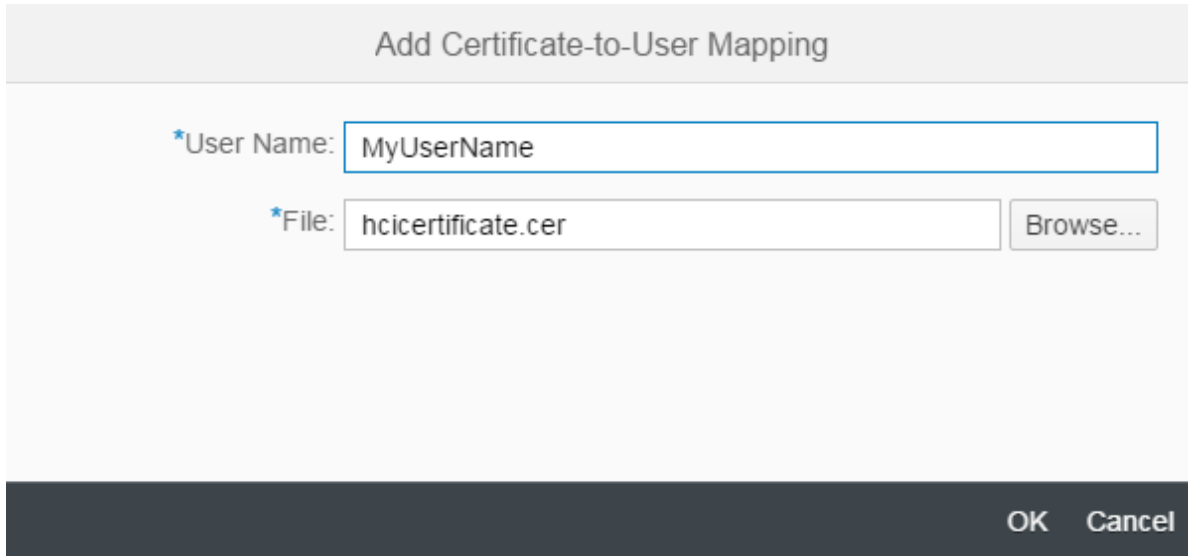
To sort and filter the content of the table, choose **Table Settings** (⚙️). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

The search allows you to filter specific certificate-to-user mappings by providing parts of their name.

Actions

To add a new artifact, perform the following steps:

1. Choose **Add**.
2. Specify the **User Name** and (next to the **File** field) click **Browse** and search for the certificate file (.cer file) on your computer.



3. Choose **OK**.

To edit an existing Certificate-to-User Mapping artifact, perform the following steps:

1. Select the artifact in the table and choose **Edit**.
2. You have the options to change the User Name, specify (select) another certificate, or to change both attributes.

You can also delete an artifact.

Related Information

[X.509 Certificates](#)

[Client Certificate Authentication and Certificate-to-User Mapping \(Inbound\), Neo Environment](#)

Managing Access Policies

Define access policies to restrict the access to integration artifacts and data processed and stored by them.

Access Policies

In SAP Cloud Integration , user permissions are granted in such a way that all tasks can be performed on all artifacts and data. Access policies provide a way to restrict access to selected artifacts and their data.

See: [Access Policies](#).

→ **Tip**

To learn more how access policies work, check out some examples at [Access Policies Examples](#).

Define Access Policies

To define an access policy, you go to **Monitor** and select the **Access Policies** tile.

Here, you define the set of artifacts that are to be protected by the access policy (for example, a specific set of integration flows).

An access policy can contain multiple artifact references, each artifact reference defining an artifact type and a condition for the name or ID of the artifacts to be protected.

See: [Defining Access Policies](#)

Provide User Access

To give dedicated users access to the artifacts protected by the access policy, you define a role and associate it with the access policy (using cockpit).

Only users that are assigned to that role can access the artifacts protected by the access policy.

See: [Creating Custom Roles for Access Policies, Neo Environment](#)

See: [Creating Custom Roles for Access Policies, Cloud Foundry Environment](#)

i Note

You can also retrieve access policies (read and write operations) by an OData V2 application programming interface (API). You find the API and its documentation on SAP Business Accelerator Hub at: [Security Content](#).

Access Policies

With access policies you can restrict the access to integration artifacts and to the data stored and processed by them.

With access policies, you can protect access to certain sets of artifacts. You can define artifact references for:

- A package – by which all artifacts of the integration package and the package actions are protected.
- A specific artifact type – for example, script collection or integration flow.

If access to at least one artifact of an integration package is restricted, you can't export, publish, or delete the integration package.

Access policies affect artifact access in the following dimensions:

→ Remember

With access policies in place, an unauthorized user is still allowed to view a protected resource.

Dimension	Restricts ...
Design time artifact	Operations you can perform in the Design section on the artifact (such like creating, uploading, editing, saving, deploying, or deleting the artifact) Simulation of artifacts is also restricted.

Dimension	Restricts ...
Deployed artifact	Operations you can perform on the deployed artifact under ► Monitor ► Manage Integration Content ► (such like restarting or undeploying the artifact or changing the log level for monitoring)
Data stored and processed by the artifact at runtime	Access to data collected or created during the execution of all integration artifacts - Monitoring data (such like message processing log attachments or business data available when Trace log level is selected) - Data stored in data stores, variables, and message queues i Note When you've restricted access to a specific integration flow (with artifact reference for Integration Flow artifact type) that contains data store operations or variables, only access to local data stores or variables (associated with this integration flow) is limited. If you like to restrict access to global data stores, global variables and message queues, you need to define an access policy with an artifact reference specified as Global Data Store, Global Variable, or Message Queue. ⚠ Caution Custom header properties are not protected by access policies. Note that custom header properties are not supposed to contain sensitive data.

Package-Level Access Policy

An access policy for an integration package limits:

- Operations on the integration package (such like exporting the package)
- Operations on all artifacts contained in the package

If an access policy for an integration package is created, users without the corresponding role can't access any data stored at runtime (for example, MPL attachments, integration flow tracing data, data store or variables entries).

Such users also can't perform any activities on the deployed artifact (for example, restarting an integration flow or changing the log level).

Artifact-Level Access Policy

An access policy for an artifact limits:

- Operations on the specific artifact
- Operations you can perform on the deployed artifact under ► **Monitor** ► **Integrations and APIs** ► **Manage Integration Content** ► (such like restarting or undeploying the artifact or changing the log level for monitoring).

Things to Remember

- Access policies cover user interface-based and API-based access to the artifact.
- If access policies form an intersection, meaning that an artifact is protected by multiple different access policies, the following rule applies: If the user has a role associated with at least one of the access policies, the user has access to the artifact.

In particular, access policies for integration packages and for specific artifact types (such like integration flows and script collections, for example) co-exist in the following terms:

Assume that you've defined an access policy for a specific integration package. Furthermore, the integration package contains an artifact for which you've defined another access policy. In this case, a user with necessary role for the package-level access policy can perform operations on all artifacts, including the artifact that is protected by an additional access policy.

→ Tip

To learn more how access policies work, check out some examples at [Access Policies Examples](#).

Access Policies Examples

Learn more about access policies based on examples.

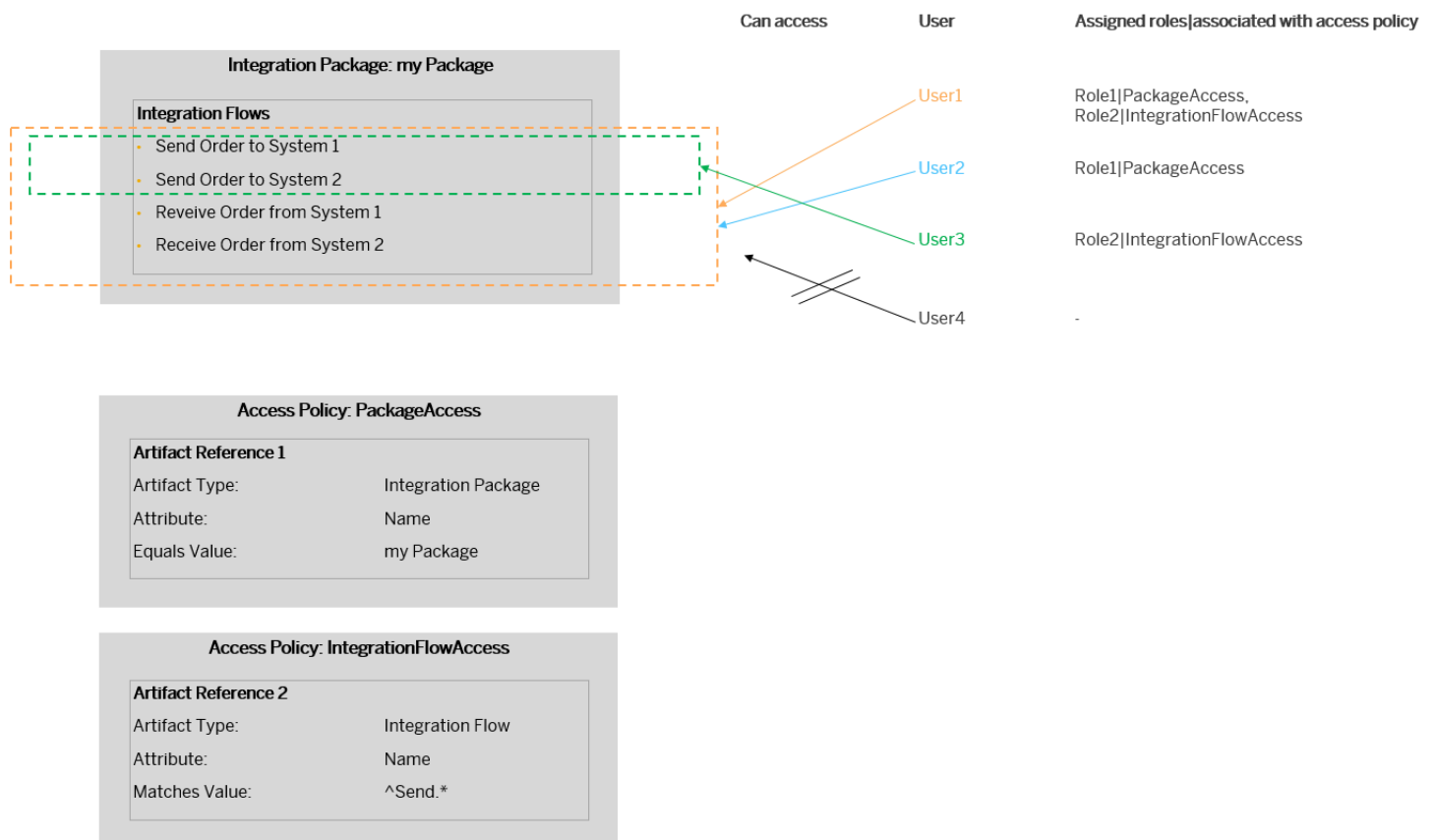
Access Policy for Integration Package

This example demonstrates an access policy for an integration package. Furthermore, this example demonstrates how access policies defined for integration packages and access policies defined for specific artifacts within the same package interact with each other.

Let's assume that two access policies are defined.

- Access policy **PackageAccess** (associated with role **Role1**) protects all artifacts of integration package **my Package**.
- Access policy **IntegrationFlowAccess** (associated with role **Role2**) protects all integration flows with a name starting with **Send**.

The diagram shows how both access policies together affect the access capabilities of individual users with different role assignments.



To now provide targeted access to the protected artifacts, the tenant administrator defines dedicated roles and assigns them to different users.

The individual users have the following access permissions:

User	Assigned role	Role associated with access policy	Can Access ...
User1	Role1 Role2	PackageAccess IntegrationFlowAccess	All integration flows of the integration package.
User2	Role1	PackageAccess	All integration flows of the integration package. User2 has role Role1 that is associated with access policy PackageAccess. Therefore, User2 can access all artifacts of the <i>protected</i> integration package.
User3	Role2	IntegrationFlowAccess	All integration flows that are protected by access policy IntegrationFlowAccess, but no additional artifacts. User3 has role Role2 that is associated with access policy IntegrationFlowAccess. Therefore, they can access the integration flows protected by this access policy. However, the other integration flows are

User	Assigned role	Role associated with access policy	Can Access ...
			protected from this user by access policy PackageAccess .
User4	(no role assigned)	n.a.	No artifact or data illustrated in the diagram.

Interaction of Access Policies Defined for Integration Flows and Data Stores

Integration flows can use [Data Store Operations](#) steps to store data in local or global data stores at runtime (see [Data Storages](#)).

Access policies protect both operations that you can perform with artifacts (such as saving an artifact) and data stored and processed by these artifacts. Therefore, access policies for integration flows and data stores are interconnected in the following way:

- An access policy defined for an integration flow protects also data stored in local data stores (and local variables) associated with this integration flow.
- This access policy, however, has no impact on any global data store defined on the tenant. Note that a global data store can be accessed to by multiple integration flows. The same applies for message queues.

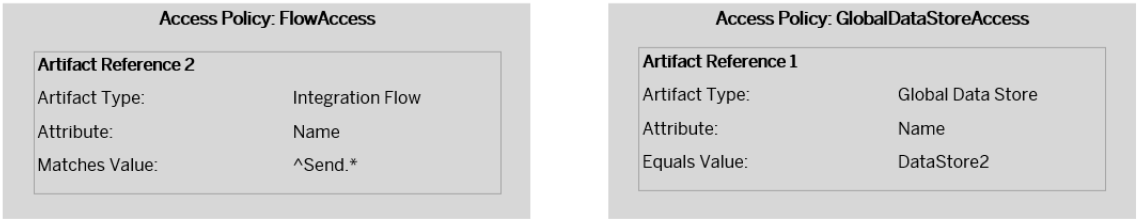
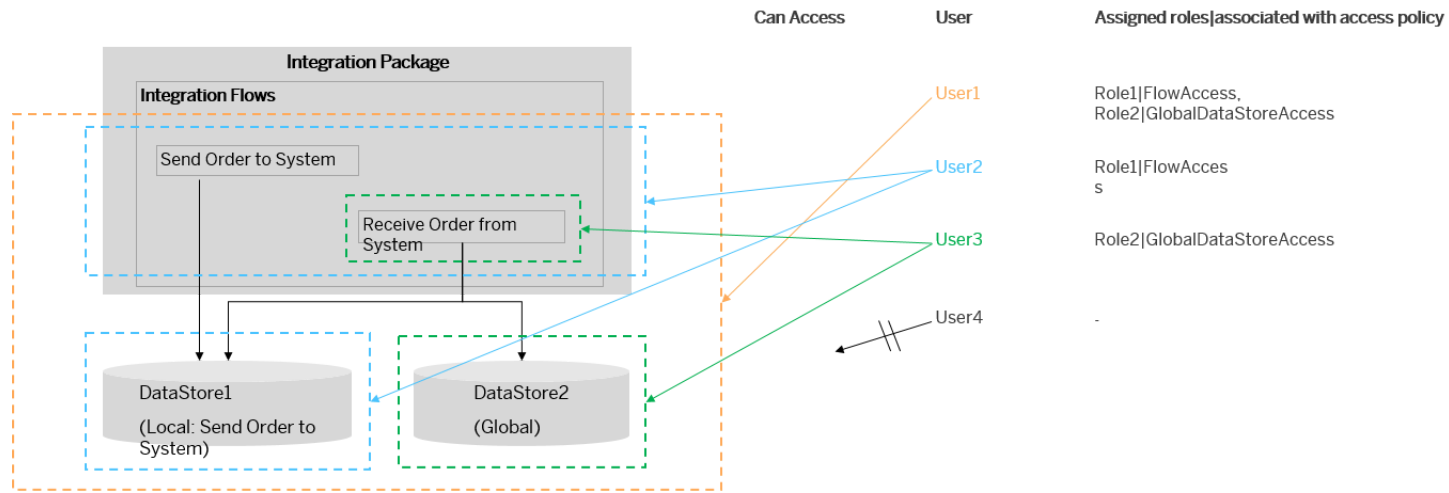
The following example illustrates this use case.

Assume that two integration flows are deployed on the tenant, and there's a global and a local data store. The local data store is associated with integration flow [Send Order to System](#).

Let's assume that two access policies are defined on the tenant.

- Access policy [FlowAccess](#) (associated with role [Role1](#)) protects all integration flows with a name starting with [Send](#).
- Access policy [GlobalDataStoreAccess](#) (associated with role [Role2](#)) protects the global data store (with the name [DataStore2](#)).

The diagram shows how both access policies together affect the access capabilities of individual users with different role assignments.



To now provide targeted access to the protected artifacts, the tenant administrator defines dedicated roles and assigns them to different users.

The individual users have the following access permissions:

User	Assigned role	Role associated with access policy	Can access ...
User1	Role1 Role2	FlowAccess GlobalDataStoreAccess	All integration flows and the content stored in all data stores.
User2	Role1	FlowAccess	All integration flows and the content stored in the local data store. User2 can access both integration flows, including the one that is protected by access policy FlowAccess. The reason for this: User2 has Role1 that is associated with access policy FlowAccess. An integration flow-related access policy also covers data stored and processed by the protected integration flow. Therefore, User2 has also access to the content of the local data store used by integration flow Send Order to System. However, as User2 doesn't have Role2 (associated with access

User	Assigned role	Role associated with access policy	Can access ...
			policy GlobalDataStoreAccess), the global data store isn't accessible for it.
User3	Role2	GlobalDataStoreAccess	The integration flow that is not protected by access policy FlowAccess and the content of the global data store. Integration flow Send Order to System is protected by access policy FlowAccess. As User3 doesn't have the role associated with this access policy, they are unable to access this integration flow. Although the content of the global data store is protected by access policy GlobalDataStoreAccess, User3 is able to access its content. The reason is: they have Role2 that is associated with access policy GlobalDataStoreAccess.
User4	(no role assigned)	n.a.	No artifact or data illustrated in the diagram.

Defining Access Policies

Define an access policy to specify which artifacts are to be protected.

Context

To create and edit access policies, your user must be assigned the authorization group [AuthGroup.Administrator](#) (Neo environment).

To create and edit access policies, your user must be assigned the role collection [PI_Administrator](#) (Cloud Foundry environment).

To view and define access policies, go to ► [Monitor](#) ► [Integrations and APIs](#) ► [Manage Security](#) ► and select the [Access Policies](#) tile.

You can use the [Runtime](#) dropdown to view a list of all available policies in specific runtimes. For more information about access policies, see [Access Policies](#).

To create new access policies, perform the following procedure:

Procedure

1. Select [All](#) from the [Runtime](#) dropdown menu.
2. On the [Access Policies](#) table, choose [Create](#).

3. Provide the following details:

Create Access Policy

Attribute	Description
Role Name	Enter a role name (mandatory). (Neo environment) The role name is the name of the associated custom role that you've to create in your subaccount (see Creating Custom Roles for Access Policies, Neo Environment). (Cloud Foundry environment) When creating the associated custom role in your subaccount, make sure that you specify the same string under Values (see Creating Custom Roles for Access Policies, Cloud Foundry Environment).
Description	Enter a description for the role specified (optional).
Runtimes	Select the runtime where you want to create a new access policy.

4. Choose **Create**. The access policy is added to the list.

You can filter the list by searching role names and descriptions, and sort the list by role name. To **Edit** or **Delete** an access policy, choose the corresponding icon on the **Actions** column.

5. Select the access policy from the list to manage the policy's artifact references.

6. On the **References** table, choose **Create**.

7. Provide the following parameters:

Create Artifact Reference

Attribute	Description
Name	Enter a name for the artifact reference (mandatory).
Description	Enter a description (optional). Example: If you choose the operator Matches, describe the set of integration flows selected by the regular expression (such as: integration flows processing data for German employees).
Artifact Type	Select an artifact type from the list. The default setting is Integration Flow. There are the following options: Integration Package Select to protect all artifacts of a certain integration package (by selecting the Integration Package artifact type). If you define an artifact reference for a specific integration package, all artifact types are covered. Access policies for integration packages and for specific artifact types (such as integration flows and script collections) co-exist as described under Access Policies. i Note

Attribute	Description
	Some artifacts in a package can have pre-existing access policies at the artifact level. To ensure continuity of service, the current action doesn't affect these pre-existing access policy protections. Consider manually cleaning up any existing access policies before creating a new one at the package level. ◦ Integration Flow ◦ API ◦ OData API ◦ REST API ◦ SOAP API i Note Access policies apply to the listed API artifact types in the same way that they apply to integration flows. ◦ Script Collection ◦ Value Mapping ◦ Message Mapping ◦ Message Queue ◦ Global Data Store ◦ Global Variable ◦ Data Type ◦ Message Type i Note To protect the content of local data stores or local variables, you need to define an access policy for the integration flow associated with the local data store or variable (using the Integration Flow artifact type). Global data stores and global variables must be protected by the artifact types Global Data Store and Global Variable respectively. For more information, see Access Policies Examples.
Attribute	Choose one of the following options: ◦ Name To define the artifact reference based on the name. ◦ ID To define the artifact reference based on the technical name or: identifier.
Operator	◦ If you select Equals, you must enter the exact name or ID of the integration artifact.

Attribute	Description
	If you select Matches, the value has to be a valid Java regular expression (not available for Integration Package type). For more information about Java regular expressions, see Pattern (Java SE 23 & JDK 23) .
Value (only when as operator you've selected Equals)	Enter the ID or the Name of the integration artifact.
Expression (only when as operator you've selected Matches)	Enter a regular expression for ID or Name.  Example For the regular expression <code>myName.*</code>, the period (.) means any character, and the asterisk (*) means zero or more times. This expression matches any value that begins with the exact text <code>myName</code> and can have any characters after it, including none. For example: <code>myName</code>, <code>myName2</code>, <code>myName_report</code>. i Note JMS queues, global variables, and global data stores support only names.

8. Once the configuration is completed, choose **Create**. You can create additional artifact references if you want to protect additional artifact types.

To **Edit** or **Delete** an artifact reference, choose the corresponding icon on the **Actions** column.

Replicating Access Policies

Context

You can replicate access policies across runtimes, allowing them to coexist in multiple runtimes simultaneously.

To replicate policies, follow these steps:

Procedure

1. Select **All** from the **Runtime** dropdown menu.
2. In the **Access Policies** table, search for the access policy you want to copy into different runtimes.
3. Choose the  edit icon from the policy's **Actions** column.
4. In the dialog box that appears, open the **Runtimes** dropdown and select one or more runtimes where you want to copy the access policy.
5. Choose **Update**. In the **Access Policies** table, the **Runtimes** column updates to show the number of runtimes assigned to the policy.
6. Choose the icon on the **Runtimes** column to check the reconciliation status of your policies. The status can be **Fail**, **Success**, or **Pending**.

i Note

If the runtime location where you want to replicate access policies is offline, the newly defined access policies are temporarily held back, and only apply to design time artifacts. Once the designated runtime comes back online, the

access policies are automatically copied to it. Until this reconciliation at the runtime location is complete, the status in the reconciliation column remains as **Pending**.

Creating Custom Roles for Access Policies, Neo Environment

Create custom roles to define access policies in the Neo environment.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Provide Access Using Custom Roles

Access policies in the Neo environment are coupled with custom user roles. Thus, to grant users access to different integration artifacts and their associated data you have to create a custom role.

Go to SAP BTP cockpit and follow the procedure described in [Managing Roles](#).

Caution

The name of the custom role must match the name of the access policy.

i Note

- You must be an administrator to create roles.
- The length of the custom role name is limited to a maximum of 200 characters. If this limit is exceeded, the custom role can't be used in an access policy. Leading and trailing spaces aren't allowed.
- Assign the role to users who are to get access once the access policies are active.

i Note

Newly assigned roles work only after login renewal.

Related Information

[Tasks and Permissions](#)

Creating Custom Roles for Access Policies, Cloud Foundry Environment

Define a custom role to be associated with an access policy.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Access policies in the Cloud Foundry environment are associated with roles based on template [CustomRoleTemplate](#) and assigned via role collections. To grant users access to different integration artifacts and their associated data, perform the following steps:

1. In cockpit, select your subaccount.
2. Create a new role as described in [Create Roles for Applications Using Existing Role Templates](#).

Specify the parameters in the following way:

- As **Role Template** select **CustomRoleTemplate**, as **Application** select it.
- Specify any role name (we recommend to include the name of the access policy).
- For **custom_role** select **Static**.
- Under **Values** enter the value you've specified as **Role Name** when creating the access policy (defined in the **Monitor** section, see [Defining Access Policies](#)).

i Note

After you've specified these parameters, you need to confirm your entries by pressing Enter on your keyboard. If not, you can't select **Next** to proceed with the role creation.

- (Optional) Select the available role collections, if you want to grant access to the artifacts referenced by the access policy specified.

3. Create a new role collection and assign the role defined in the previous step to it (see [Define a Role Collection](#)).

Alternatively, you can select an existing role collection and assign the role defined in the previous step to it.

4. Assign the role collection containing the access policy role to the users or groups needing access to referenced artifacts.
See: [Assign Role Collections to Users or User Groups](#).

i Note

Newly assigned roles work only after login renewal.

The new role collection is assigned to your user and you've access to the restricted integration artifacts and their associated data.

For more information about roles and role collections see:

- [Create Roles for Applications Using Existing Role Templates](#)
- [Assign Role Collections](#)

Related Information

[Tasks and Permissions](#)

Managing JDBC Material

Learn how JDBC Materials supports Cloud Integration for interacting with your database server.

Use JDBC Data Source to establish connection with databases and JDBC Drivers helps you to upload and install third-party compliant JDBC type-4 drivers to connect with third-party databases.

Related Information

[Managing JDBC Data Sources](#)

[Configure JDBC Drivers](#)

Managing JDBC Data Sources

The **JDBC Data Sources** allows you to create and manage a cluster of artifact connections to interact with a database (DB). Each data source contains information on database type, and database-specific configuration parameters.

Create a data source to handle a connection setup to a cloud or on-premise database. For the list of supported databases, see [JDBC Receiver Adapter](#).

When you select a **JDBC Data Source**, the details are displayed to the right of the pane. The header area provides the following information on:

- **Configuration Parameters**
Displays information corresponding to a specific database such as **Database Type**, **User**, **Database ID** and so on.
- **Logs**
Displays information recorded during deployment of the data source. If the deployment fails, it displays the error details along with other technical information.

Prerequisites to Configure Data Source for On-Premise Database

Note the following when configuring the connection between a cloud application an an on-premise system using SAP Cloud Connector: The internal host and internal port configurations are different compared to virtual host and virtual port.

1. Log in to your Cloud Connector subaccount, to establish link between your tenant and the on-premise database.
2. Enable your cloud application to access a back-end system. For more information, see [Configure Access Control](#).
3. Connect your cloud application to an on-premise system. For more information, see [Consuming the Connectivity Service](#).

If you choose to access SAP HANA Platform database on SAP BTP, see [Configure a Service Channel for an SAP HANA Database](#).

After you complete the above configuration, now create a JDBC data source for your on-premise database.

Adding or Editing a JDBC Data Source

1. Go to the **Monitor** view and choose the tile **JDBC Material**.
2. Choose **JDBC Data Sources** tab and perform one of the following steps:
 - Choose **Add** for creating a new data source.
 - Choose **Edit** to modify an existing data source.

i Note
You need not redeploy an integration flow after editing the data source.

3. Specify the following attributes:

Database-Specific Configuration Parameters	
Field	Description
Name	Define a name for the data source. i Note

Field	Description
	You are allowed to enter only alphanumeric characters. Special characters and white space aren't supported.
Description	Provide a detailed description of the data source.
Database Type	Select one of the supported database types. For more information, see JDBC Receiver Adapter. i Note JDBC drivers that are uploaded to provide connectivity to databases managed by third party are shown in the dropdown.
User	Enter the username corresponding of the target database.
Password	Enter the password corresponding of the target database. i Note Re-enter the same password in the Repeat Password field to confirm that the provided password is correct.
Database ID (only available for SAP ASE Service (Neo), SAP HANA Service (Neo) databases that are managed by SAP)	Specify the ID of the target database.
Access Token (only available for SAP ASE Service (Neo), SAP HANA Service (Neo) databases that are managed by SAP)	Specify the access token generated for this application and that is used to identify the user that accesses the databases. SAP BTP provides the token to access SAP database. For more information for managing access, see Managing Access to Databases for Other Subaccounts .
JDBC URL	Specify the database connection URL that the JDBC driver uses to connect with the cloud or on-premise database. For more information, see JDBC Receiver Adapter.
Location ID (for on-premise database only)	Identifies the location of this Cloud Connector for a specific subaccount. The location ID must be unique per subaccount and should be an identifier that can be used in a URI. To route requests to a Cloud Connector with a location ID, the location ID must be configured in the respective destinations. If you don't specify any value for Location ID, the default is used.
Enable Connection Pool	Select to activate connection pooling and configure connection pool settings. i Note The Maximum number of JDBC data sources that can be created depends on whether connection pooling is enabled. Without connection pooling, up to 49 data sources can be created. When Connection Pooling is enabled, up to 500 data sources can be created in total across both cloud and

Field	Description
	on-premise databases, with a maximum of 200 data sources allowed for on-premise database connections .
i Note The following fields appear only when "Enable Connection Pool" is checked.	
Maximum Pool Size	The upper limit on the number of database connections maintained by the connection pool, including active and idle connections. It controls peak resource usage and prevents database server overload. - Default value: 10 - Valid range: 1 to 10
Minimum Idle Connections	The minimum number of database connections kept idle in the connection pool, ensuring immediate availability for new requests. - Default value: 1 - Valid range: 0 to maximum pool size
Idle Timeout	The maximum duration a database connection may remain idle in the pool before being eligible for cleanup. - Default value: 600000 ms - Valid range: 10000 ms to unlimited
Connection Timeout	The maximum wait time to obtain a database connection from the pool when all connections are in use. - Default value: 30000 ms - Valid range: 250 ms to unlimited

4. Deploy the data source.

i Note

To undeploy a data source, select it from the list and choose **Undeploy**. The system displays a warning and requires you to confirm the task. After confirmation the data source is undeployed and removed from the list.

Quick Actions

Field	Description
 <i>(Settings)</i>	Helps to sort and filter the content of the table. You can define how the table entries are to be sorted (in ascending or descending order). You can also filter the table entries for certain attributes.
 <i>(Reload)</i>	Reloads the table content.

Related Information

[Configure JDBC Drivers](#)

Configure JDBC Drivers

Learn how to upload and deploy JDBC type-4 compliant third-party drivers on Cloud Integration.

JDBC connectivity provides a mechanism for integration flows deployed on Cloud Integration tenants to connect with Amazon RDS (SQL Server and Oracle) and HANA-asS databases hosted on AWS using JDBC drivers. The tenant administrator configures JDBC drivers to enable you to establish connection to a database managed by a third-party vendor. Your database vendor should provide the driver or download it from their official website. We highly recommend that you work with the latest version of the third-party JDBC driver. All uploaded drivers undergo a security validation before being deployed on a tenant.

i Note

Cloud Services may include features that permit third-party code or tools to be downloaded/uploaded into Customer's Cloud Service account by (i) Customer or (ii) by SAP on behalf of and at the request of Customer from a non-SAP service for which Customer has a license, to facilitate the exchange of data or enable or improve other data processing activities. Such third-party code or tools are Customer Data. The code or tools may not be used to connect the Cloud Service or other software/cloud services to a third-party database licensed from SAP under a run-time license for use with specified SAP Applications. Customer must license a full-use license from the applicable database vendor to use such code or tools to replicate data from such database into the Cloud Service.

Driver Name	Supported Versions
Microsoft JDBC Driver for SQL Server ➡	• msSQLDriver12.4.0 ➡ • msSQLDriver11.2.3 ➡ • msSQLDriver8.2.2 ➡ • msSQLDriver8.4.1 ➡
Oracle JDBC Driver ➡	• oracleDB11.2.0.4 ➡ • oracleDB12.2.0.1 ➡ • oracleDB19.11.0.0 ➡ • oracleDB19.16.0.0 ➡
IBM DB2 JDBC Driver	• db2Driver11.5.4 ➡ • db2Driver11.5.5 ➡
MariaDB Connector ➡	mariadb-connector-j-3.3.3 ➡

You can create Amazon Relational Database Service (Amazon RDS) based data source that supports the following database engines:

- Microsoft SQL Server
- Oracle

Before consuming the drivers in runtime, you must deploy the uploaded driver as a data source to establish a connection with above mentioned databases. For more information on deploying a data source, see [Managing JDBC Data Sources](#).

Deploying JDBC Drivers

1. Choose the **JDBC Driver** tab, and then choose **Add** for uploading a driver from your file system.
2. Specify the following attributes:

Driver-Specific Parameters

Field	Description
Database Type	Select one of the database types that are supported: ◦ Microsoft SQL Server ◦ Oracle ◦ DB2 ◦ MariaDB
JDBC Driver File	Browse to choose the driver file from your file system. i Note Only JAR (*.jar) file formats are supported.

3. Deploy the driver.

Quick Actions

Field	Description
 <i>(Settings)</i>	Helps to sort and filter the content of the table. You can define how the table entries are to be sorted (in ascending or descending order). You can also filter the table entries for certain attributes.
 <i>(Reload)</i>	Reloads the table content.

Managing User Roles, Cloud Foundry Environment

The user role monitor allows a tenant administrator to manage user roles which then can be used during inbound authorization of an integration flow execution.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

How to Access the User Roles Monitor

You open the area with the following action:

In the **Monitor** view, select the **User Roles** tile in the **Manage Security** section. User roles defined by SAP and by the tenant administrator are displayed.

UI Overview

A list of user roles are displayed in a table. For each role, the following attributes are displayed:

Attributes of Custom Role-Related Entries

Attribute	Description
Name	Display name of the user role i Note When creating a new user role, be aware of the following allowed characters in user role names: a-z, A-Z, 0-9, _ (underscore), . (period).
Description	Description of the user role.
Owner	The owner of the entry: Tenant Administrator The entry is owned by the tenant administrator on customer side. SAP The entry is owned by SAP.
Actions	Choose one of the two buttons to perform an action. The available actions are: Edit role (✎) On the subsequent screen, you can edit the description of the selected user role. Delete role (🗑)

i Note

The technical user roles **ESBMessaging.send** and **API.invoke** is predefined by SAP and cannot be removed or changed.

i Note

There is no default role collection available for the **API.invoke** role. Therefore, the tenant administrator must create a custom role collection that includes this role and make it available for assignment to IDP users in order to support the IDP user-based flow.

To sort and filter the content of the table, choose **Table Settings** (⚙) displayed above the table. On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

You also have the option to refresh the table by selecting **Reload content** (↻).

Actions

The following actions are available:

- **Add**

After you have chosen **Add**, a subsequent input box appears where you need to specify the role name and the role description. Only the role description is editable afterwards.

- **Download to JSON**

A JSON file of the selected user roles is downloaded. You need to import the JSON file in the **Process Integration Runtime** tile to create a service instance in the Cloud Foundry environment that is associated with a user role. Only then, you can call an integration flow endpoint and ensure the processing of the integration flow on the runtime node.

You can either use a custom role that you created, or use the predefined roles **ESBMessaging.send** or **API.invoke**. The JSON content of the role **ESBMessaging.send** is:

Sample Code

```
{
  "roles": [
    "ESBMessaging.send"
  ]
}
```

Similarly, the JSON content of the predefined role **API.invoke** is:

Sample Code

```
{
  "roles": [
    "API.invoke"
  ]
}
```

Instead of downloading the JSON file, you can also use **Copy selected roles in JSON format to the clipboard** () at the beginning of the table. In the next step, you can use the copied format to create, as an example, a new service instance.

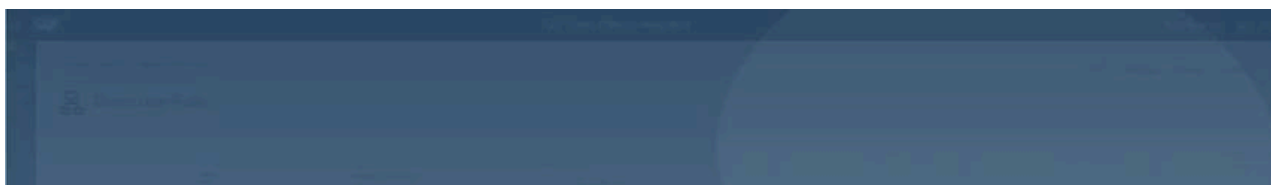
Next Steps

For a detailed instruction on how to create a service instance and a service key, visit [Creating Service Instance and Service Key for Inbound Authentication](#).

Watch the next video to know more about a basic inbound authentication end-to-end scenario.

i Note

This video was created before our branding changes related to SAP technology were announced in January 2021.





[Open this video in a new window](#)

Related Information

[Basic Authentication with clientId and clientsecret for Integration Flow Processing](#)

Performing Connectivity Tests

You can test the connectivity to a receiver system.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) 📄.

To access the tests, select **Connectivity Tests** tile in **Manage Security** section of **Overview** page. Different types of connectivity can be tested by selecting the corresponding tab.

Each tab offers a **Request Form** and a **Send** button. The response shows a message plus additional connectivity type specific information.

Related Information

[SSH Connectivity Tests](#)

[SMTP Connectivity Tests](#)

[TLS Connectivity Tests](#)

[IMAP Connectivity Tests](#)

[POP3 Connectivity Tests](#)

[AMQP Connectivity Tests](#)

[Cloud Connector Connectivity Tests](#)

[Kafka Connectivity Tests](#)

TLS Connectivity Tests

When you've chosen the TLS connection, the test tool checks the following:

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

- if the receiver (host) is reachable for the tenant.
- if the keystore is deployed correctly and contains those keys that are required for the specified authentication method during TLS handshake.

To perform the TLS connectivity test, you need to specify the following settings:

TLS Connectivity Test Options

Attribute	Description
Host	Add the host name of the receiver. The host name must not contain any path or schema, for example, <code>https://</code>). In particular, you must not add a URL as the host name.
Port	Add the port that is to be used for outbound communication. Standard port is 443.
Authenticate with Client Certificate (optional)	Choose this option if the client is to be authenticated against the receiver (server) during the TLS handshake (a mutual authentication). If you choose this option, you can also specify the key to be used for the test.
Alias (only if Authenticate with Client Certificate has been chosen)	Add the alias that identifies the relevant key pair for client certificate authentication. In case the receiver requires that during the TLS handshake also the tenant (client) is authenticated against the receiver (server), you can configure the connection test accordingly (choosing the Authenticate with Client Certificate option). In that case, add the Alias (to indicate which key is to be used from the tenant keystore). For the tenant to be able to authenticate itself as the client against the receiver, a suitable key has to be available in the deployed keystore. With the Alias field, you can narrow down further the check that a specific key is being used (for example, a key that matches the configuration settings in the corresponding adapter).
Include new SAP Key (only if Authenticate with Client Certificate has been chosen)	Choosing Include new SAP Key allows you to execute the connectivity test with the new key pair from the New SAP Keys keystore. Using this option, you can test the backend connectivity before activating the new SAP key. For further information, see Activating a New Key Pair Provided by SAP. i Note

Attribute	Description
	The connectivity test checks if the TLS handshake uses the key certificate pair. However, it doesn't validate any authorizations in the target system. If Include new SAP Key is selected and the specified certificate is not available in the New SAP Keys keystore, the system uses the certificate from the Current keystore as fallback option.
Validate Server Certificate Required	Allows you to validate the server certificate. When you've chosen the Validate Server Certificate Required option (which is the default setting), the following checks are executed: • If the server certificate belongs to the server the client connects to. • If the certificate is signed by an instance the client trusts. If it wasn't successful and there's an error message, you can unselect the Validate Server Certificate Required option.

If the connectivity test was successful, you get the information about the different checks chosen when sending the request. Afterwards, the **Server Certificates** are displayed with the additional option to download the certificate chain or to add the root CA certificate directly to the tenant keystore.

Action	Description
Download	Option to save the Server Certificate Chain and add it to your trusted store.
Add to keystore	The Add to keystore option is only available if the server response contains a valid root CA certificate. If you use this option, you can directly upload the root CA certificate to your tenant's keystore. i Note If the server response does not contain the entire certificate chain, the Add to keystore option is not available. To establish the trust relation to the server, the missing intermediate certificates, as well as the root CA certificates must be available in the tenant keystore. If the intermediate certificates (there can be more than one) or the root CA certificate are missing, you have to get these certificates yourself and upload them to the tenant keystore. For additional information about Certificate Chains, see: Certificate Chains ⚠ Caution If you add a certificate to the keystore via Add to keystore, this certificate is a trusted certificate for the SAP Cloud Integration system, which means that it trusts any server certificate signed by this certificate. Make sure that you only add certificates from a trustworthy source and check the fingerprint that is displayed in the ensuing dialog.

SSH Connectivity Tests

Context

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

When you have selected the SSH connection type, the test tool checks if the SSH outbound connection reaches the associated SFTP server.

Depending on the chosen authentication, the following is checked by the test:

- If the server (host) is reachable for the tenant
- If the configured `known_hosts` file deployed on the tenant contains the certificate of the SSH server.

Procedure

1. In the **Manage Security** section, select **Connectivity Tests**.
2. Select **SSH**.
3. Define the attributes for the test.

SSH Connection Test Options

Attribute	Description
Host	Enter the host name of the receiver. The host name must not contain any path or schema (for example, <code>myHost.org</code>). In particular, you must not enter a URL as the host name.
Port	Enter the port that is to be used for outbound communication. Standard port is 22.
Proxy Type	Select a proxy type from the drop-down list: ◦ Internet - or ◦ On-Premise For more information, see Using SAP Cloud Connector with Cloud Integration Adapters
Location ID (only available if On-Premise is selected for Proxy Type)	To connect to a Cloud Connector instance associated with your account, enter the location ID that you 've defined for this instance, in the destination configuration on the cloud side
Timeout (ms)	Specifies a timeout (in milliseconds) after which the connection to the server (host) should be terminated. The default value is 10.000 ms.
Authentication	There are the following options: ◦ None

Attribute	Description
	SFTP server does not require any authentication of the calling component. Public Key SFTP server authenticates the calling component (tenant) based on a public key. If selected, a user name (specified under User Name) and the key (specified under Private Key Alias) are evaluated by the system to authenticate the tenant against the SFTP server. User Credentials SFTP server authenticates the calling component (tenant) based on the user name and password. To make this configuration setting work, you need to define the user name and password in a User Credential artifact and deploy the artifact on the tenant. Dual SFTP server authenticates the calling component (tenant) with two authentication methods: based on a public key and based on user credentials. At runtime, the system evaluates the values of additional parameters in the following way: - For the authentication step based on user credentials: Credentials from the deployed artifact with the name given by the Credential Name parameter are evaluated by the system to authenticate the tenant against the SFTP server. - For the authentication step based on public key: User name contained in the deployed artifact with name given by the Credential Name parameter and the key identified by the Private Key Alias parameter are evaluated by the system to authenticate the tenant against the SFTP server.
User Name	(Only if Public Key or Dual is selected as Authentication) Enter the ID of the user under which the tenant calls the SFTP server.
Private Key Alias	(Only if Public Key or Dual is selected as Authentication) Alias to identify the private key in the keystore used for the communication with the SFTP server.
Credential Name)	(only if User Credentials or Dual is selected as Authentication) Name of User Credentials artifact deployed on the tenant. This artifact contains user name and password that are used for authentication at the server.
Host Key Verification	(Only if Public Key, User Credentials, or Dual is selected as Authentication).

Attribute	Description
	Verifies the host key. ◦ Against Tenant Host key is verified against known_hosts artifact deployed on tenant. ◦ Against Partner Directory Host key is verified against known hosts defined in Partner Directory.. i Note You can also load a known_hosts file from the Partner Directory. To point to the Partner Directory content, you need to set the following property in the integration flow before calling the SFTP receiver adapter: SAP_FtpPdUri You can use, for example, a Content Modifier for this purpose. The value of the property needs apply to the following format: pd:partnerId:parameterId:Binary ◦ Off i Note It is recommended to use this option with particular caution when you have selected User Credentials as Authentication because in that case you take the risk that you forward the password to an unverified server.
Check Directory Access	Only if Public Key or User Credentials is selected as Authentication) Select this option if you like to check the access to the target directory. If you leave the Directory field empty, the default directory is your home directory. ! Restriction You can only specify sub-directories of the home directory.

→ Tip

You can also use the connectivity test to get the host key of the SFTP server.

- Enter the host name of SFTP server.
- For **Authentication** select **None**.
- Choose **Send**.

If the host name of SFTP server is correct, the test provides the message **Successfully reached host at xxx:22**.

d. Choose **Copy Host Key**.

This key is required to specify the known hosts file (to be uploaded as **Known Hosts (SSH)** artifact).

See:

- [Maintaining SSH Known Hosts for SFTP Connectivity](#)
- [Deploying an SSH Known Hosts Artifact](#)

Related Information

[Setting Up Outbound SFTP Connections \(Details\)](#)

FTP Connectivity Tests

You can perform FTP connectivity tests to check the settings required by the FTP adapter.

FTP Connection Test

Attribute	Description
Host	Enter the host name of the receiver. The host name must not contain any path or schema (for example, <code>https://</code>). In particular, you must not enter a URL as the host name.
Port	Choose or enter the port that is to be used for outbound communication. Standard port is 21 (FTPES/FTP) or 990 (FTPS) depending on the network protocol.
Proxy Type	Select a proxy type from the drop-down list: • Internet or • On-Premise
Location ID	Only if On - Premise is selected as Proxy Type. To connect to an SAP Cloud Connector instance associated with your account, enter the location ID that you've defined for this instance, in the destination configuration on the cloud side
Encryption	Select the encryption type: • Explicit FTPS (none) • Implicit FTPS • Plain FTP If you select Plain FTP, there is no valid server certificate.

Attribute	Description
Authentication	Choose which mechanism is to be used to authenticate against the server. Possible values are: None No authentication is attempted. No credential can be chosen. User Credentials
Credential Name	(Only if User Credentials has been chosen for Authentication) Choose the name of a deployed credential to use for authentication.
Valid Server Certificate	Allows you to validate the server certificate. When you've selected the Valid Server Certificate option (which is the default setting), the following checks are executed: - If the server certificate belongs to the server the client connects to, - If the certificate is signed by an instance the client trusts. If it wasn't successful and there's an error message, you can unselect the Validate Server Certificate option.
Check Directory Access	You can check whether you can access a directory.
Directory	Select this option if you like to check the access to the target directory. If you leave the Directory field empty, the default directory is your home directory.

SMTP Connectivity Tests

You can perform SMTP connectivity tests to check the settings required for configuring the receiver mail adapter.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

SMTP Connection Test

Attribute	Description
Host	Enter the host name of the receiver. The host name must not contain any path or schema (for example, <code>https://</code>). In particular, you must not enter a URL as the host name.
Port	Choose or enter the port that is to be used for outbound communication. Standard port is 587 (SMTP /STARTTLS) or 465 (SMTPS) depending on the network protocol.

Attribute	Description
Proxy Type	Select a proxy type from the drop-down list: - Internet or - On-Premise
Location ID	Only if On-Premise is selected as Proxy Type. To connect to an SAP Cloud Connector instance associated with your account, enter the location ID that you've defined for this instance, in the destination configuration on the cloud side
Protection	Select the protection type: - Off (none) - SMTPS i Note The secure protocol is only available for the Proxy Type Internet. - STARTTLS Mandatory - STARTTLS Optional
Authentication	Choose which mechanism is to be used to authenticate against the server. Possible values are: None No authentication is attempted. No credential can be chosen. Encrypted User/Password The user name and password aren't sent in plain text to the server. This authentication mechanism is secure even without a secure connection. Plain User/Password The user name and password are sent in plain text. Only use this option together with SSL or TLS, as otherwise an attacker could obtain the password.
Credential Name	(Only if User Credentials has been chosen for Authentication) Choose the name of a deployed credential to use for authentication.
Validate Server Certificate	Allows you to validate the server certificate (only when Protection is not Off). When you've selected the Validate Server Certificate option (which is the default setting), the following checks are executed: If the server certificate belongs to the server the client connects to,

Attribute	Description
	If the certificate is signed by an instance the client trusts. If it wasn't successful and there's an error message, you can unselect the Validate Server Certificate option.
Check Mail Addresses	Enter sender and optionally receiver email address.
From	If you have selected Check Mail Addresses , enter the sender's email address
To	If you have selected Check Mail Addresses , enter the receiver's email address

If the connectivity test was successful, you get the information about the different checks. The server certificates are displayed. A download option is available allowing you to save and add them to your trust store.

IMAP Connectivity Tests

When you have chosen the IMAP (Internet Message Access Protocol) connection, the test tool checks the following:

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

- If the receiver (host) is reachable for the tenant

IMAP Connectivity Test Options

Attribute	Description
Host	Enter the host name of the receiver. The host name must not contain any path or schema, for example, <code>https://</code>). In particular, you must not enter a URL as the host name.
Port	Enter the port that is to be used for outbound communication. Standard port is 143 (IMAP/STARTTLS) or 993/IMAPS
Proxy Type	Select a proxy type from the drop-down list: - Internet or - On-Premise
Location ID	Only if On-Premise is selected as Proxy Type. To connect to an SAP Cloud Connector instance associated with your account, enter the location ID that you've defined for this instance, in the destination configuration on the cloud side
Protection	Select the protection type: Off (none)

Attribute	Description
	- IMAPS i Note The secure protocol is only available for the Proxy Type Internet. - STARTTLS Mandatory - STARTTLS Optional
Authentication	Choose the mechanism to authenticate against the server. Possible values are: Encrypted User/Password The user name and password aren't sent in plain text to the server. This authentication mechanism is secure even without a secure connection. Plain User/Password The user name and password are sent in plain text. Use this option only together with SSL or TLS, as otherwise an attacker could obtain the password.
Credential Name	Enter a credential name or select one from the drop-down list.
Validate Server Certificate	Allows you to validate the server certificate (only when Protection is not <code>Off</code>). When you've selected the Validate Server Certificate option (which is the default setting), the following checks are executed: - If the server certificate belongs to the server the client connects to, - If the certificate is signed by an instance the client trusts. If it wasn't successful and there's an error message, you can unselect the Validate Server Certificate option.
List Folders or for Check Mailbox Content	If you check List Folders, you get a list of mail folders. If you select Check Mailbox Content, the folders are checked and the total number of mails and the number of unread mails is displayed.
Folder (mandatory)	If you select Check Mailbox Content , you can specify a folder.

If the connectivity test was successful, you get the information about the different checks. The server certificates are displayed. A download option is available allowing you to save and add them to your trust store.

POP3 Connectivity Tests

When you have chosen the POP3 (Post-Office_Protocol) the test tool checks the following.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

IMAP Connectivity Test Options

Attribute	Description
Host	Enter the host name of the receiver. The host name must not contain any path or schema for example, <code>https://</code>). In particular, you must not enter a URL as the host name.
Port	Enter the port that is to be used for outbound communication. Standard port is 110 or 995
Proxy Type	Select a proxy type from the drop-down list:(- Internet or - On-Premise
Location ID	Only if On - Premise is selected as Proxy Ttype. To connect to an SAP Cloud Connector instance associated with your account, enter the location ID that you 've defined for this instance, in the destination configuration on the cloud side
Protection Thi	Select the protection to be used, from the drop-down list. i Note STARTTLS is not supported for port 995. For port 110, STARTTLS is supported and checked. Select the protection type: - Off (none) - POP3S i Note The secure protocol is only available for the Proxy Type Internet. - STARTTLS Mandatory - STARTTLS Optional
Authentication	Choose which mechanism is to be used to authenticate against the server. Possible values are: Encrypted User/Password The user name and password are not sent in plain text to the server. This authentication mechanism is secure even without a secure connection.

Attribute	Description
	Plain User/Password The user name and password are sent in plain text. You should only use this option together with SSL or TLS, as otherwise an attacker could obtain the password.
Credential Name (mandatory)	Enter a credential name or select one from the drop-down list.
Validate Server Certificate	Allows you to validate the server certificate (only when Protection is not Off). When you have selected the Validate Server Certificate option (which is the default setting), the following checks are executed: - If the server certificate belongs to the server the client connects to - If the certificate is signed by an instance the client trusts If it was not successful and there is an error message, you can unselect the Validate Server Certificate option.
Check Mailbox Content	If you check Check Mailbox Content the mailbox is checked and the total number of mails in the inbox is displayed.

If the connectivity test was successful, you get the information about the different checks. The server certificates are displayed. A download option is available allowing you to save and add them to your trust store.

AMQP Connectivity Tests

If you choose the AMQP (Advanced Message Queuing Protocol), the test tool checks if the connection is successful or not.

1. To perform the AMQP connectivity test, you need to specify the following settings:

AMQP Connectivity Test Options

Attribute	Description
Transport Protocol	Select a transport protocol from the drop-down list. You can choose between - TCP and - WebSocket Default value is TCP
Host	Enter the host of the messaging server (mandatory).
Port	Enter the port of the messaging server (mandatory). Default values are: 5671 for TCP or 443 for WebSocket

Attribute	Description
Path	Additional attribute for the WebSocket protocol. Access path of the messaging server. Default value is <i>empty</i> .
Proxy Type	Select a proxy type from the drop-down list: - Internet or - On-Premise
Location ID	Only if On-Premise is selected as Proxy Type . To connect to an SAP Cloud Connector instance associated with your account, enter the location ID that you've defined for this instance, in the destination configuration on the cloud side
Connect with TLS	Enable the secure connection via TLS (Transport Layer Security). Default value: checkbox selected.
Validate Server Certificate	i Note Only applies if you've selected Connect with TLS. Allows you to validate the server certificate. When you've selected the Validate Server Certificate option (which is the default setting), the following checks are executed: - If the server certificate belongs to the server the client connects to, - If the certificate is signed by an instance the client trusts If it wasn't successful and there's an error message, you can unselect the Validate Server Certificate option.

2. Press **Send**.

If the connectivity test was successful, you get the information about the different checks. The server certificates are displayed. A download option is available allowing you to save and add them to your trust store.

Kafka Connectivity Tests

If you choose the Kafka Connectivity test, the test tool checks if the connection is successful or not.

To perform the Kafka connectivity test, you need to specify the following settings:

Kafka Connectivity Options

Parameter	Description
Host	Enter the host name of the messaging server.
Port	Enter the port name of the messaging server.

Parameter	Description
Time out (in ms)	Specify the maximum amount of time the client waits for a response from the kafka broker. Default value is: 10000 ms
Authentication	Select the type of authentication for connecting to the broker. SASL is selected by default. • SASL • Client Certificate (security protocol: SSL) • Connect with TLS
SASL Mechanism (only if SASL is selected for Authentication)	Select the SASL mechanism: • PLAIN • SCRAM-SHA-256
Connect with TLS (only if SASL is selected for Authentication)	Select this option to switch between PLAIN and SCRAM-SHA-256 .
Credential Name (only if SASL is selected for Authentication)	Enter the credential name of the username-password pair specified during the deployment of the user credential on the tenant.
Private Key Alias (only if Client Certificate is selected for Authentication)	Enter the private key alias.

2. Press **Send**.

If the connectivity test was successful, you get the information about existing topics in the broker. This list is limited to up to 200 topics. If there are more than 200 topics, only the first 200 are returned.

For more information about the Kafka sender adapter see: [Configure the Kafka Sender Adapter](#)

Cloud Connector Connectivity Tests

When you have chosen the Cloud Connector, the test tool checks whether the Cloud Connector has been configured and can be reached.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

To use the Cloud Connector, you must have installed and configured it as described in [Outbound: SAP Cloud Connector](#)

To perform the connectivity test for the Cloud Connector enter a **Location ID** and choose **Send**.

i Note

The location ID identifies a specific Cloud Connector and is unique per subaccount. A cloud connector with an *empty* location ID can also exist.

If the system could not connect to the cloud connector, check the configuration and whether the location ID exists.

Manage Stores

The Manage Stores section allows you to manage various temporary data storages on your tenant.

Related Information

[Managing Data Stores](#)

[Managing Variables](#)

[Managing Message Queues](#)

[Managing Number Ranges](#)

[Managing Partner Directory Entries](#)

Managing Data Stores

Context

The **Manage Stores** section provides an overview of storages that are temporarily used to persist data of different kind during message processing.

Caution

The monitor displays a maximum of 100000 data store entries per data store. If the data store contains more than this limit, search results may be incomplete, and not all relevant entries may be displayed. Keep this limitation in mind when interpreting search results.

If you need to search through more than 100000 records, we recommend using the [Message Stores API's](#)  **Data Stores** resource for efficient querying and retrieval (see also [Message Stores](#)).

Procedure

1. Choose **Monitor**.
2. Choose the **Data Stores** tile.
3. On the left pane of **Manage Data Stores** the name of the data store is displayed.

Below the data store name the following information is displayed:

- The name of an integration flow - in case the the data store is used by one integration flow (local data store)
- The string **Global** - in case the data store is shared across all integration flows deployed on the tenant (global data store)

Furthermore, the number of entries available in the data store and (if applicable) the number of overdue entries is shown for each data store.

4. On top of the list of data stores, you can access the following functions:

Table Settings	Description
Filter by Name	Allows you to filter for data store names that contain a certain string.
Sort By	Allows you to sort the messages either in ascending or descending order. You can select one of the following sorting criteria: ◦ Name ◦ Visibility ◦ Total Number of Entries ◦ Number of Overdue Entries
Filter By	Allows you to filter by message using the following sorting criteria: ◦ Status: Filters data stores with or without overdue entries (you can select either one or both options). ◦ Visibility: Filters the data stores by the global or technical name of the integration flow. .

5. On the right side of the screen, details about the data store selected from the left side are shown.

The following attributes are displayed for each data store entry:

Attribute	Description
ID	ID of the data store entry The data store entry ID can be specified by the integration flow developer when designing the data store Write step (Entry ID parameter) or, in case not specified during integration flow design, a unique ID is generated by the system when the entry is written to the data store at runtime.
Message ID	ID of the message processing associated with the data store entry
Status	Status of the data store entry. Possible values are: ◦ Waiting: Status after the entry has been written into the data store (and is waiting to be consumed by another integration flow) ◦ Overdue: Status in case the entry has not been consumed in the specified due time
Due At	Date and time the entry is to be consumed (as configured in the data store Write step, Retention Threshold for Alerting parameter)
Created At	Date and time the entry has been created
Retain Until	Date and time when the data store entry will be deleted (as specified in the data store Write step, Expiration Period

Attribute	Description
	parameter)

6. Use the following options available on the header above the table to manage the entries:

Function	Description
Filter by ID	You can filter the list by providing a search string in the search field. The list displays all entries that contains the search strings in either entry ID or message ID.
Delete	Deletes one or multiple data store entries. To select multiple entries, you need to select the Multiselect Mode first (see below).
Reload Content	Refreshes the data store entry list.
Table Settings	Allows you to sort and filter the entries.
Multiselect Mode	You can use this option to delete multiple entries or to download a single entry .

7. For each entry, select ... *Actions* from the **Actions** column for the following options:

Action	Description
Download	Downloads an entry to your computer. You can download only one entry at a time.
View Payload	Downloads the ZIP file, and opens a dedicated dialog that displays the message body, headers, properties and attachments. The dialog also provides the Download (the original ZIP file) and the Copy Body Content actions, so that you can keep working with the entry without leaving the page. i Note The viewer displays XML, JSON, and string content. If the payload contains binary content, it is not displayed in the viewer. You can still download it to view its contents.

Related Information

[Define Data Store Operations](#)

Managing Variables

The **Variables** view allows you to monitor variables used in integration flows.

→ **Tip**

To check which roles have permission to monitor variables, see [Tasks and Permissions for Cloud Integration](#).

Choose **Monitor**.

Choose the **Variables** tile in the **Manage Store** section. You get an overview of the existing variables, with the following attributes.

Table Settings	Description
Name	The variable name is defined in the integration flow.
Visibility	A variable can be globally visible across all deployed integration flows of the tenant or be used only by one integration flow.
Integration Flow	Displays the ID of the integration flow the variable is used in.
Updated At	Shows date and time when the variable content was last updated.
Retain Until	Displays date and time until the variable is still available. The retention time is updated along with any update of the variable.
Actions	You can download the variable content or delete the variable.

You can filter in the table either by variable name or integration flow.

By clicking on the variable name in the table, you can see its content. If the variable content is not defined as a string value, its content cannot be displayed and a message is shown. You can also download the variable by choosing **Download**. If you choose to save the variable, the system creates a .zip file, containing the header properties file.

i Note

If you want to delete a variable, check that this variable is no longer in use, as there is no "where used list" available.

Managing Message Queues

You can monitor queues that are active for a tenant.

The following adapters can store messages temporarily in queues or read messages from there:

- AS2 sender adapter temporarily stores messages during runtime processing in JMS queues.

See:

[Configure the AS2 Sender Adapter](#)

- JMS receiver adapter writes messages to JMS queues and the JMS sender adapter reads messages from JMS queues.

See:

[Configure the JMS Sender Adapter](#)

[Configure the JMS Receiver Adapter](#)

- XI sender and receiver adapter can temporarily store messages for exactly-once (EO) scenarios (if **JMS Queue** is selected as **Temporary Storage** option).

See:

[Configure the XI Sender Adapter](#)

[Configure the XI Receiver Adapter](#)

To open the **Manage Message Queues** screen, perform the following steps:

- 1. Go to **Monitor** .
- 2. Choose the **Message Queues** tile under **Manage Stores**.

i Note

The message queue service is only available if Enterprise Messaging is activated and a JMS Message broker is provisioned. (See: [Activating Enterprise Messaging](#))

Manage Message Queues Screen

The **Manage Message Queues** screen has a header and two main areas: an overview table on the left and a detailed content table on the right right.

Section	More Information
Header of the screen In the JMS Resources information box below the header, you can see the status of the current JMS resource usage. i Note Storage capacities of a queue have a limit.	Checking JMS Resources.
Queues Active queues are displayed in the master table on the left side of the screen.	Managing Queues.
Messages Clicking a queue shows the messages stored in that queue on the right side of the screen.	Managing Messages Stored in Queues.

General Actions

You can perform the following general actions that affect all queues:

General Actions

Attribute	Description
Check 	Show unused and missing queues. The check results show the following: • Queues that aren't used in any deployed integration flow. • Queues that are referenced by integration flows but don't exist because they have been deleted by mistake. You can delete queues that aren't needed and generate missing queues by redeploying the integration flow.

Attribute	Description
Reload (↻)	Reload the table.
Sort (↑↓)	Sort the table entries according to the displayed sorting criteria.

Related Information

- [Cloud Integration – Checks in JMS Message Queue Monitor](#)
- [Cloud Integration – Configure Dead Letter Handling in JMS Adapter](#)
- [AS2 Adapter](#)
- [JMS Adapter](#)

Checking JMS Resources

Storage capacities of a queue have a limit.

In the **JMS Resources** information box, you can see the status of the current JMS resource usage.

- **Ok** (all resources show status **Ok**)
- **Critical** (at least one of the resources with status **Critical**)
- **Exhausted** (at least one of the resources with status **Exhausted**)

Click **Details** to find out more details.

JMS Resource Details

Number of Queues	Number of queues already in use in reference to the maximum number of queues available. This row illustrates the remaining resources: • Ok (green): multiple queues remaining. • Critical (orange): only one or no queue remaining. i Note For more information on how to increase the JMS resources, check out JMS Resource Limits and Optimizing their Usage.
Capacity	Data volume (in MB) already used compared to the maximum data volume available. This row illustrates the remaining resources: • Ok (green): Informs the user that the total number of MBs available is > 20%. • Critical (orange): Informs the user that the total number of MBs available is < 20%. • Exhausted (red): Informs the user that the total number of MBs has reached 95% of the overall available storage.

Queue Usage	Displays a summary of the individual statuses of the respective queues. i Note The color changes according to the total queues' statuses. • Ok (green): all queues display the status Ok. • Critical (orange): min. 1 queue with status Warning. • Exhausted (red): min. 1 queue with status Error.
Transactions	Consumers, providers, and an open transaction are always required to process a message at runtime. If a message is stored to the JMS queue (for example, by a JMS receiver channel), a provider is required. For polling a message from the JMS queue, a consumer is required (for example, a JMS sender channel). If the JMS Resources display the status Critical, you need to optimize the usage of transactions in consumers and providers. See: • Define Proper Transaction Handling • Cloud Integration – JMS Resource and Size Limits  (SAP Community blog)
Providers	
Consumers	
Runtime Nodes	Number of runtime nodes (aka worker nodes) in use.
Tenant Management Nodes	Number of tenant management nodes in use.

More Information

For more information on the limits and how to increase storage capacity, check out:

- [JMS Resource Limits and Optimizing their Usage](#)
- [JMS Resource Limits and Optimizing their Usage](#) (SAP Community blog)

Managing Queues

Monitor and manage the active queues.

Manage Message Queues

All message queues that are active for the tenant are displayed in a table. For each queue, the following information is displayed:

Message Queue Overview

Attribute	Description
Name	Name of JMS queue.
Access Type	Type of access to the JMS queue which can be exclusive or non-exclusive.

Attribute	Description
Entries	Number of messages stored in queue.

To sort and filter the content of the table, choose **Table Settings** (⚙️). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

The search allows you to filter specific queues by providing parts of their name.

Actions to Perform on Single Queue

For a dedicated queue, choose ... *Actions* to perform the following actions:

Actions

Action	Description
Retry (only for non-exclusive queues)	Triggers a retry of all messages in the selected queue. i Note You can't retry stopped queues.
Usage	Displays the data volume (in MB) already used compared to the maximum data volume available (per sub queue). There are the following sub queues: Processing Queue Processes the messages (as, for example, are stored by a JMS receiver and consumed by a JMS sender adapter). Error Queue (only for non-exclusive queues) Temporarily stores messages in an error state until the next configured retry. No messages are consumed from this queue. Instead of this, the messages are taken back to the processing queue when the configured retry interval has been reached. Chunking Queue Messages with a size larger than 5 MB are split into one parent message of 5-MB size and multiple chunks. The parent message is stored in the processing queue while the multiple chunks are stored in the chunking queue (see also JMS Resource Limits and Optimizing their Usage). This screen provides the following functions: Configure Size Allows you to configure the maximum size of the selected queue. For more information and examples how to configure queue sizes, check out the SAP Community blog Cloud Integration – JMS Resource and Size Limits 📄. Stop (only available for started queues)

Action	Description
	Use this function to take out a queue from processing. That way, you can avoid that the queue runs at full capacity. • Start (only available for stopped queues) Use this function to start a queue manually.
Configure Size	Change the maximum size of the sub queues. For more information and examples how to configure queue sizes, check out the SAP Community blog Cloud Integration – JMS Resource and Size Limits .
Where-Used	Displays the integration flows in which a queue is used, and specifies whether the integration flows write to or consume a queue, or both. If you select the link to the integration flow, it opens in read-only mode and you can check the scenario that is using the queue. During the operation of your scenarios, you may find that messages are piling up in one queue, and you want to get more details so that you can analyze why the messages aren't being processed.
Move	Use to manually trigger the process of moving all messages from the source queue to another target queue. → Remember This operation moves the messages that are currently in the source queue and a limited number of messages that arrive while the move is in progress. The move is bounded and stops when that limit is reached, preventing it from running indefinitely during a fast inflow of messages. Any messages remaining in the source queue after the operation are not moved automatically. After the move completes, refresh the Queues table to confirm whether messages remain in the original queue. Trigger a new move if needed.
Delete	Deletes the selected queue and all messages stored in the queue. i Note You can't delete any queues that are (still) in use by one or more integration flows. Undeploy the integration flow first and then continue with deleting the queue. You can't delete stopped queues.

Managing Messages Stored in Queues

Monitor and manage messages stored in a queue.

When you select a message queue, the messages stored in that queue are displayed.

The following information is displayed for each message shown:

i Note

The message queue details view doesn't always show all messages of a queue in a table. Depending on the number and the size of the messages, some messages might not be displayed.

Message Details

Attribute	Description
JMS Message ID	ID of the message in the JMS queue. This parameter is a unique hex-encoded ID generated by the runtime when the entry is written to the JMS queue. You can copy the JMS Message ID by choosing  <i>Copy</i>.
Message ID	ID of the message processing log.
Status	The following values are possible: • Waiting The message is still waiting to be processed. • Overdue The message hasn't yet been processed and the deadline (Due At date) has passed. • Failed An attempt was made to process the message but an error occurred. • Blocked The message was involved in multiple node crashes and was therefore not processed. The message isn't automatically restarted as it was probably this message that caused the node to crash. You can see that the status is Blocked, and also that the Retry At column is empty. Check the message size and the integration flow. The message may be too large or there could be an error in the integration flow. You can then either delete the message or perform a manual. Make sure that the problem that caused the crash in the integration flow has been resolved. Maybe the size of the virtual machine has to be increased.
Due At	Due date for the message The due date is calculated as follows: The number of days configured as the Retention Threshold for Alerting in the adapter that the message originates from is added to the date on which the message is created. When the due date is reached, the status of the message is set to Overdue.
Created At	Date when the message was stored in the queue.
Retry Count (for non-exclusive queues)	Number of retries of the message.
Next Retry On (for non-exclusive queues)	The date when the next retry is scheduled.
Retain Until (for non-exclusive queues)	Date until which the message is retained. Once this date is reached, the message is automatically deleted.
Sender	Name of the sender made available by the SAP_Sender header.

Attribute	Description
Receiver	Name of the receiver made available by the SAP_Receiver header.
Application Message Type	Application-specific message type set by the SAP_MessageType header.
Application Message ID	The identifier set by the SAP_ApplicationID header.
Correlation ID	The identifier used to correlate different messages with each other that are processed jointly in the context of an integration scenario.
Custom Status	The identifier set by the SAP_MessageProcessingLogCustomStatus header, which you can use to monitor and diagnose message processing outcomes.

You can perform the following actions for the messages you select:

Action	Description
Move	Move up to 100 selected messages to any Exclusive or Non-Exclusive queue. i Note When you select more than 100 messages, only the first 100 messages are moved. i Note The message order is not maintained when moving messages from an Exclusive queue.
Retry (for non-exclusive queues)	Triggers retry of the selected message. To be able to trigger a message retry, you need the following authorization: AuthGroup.Administrator or AuthGroup.IntegrationDeveloper (in the Neo environment) PI_Integration_Developer or PI_Administrator (in the Cloud Foundry environment)
Delete	Deletes the selected message. To be able to delete a message, you need the following authorization: AuthGroup.Administrator (in the Neo environment) PI_Administrator (in the Cloud Foundry environment)

For each message displayed on the Messages table, select ... Actions from the Actions column for the following options:

Action	Description
Download	Downloads the message with attachments as a zip file. The zip file has four entries with the following names: - attachments - Folder that contains message attachments (as .txt files) - body

Action	Description
	Contains the message body. <code>properties.prop</code> Contains the exchange properties. <code>headers.prop</code> Contains the headers. i Note If a JMS message has several attachments, a zip entry is generated for each attachment. The names of the attachment entries have the following form: <i>attachment_<File Name / Content ID / UUID></i> <i><File Name / Content ID / UUID></i> means that either the file name, the content ID header, or a UUID is used as the name component, depending on whether the value for the file name or the content ID is available. - If there's no attachment, the zip file doesn't contain an attachment entry. - The naming convention for the entry with the message body is <i>body_<Content ID></i>, where <i>_<Content ID></i> is only included if there's a content ID header for the body. - The entries <code>exchange.properties</code> and <code>header.properties</code> are only included if the JMS message has properties or headers. To be able to download a message, you need the following authorization: <code>AuthGroup.BusinessExpert</code> (in the Neo environment) <code>PI_Business_Expert</code> (in the Cloud Foundry environment)
View Payload	Downloads the ZIP file, and opens a dedicated dialog that displays the message body, headers, properties, and attachments. The dialog also provides the Download (the original ZIP file) and the Copy Body Content actions, so that you can keep working with the entry without leaving the page. i Note The viewer displays XML, JSON, and string content. If the payload contains binary content, it is not displayed in the viewer. You can still download it to view its contents.

JMS Resource Limits and Optimizing their Usage

The JMS messaging instance has limited resources. You can increase the capacity of these resources to a certain extend.

Default and Maximum JMS Resources

i Note

The JMS messaging instance is used in asynchronous messaging scenarios with the JMS, AS2, AS4, and XI adapter.

For SAP Integration Suite (all service plans) and for existing SAP Cloud Integration (Enterprise Edition) licenses, there are the following default and maximum queue capacities.

	Default JMS resources	Maximum JMS resources to get by self-service
Number of JMS queues	30	100
Queue Capacity	9.3 GB	30 GB
Number of transactions	150	500
Number of consumers	150	500
Number of providers	159	500

i Note

Capacity of one JMS queue:

Resources such like number of transactions, message volume, and so forth, can be calculated from the resources associated with one single JMS queue.

One JMS queue comes with the following resources:

Queue capacity	300 MB
Transactions	5
Consumers	5
Providers	5

Independent of the allocated JMS resources, the following applies:

- The maximum capacity for one queue is 95% of the total queue capacity.
- 256 messages can be processed in one transaction.

This restriction is relevant when you have designed integration flows that split messages where multiple split messages are executed in the same transaction. If a message is bigger than 5 MB, before storing it in the JMS queue, Cloud Integration splits it internally into smaller chunks.

This means: The maximum message size (including attachments) that can be stored in a JMS queue is 1280 MB (256 * 5 MB). This limit is given by the fact that maximum 256 messages can be processed in one transaction (see note above).

For more guidance and calculation examples, check out the following SAP Community blog: [Cloud Integration – JMS Resource and Size Limits](#) (section *Numbers of Messages in one Transaction*).

- Headers and exchange properties defined in the integration flow before the message is saved to the queue must not exceed 4 MB in total.

How to Increase JMS Resources

You can increase the capacity up to 100 queues be self-service, as described in the table below.

License	Increase up to 100 queues by self-service	Increase to > 100 queues
SAP Integration Suite	To get up to 100 queues, for the Cloud Integration capability, increase the JMS queues number. See the following SAP Community blog: Integration Suite – Simplified Onboarding and Provisioning (section <i>Increasing the JMS Message Broker resources</i>)	Check out 2589823 .

How Cloud Integration Manages Queues

The JMS sender or receiver adapter generates a message queue during deployment.

Unused message queues are not deleted automatically. Since queues are limited resources, you should manually delete any queues which are not used by your integration flows any more.

Information on deleted message queues is available in the audit log.

Caution

There are certain limitations related to transactional behavior when using this adapter type together with **Data Store Operations** steps, **Aggregator** steps, or global variables. For more information, visit the following SAP Community blog: [Cloud Integration – How to configure Transaction Handling in Integration Flow](#).

Note

The JMS adapter stores only simple data type headers (primitive data types or strings).

Supported Headers: [Headers and Exchange Properties Provided by the Integration Framework](#)

JMS Consumers and Providers

Processing JMS messages always requires a consumer (JMS sender adapter) or provider (JMS receiver adapter) and a transaction. In critical resource situations (as analyzed by the **Manage Queues** monitor), you can optimize the usage of transactions based on the following calculations.

	Min. No. of ...	Max. No. of ...
Consumers	(no. of runtime nodes) x (no. of JMS queues)	(no. of runtime nodes) x (no. of JMS queues) x (no. of concurrent processes) (where the value of the latter argument is derived from the sender channel parameter Number of Concurrent Processes)
Providers	Number of providers for a tenant cannot be calculated (depends on the sender system). The no. of providers is equal to the no. of parallel sender calls.	
Transactions	(min. no. of consumers) + (no. of providers)	(max. no. of consumers) + (no. of providers)

i Note

Runtime nodes are also referred to worker nodes.

i Note

- Number of JMS queues: Note that when the first integration flow that uses a JMS queue is deployed, a queue is created.
- To find out the number of runtime nodes or the number of tenant management node (only available when you operate SAP Cloud Integration in the Neo environment), open the tile **Message Queues** (in the **Monitor** section of the Web UI under **Manage Stores**) and click **Details** in the information box below the header.
- Transactions are distributed dynamically to providers and consumers.
- To increase the number of queues, raise an incident (component **BC-CP-EM-MES**).

For more information, read the SAP Community blog: [Cloud Integration – JMS Resource and Size Limits](#).

Related Information

[Managing Message Queues](#)

Managing Number Ranges

The topic provides an overview of number ranges related artifacts.

You choose the **Number Ranges** tile in the **Manage Stores** area to view the artifacts with the corresponding status.

Number Ranges Overview

While sending out a document in case of EDI processing, a unique interchange number must be added to each document. In order to add such an interchange number you can use the Number Range Object. In case of EDIFACT the outgoing EDIFACT messages should have Interchange Control Reference with length 1 to 9 digits, with minmum length being 1 digit and maximum length ranging to 9 digits.

Visit the [blog](#), to understand how to consume Number Ranges in EDI message processing.

A list of number ranges is displayed in a table. For each artifact, the following attributes are displayed:

Attributes of Number Ranges Artifacts

Attribute	Description
Name	Displays name of the artifact
Minimum Value	Minimum value of the artifact should be greater than or equal to 0.
Maximum Value	Maximum value of the artifact should be less than 15 digit.
Next Value	Displays a value that can be used the next time you invoke this artifact. i Note

Attribute	Description
	The Next Value shown in the UI corresponds to the Current Value in the API. The difference is only in terminology.
Field Length	Displays the number of digits for the current value. If the value of maximum value attribute is 100 then the field length should be greater than 2, if the value of maximum value attribute is 99999 then the field length should be greater than 4 and so on. If the value of this attribute is 4 and the value of current value attribute is 7, then the number range is reflected as 0007 and not just 7. This refers to the concept of padding '0's to the value. If the value of this attribute is 0 then the value of current value attribute is flashed as it is with no padding on the number range. The maximum value allowed for this attribute is 14.
Rotate	If this attribute is set and the number range reaches specified maximum value , then the current value resets to specified minimum value .
Deployed By	Displays the user id of the user who deployed the artifact.
Deployed On	Displays the time when the artifact was deployed.

To sort and filter the content of the table, choose **Table Settings** (⚙️). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

Actions

- To create or deploy a new artifact, choose **Add**.
- To edit and redeploy an existing artifact, select the artifact in the table and choose **Edit**.
- To undeploy an artifact, select the artifact in the table and choose **Undeploy**.

Adding a Number Range

- To create or deploy a new artifact, choose **Add** and in the **Add Number Ranges** dialog, provide the following:

Field	Description
Name	A unique name for the number range artifact. Must not already exist on any of the selected runtimes. i Note The following name validation rules apply when deploying to multiple runtimes: ◦ If you select runtimes first and then enter a name, the system checks whether the name already exists on any of the selected runtimes. If a conflict is found, a duplication error is displayed.

Field	Description
	If you enter a name first and then open the Runtimes dropdown, any runtime on which that name already exists is greyed out and cannot be selected.
Description	An optional description of the artifact.
Runtimes	One or more runtime nodes to deploy the artifact to. The field is pre-filled with the runtime currently selected from the dropdown on the page. You can add further runtimes, including Cloud Integration and any active Edge Integration Cell nodes.
Maximum Value	The maximum value of the number range. Must be fewer than 15 digits.
Minimum Value	The minimum value of the number range. Must be greater than or equal to 0.
Rotate	If selected, the current value resets to the minimum value when the maximum value is reached.

- Choose **Add**.

Managing Partner Directory Entries

Manage Partner Directory entries that can be used to parameterize integration flows.

The Partner Directory facilitates the management of integration scenarios encompassing communication networks with numerous partners. In these scenarios, integration flows are parameterized with partner-specific parameters that are centrally stored in the Partner Directory. At runtime, integration flows can read information from the Partner Directory, based on the partner-specific values contained in the incoming request or the payload of the inbound message. This *dynamic* usage of integration flow parameters provides the advantage of adding new partners to the network without the need to edit and redeploy any integration flow, allowing for the extension of the business partner network without any downtime of the scenario.

→ Tip

Integration flows can access data from the Partner Directory through actions like the Script step.

For more comprehensive details on the concept and entities stored in the Partner Directory, see [Parameterizing Integration Flows Using the Partner Directory](#).

To view examples of integration flows using the Partner Directory, consult the integration flow design guidelines at [Use the Partner Directory Appropriately](#).

You can also use the Cloud Integration OData API to create, edit, and display the content of the Partner Directory (see: [Partner Directory](#)).

With this feature, you can manage the contents of the Partner Directory by creating new entries, as well as editing and deleting existing ones.

To access the Partner Directory:

1. Navigate to **Monitor**.

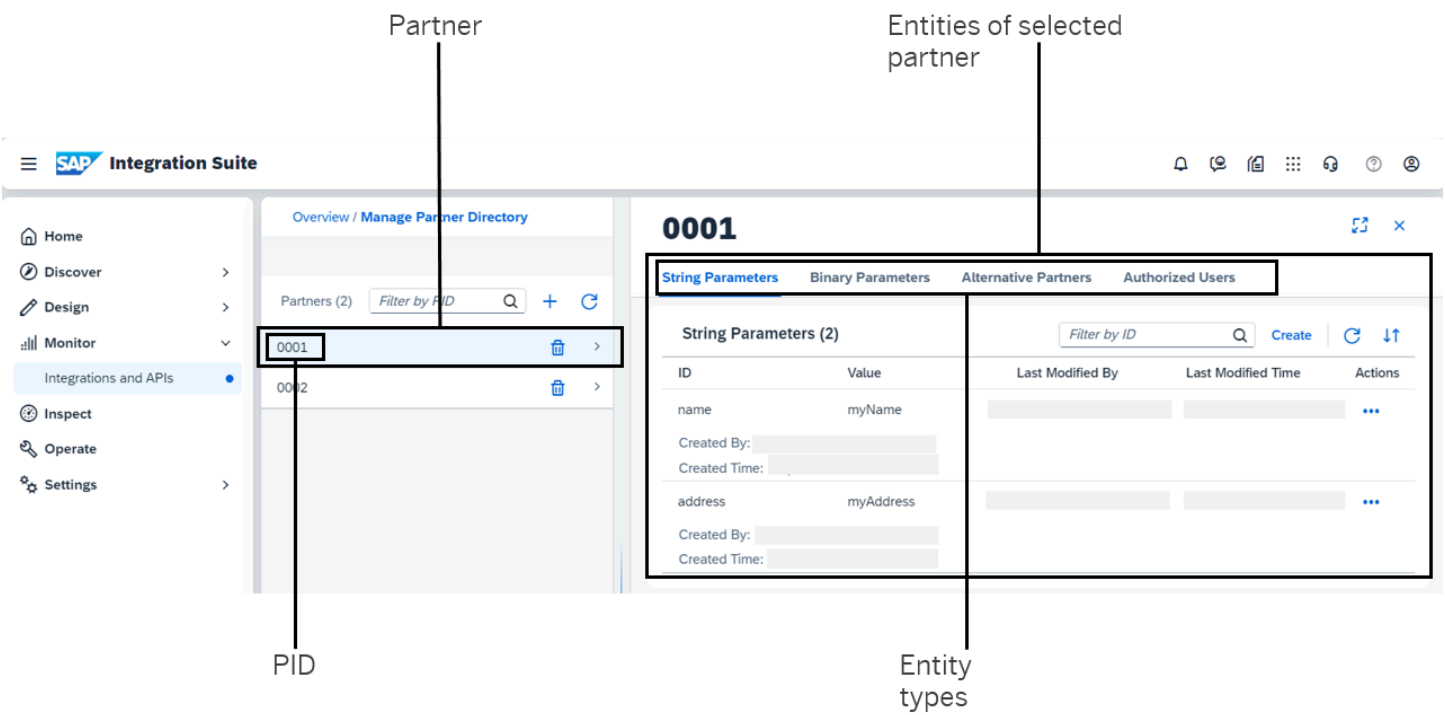
2. Choose **Partner Directory**.

If partners are defined, the screen displays a master table featuring the list of partners.

3. Clicking on a partner in the list reveals the entities defined for the partner in another table on the right.

Screen Components

Each partner is identified by a partner ID (**PID**). When a partner is selected, different entity types are accessible through different tabs in the second table. For each entry, administrative information is displayed, including the user who created and modified the entry, and the timestamp of the entry's creation or last modification. Depending on the entity type, additional attributes are displayed for an entry (see: [Partner Directory Entity Types](#)).



Functions

The following operations can be performed on partners and partner entities:

Object	Operation	More Information
Partner	Add	Add Partner
	Delete	Delete Partner
Partner entity (such like string parameter, binary parameter, alternative partner, or authorized user)	Create	Create Partner Directory Entity
	Edit	Edit Partner Entity
	Copy	Copy Partner Entity
	Delete	Delete Partner Entity

Partner Directory Entity Types

Learn more about tha Partner Directory entity types.

Each entry in the Partner Directory is identified by a partner ID **PID** which is unique within the Partner Directory.

The **PID** may consist of alphanumeric characters (A-Z, a-z, 0-9), as well as the following special characters: '-', '_', '~', '<', '>', and '@'.

The Partner Directory has different entity types. In addition to the **PID**, each entry comes with additional attributes. The set of additional attributes and their meaning depend on the entity type:

Entity Type	Specific Attributes
String Parameters Use this entity type to define a user string parameter associated with a partner. Use this content type if you like to represent the partner by textual data, such as words, sentences, or any other kind of character sequence. i Note A Partner Directory parameter has a value and a type. The currently supported predefined types are Binary and String.	ID An identifier for the string parameter. The ID of this entity may consist of alphanumeric characters (A-Z, a-z, 0-9), as well as the following special characters: '-', '_', '~', '<', '>', and '@'. Value The value associated with the string parameter. A string parameter has a maximum length of 4000 characters.
Binary Parameters Use this entity type to define a binary parameter associated with a partner. Use this content type if you like to represent the partner by binary data, such as files, or any other non-textual data. A binary parameter has a maximum size of 1,5 MB. i Note A Partner Directory parameter has a value and a type. The currently supported predefined types are Binary and String.	ID An identifier for the binary parameter. The ID of this entity may consist of alphanumeric characters (A-Z, a-z, 0-9), as well as the following special characters: '-', '_', '~', '<', '>', and '@'. Value The value (file) associated with the binary parameter. For this entity type, you can upload a file from your computer. The parameter Value of this entity can have a maximum length of 262144 bytes. Content Type Specifies the type of content, for example, xml or json. The following content types are supported (defined by the ContentType parameter): xml, xsl, xsd, json, text, zip, gz (for GZIP files), zlib, crt (for DER encoded X.509 certificates). Encoding There are the following options: US-ASCII ISO-8859-1 UTF-8 UTF-16BE

Entity Type	Specific Attributes
	- UTF-16LE - UTF-16
Alternative Partners Use this entity type to define an alternative identifier of a partner. i Note Examples for use cases where an entity can be defined by different identifiers: - A bank can be identified by the German bank code number (Bankleitzahl or: BLZ) or the international Bank Identifier Code (BIC). - A person can be identified in many different ways, for example, by their customer number at a mail order company, or their name or employee number at their place of work. To able to identify a person uniquely, it isn't sufficient to just know the customer number, for example, you must know that this number is a customer number. Only once the frame of reference (the identification scheme Customer Number) is also known can a person be identified uniquely (by using the number). In the example, the agency Mail Order Company manages the identification scheme Customer Number. Different customers are issued unique names (customer numbers) within this scheme.	Agency Abbreviation for the organization that defines the identification scheme and that issues names for the partner to be identified (maximum length of 120 characters). Scheme Reference framework within which partners are uniquely identified by names (maximum length of 120 characters). ID Identifier for the partner (issued by the agency) (maximum length of 225 characters). This is an ID or name that identifies the partner within the given scheme. i Note Restriction: - The triple Agency, Scheme, Id must be unique. - The triple Agency, Scheme, Pid must be unique (only then you can look up the alternative partner Id from such a triple).
Authorized Users Use this entity type to define a user with which a partner sender system can log in into Cloud Integration. This entity is required for the authorization of the partner sender system when calling Cloud Integration. An Authorized Users entry provides information on the authorized message sender and the unique Partner ID (PID) linked to the partner in the Partner Directory. If the partner uses HTTPS with client certificate authentication to connect to Cloud Integration, a certificate-to-user-mapping is required (in the Neo environment).	User The user authorized to send messages.

Add Partner

Add a new partner to the Partner Directory.

When you create a partner, you must specify a partner ID (PID) first that identifies the partner in the Partner Directory. For each partner (with a given PID), you can then create multiple entries for different entities (such like string parameter, binary parameter, alternative partner, and authorized user).

1. Go to **Monitor**.
2. Choose **Partner Directory**.

- 3. Select **+** (**Add**).
- 4. On the **Create Partner Entry** dialog, specify the **PID**.

This ID identifies the partner within the Partner Directory.

Make sure that the entered value is unique within the Partner Directory.
- 5. Specify the **Entity Type**.
- 6. Specify the remaining parameters. Which parameters are available, depends on the chosen entity type (see also: [Partner Directory Entity Types](#)).

Entity Type	Specify the Following Attributes
String Parameters	◦ ID ◦ Value
Binary Parameters	◦ ID ◦ Value ◦ Content Type ◦ Encoding Only available if for Content Type one of the following options has been selected: xml, xsl, xsd, json, text.
Alternative Partners	◦ Agency ◦ Scheme ◦ ID
Authorized Users	User

- 7. Select **Create**.

With the new partner, you have also created the first partner entity.
- 8. If you like to add more entities for the partner, click and repeat the previous steps.

Delete Partner

Delete a partner from the Partner Directory.

With this step, you delete a partner and all its entities.

- 1. Go to **Monitor** .
- 2. Choose **Partner Directory**.
- 3. Ffrom the master table, select the partner you like to delete.
- 4. Select the recycle bin icon (**Delete**).

- On a subsequent screen, confirm this step by entering DELETE.
- Select **Delete**.

Create Partner Directory Entity

Create an entity for an existing partner.

- Go to **Monitor**.
- Choose **Partner Directory**.
- In the master table, select a partner.
- In the detailed table at right, choose the tab with the type of the entity that you like to create (for example **String Parameters**).
- Select **Create**.
- On the **Create <Entity Type>** screen, enter the attributes of the entity. Which attributes to specify depends on the entity type, as shown in table below (see also: [Partner Directory Entity Types](#)).

Entity Type	Specify the Following Attributes
String Parameters	PID Is already prefilled but can be changed. ID Value
Binary Parameters	PID Is already prefilled but can be changed. ID Value Content Type Encoding Only available if for Content Type one of the following options has been selected: xml, xsl, xsd, json, text.
Alternative Partners	PID Is already prefilled but can be changed. Agency Scheme ID
Authorized Users	PID

Entity Type	Specify the Following Attributes
	Is already prefilled but can be changed. o User

7. Select [Create](#).

Edit Partner Entity

Edit an existing partner entity.

1. Go to [Monitor](#) .
2. Choose [Partner Directory](#).
3. In the master table, select a partner.
4. In the detailed table at right, select first a tab for the entity type (for example, [String Parameters](#)) and then entry.
5. Choose [Actions](#) [Edit](#) .
6. On the [Edit <Entity Type>](#) screen, edit the attributes of the entity. The editable attributes depend on the entity type, as shown in table below (see also: [Partner Directory Entity Types](#)).

Entity Type	Edit the Following Attributes
String Parameters	Value
Binary Parameters	- o Value - o Content Type - o Encoding Only available if for Content Type one of the following options has been selected: xml, xsl, xsd, json, text.
Alternative Partners	PID Specify the new partner for which the alternative partner should apply.
Authorized Users	PID Specify the new partner for which the authorized user should apply.

7. Choose [Edit](#).

Copy Partner Entity

Copy a Partner Directory entity.

The main use case of copying a Partner Directory entity is to use it in the context of another partner (PID).

1. Go to **Monitor**.
2. Choose **Partner Directory**.
3. In the master table, select a partner.
4. In the detailed table at right, select first a tab for the entity type (for example, String Parameters) and then entry.
5. Choose **Actions > Copy**.
6. On the **Copy <Entity Type>** screen, edit the attributes of the entity. The editable attributes depend on the entity type, as shown in table below (see also: [Partner Directory Entity Types](#)).

Entity Type	Edit the Following Attributes
String Parameters	◦ PID Specify the new partner for which the authorized user should apply. ◦ ID Is predefined but can be changed. ◦ Value Is predefined but can be changed.
Binary Parameters	◦ PID Specify the new partner for which the authorized user should apply. ◦ ID Is predefined but can be changed. ◦ Value Is predefined but can be changed. ◦ Content Type Is predefined but can be changed. ◦ Encoding Is predefined but can be changed. Only available if for Content Type one of the following options has been selected: xml, xsl, xsd, json, text.
Alternative Partners	◦ PID ◦ Agency Is already prefilled but can be changed. ◦ Scheme Is already prefilled but can be changed. ◦ ID

Entity Type	Edit the Following Attributes
Authorized Users	◦ PID ◦ User Is already prefilled but can be changed.

7. Choose **Copy**.

Delete Partner Entity

Delete an entity of a partner from the Partner Directory.

1. Go to **Monitor**.
2. Choose **Partner Directory**.
3. In the master table, select a partner.
4. In the detailed table at right, select first a tab for the entity type (for example, String Parameters) and then entry.
5. Choose **Actions > Delete**.
6. On a subsequent screen, confirm this step by entering DELETE.
7. Select **Delete**.

Access Logs, Neo Environment

The **Access Logs** section allows you to monitor audit logs (resulting from system changes) and to analyze errors that occurred during inbound HTTP processing (and documented in system log files).

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

Related Information

[Monitoring Audit Log, Neo Environment](#)

[Monitoring System Log Files, Neo Environment](#)

Monitoring Audit Log, Neo Environment

The audit log contains information on system changes. These events can be for example the deployment of an integration flow as well as a configuration change.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

You view the audit log by clicking the tile **Audit Log** in the **Manage Security** area.

i Note

To view the content of the log, you need the authorization `AuditLog.Read`.

You can control the display of the messages by adjusting the filter setting **Time Range**.

You can choose from the following preset time ranges:

- All
- Past Hour
- Past 24 Hours
- Past Week
- Custom

i Note

The audit log data retention time in the database is 30 days.

You can retrieve the following information from the audit log list and filter the entries by **Object Name**, **User** or **Source**

Audit log List

Attributes	Description
Time	Displays the time of the event.
Action	Displays the action performed on the system such <i>Create</i> or <i>Delete</i>
Object Type	Displays the object type such as <i>IntegrationFlow</i> on which the action was performed on.
Object Name	Displays the object name such as the integration flow name.
User	Displays the user who triggered the action.
Source	Displays the IP address of the source that issued the action.

i Note

If an SAP user triggers the changes, **User** and **Source** are displayed as *SAP* in the audit log list.

You can also sort the audit log list by **Time**, **Action**, **Object Type** or **Object Name**.

i Note

The audit log retrieves a maximum of 1000 entries from the data base. If you have more than 1000 entries in the selected time range, you will be prompted to adjust your filter settings accordingly.

Related Information

Monitoring System Log Files, Neo Environment

This section contains information on system log files. These log files can be either HTTP access files or default trace files.

i Note

This information is relevant only when you use SAP Cloud Integration in the Neo environment.

→ Remember

This component or some of its features might not be available in the Cloud Foundry environment. For more information on the limitations, see SAP Note [2752867](#) .

You view the system log files by clicking **System Log Files** in the **Access Logs** area.

You can retrieve the following information from the system log file list:

Log Files

Attributes	Description
Name	Displays the log file name
Log Type	Displays 2 different log types the CP default trace or the HTTP access log
Updated At	Displays the date of the last update
Size	Displays the file size
Actions	You can either download the file by selecting  or get the URL by selecting  .

You can filter the log files by names and sort the list either by **Name** , **Updated At**, or file **Size**. You can either download a specific file or get the file URL. The file URL allows you to send it per mail to allow further analysis on another computer for example. In both cases, the system provides a packed log file (.zip file).

i Note

The log file retention time is 7 days.

If you select **Collections**, you get the most recent log files for each runtime node. You can either download the collection or get the URL and in both cases the system provides packed log files (.zip files).

Access Logs, Cloud Foundry Environment

The **Access Logs** section allows you to monitor and to analyze errors that occurred during inbound HTTP processing (and documented in system log files)

⚠ Caution

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

You can access the system logs only if the Cloud Logging Service is available for your Cloud Integration application. This feature is not yet enabled on all data centers.

Related Information

[Monitoring System Log Files, Cloud Foundry Environment](#)

Monitoring System Log Files, Cloud Foundry Environment

This section contains information on system log files. These log files can be either HTTP access files or default trace files

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

You view the system log files by clicking [System Log Files](#) in the [Access Logs](#) area.

You can retrieve the following information from the system log file list:

Log Files	
Attributes	Description
Name	Displays the log file name.
Log Type	Displays 2 different log types the trace log or the HTTP access log.
Updated At	Displays the date of the last update.
Entries	Displays thenumber of entries in the log file.
Actions	You can either download the file by selecting ⬇ or get the URL by selecting 📄.

You can filter the log files by names and sort the list either by [Name](#) , [Updated At](#), or [Entries](#). You can either download a specific file, or get the file URL. The file URL allows you to send it per mail, to allow further analysis on another computer for example. In both cases, the system provides a packed log file (.zip file).

i Note

The log file retention time is 30 days.

If you select [Collections](#), you get the most recent log files. You can either download the collection or get the URL and in both cases the system provides packed log files (.zip files).

External Logging Cloud Foundry Environment

The external logging feature enables customers to send message processing logs to an external system, independently of available database storage. This section describes how to enable it on your tenant.

Context

 Caution

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

The major customer use cases for the external logging feature are the integration into existing monitoring solutions, or the consolidation of logs of multiple tenants. You can use it for high-throughput scenarios (currently limited by our systems and databases) and high volume queries. And it is also suitable for advanced analysis, as well as for correlation with business data and long-term archiving.

 Caution

Avoid transmitting message payloads or other large data volumes to an external logging system through custom headers. This may affect the performance of the data replication to the external logging system.

The external logging feature supports the following **target systems**:

- [Splunk](#). You can send logs to Splunk through the HTTP Event Collector (HEC) API.
- [JSON](#). You can send logs as JSON objects to any HTTP endpoint. For example, you can integrate with custom log processing pipelines.
- [OTEL](#). You can send logs as OpenTelemetry Protocol (OTLP) log records over HTTP to any OTLP-compatible receiver.

Select the target system by setting the **TargetSystem** destination property to one of the values listed above.

! Restriction

You can assign only one type of destination as a target system at a time.

Prerequisites

To activate external logging in the Cloud Foundry Environment on your tenant, you have to fulfill the following prerequisites:

- You are tenant Admin with the role *AuthGroup_Administrator* as well as the role *ExternalLoggingActivate*.
- Use the *activateExternalLogging* function of the official OData API, see [Message Processing Logs](#) .

To access APIs clients, see [Setting Up Inbound HTTP Connections \(for API Clients\)](#). Create a new role collection and include the following roles: *ExternalLoggingActivationRead* and *ExternalLoggingActivate*.

Procedure

To create a destination:

1. Go to your subaccount, and under **Connectivity**, choose **Destinations**.
2. Select the context level where you want to create destinations.
3. Choose **Create**, and then choose **From Scratch** to create a new destination.
4. Configure the following parameters:

Mandatory Destination Properties

Property (mandatory with *)	Value or Description
-----------------------------	----------------------

Property (mandatory with *)	Value or Description
Name*	The name must be <i>CloudIntegration_MonitoringDataConsumer</i> .
Type*	The type HTTP is used for both HTTP and HTTPS connections.
URL*	The URL of the target system endpoint.
Proxy Type *	Choose Internet for cloud-hosted target systems, and On Premise for on-premises installations accessed through the Cloud Connector.
TargetSystem *	Identifies the type of target system. Choose one of the following: • Splunk • JSON • OTEL

You can add additional properties by using the **New Property** option.

! Restriction

If you want to add additional properties, you must enter them manually.

Additional Properties

Property	Value or Description
CompressData	If set to true (default), the system compresses HTTP request bodies using GZIP before transmission.
TrustAll	If set to true , the system disables SSL certificate verification. i Note Use this property with caution.
TrustStoreLocation	Path or name of a trust store for SSL verification. Supported formats include .jks, .p12, .pfx, or a raw PEM certificate.
TrustStorePassword	Password for the trust store specified in TrustStoreLocation .
CloudConnectorLocationId	Logical location ID for the Cloud Connector instance. This is required when the system is configured with multiple Cloud Connector instances.
MaxBatchSize	Maximum number of bytes per write batch.
MaxBatchCount	Maximum number of log events per write batch.
DateFormat	Format string for human-readable timestamps in log events. i Note The default format is yyyy-MM-dd HH:mm:ss.SSS.
SubValuesInline	If set to true, the system adds values such as custom header attributes and adapter attributes as nested JSON structures within

Property	Value or Description
	the base Run event. This property is independent of SubValuesExtra. The default is false .
SubValuesExtra	If set to true (default), the system adds values such as custom header attributes and adapter attributes as separate stand-alone events following the base Run event. This property is independent of SubValuesInline.

i Note

Each target system has its own set of additional properties. For more details, see the sections for [Splunk](#), [JSON](#), and [OTEL](#).

Enabling External Logging

To enable external logging, send a POST call to the following URL: `https://path-to-odata-api/api/v1/activateExternalLogging[?defaultLogLevel=(NONE|INFO|ERROR)]`, where the *path-to-odata-api* is specific to your environment.

The optional parameter `defaultLogLevel` specifies the external logging level applied to existing and newly deployed integration flows. If this parameter is omitted or is **NONE**, then external logging doesn't take place, unless explicitly activated for a specific integration flow. If the log level is set to **INFO** or **ERROR** instead, MPLs for existing integration flows and newly deployed integration flows are immediately sent to the external system, unless the setting is explicitly changed for a specific integration flow. You can change the default setting anytime by calling the activation API again. If external logging is already active, it takes up to 5 minutes for the new default level to take effect.

If the enabling of the external logging function is successful, you get a response code 200 and a message of success.

To check whether external logging is currently configured on a tenant, use the OData API allowing to query the message processing logs. Send a *GET* call to the URL: `https://path-to-odata-api/api/v1/ExternalLoggingActivationStatus('tenant-name')`, where *path-to-odata-api* is specific to your environment. The parameter `tenant-name` doesn't specify the name of the tenant for which you check the status of the external logging function. The information about the tenant is taken from the security token used for the request.

You've to reload the **Monitor** to be able to see the external logging level options for your integration flows. If the enablement isn't successful, the system throws an error code as well as an error message.

If the external logging is activated,

1. Go to the **Manage Integration** section.
2. Choose the integration flow you want to have the external logging for.
3. In the **Log Configuration** section, in addition to the standard **Log Level** you can set the **External Log Level**.
4. You've 2-log level options to choose from: **Info** and **Error**.

After the successful activation of the external logging, you can deploy an integration flow and monitor it as usual.

The MPLs are sent to the the configured external system. The MPLs sent to the external destination aren't stored in an SAP database, and SAP can't access them. Internal and external log levels can be set independently.

i Note

If an integration flow is undeployed, the external logging level is lost. You've to reconfigure the external logging level after redeployment of the integration flow.

Disabling External Logging

After activation, you can cancel the logging to an external destination by deactivating the external logging feature. To disable external logging, send a POST call to the following URL <https://path-to-odata-api/api/v1/deactivateExternalLogging>, where the *path-to-odata-api* is specific to your environment. If the disabling of the external logging function is successful, you get a response code 200 and a message of success.

After deactivation, no MPLs are sent to the external system. Logging configuration for deployed integration flows is retained and once activated again, the configured external logging level is restored to the value before deactivation.

External Logging Status Information API

We provide a status API to monitor the current status of external logging or diagnose errors in case external logging is disabled.

To retrieve status information, send a GET call to the following URL: <https://path-to-odata-api/api/v1/ExternalLoggingEvents> where the *path-to-odata-api* is specific to your environment. If external logging is activated, it returns a list of events that occurred in the last 14 days for these possible events:

- External Logging was activated.
- External logging was deactivated.
- Errors causing failures in sending data to an external system.
- Resuming sending data to an external system once an error is cleared.

If there are errors, the event carries the error reason as well as the error details received from the external system (if available).

Splunk

When you set **TargetSystem** to **Splunk**, the system sends log events to Splunk through the HTTP Event Collector (HEC) API. The destination URL must point to the Splunk HEC base URL without including the endpoint path.

❖ Example

For a Splunk Cloud Platform host named myhost, use: <https://http-inputs-myhost.splunkcloud.com>.

For details on enabling and configuring the HTTP Event Collector, see [Set up and use HTTP Event Collector in Splunk Web](#) ➦.

Authentication for Splunk requires an HEC token. You can provide the token in one of the following ways:

- By using basic authentication. Set **Authentication** to **Basic Authentication**, **User** to any value (such as Splunk), and **Password** to the HEC token value.
- By using the **AuthHeader** property. Set **AuthHeader** to the value **Splunk <token>** (the word Splunk, followed by a space, then the token value).

The Splunk target system supports both HTTP and HTTPS connections using direct connections or connections through the Cloud Connector. For authentication, it only supports **None** (when the HEC token is provided through the **AuthHeader** property) or **Basic Authentication** (when the HEC token is provided through the **Password** property). The system doesn't support any other authentication methods in the destination configuration.

You can add the following additional properties for Splunk using the **New Property** option:

Property	Value or Description
IndexName	Enter an index name.
Source	Any value, used to fill in the source attribute Splunk events.
Host	Any value, used to fill in the host attribute Splunk events.
AuthHeader	Custom value for the HTTP authorization header. For example: Bearer <token> .
SourceType	If the source type is specified, it's set to the defined value in Splunk events.
DynamicSourceType	If set to true, the source type is set to a value matching one of the following event types: • Run for MPL runs • CustomHeader for custom header events • AdapterAttribute for adapter attribute events
SourceTypePrefix	If DynamicSourceType is true, and the SourceTypePrefix property is, the property value is prepended to the source type.  Example For a prefix of Custom and an MPL run event, the source type is to CustomRun.
Endpoint	The HEC endpoint path appended to the base URL. The default value is /services/collector/event

JSON

When you set **TargetSystem** to **JSON**, the system serializes log events as JSON objects and sends them via HTTP POST to the configured URL. This target system is suitable for integrations with custom log processing pipelines and any HTTP endpoint that accepts JSON payloads.

The JSON target system supports HTTP and HTTPS destinations. You can use both direct connections and connections through the cloud connector. The system supports all authentication methods offered in the destination configuration.

You can add the following additional properties for the JSON target system via **New Property**:

Property	Value or Description
JSONBatchType	Controls how the system sends multiple log events in a batch. This property takes effect only when MaxBatchCount is greater than one. Possible values: • single (default): the system sends each log event in a separate HTTP POST request. • array: the system sends all log events in a batch in a single request as a JSON array: [{...}, {...}].

Property	Value or Description
	sequence: the system sends log events in a single request, concatenated and separated by the delimiter configured in <code>JSONSequenceDelimiter</code>. This follows the newline-delimited JSON (NDJSON) style.
<code>JSONSequenceDelimiter</code>	The delimiter inserted between JSON objects when you select sequence for <code>JSONBatchType</code> . The default is <code>\n</code> (newline). You can use escape sequences: <code>\n</code> , <code>\r</code> , <code>\r\n</code> .
<code>JSONFormatting</code>	Controls the formatting of individual JSON objects. Possible values: standard (default): compact JSON as produced by the JSON serializer. pretty: JSON formatted with two-space indentation for easier reading. compact: single-line JSON with all embedded newlines removed.

OTEL

When you set **TargetSystem** to **OTEL**, the system sends message processing logs as log records that follow the OpenTelemetry Protocol (OTLP) Logs format. The system transmits these logs via HTTP POST to the configured URL, using the OTLP/HTTP JSON encoding. This target system works with any OTLP-compatible log backend. Use it with backends such as OpenTelemetry Collector, Grafana Loki, or other OTLP receivers.

The system wraps each batch of log records in the standard OTLP `resourceLogs` structure. It sets the `service.name` resource attribute to identify the source of the logs. Each log record contains the following information:

- timeUnixNano**. The time that message processing stops, in nanoseconds.
- observedTimeUnixNano**. The time that message processing stops, plus the transmission lag, in nanoseconds.
- severityNumber**. The number **17** represents an error, including failed/aborted/cancelled/escalated messages. Number **9** (Info) is for all other statuses.
- body**. The complete message processing log as a JSON string.

The **OTEL** target system supports HTTP and HTTPS destinations. It supports both direct connections and connections through the cloud connector. It also supports all authentication methods available in the destination configuration.

You can add the following additional properties for the **OTEL** target system via **New Property**:

Property	Value or Description
<code>OTELServiceName</code>	The value of the <code>service.name</code> resource attribute. The default value is <code>MessageProcessingLogs</code>
<code>OTELScopeName</code>	The name of the instrumentation scope. The default value is <code>MPL</code> .
<code>OTELScopeVersion</code>	The version of the instrumentation scope. The default value is <code>1.0.0</code>

i Note

The OTEL target systems sends an empty POST body ({}) during connection validation. It checks whether the response contains `partialSuccess` to confirm the endpoint is an OTLP receiver. If the response does not contain `partialSuccess`, the connection test still succeeds. However, the system warns users that the server may not be a valid OTLP receiver.

Possible Errors During External Logging Activation

When activating external logging, different errors can occur.

Find below the different error situations and the error messages senr back by the OData API.

Error Situation	Error Message and Response Body
No destination with the name <code>CloudIntegration_MonitoringDataConsumer</code> was found in the subaccount's destination configuration.	Retrieving Destination failed. Destination with name <code>CloudIntegration_MonitoringDataConsumer</code> does not exist.
An error ocured with the configuration of the destination. The error is returned in the <code><DESTINATION_ERROR></code> element. Common issues are missing mandatory parameters and invalid values in free-form parameters.	<pre>{ "result": "ERROR", "message": "<DESTINATION_ERROR>", "infos": [{ "description": "Read Tenant Information", "result": "SUCCESS", "dataSent": "", "dataReceived": "" }, { "description": "Validating Connection Settings", "result": "ERROR", "dataSent": "", "dataReceived": "<DESTINATION_ERROR>" }] }</pre>
A problem occurred when connecting to the Splunk system. The error is returned in the <code><ERROR_MESSAGE></code> element. Common errors include DNS issues, configuration issues with Cloud Connector and firewall configuration problems.	<pre>{ "result": "ERROR", "message": "<ERROR_MESSAGE>", "infos": [{ "description": "Read Tenant Information", "result": "SUCCESS", "dataSent": "", "dataReceived": "" }, { "description": "Validating Connection Settings", "result": "SUCCESS", "dataSent": "", "dataReceived": "All Settings Valid" }] }</pre>

Error Situation	Error Message and Response Body
	<pre> "description": "Instantiating Selected Adapter", "result": "SUCCESS", "dataSent": "", "dataReceived": "Adapter Created" }, { "description": "Calling OPTIONS on endpoint", "result": "ERROR", "dataSent": "OPTIONS /services/collector/event", "dataReceived": "<ERROR_MESSAGE>" }] }</pre>
A problem occurred when sending events to the target Splunk system after a successful configuration. The error message is returned in the <SPLUNK_ERROR> element. This message typically indicates a problem with the Splunk target instance or a mismatch of Splunk and destination configuration.	<pre> { "result": "ERROR", "message": "<SPLUNK_ERROR>", "infos": [{ "description": "Read Tenant Information", "result": "SUCCESS", "dataSent": "", "dataReceived": "" }, { "description": "Validating Connection Settings", "result": "SUCCESS", "dataSent": "", "dataReceived": "All Settings Valid" }, { "description": "Instantiating Selected Adapter", "result": "SUCCESS", "dataSent": "", "dataReceived": "Adapter Created" }, { "description": "Calling OPTIONS on endpoint", "result": "SUCCESS", "dataSent": "OPTIONS /services/collector/event", "dataReceived": "200" }, { "description": "Polling Splunk HTTP Event Collector Endp", "result": "ERROR", "dataSent": "POST /services/collector/event", "dataReceived": "<SPLUNK_ERROR>" }] }</pre>

Manage Locks

The Manage Locks section lets you manage various locked entities on your tenant.

[Message Locks](#)

This section allows you to display and manage lock entries that are created (in the in-progress repository) to avoid the same message being processed several times in parallel (for example, by different runtime nodes).

[Designtime Artifact Locks](#)

As tenant administrators, use this self-service capability to view and unlock the integration artifacts and packages in the tenant that are locked by the users of the tenant.

Message Locks

This section allows you to display and manage lock entries that are created (in the in-progress repository) to avoid the same message being processed several times in parallel (for example, by different runtime nodes).

→ **Tip**

Example:

Several runtime nodes try to read a file from an SFTP server (through SFTP sender channels).

To prevent double processing, a lock entry is written to the in-progress repository each time a file is processed by a runtime node. As long as this lock entry exists, no other component can access the file. After message processing, the lock is removed by the runtime.

In certain situations (for example, a runtime node crashes because of an out-of-memory error), the message is retried after the node is restarted until the expiration time is reached. In this case, lock entries could remain in the in-progress repository and block subsequent message processing. You can use the **Manage Locks** view to analyze the situation and manually delete lock entries, if necessary, to reprocess the message.

If you choose the **Message Locks** tile under **Manage Locks** tile in the **Monitor** application, a list of locks is displayed. Be aware that only the most recent 500 lock entries are displayed.

The following information is shown for each lock entry:

Attribute	Description
Component	Component that wrote the lock entry AS2, JMS, Mail, SFTP, and XI adapters can write locks.
Source	Component that caused the lock Example for SFTP: user_sftp@ld1234.mycompany.corp
Name	Name of the Lock
Entry	Content of the lock entry Example for SFTP: directory1/dir2/test.xml If the lock relates to an SFTP connection, the Entry field shows a file name (of the file that the SFTP adapter tries to read from the SFTP server). This parameter provides a link to the message in the Managing Message Queues monitor.

Attribute	Description
	To access the associated JMS Message, click ID. The Queue Monitor opens and loads the JMS Message. This is only possible for JMS locks.
Created at	Time when the lock was created This is the time when a runtime node tried to process the message for the first time.
Expires at	Time when lock entry expires If the message is retried but cannot be processed successfully before the expiration time, processing is stopped. In this case, the lock entry has to be removed manually to enable further processing.

You can also search for table entries (search field).

To sort and filter the content of the table, choose **Table Settings** (⚙️). On the subsequent screen, you can define how the table entries are to be sorted (by specifying an attribute and whether the entries are to be sorted for that attribute in ascending or descending order). You can also filter the table entries for certain attributes.

To remove the lock entry and retrigger message processing, select the entry and choose **Release**.

⚠️ Caution

Before releasing a lock entry, make sure you do a careful problem analysis. In particular, make sure that you've understood how the lock entry in question relates to the actual problem you are trying to solve. Careless usage of the release lock function can lead to data inconsistencies or other serious problems.

For example, in the case of an SFTP connection the lock entry is a file name (including the file path). In this case, check whether the problem is related to the file that is to be processed through SFTP.

Another indicator is the duration of the lock (time that has passed since the time specified under **Created at**).

Related Information

[Cloud Integration – Configure Dead Letter Handling in JMS Adapter](#) 
[JMS Adapter](#)

Designtime Artifact Locks

As tenant administrators, use this self-service capability to view and unlock the integration artifacts and packages in the tenant that are locked by the users of the tenant.

There could be scenarios where a user locked a design time artifact and didn't release it. For example, an integration developer could have locked an integration flow to update it as per the revised business logic. But if the user is unavailable for various reasons and there's an urgent business need to edit the artifact, the tenant administrators can release the artifacts. The tenant administrators can unlock the artifact anytime after it was locked by a user.

Choose **Monitor** > **Designtime Artifact Locks**  to see the list of design time artifacts that are in locked status. All users of the tenant can view the locked artifacts. Only the tenant administrators can unlock the artifacts using the  (*unlock*) icon. After confirming the unlock action, the artifact gets unlocked and removed from the list.

The following information is shown for each locked artifact:

Attribute	Description
Name	Name of the artifact that is locked.
Type	Type of the artifact that is locked, like integration flow, REST API, or integration package.
Package	For integration artifacts, the name of the package to which the locked artifact belongs.
Locked At	Time when the user locked the artifact.
Locked By	Name of the user who locked the artifact.

Guidelines and Best Practices for Message Monitoring

The Operations section provides various ways to monitor message processing.

To efficiently monitor message processing and analyze runtime errors, you can follow certain best practices.

Related Information

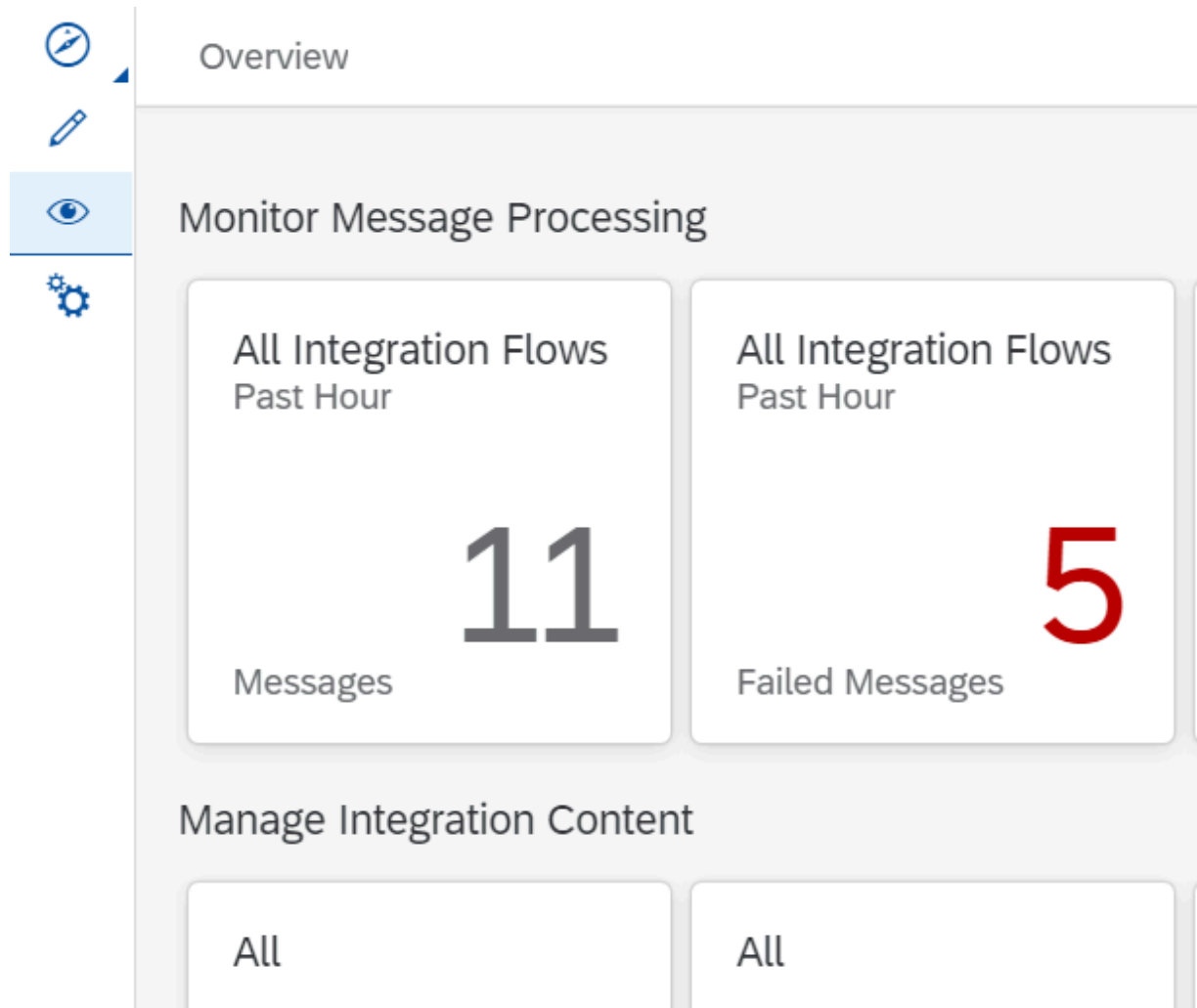
- [Checking Out the Message Processing Log](#)
- [Tracing the Execution of an Integration Flow](#)
- [Checking the Default Trace \(in the Neo Environment\)](#)
- [Blog: Monitoring Tools for Cloud Integration capability of SAP Integration Suite](#)
- [Blog: Alerting for Cloud Integration, Capability of SAP Integration Suite](#)

Checking Out the Message Processing Log

In the **Monitor** section, you can check out the status of processed messages.

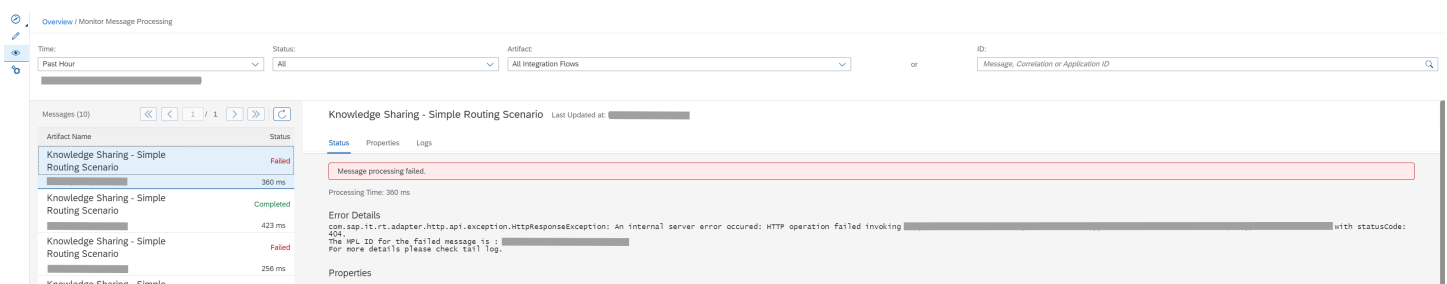
In particular, you can check if message processing has failed for dedicated integration flows.

To do that, open **Monitor** and select a tile under **Monitor Message Processing**.



On clicking a tile, a dialog opens where you can select for individual integration flows.

When you click an integration flow for which message processing failed, a screen with more details opens. For failed messages, an error message is displayed which help already to analyze the issue.



If this information isn't sufficient to analyze the error, you can trace integration flow processing as described under [Tracing the Execution of an Integration Flow](#).

Tracing the Execution of an Integration Flow

If you like to find out in detail how a message is processed and transformed at runtime, you can run the related integration flow with the **Trace** log level.

This is also important if the error information in the message processing log is not sufficient and you like to investigate in more detail where during message processing an error occurred.

Setting this log level for an individual integration flow, you can collect the message content exchanged at runtime. That way, you can understand in more detail the transformation process of a message for an integration flow.

i Note

Monitoring an integration flow for which log level **Trace** is configured is also referred to as *tracing*.

Tracing allows you to log the following parts of the message in the processing pipeline at each integration flow step:

- Payload
- Header
- Properties

You (in the role of an integration developer or tenant administrator) explicitly need to enable tracing for each integration flow individually.

How to Change Log Level to Trace

Perform the following steps:

1. Deploy the integration flow.
2. Open the **Operations** view (**Monitor** section) and select tile **All** under **Manage Integration Content**.
3. On the next screen, select the integration flow that you like to trace.
4. For **Log Level**, select **Trace**.

The screenshot shows the 'Manage Integration Content' interface. On the left, a list of integration flows is shown, all with a status of 'Started'. The main panel displays details for 'Knowledge Sharing - Simple Routing Scenario'. The 'Log Configuration' section shows a dropdown menu for 'Log Level' with options: Info, None, Info, Debug, and Trace. The 'Trace' option is highlighted with a red box.

5. Select **Change** to confirm your setting.
6. Trigger the processing of the integration flow.

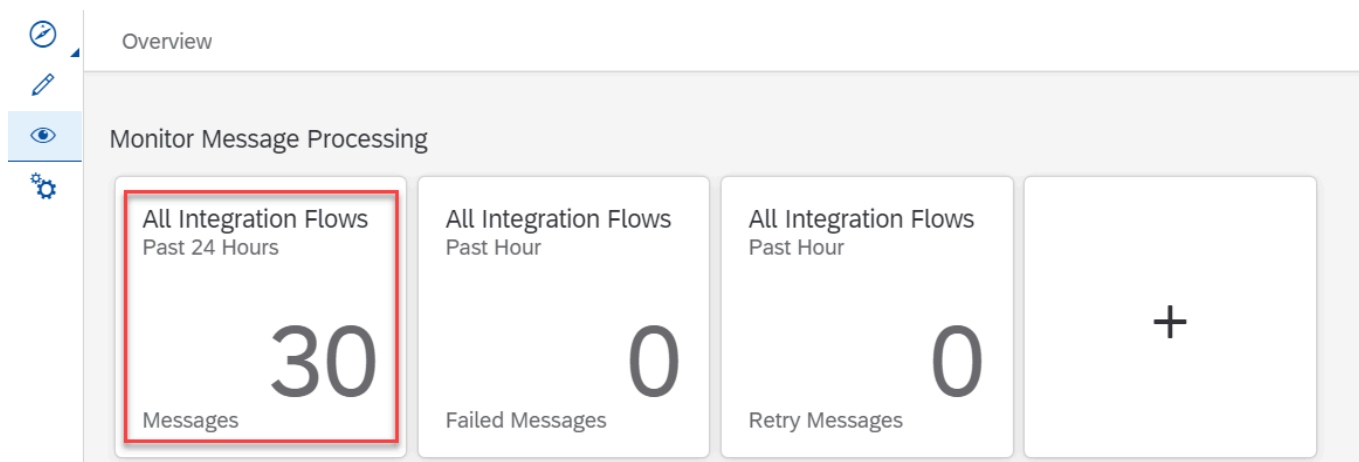
i Note

Log level **Trace** expires after a certain time (default value: 10 minutes) and switches back to the log level set before. The trace data is cleared from the tenant storage after 60 min.

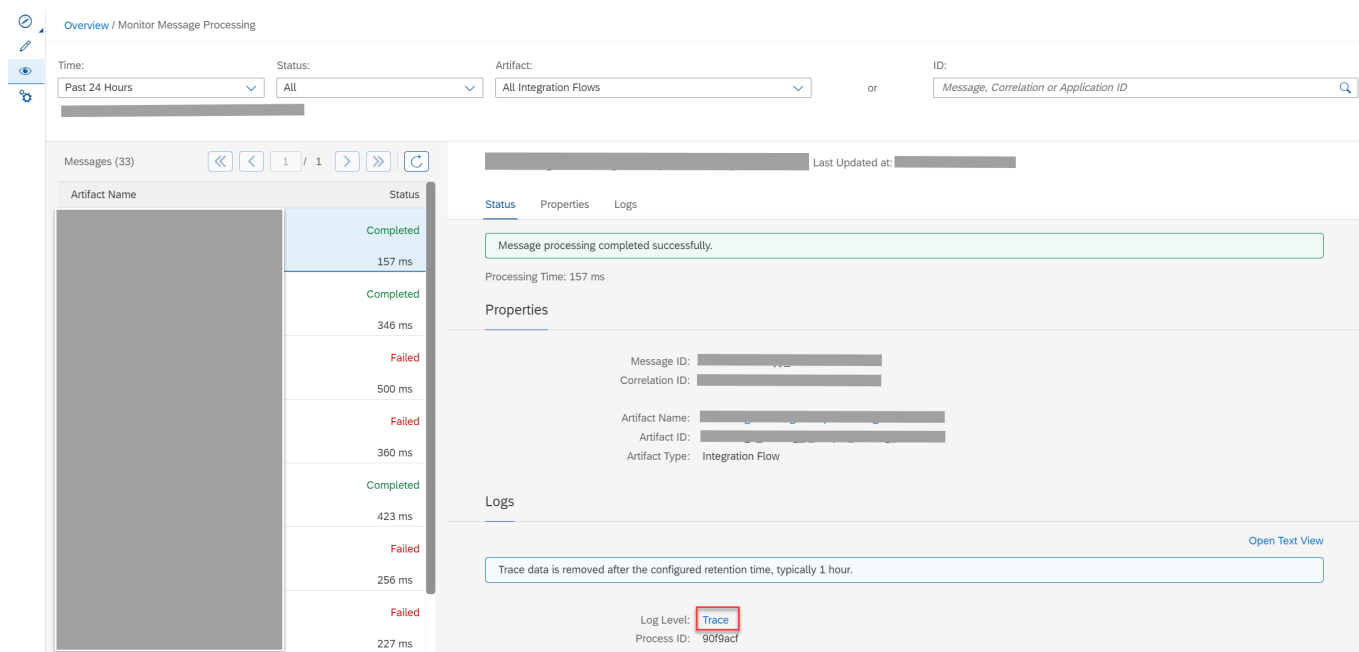
How to View the Trace

After the integration flow has been processed again, perform the following steps:

1. Open the **Operations** view (**Monitor** section) and select tile **All Integration Flows (Past 24 Hours)** (or another tile providing access to the monitoring data of the integration flow) under **Monitor Message Processing**.



2. Select the integration flow and on the right side of the screen click the link **Trace**.

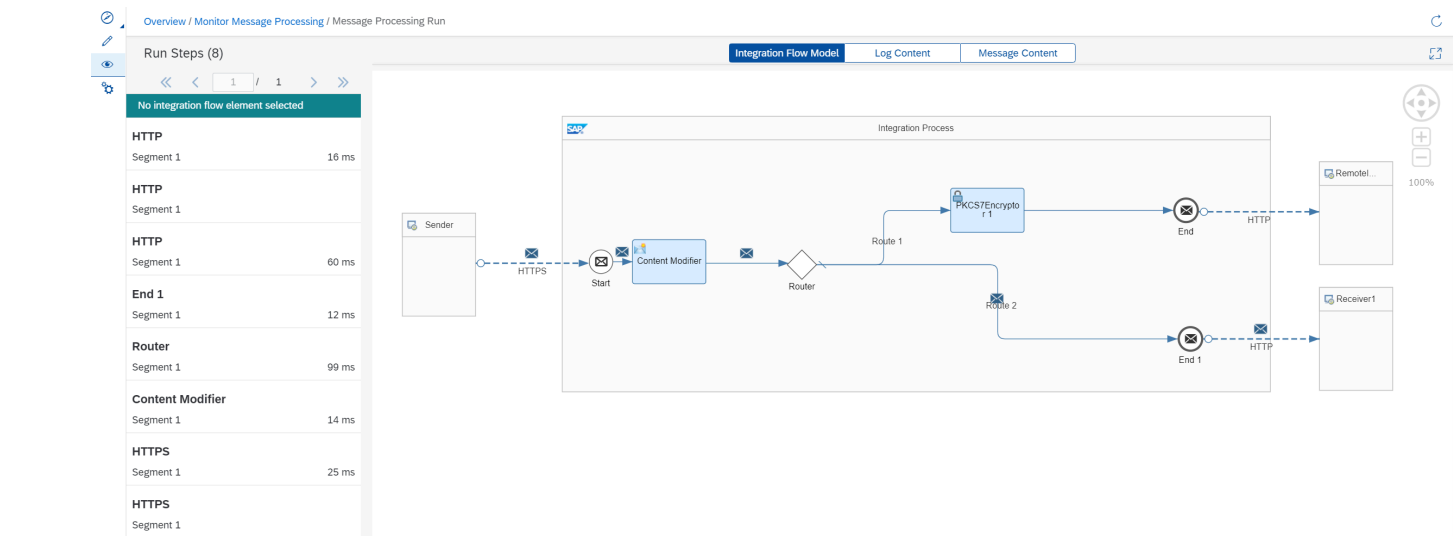


3. The integration flow model and the trace details are shown.

The names of the integration flow steps (as specified during design time) are shown on the left-hand side. They are sorted based on execution in descending order by default: The last executed step is the first entry in the list.

In case of large integration flows, the list contains multiple pages, again in the same order, the first page containing the last executed steps.

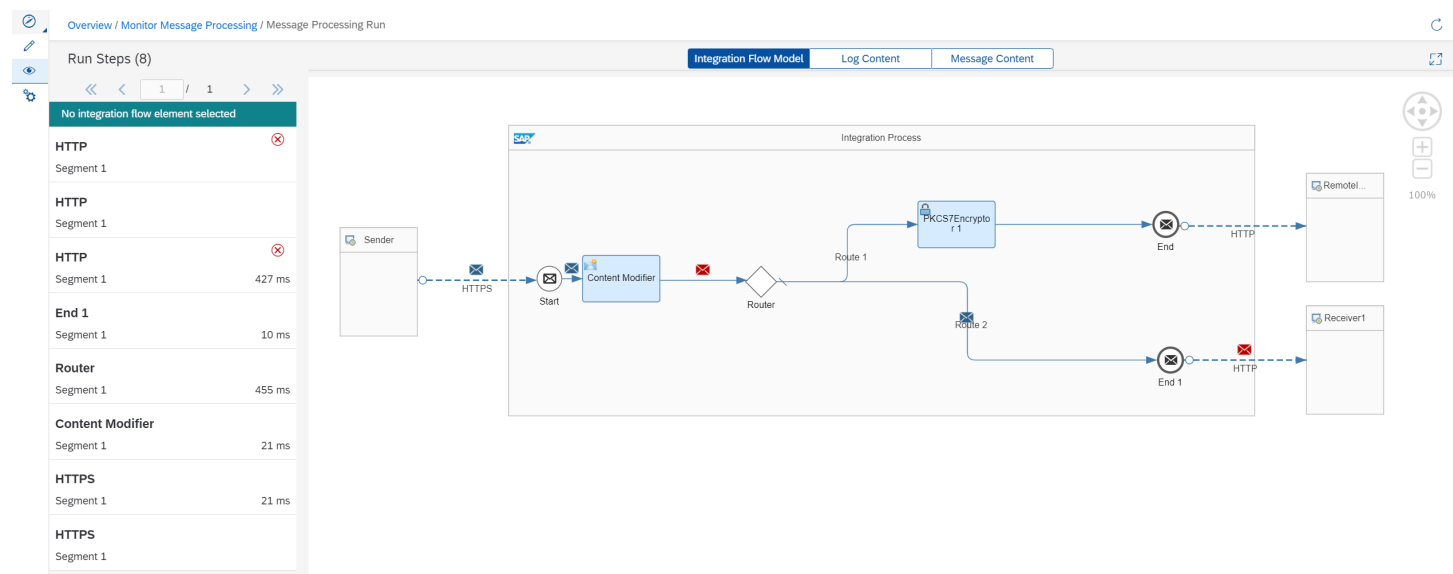
The integration flow model is displayed on the right-hand side.



The envelope icon indicates the message's traversal path:

Icon	Description
	Successfully processed step with trace data available.
	Error in processed step with trace data available.
	Successfully processed step with no trace data available (for log level Info or Debug, or if the trace retention period (default 60 min) is exceeded).
	Error in processed step with no trace data available.

The following screenshot shows the tracing data if the message passed the lower route of the integration flow and sending the message to the receiver resulted in an error.



To view the message content, select tab **Message Content** above the integration flow model.

→ Tip

There is also a log level **Debug** that provides more detailed information on message processing. When configured, it also shows the individual integration flow steps executed during a message processing run. The difference to **Trace** log level is that:

- Log level **Debug** expires after a certain time.

- Log level **Debug** does **not** provide the message content (payload, header, and properties).

i Note

It is recommended to set log levels **Trace** or **Debug** only in a development or test environment.

More Information

[Setting Log Levels](#)

[Cloud Integration – Setting the Log Level for Message Processing](#) 

[Troubleshooting Message Processing in the CPI Web Application](#) 

[Cloud Integration – Audit Log in Web UI Monitor](#) 

[Cloud Integration – Enabling Trace for Message Processing](#) 

[Enhancements to Message Processing Log Viewer in the Web Application](#) 

[How to analyze runtime issues](#) 

SAP note [2665258](#) 

Checking the Default Trace (in the Neo Environment)

To analyze runtime errors, you can check the default trace.

To do that, open the **Operations** view and click the **System Log Files** tile. Check out the latest files starting with `ljs_trace_` (which you can also download to your computer). You notice that there are multiple entries with a similar same time stamp (**Updated At**).

i Note

One log is written for each runtime node started on the tenant.

More information:

[Monitoring System Log Files, Neo Environment](#)

[Cloud Integration – System Log Download](#)  (SAP Community blog)

For a detailed instruction how to analyze the default trace in the context of inbound connection issues, check out the following SAP Community blog: [Cloud Integration – How to Setup Secure HTTP Inbound Connection with Client Certificates](#) .

Monitoring Scenarios that Include Message Queues

Learn more about monitoring scenarios that include message queues and possible error situations.

JMS Message Processing and Quality of Service

When monitoring message processing of integration flows that contain JMS message queues (in the [Monitor Message Processing](#) area), you notice that in most cases each message is processed exactly once. However, in rare cases a message processing log suggests that there's an inconsistency indicating that a message has been processed more than once from a message queue. If Cloud Integration sends this message to a receiver system (for example, using a SOAP channel) and the receiver system isn't prepared for this, it can lead to problems.

The JMS broker doesn't provide exactly once processing. For non-exclusive queues, there are cases when a message is processed more than once and even in parallel by different consumers. This can happen, for example, when during processing from the JMS message queue the connection to the JMS broker gets lost. The reason for this is that when the connection is interrupted, the lock on the message held by the broker is released (so the message becomes available for redelivery).

For exclusive queues, retry and inconsistencies are handled differently, and exactly once in order processing can be achieved if the connection remains healthy.

i Note

There are two different queue types that affect how messages are processed.

- Non-Exclusive Queues: Allow multiple consumers to process messages in parallel, which can lead to messages being processed more than once if connections are lost or interrupted.
- Exclusive Queues: Restrict message processing to a single consumer at a time, ensuring ordered processing.

Duplicate Processing and Receiver Impact

When a message is processed by one consumer (an integration flow deployed on a Cloud Integration worker node), the JMS broker writes a lock entry that makes sure that no other consumer starts processing the same message. Exclusive queues inherently handle this with ordered processing, but for non-exclusive queues, if the connection between the consumer (Cloud Integration) and the JMS broker is interrupted, the lock is released. In this case, another consumer can begin processing the same message.

For example, consider the following scenario:

- Consumer 1 starts processing a message from the message queue. During this step, the connection to the JMS broker gets lost.
- Because Consumer 1 never acknowledges the message after the connection is lost, the JMS broker can redeliver it to another consumer.
- Consumer 1 doesn't have any information that the connection was lost and still tries to process the message.
- During this period, another consumer (Consumer 2) starts to process the same message.

If the receiver adapter used to send the final message to another system does not support exactly once message processing, such behavior can lead to inconsistencies.

If the receiver adapter does support exactly once processing, the JMS broker may still deliver the message more than once, but the receiver ensures that each message is processed only once. This means that, from the receiver's perspective, the application data remains consistent even if duplicates are sent.

However, configuring exactly once processing on the JMS adapter alone does not guarantee end-to-end exactly-once delivery — broker redelivery and receiver-side semantics (for example, deduplication or idempotent processing) also matter.

Furthermore, the functionality to reprocess a message builds on the assumption that JMS processes a message exactly once. If, in rare cases, this assumption is incorrect, runtime retries may not behave as configured. In such a case, based on the JMS processing Cloud Integration *can assume* a worker-node failure (which can happen, for example, caused by an out-of-memory situation) and apply its platform retry logic. In that case, the retry interval is set to 18 minutes.

i Note

For exclusive queues, the retry is managed by the JMS broker and doesn't utilize separate error queues as in non-exclusive queues.

To mitigate such problems, use an adapter that supports exactly once processing where possible (for example, the XI adapter). For more information on the different options to design integration scenarios with exactly once processing, see: [Quality of Service Exactly Once](#)

Retry Process

If messages fail during processing from a queue, a retry of the message is triggered. The JMS sender adapter allows you to configure message retry, for example, to set a specific retry interval (see [Configure the JMS Sender Adapter](#)).

However, in specific cases, the message is processed differently than expected from the retry configuration.

Note the following aspects related to retry processing for non-exclusive queues:

- **Normal retry**

In general, JMS transactions are used to process messages. A retry from the JMS broker is performed when a rollback occurs.

The JMS transaction is finished after the message processing log (MPL) status is set. If an error occurs in the commit or rollback phase of the JMS session, the MPL status is adapted in most cases. However, some cases aren't covered. In these cases, the message status is set to **Completed**. That means that retries can occur even if the log shows success.

- **Broker retries**

The JMS broker immediately retries the message. This behavior can't be changed. Since this retry is too fast for almost all applications in non-exclusive setups, it isn't used directly in Cloud Integration.

If the retry that moves messages to the error queue fails, the JMS broker starts to retry the message with a very high frequency. This can occur if a transacted session isn't available due to a JMS resource shortage. Usually, this resource shortage occurs only for a very short time because sessions are continuously released during message processing.

- **Re-delivery / error queue**

The property `JMSRedelivered` indicates that a retry was performed by the JMS broker. If the `JMSRedelivered` property is set, the message is stored in a separate error queue (see [Managing Queues](#)). For the sending of messages into the error queue, a separate message processing log is written.

The setting of property `JMSRedelivered` by the JMS broker isn't accurate. The JMS specification contains a paragraph that says: *If a client receives a message with the JMSRedelivered indicator set, it is likely, but not guaranteed, that this message was delivered but not acknowledged in the past.*

If the JMS broker sends a message to the JMS client and receives an error, it isn't sure that the message did not reach the client. There's a specific degree of freedom for the broker about when to set this property and when not. Therefore, it's possible that a message is flagged as redelivered and moved to the error queue even though it hasn't reached the JMS client before.

There are also cases of early message delivery to the client application although it hasn't started to process this message. If this client is in a shutdown phase (for example, downscaling, integration flow un-deployment or re-deployment, software update) it can happen that it ignores this message entirely and not acknowledges its retrieval. The broker has to resend this message to the next consumer, but it flags it as a redelivery (which is the correct state from the broker's perspective).

- **Cloud Integration retry job**

An automatic job runs every minute and sequentially goes through all non-empty error queues to move their messages back to the processing queue according to the configured redelivery configuration in the adapter. In this case, the messages are added to the end of the queue. If the processing queue contains many messages, an added message isn't processed immediately.

The timeout is set to 5 seconds. This means that, after the last message in a non-empty queue, it takes 5 seconds until the next queue is scanned.

To summarize, the following situations can occur for non-exclusive queues:

- In rare cases, the MPL doesn't reflect all details of how the message has been processed during JMS retries.
- The retry times can deviate from the configured values, depending on the situation.
- There can be retries although no errors are visible in the MPL.
- If using many queues, additional configurations can be necessary to make sure that the retry intervals are realized as configured in the channel.

There can also be combinations of issues with the reprocessing of messages and issues related to the quality of service, which may be mitigated by using the relevant adapter.

Archiving Data, Cloud Foundry Environment

The Archiving feature allows you to store message processing logs in an archiving system such as a Content Management System (CMS).

i Note

These instructions are relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

! Restriction

Availability of this feature depends upon the SAP Integration Suite service plan that you use. For more information about different service plans and their supported feature set, see SAP Note [2903776](#) .

To use the archiving feature, you need to [Configure a Destination](#), [Enable Archiving](#) and [Configure Archiving Settings](#).

Monitoring data are stored per default 30 days, see [Data Storage Security](#). If you want to store data for a longer period, you can use the archiving feature. To archive data, you need to set up an archiving system, such as a Content Management System. The archiving system must support the CMIS API standard version 1.1.

! Restriction

Since the CMS system isn't part of the SAP delivery, you're responsible for monitoring the CMS-server. You must ensure that the CMS-Server is always accessible, and that you react on occurring issues. You can monitor the progress of the data archiving via the Monitoring API, see: [Key Performance Indicators for Archiving Runs](#)

You can archive data for **persist step content**, **adapter payloads**, and **attachments**. Sensitive data contained for example in an authorization header will be masked during archiving.

i Note

Payload recording is currently only supported for the following adapters: AS2, ELSTER, OData, HTTPS, MAIL, HTTP; RFC, SFTP, SuccessFactors, SOAP, XI, IDoc.

A message processing log (MPL) is archived in the CMS by storing new entries having message processing log (MPL) related properties, as well a .zip file. The properties include values from the MPL header. The .zip file contains the payload of the processed messages.

For every step of an integration flow, where data of the type: persist step content, adapter payloads, and attachments are written, there's a folder named after the step-id. This folder contains the files, and the files contain the data.

Example: For attachments, the structure is: a folder named after the step-id in which the attachment is written, containing 1 file per attachment. The name of the file is the name of the attachment.

i Note

Even though the **Persist** step is saving the data encrypted, the data isn't encrypted when stored in the CMS for archiving.

→ Tip

If you're using the archiving feature for the first time, we recommend trying out the feature with a few testing integration flows, in order to establish a successful setup before using it for integration flows in production. We further recommend checking the zip-files uploaded for the testing integration flows to verify that all the information required for your archiving use case is contained in the zip file. Since unnecessary information puts extra load on the runtime database, we recommend that you archive only the necessary information. For example, if you use **Receiver** or **Sender Message** recording, it's not necessary to also archive **Persisted Messages**.

i Note

To some extent, the performance of archiving can be tweaked with internal parameters maintained by our Operations Team only. If you find in your tests that the performance of archiving isn't sufficient for your use case, open a ticket with SAP Support. This ticket allows us to investigate whether we can reach the desired performance by tweaking the parameters.

Related Information

[Configuring Destination](#)

[Enable Archiving in the Cloud Foundry Environment](#)

[Configuring Archiving Settings](#)

Configuring Destination

To configure a connection to your CMS system for data archiving, you've to define a destination in the subscriber account of your tenant and follow the steps described in "Enable Archiving in the Cloud Foundry Environment".

Create a new **Destination** pointing to the CMS System. Fill in the properties stated in the table:

Property	Value or Description
----------	----------------------

Property	Value or Description
Name	CloudIntegration_LogArchive i Note You can't change the destination name.
Type	Type of protocol must be HTTP.
Description	Description of the destination
URL	Add the URL : For Proxy Type Internet - The CMS Repository For Proxy Type OnPremise - The CMS Repository and replace host and port by the virtual host and port configured in the SAP Cloud Connector. The required protocol in the URL is HTTP. This protocol isn't a security risk, as you can specify HTTPS in the SAP Cloud Connector configuration and thus ensure that the route from cloud connector on the on-premise system is encrypted. The connection between cloud connector and application VM in the cloud is always encrypted, as shown in: Configuration of On-Premise Systems i Note The default binding is the browser-binding. Unless you specify AtomPub as BindingType, you've to specify the URL for the browser binding. WebServices binding isn't supported.
Proxy Type	- Internet /OnPremise - Internet for the direct connection to CMS - OnPremise for Cloud Connector use i Note Depending on the connection you expose to your repository, you have to use Cloud Connector because direct connection to the repository isn't always possible.
Authentication	BasicAuthentication or OAuth2ClientCredentials i Note Until now, only client credentials grant is supported for Oauth authentication.
User	The user authorized for the CMS repository for BasicAuthentication. The user who connects with CMS repository requires several permissions. The permissions must include: cms : read

Property	Value or Description
	• <code>cms:write</code> • The query capability <code>metadataonly</code>
Password	The user's password for BasicAuthentication.
Client ID	Client ID for the OAuth2ClientCredentials authentication
Client Secret	Client Secret for the OAuth2ClientCredentials authentication without mTLS (mutual TLS).
Use mTLS for Token Retrieval	Select the checkbox, if you want to use mTLS (mutual TLS) for OAuth2ClientCredentials authentication. This is only applicable for proxy type Internet
Token Service Key Store Location	Specify the value, if you want to use OAuth2ClientCredentials authentication with mTLS. We require a keystore to be uploaded via the "Upload and Delete Certificates" link (see screenshot). This keystore is used for mTLS and contains the authentication data for the client. The keystore has to meet the following requirements: 1. The keystore must be in the JKS format. 2. The keystore must contain exactly one keypair, containing the private key and the certificate chain required for the client. 3. The keystore and key pair must be password protected with the same password. After uploading the keystore, enter a name by selecting it from the dropdown list.
Token Service Key Store Password	Password for the keystore and the key pair contained in the keystore when using OAuth2ClientCredentials with mTLS.
Token Service URL Type	You can use both values <code>Dedicated</code> and <code>Common</code> for OAuth2ClientCredentials authentication.
Token Service URL	The authentication URL grants the OAuth token for OAuth2ClientCredentials authentication. For the proxy-type <code>OnPremise</code>, you can only specify a URL with the http protocol. It seems to be a security risk, and is inconvenient if your URL uses the https protocol. But internally, we replace "http" with "https" in the URL used to call the token. We assume a URL with https for the proxy-type <code>OnPremise</code> and authentication type <code>OAuth2ClientCredentials</code> in a URL specified with http. For example: if you call the URL "http://xxx" in the destination, it's internally replaced by "https://xxx".
Token Service User	This value can be left empty. (We don't use it yet)
Token Service Password	This value can be left empty. (We don't use it yet)

Additional Properties	Value or Description
ArchiveCompletedMessagesOnly (optional)	If you define this parameter and set its value to TRUE, or true, only archiving relevant MPLs with status COMPLETED are archived. If this parameter isn't specified, or specified with any value other than TRUE or true, the MPL status isn't considered, except status DISCARDED. All archiving relevant MPLs are archived, except MPLs with status DISCARDED.  Caution If you've set the parameter with the value TRUE, or true, and you run an archiving job with this setting, the run changes the archiving relevance of MPLs. In this case, these particular MPLs with another status than COMPLETED aren't archived anymore by further job runs. They'll be deleted by one of the next runs of the <i>Monitoring Data Cleanup Job</i>.
BindingType (optional)	Define this parameter only if you want to use the AtomPub-Binding instead of the Browser-Binding to connect to the repository. In this case, specify AtomPub as parameter value and ensure that the URL property has the correct URL for the AtomPub-Binding. If you use SAP Cloud Connector, ensure that the URL for the AtomPub-Binding URL has the URL property value. If you use AtomPub-Binding, the AtomPub URL must be displayed as an accessible resource in the SAP Cloud Connector configuration.
ClientCompression (optional)	If you set this parameter value to true or TRUE, the requests sent to the CMIS-Repository are compressed using gzip. This procedure improves the application performance if the repository can handle compressed requests. Before activating this parameter, check if your repository can handle compressed requests. We recommend using this parameter only if the performance observed, isn't acceptable for your usage scenario.
ConnectTimeout (optional)	With this parameter, you set the HTTP connect timeout for your CMIS client. The value is a positive whole number and specifies the timeout in milliseconds. i Note The timeout default value is 30000 milliseconds. The minimum value is 5000 milliseconds and the maximum value 100000 milliseconds. You can only specify values within this range. For values lower than the minimum, the minimum is set, and for values higher than the maximum, the maximum is set. Use this parameter if you observe frequent failures during testing because the timeout value being too low, or if the requests take too long to time out.
CookiesForAuthentication (optional)	If you set this parameter value to true or TRUE, cookies are used to transmit authentication information. This action improves the application performance, but doesn't apply to all repositories. We recommend using this parameter only if the performance observed, isn't acceptable for your usage scenario.

Additional Properties	Value or Description
CustomAttributeAndTypeNamesMapping (optional)	Use this parameter if you want to have your own document type, name, and document attributes in your CMS repository, differing from the ones provided by SAP. The parameter value has to be a JSON string, as shown in the example: ⌵ Sample Code <pre>{ "mpl:message": "sapcpi_mpl_message", "mpl:messageGuid": "c_mpl_message_guid", "mpl:correlationId": "c_mpl_correlation_id", "mpl:messageId": "c_mpl_message_id", "mpl:messageType": "c_mpl_message_type", "mpl:logStart": "c_mpl_log_start", "mpl:logEnd": "c_mpl_log_end", "mpl:sender": "c_mpl_sender", "mpl:receiver": "c_mpl_receiver", "mpl:iFlowComponentType": "c_mpl_iflow_component_type", "mpl:iFlowName": "c_mpl_iflow_name", "mpl:iFlowSymbolicName": "c_mpl_iflow_symbolic_name", "mpl:status": "c_mpl_status", "mpl:logLevel": "c_mpl_log_level", "mpl:statusCustom": "c_mpl_status_custom", "mpl:messageSize": "c_mpl_message_size", "mpl:transactionCount": "c_mpl_transaction_count" }</pre> The keys in the JSON are the names as provided by SAP, the values are the names you want to use instead, in the type you create. The name of the type you create is the value of <i>mpl:message</i> in the JSON. In the sample code, the type name is <i>sapcpi_mpl_message</i>. i Note The archived values of <i>mpl:sender</i> and <i>mpl:receiver</i> have a maximum length of 40 character. If the sender or receiver value of a message processing log exceeds 40 characters, the system truncates the value to 40 characters in the archive.
DirectConnection (optional)	Defines how the session to the CMS repository is created. If you don't specify the parameter or if the parameter value is set to any other value than <code>false</code> or <code>FALSE</code>, the session is created by using the <code>RepositoryId</code> in the call for the session. If the parameter value is set to <code>false</code> or <code>FALSE</code>, the system creates a list of all existing repositories and then uses the repository of the specified ID to create the session. i Note Use this parameter only if you browse out during testing, that the default connection doesn't work.
PathPrefix (optional)	The default format for data archiving paths is year/month/day, using the log end date of the MPL to determine this date. The data archiving path is created under the root folder. The log end date is stated in UTC.

Additional Properties	Value or Description
	Use the <code>PathPrefix</code> parameter if you want to create the archiving path in a specific folder. Specify the path from the root folder down to the specific folder. Use <code>/</code> to start the path with and to separate its segments. If you don't create the folder under the <code>PathPrefix</code> in advance, the archiving job creates this folder, provided the user-defined in the destination has the required permissions to create the path specified in the <code>PathPrefix</code> parameter.
ReadTimeout (optional)	With this parameter, you set the HTTP read timeout for your CMIS client. The value is a positive whole number and specifies the timeout in milliseconds. i Note The timeout default value is set to 600000 milliseconds. The minimum value is 5000 milliseconds and the maximum value 1000000 milliseconds. You can only specify values within this range. For values lower than the minimum, the minimum is set, and for values higher than the maximum, the maximum is set. Use this parameter if you observe frequent failures during testing because the timeout value being too low, or if the requests take too long to time out.
RepositoryId	ID of the CMS Repository. Specify the repository Id and not the repository name.
ResponseCompression (optional)	If you set this parameter value to <code>true</code> or <code>TRUE</code>, your CMIS client requests that the repository compresses the responses sent. This procedure improves the application performance if the repository supports compression of responses. Before activating this parameter, check if your repository supports response compression. We recommend using this parameter only if the performance observed, isn't acceptable for your usage scenario.

❖ Example

These screenshots are for illustrative purposes only. You can use all additional properties independently from the authentication type.

- This example shows an archiving destination that uses OAuth2 client credentials with the required properties (without mTLS):

Destination Details

Source:
New destination from scratch

Main Properties

Name: *

CloudIntegration_LogArchive

Type: *

HTTP

Description:

Connection Data for Archiving with SAP Document Service

Proxy Type: *

OnPremise

URL: *

Location ID:

Authentication

Authentication: *

OAuth2ClientCredentials

Client ID: *

☐ Use mTLS for token retrieval

Client Secret:

☐ Set empty

Token Service URL: *

Token Service URL Type: *

Common

☐ Use Basic credentials for Token Service

Additional Properties

Key:

RepositoryId

Value:

Add Property

- This example shows an archiving destination that uses OAuth 2.0 client credentials and includes the required properties with mTLS:

Destination Details

Source:
New destination from scratch

Main Properties

Name: *

CloudIntegration_LogArchive

Type: *

HTTP

Description:

Connection Data for Archiving with SAP Document Service

Proxy Type: *

Internet

URL: *

Authentication

Authentication: *

OAuth2ClientCredentials

Client ID: *

☒ Use mTLS for token retrieval

Token Service Key Store Location: *

Token Service Key Store Password:

☐ Set empty

Token Service URL: *

Token Service URL Type: *

Dedicated

☐ Use Basic credentials for Token Service

Additional Properties

Key:

RepositoryId

Value:

Add Property

- Example of an archiving destination with BasicAuthentication and all additional properties:

Destination Details

Source:
New destination from scratch

Main Properties

Name: *

CloudIntegration_LogArchive

Type: *

HTTP

Description:

Connection Data for Archiving with SAP Document Service

Proxy Type: *

OnPremise

URL: *

Location ID:

Authentication

Authentication: *

BasicAuthentication

Token Service Key Store Location: *

Token Service Key Store Password:

☐ Set empty

User: *

Password:

☐ Set empty

This is custom documentation. For more information, please visit [SAP Help Portal](#).

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Additional Properties

Key:	ArchiveCompletedMessagesOnly	Value:		
Key:	BindingType	Value:		
Key:	ClientCompression	Value:		
Key:	ConnectTimeout	Value:		
Key:	CookiesForAuthentication	Value:		
Key:	CustomAttributeAndTypeNamesMapping	Value:		
Key:	DirectConnection	Value:		
Key:	PathPrefix	Value:		
Key:	ReadTimeout	Value:		
Key:	RepositoryId	Value:		
Key:	ResponseCompression	Value:		

Add Property

Related Information

[Enable Archiving in the Cloud Foundry Environment](#)

Using SAP Cloud Connector

You need to configure the SAP Cloud Connector to point to your content management system (CMS).

For more details about the SAP Cloud Connector, see: [Outbound: SAP Cloud Connector](#).

i Note

The location ID is currently not supported.

In the System Mapping, set **Host in Request Header** to Use Internal Host.

Example of system mapping pointing to a CMS system:

Virtual host and port cannot be edited

Back-end Type:Non-SAP System

Protocol:HTTPS

Virtual Host:chemistry

Virtual Port:8080

*Internal Host:

*Internal Port:443

SAProuter:

Principal Type:None

CNC Destsec Name:

Host In Request Header:Use Internal Host

Check Internal Host:

Save

Cancel

For security reasons, set **Virtual Host** and **Virtual Port** values different to **Internal Host** and **Internal Port** values. With setting identical values for internal host and port and virtual host and port, you save a direct connection to the repository in the destination. This direct connection can lead to attacks on the CMS, if the destination is known by unauthorized persons.

The screenshot shows the 'Edit Resource' form. The 'Access Policy' section is highlighted with an orange box. It contains two radio button options: 'Path Only (Sub-Paths Are Excluded)' and 'Path And All Sub-Paths'. The 'Path And All Sub-Paths' option is selected. Other fields include 'URL Path', 'Active' (checked), 'WebSocket' (unchecked), and 'Description'.

In the [Edit Resource](#) screen, set the [Access Policy](#) to [Path And all Sub-Paths](#).

Enable Archiving in the Cloud Foundry Environment

To enable data archiving on a tenant in the Cloud Foundry environment, use the OData API.

Before You Start

You can configure the content metadata that your document management system uses to archive message processing logs (MPL).

You can modify several properties before enabling archiving. Once archiving is enabled, you can't change these properties. Expand the code section below to view the full metadata.

! Restriction

Depending on your document management system, only some properties may be supported. For example, OpenText DMS doesn't support `orderable: true`.

Code

```
{
  "id": "mpl:message",
  "localName": "message",
  "localNamespace": "mpl",
  "displayName": "MPL Message",
  "queryName": "mpl:message",
  "description": "MPL Message",
  "baseId": "cmis:document",
  "parentId": "cmis:document",
  "creatable": true,
```

```

"fileable": true,
"queryable": true,
"fulltextIndexed": false,
"includedInSupertypeQuery": true,
"controllablePolicy": false,
"controllableACL": true,
"versionable": false,
"contentStreamAllowed": "allowed",
"propertyDefinitions": {
  "mpl:messageGuid": {
    "maxLength": 32,
    "id": "mpl:messageGuid",
    "localName": "messageGuid",
    "localNamespace": "mpl",
    "displayName": "mpl:messageGuid",
    "queryName": "mpl:messageGuid",
    "description": "Guid of message",
    "propertyType": "string",
    "cardinality": "single",
    "updatability": "oncreate",
    "inherited": false,
    "required": true,
    "queryable": true,
    "orderable": true,
    "openChoice": false
  },
  "mpl:correlationId": {
    "maxLength": 120,
    "id": "mpl:correlationId",
    "localName": "correlationId",
    "localNamespace": "mpl",
    "displayName": "mpl:correlationId",
    "queryName": "mpl:correlationId",
    "description": "correlationId",
    "propertyType": "string",
    "cardinality": "single",
    "updatability": "oncreate",
    "inherited": false,
    "required": true,
    "queryable": true,
    "orderable": true,
    "openChoice": false
  },
  "mpl:messageId": {
    "maxLength": 120,
    "id": "mpl:messageId",
    "localName": "messageId",
    "localNamespace": "mpl",
    "displayName": "mpl:messageId",
    "queryName": "mpl:messageId",
    "description": "messageId",
    "propertyType": "string",
    "cardinality": "single",
    "updatability": "oncreate",
    "inherited": false,
    "required": true,
    "queryable": true,
    "orderable": true,
    "openChoice": false
  },
  "mpl:messageType": {
    "maxLength": 100,
    "id": "mpl:messageType",
    "localName": "messageType",
    "localNamespace": "mpl",
    "displayName": "mpl:messageType",
    "queryName": "mpl:messageType",
    "description": "messageType",
    "propertyType": "string",
    "cardinality": "single",
    "updatability": "oncreate",
    "inherited": false,
    "required": true,
    "queryable": true,
    "orderable": true,
  }
}

```

```

        "openChoice": false
    },
    "mpl:logStart": {
        "id": "mpl:logStart",
        "localName": "logStart",
        "localNamespace": "mpl",
        "displayName": "mpl:logStart",
        "queryName": "mpl:logStart",
        "description": "logStart",
        "propertyType": "datetime",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:logEnd": {
        "id": "mpl:logEnd",
        "localName": "logEnd",
        "localNamespace": "mpl",
        "displayName": "mpl:logEnd",
        "queryName": "mpl:logEnd",
        "description": "logEnd",
        "propertyType": "datetime",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:sender": {
        "maxLength": 40,
        "id": "mpl:sender",
        "localName": "sender",
        "localNamespace": "mpl",
        "displayName": "mpl:sender",
        "queryName": "mpl:sender",
        "description": "sender",
        "propertyType": "string",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:receiver": {
        "maxLength": 40,
        "id": "mpl:receiver",
        "localName": "receiver",
        "localNamespace": "mpl",
        "displayName": "mpl:receiver",
        "queryName": "mpl:receiver",
        "description": "receiver",
        "propertyType": "string",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:status": {
        "maxLength": 40,
        "id": "mpl:status",
        "localName": "status",
        "localNamespace": "mpl",
        "displayName": "mpl:status",
        "queryName": "mpl:status",

```

```

        "description": "status",
        "propertyType": "string",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:logLevel": {
        "maxLength": 10,
        "id": "mpl:logLevel",
        "localName": "logLevel",
        "localNamespace": "mpl",
        "displayName": "mpl:logLevel",
        "queryName": "mpl:logLevel",
        "description": "logLevel",
        "propertyType": "string",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:statusCustom": {
        "maxLength": 40,
        "id": "mpl:statusCustom",
        "localName": "statusCustom",
        "localNamespace": "mpl",
        "displayName": "mpl:statusCustom",
        "queryName": "mpl:statusCustom",
        "description": "statusCustom",
        "propertyType": "string",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:messageSize": {
        "id": "mpl:messageSize",
        "localName": "messageSize",
        "localNamespace": "mpl",
        "displayName": "mpl:messageSize",
        "queryName": "mpl:messageSize",
        "description": "messageSize",
        "propertyType": "integer",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    },
    "mpl:transactionCount": {
        "id": "mpl:transactionCount",
        "localName": "transactionCount",
        "localNamespace": "mpl",
        "displayName": "mpl:transactionCount",
        "queryName": "mpl:transactionCount",
        "description": "transactionCount",
        "propertyType": "integer",
        "cardinality": "single",
        "updatability": "oncreate",
        "inherited": false,
        "required": true,
        "queryable": true,
        "orderable": true,
        "openChoice": false
    }

```

```

    },
    "mpl:iFlowComponentType": {
      "maxLength": 40,
      "id": "mpl:iFlowComponentType",
      "localName": "iFlowComponentType",
      "localNamespace": "mpl",
      "displayName": "mpl:iFlowComponentType",
      "queryName": "mpl:iFlowComponentType",
      "description": "iFlowComponentType",
      "propertyType": "string",
      "cardinality": "single",
      "updatability": "oncreate",
      "inherited": false,
      "required": true,
      "queryable": true,
      "orderable": true,
      "openChoice": false
    },
    "mpl:iFlowName": {
      "maxLength": 150,
      "id": "mpl:iFlowName",
      "localName": "iFlowName",
      "localNamespace": "mpl",
      "displayName": "mpl:iFlowName",
      "queryName": "mpl:iFlowName",
      "description": "iFlowName",
      "propertyType": "string",
      "cardinality": "single",
      "updatability": "oncreate",
      "inherited": false,
      "required": true,
      "queryable": true,
      "orderable": true,
      "openChoice": false
    },
    "mpl:iFlowSymbolicName": {
      "maxLength": 150,
      "id": "mpl:iFlowSymbolicName",
      "localName": "iFlowSymbolicName",
      "localNamespace": "mpl",
      "displayName": "mpl:iFlowSymbolicName",
      "queryName": "mpl:iFlowSymbolicName",
      "description": "iFlowSymbolicName",
      "propertyType": "string",
      "cardinality": "single",
      "updatability": "oncreate",
      "inherited": false,
      "required": true,
      "queryable": true,
      "orderable": true,
      "openChoice": false
    },
    "mpl:tenantId": {
      "maxLength": 150,
      "id": "mpl:tenantId",
      "localName": "tenantId",
      "localNamespace": "mpl",
      "displayName": "mpl:tenantId",
      "queryName": "mpl:tenantId",
      "description": "tenantId",
      "propertyType": "string",
      "cardinality": "single",
      "updatability": "oncreate",
      "inherited": false,
      "required": false,
      "queryable": true,
      "orderable": true,
      "openChoice": false
    }
  }
}

```

To configure the type definition, follow these steps:

i Note

If you don't need to change the metadata, you can skip these steps. Enabling archiving automatically deploys the type definition.

1. Make the necessary changes to the metadata.
2. Log in to your document management system's repository using WorkBench. For more information, see [Utilizing CMIS Workbench together with BTP DMS](#) .
3. Navigate to **Type** in the top menu bar.
4. Select **CMIS Document** (cmis:document).
5. Choose **Create Type** in the top menu bar.
6. Upload the metadata as a JSON file.

i Note

SAP doesn't track configuration changes made on the customer side and isn't responsible for any changes to the properties listed above.

Procedure

To enable data archiving on your tenant, use the function *activateArchivingConfiguration* of the **Message Processing Logs** OData API (at: <https://api.sap.com/api/MessageProcessingLogs> , select **Data Archiving** resource).

A successful activation call performs the following steps:

- Checks if a destination with the correct name exists.
- Activates the archiving job.
- Enables configuration of the archiving settings on the user interface.

On creation of the archiving job, the system defines a random start time for it.

→ Remember

To avoid job execution errors, ensure to have the destination and (if you use it) the Cloud Connector configured correctly.

i Note

To call the archiving function, you need the role *DataArchiving.Activate*.

i Note

For authorization to the following APIs, follow the instructions provided under [Setting Up Inbound HTTP Connections \(for API Clients\)](#).

Make sure that your user is assigned to the *DataArchiving.Read* or *DataArchiving.Activate* role, depending on the API call you want to make.

To enable archiving, send a *POST* call to the URL : *https://path-to-odata-api/api/v1/activateArchivingConfiguration*, where *path-to-odata-api* is specific to your environment.

If enabling the archiving function is successful, you get a response code **200** and a message of success. After a successful activation call, you can configure archiving for integration flows in the **Monitor** section. To do so, you've to reload the **Monitor** section to be able to see the configuration options for your integration flows.

If the enablement isn't successful, the system throws an error code as well as an error message.

To check whether archiving is currently configured on a tenant, use the OData API allowing to query the message processing logs. Send a *GET* call to the URL: *https://path-to-odata-api/api/v1/ArchivingConfigurations('tenant-name')*, where *path-to-odata-api* is specific to your environment.

i Note

To call this function, you've to have the role *DataArchiving.Read*.

i Note

The parameter `tenant - name` **doesn't** specify the name of the tenant for which you check the status of the archiving function.

The information for which tenant you want to check the archiving status, is taken from the security token used for the request.

Related Information

[HTTP Calls and URI Components](#)

Configuring Archiving Settings

You can configure the archiving settings via the **Integration Content Monitor** for each integration flow.

Archive Data
Configure

Sender Channel Messages

Log Attachments

Prerequisites

The archiving feature is enabled.

i Note

To be able to read and set the archiving configuration, you must either be an **Integration Developer**, a **Tenant Administrator**, or a **Business Expert** or have the role `TraceConfigurationEdit` assigned.

Use **Configure** to modify the current settings. Once done, save or discard your changes.

Archiving is available for the following options:

- **Sender Channel Messages**: Messages received from Sender and responses returned to Sender.
- **Receiver Channel Messages** : Messages sent to Receiver and responses returned by Receiver.

- **Persisted Messages:** Messages stored via `Persist` flow step.
- **Log Attachments:** Message Processing Log attachments.

Carefully consider the integration flows you want to activate the archiving for, and the configuration needed for these integration flows. With each additional setting, the system gathers more data in the `.zip` file to be uploaded, and increases the volume of data stored in the CMS. Also consider that **Sender Channel Messages** and **Receiver Channel Messages** are stored in the database in addition to the data already stored for the MPL.

Use payload recording to record incoming and outgoing payloads. Since the same message can exist multiple times for sender and receiver channels, it's sufficient to record them once at one of the channels for archiving purposes. JMS and ProcessDirect Adapter payloads are only used within SAP environments, enabling payload recording for these adapters isn't recommended.

Additional Recommendations

- If you've integration flows calling each other, then carefully decide on recording payloads of sender and receiver adapter to avoid storing too much data.
- If you archive adapter payloads, review your integration flow design, to check if you can remove any attachments writings or use `Persist Step` components.
- To be able to record the incoming and outgoing payloads in the integration flow as early as possible, enable the **Sender** and **Receiver Channel** recording.

Conditions for Archiving Message Processing Logs

To be archived, all the MPLs have to fulfill the following conditions:

1. The MPLs are defined as "relevant for archiving". This condition is true, if at least one of the archiving configuration settings were active for the integration flow producing the MPL, at the execution time of the integration flow.
2. The MPLs are at least `x` days old. `x` is the number of days after which an MPL is to be archived. The age of an MPL is determined by its log end date. The default value for `x` is 7.
3. The MPLs aren't in status `DISCARDED`. MPLs in status `DISCARDED` aren't relevant for archiving.

i Note

Message Processing Logs with all other status values are archived if they fulfill conditions 1 and 2.

i Note

The archived values of `mpl:sender` and `mpl:receiver` have a maximum length of 40 characters. If the sender or receiver value of a message processing log exceeds 40 characters, the system truncates the value to 40 characters in the archive.

⚠ Caution

If you've selected the optional parameter `ArchiveCompletedMessagesOnly`, and set its value to `TRUE`, or `true`, only archiving relevant MPLs with status `COMPLETED` is archived. See: [Configuring Destination](#)

Key Performance Indicators for Archiving Runs

In addition to archiving MPLs, the system job also records specific Key Performance Indicators (KPIs) of the archiving runs.

You can use these indicators, for example, to check for recurring errors, or potential performance problems. Use the official OData API to query the KPIs.

List of Job KPIs

Key Performance Indicators	Description
RunStart	Date and time when the archiving run started.
RunDurationInMinutes	Duration of an archiving run (in minutes).
DataCollectionDurationInMinutes	Total time needed to collect the data from relevant data stores (in minutes).
DataCompressionDurationInMinutes	Total time needed to compress the uploaded data (in minutes).
DataUploadDurationInMinutes	Total time needed to upload the data to the CMS (in minutes).
MplsToBeArchived	Number of archiving relevant message processing logs (MPLs).
MplsArchived	Number of archived MPLs. MPLs already uploaded by a previous run, but not yet flagged as archived, are counted as well.
MplsArchivingFailed	Number of errors that occurred during archiving MPLs. This number includes only errors occurring during preparation of upload and upload of MPLs. Errors leading to the failure of the job run excluded. i Note The number of <code>MplsToBeArchived</code> and the number of <code>MplsArchivingFailed</code> + <code>MplsArchived</code> can differ, if the archiving job ends with status <code>FAILED</code>.
MplsArchivedUntilDate	The date until which MPLs are considered as being archiving relevant for this job run. All MPLs matching the date or older are archived.
DateOfOldestMplToBeArchivedAfterRun	Date of the oldest MPL which is still archiving relevant after the run. (If all MPLs have been archived, this value is <code>null</code>).
DataUploadedInMb	Total volume of uploaded data in megabytes
RunStatus	Status of the job run. Possible values are: <code>COMPLETED</code> and <code>FAILED</code>. i Note Status <code>COMPLETED</code> doesn't indicate, that no errors occurred during the archiving process. It only states, that the job run without any error causing the job run to end prematurely.
RunFailedPhase	Job phase in which the run failed. The job run is divided into 2 phases: <code>INITIALIZATION</code> and <code>EXECUTION</code>. i Note If the <code>RunStatus</code> is <code>COMPLETED</code>, this indicator is null. If the job fails during <code>INITIALIZATION</code>, some necessary resources during this phase, such as reading values from the destination, run into an error.

Key Performance Indicators	Description
	In this case, • Check if the configuration of your destination is correct. • Check if you can successfully connect to the repository, • If you use SAP Cloud Connector, check that the configuration of the SAP Cloud Connector is correct. Assure that the SAP Cloud Connector exposes the resources required. • Check that the credentials configured in the destination enable the connection to the repository. Assure that the user has all the necessary authorisations for the required operations on the repository. If the job fails during the EXECUTION phase, an error occurred either during retrieval, compression, or uploading of the data. In this case, • Check your repository logs for any recurring errors. • Check if the error occurs only once or if it occurs in multiple runs in succession.

How to query the Key Performance Indicators.

You can query the KPIs with the same OData API used to query the Message Processing Logs:

ArchivingKeyPerformanceIndicators with RunStart as primary key.

❖ Example

The following query is an example where the query filter for all entries with MplsToBeArchived = 5000.

Request Method: GET

Request

URL: `https://path-to-odata-api/api/v1/ArchivingKeyPerformanceIndicators?$filter=MplsToBeArchived=5000`

☰ Sample Code

```

<entry>
  ...
  <content type="application/xml">
    <m:properties>
      <d:RunStart>2021-04-30T13:37:00.243</d:RunStart>
      <d:RunDurationInMinutes>11</d:RunDurationInMinutes>
      <d>DataCollectionDurationInMinutes>7</d>DataCollectionDurationInMinutes>
      <d>DataCompressionDurationInMinutes>0</d>DataCompressionDurationInMinutes>
      <d>DataUploadDurationInMinutes>2</d>DataUploadDurationInMinutes>
      <d:MplsToBeArchived>5000</d:MplsToBeArchived>
      <d:MplsArchived>4999</d:MplsArchived>
      <d:MplsArchivingFailed>1</d:MplsArchivingFailed>
      <d:MplsArchivedUntilDate>2021-04-30T13:47:47.996</d:MplsArchivedUntilDate>
    </m:properties>
    <d:DateOfOlddestMplToBeArchivedAfterRun>
      2021-04-30T13:34:54.183
    </d:DateOfOlddestMplToBeArchivedAfterRun>
  </content>
</entry>

```

```
</d:DateOfOldestMplToBeArchivedAfterRun>
    <d:DataUploadedInMb>130</d:DataUploadedInMb>
    <d:RunStatus>COMPLETED</d:RunStatus>
    <d:RunFailedPhase m:null="true"/>
  </m:properties>
</content>
</entry>
```

Central Monitoring and Alerting Solutions

You can use the monitoring and alerting solutions SAP Cloud ALM and SAP Focused Run together with SAP Cloud Integration.

Central Monitoring and Alerting Using SAP Cloud ALM

You can use the cloud-native solution SAP Cloud ALM to monitor business scenarios and, in particular, SAP Cloud Integration tenants.

The **Monitor** application of SAP Cloud Integration allows you monitor messages processed on your SAP Cloud Integration tenant.

In addition to this *local* monitoring feature (local with regard to your tenant), you can use SAP Cloud ALM to operate your IT landscape centrally with no need to navigate to each local monitoring tool. It offers end-to-end business and IT monitoring for hybrid landscapes that brings the technical flow closer to the business context.

SAP Cloud ALM is also set as central tool for triggering alerts. Based on the raw data retrieved from the cloud applications and systems of your hybrid landscape, erroneous states can be calculated and several outbound actions triggered for proactive alerting. You can notify personas over email, chat-based applications, creating tickets to work on erroneous situations outside of SAP Cloud ALM.

i Note

SAP Cloud ALM is SAP's cloud-based tool for application lifecycle management (ALM) built natively on SAP Business Technology Platform.

The following use cases of SAP Cloud ALM for Operations are relevant for the monitoring of integration scenarios end-to-end and for the alerting in case of erroneous situations:

Use Case	Description
Integration and exception monitoring	Monitor your cloud and hybrid (cloud and on-premise) business scenarios end-to-end while focusing on exceptional and error situations. The system correlates single messages exchanged in a business scenario with each other and provides an end-to-end visibility across all involved components of a business scenario. You can see the incoming and outgoing messages for each component involved in the scenario. Business users can also identify business-driven problems and search for messages based on a specific business context attributes. Integration experts can search for messages based on technical attributes. For proactive monitoring, you can oversee all open alerts in an embedded alert inbox and collaborate with others in the issue resolution process. Additionally, you can also monitor integration-related exceptions. Examples are Cloud Integration exceptions related to artifacts that

Use Case	Description
	are deployed unsuccessfully on one of the monitored SAP Cloud Integration tenants. For further error analysis, you can drill down to detailed information on each single message. To tackle corrective measures, you can use a contextual link and navigate to the local monitoring application of SAP Cloud Integration.
Health monitoring	Understand the technical healthiness of your cloud applications and on-premise systems. This feature helps you to oversee the technical aspects of your SAP Cloud Integration tenants. You can track the validity of certificates across all tenants and monitor the JMS resources of all Cloud Integration tenants.

More information:

- [SAP Cloud ALM for Operations](#) 
- [SAP Cloud ALM for Operations - Expert Portal](#) 
- [SAP Cloud ALM on SAP Help Portal](#)

SAP Focused Run

SAP Focused Run is a solution for service providers who want to host all their customers in a central, scalable, safe, and automated environment.

SAP recommends service providers and customers with a large on premise system landscape to consider using SAP Cloud Integration with SAP Focused Run.

More information:

- [SAP Focused Run on SAP Help Portal](#)
- [SAP Focused Run](#) 
- [SAP Focused Run - Expert Portal](#) 

Inspect

Inspect the usage of integration resources caused by your active integration flows.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Each SAP Cloud Integration tenant comes with certain physical resources that are consumed by integration flows at runtime. Examples of such integration resources are the available database connections.

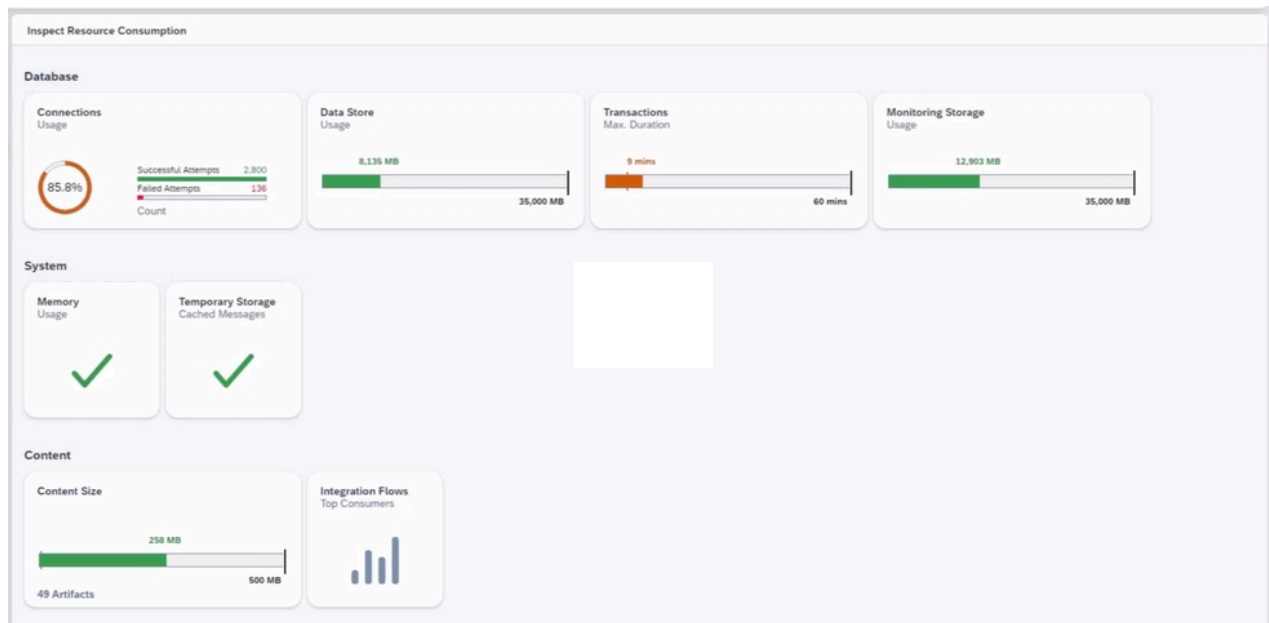
Inspect allows you to inspect the consumption of integration resources, identify those integration flows that contribute significantly to integration-resource exhaustion, and perform steps to resolve critical situations. Based on the insights, you can, for example, optimize integration flow design to overcome integration resource bottlenecks.

i Note

To use this feature, role collection `MonitoringDataRead` needs to be assigned to your user (see [Tasks and Permissions](#)).

The **Inspect** overview page displays various tiles that represent the consumption levels of different resources in your tenant. Each tile shows the overall consumption of a resource over the past 24 hours.

Watch the following video to learn how to use **Inspect** and for an overview of a typical workflow. The video uses database connection usage as an example, but the workflow also applies to any other resources.



[Open this video in a new window](#)

Inspect Database Resources

Inspect the usage of integration resources associated with the tenant database and with the database connection pool.

- Using the **Connections** tile, you can inspect resource usage of the database connections caused by integration flows.

See: [Inspect Database Connection Usage](#)

- Using the **Data Store** tile, you can inspect resource usage of the tenant database caused by integration flows using data store operations steps.

See: [Inspect Data Store Usage](#)

- Using the **Transactions** tile, you can inspect resource usage of the database transactions caused by integration flows.

See: [Inspect Database Transaction Usage](#)

- Using the **Monitoring Storage** tile, you can inspect resource usage of the monitoring database storage caused by integration flows.

See: [Inspect Monitoring Storage Usage](#)

Inspect System Resources

Inspect system resource usage caused by your active integration flows.

- Using the **Memory** tile, you can inspect resource usage of system memory caused by integration flows.

See: [Inspect System Memory Usage](#)

- Using the **Temporary Storage** tile, you can inspect the storage usage of temporary files.

See: [Inspect Temporary Storage](#)

Inspect Content Resources

Inspect content resource usage caused by your integration artifacts.

- Using the **Content Size** tile, you can inspect the file size that integration artifacts occupy.

See: [Inspect Content Size](#)

- Using the **Integration Flows** tile, you can inspect the integration flows with the highest resource consumption.

See: [Inspect Top Consuming Integration Flows](#)

→ Tip

When resource consumption becomes critical, a **Recommendations** button becomes available. You can choose this button to receive tailored suggestions and assistance. These recommendations help resolve potential issues when usage levels reach warning or critical stages.

Inspect Data Store Usage

Inspect the data store usage and message size of the tenant database for a given time period.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Each SAP Cloud Integration tenant is associated with a physical database that has a limited size. The data store is a logical storage that consumes tenant database volume. It can be accessed to by certain integration flows steps (read and write access) to read and write the message payload or variables during runtime. There are two kinds of data stores:

- Local data store. It can be accessed explicitly by a single integration flow.
- Global data store. It can be accessed by all integration flows deployed on a tenant.

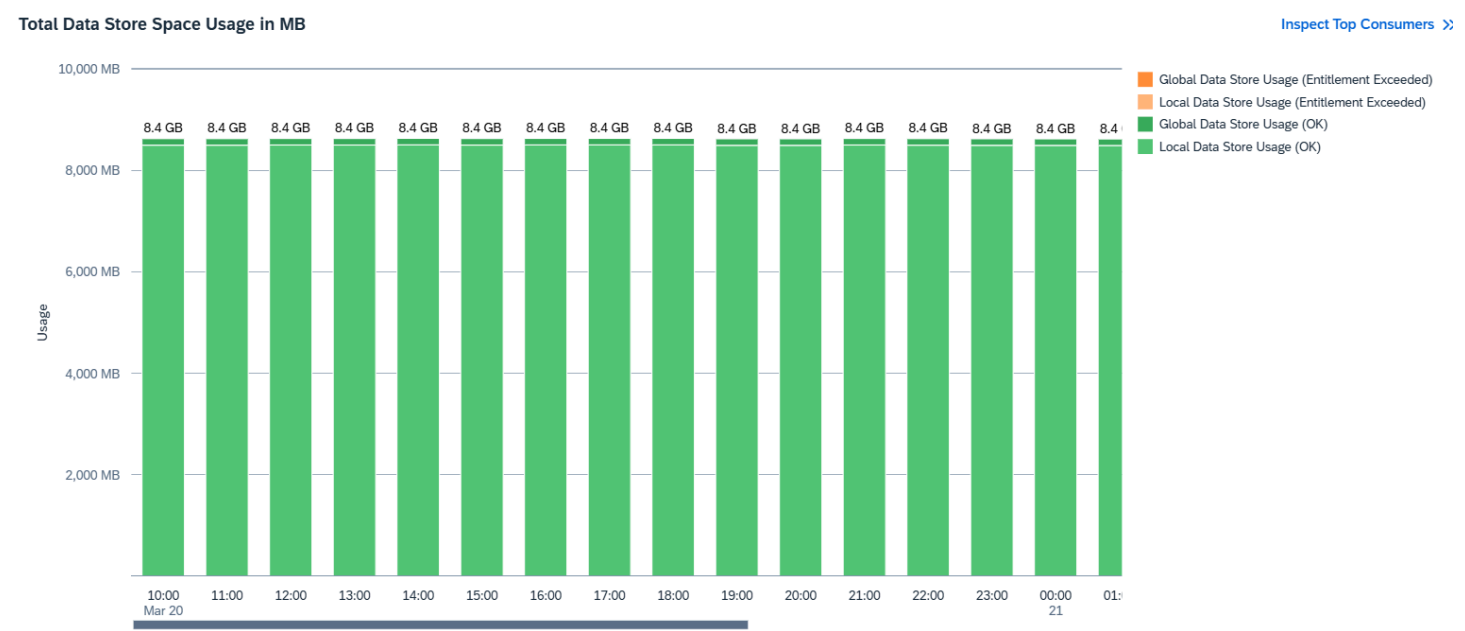
This database is used by certain steps during the runtime of integration flows. These steps allow you to write data to the data store and to read the data from it (see [Define Data Store Operations](#)). These integration flows, consequently, use tenant database volume.

The default value of the entitlement for the total data store space usage is 35 GB, as documented at [What Is SAP Cloud Integration?](#). If the entitlement is exceeded there's a risk of overloading the database storage, which might eventually cause problems in the database management system and impact message processing.

Screen Components

You can choose to display either the **Total Data Store Space Usage** or the **Data Store Message Size** in a graphical format using the **Resource** dropdown menu.

The bar chart indicates the tenant database's data store usage over a defined period, as shown in the image below:



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval.

i Note

The data store usage is plotted in a bar graph against time. The horizontal axis shows the time window, and the vertical axis indicates the total data store space used in MB.

The usage level is represented by the following elements:

Graphical Element	Data Store Usage	Meaning
Dark Green Bar	Global (OK)	Global data store usage is within the given limit.
Light Green Bar	Local (OK)	Local data store usage is within the given limit.
Dark Orange Bar	Global (Entitlement Exceeded)	Global data store usage exceeds the given limit.
Light Orange Bar	Local (Entitlement Exceeded)	Local data store usage exceeds the given limit.

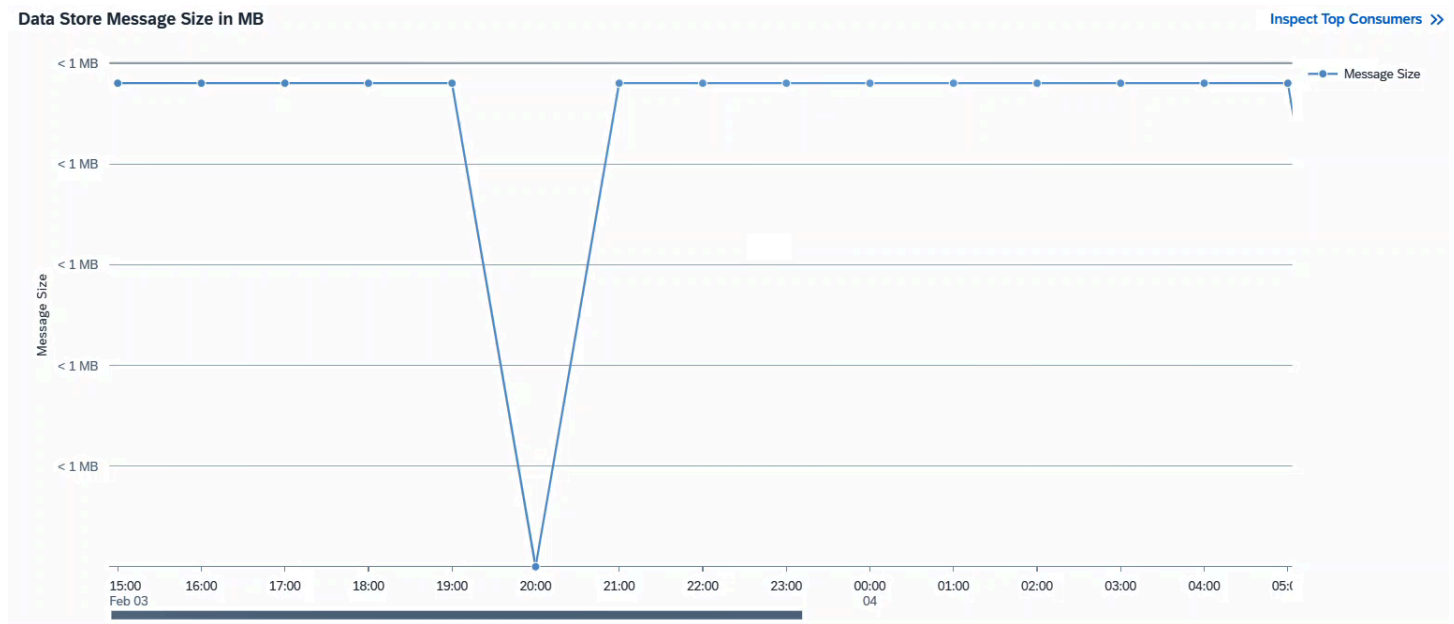
The bar height represents the average data volume (in MB) used by data stores, whether global or local. The threshold is based on the storage consumption entitlement for the tenant database.

i Note

The system reads the resource consumption every hour. This means that there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Data Store Message Size

The line chart for **Data Store Message Size** displays the size of messages stored in the data stores over a specific time period, as shown in the image below:



The line height indicates the size that messages reached at different points in time. Select one of the bullet points to view the exact message size at the chosen time.

You can use the **Message Size** filter to choose between displaying the **Average** or **Maximum** message size. The average is calculated over hourly or daily intervals, depending on the time period you select in the **Time** filter. While the average usage of data stores might seem uncritical, usage could reach warning or critical levels for short periods. To inspect this situation, select **Maximum** in the **Message Size** filter.

Functions

Choose a bar or a bullet to get more context information and access the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the selected time period. For more information, see [Monitor Message Processing](#).

- **Inspect Top Consumers**

Navigate to the **Top Data Stores** screen that allows you to inspect tenant database usage by data stores in more detail.

The target screen shows:

- Local data stores that are identified as top consumers of the tenant database for a specific time period (as selected by the **Time** parameter).
- Global data stores that are identified as top consumers of the tenant database for a specific time period (as selected by the **Time** parameter).

See:

[Inspect Top Data Stores by Usage](#)

[Inspect Top Data Stores by Message Size](#)

- **Show Data Stores**

Navigate to the **Manage Data Stores** screen for the selected time period . For more information, see [Managing Data Stores](#).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

Troubleshooting

If there's a critical situation with a data store, check if the data store is used at all in any integration scenario. Also check if you can reduce the setting for **Expiration Time**.

Check if the data stores shown as main consumers contain many overdue entries. In this case, check why these entries aren't being consumed in a timely manner, and consider deleting the overdue entries if they're no longer required, in order to free up storage space.

In order to reduce storage consumption by large data stores, consider reducing the **Expiration Period** (see [Define Data Store Write Operations](#)).

It isn't recommended to use data stores for monitoring purposes, for example, by storing all failed messages for inspection. Data stores are intended as temporary storage for transactional data.

If you're using the **Data Store Write** integration flow steps for development and testing purposes, this means, if you want to inspect how messages are being processed, use the integration flow tracing feature (see [Guidelines and Best Practices for Message Monitoring](#)).

More information:

- [Define Data Store Operations](#)
- [Store/Retrieve Messages in/from the Data Store](#)
- [Anticipate Message Throughput When Choosing a Storage Option](#)
- [Using Data Storage Features When Designing Integration Flows](#)
- [Define Transaction Handling](#)

Inspect Top Data Stores by Usage

Inspect local and global data stores by database usage (in MB) for a selected time period.

You can choose to display information about global or local data stores by using the **Data Store Visibility** menu. Local data stores are accessible exclusively by single integration flows, while global data stores are accessible by all integration flows deployed on a tenant.

The heat map displays only the data stores identified as top database volume consumers. These are sorted from highest to lowest based on database volume usage. The color shade of each cell reflects the level of database volume usage.

→ Tip

You can use the **Top Data Stores** filter to adjust the number of data stores shown on the chart. The default is six, and the maximum is 15. The total number of data stores displayed is the data stores for which there's available data.

i Note

When you navigate from the usage overview page, the time period you selected is automatically highlighted in the heat map.

Select a cell to get more context information and to get access to the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar (see [Monitor Message Processing](#)).

- **Show Data Stores**

Navigate to the **Data Stores** tile, **Manage Stores** screen, to show all data stores of the tenant (see [Managing Data Stores](#)).

- **Inspect Data Store**

Inspect resource consumption for the selected data store.

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

- **Copy Name to Clipboard**

You can also access the functions below in the heat maps that show **local** data stores:

- **Show Database Connections**

Inspect database connection usage (it can correlate with database consumption by data stores). For more information, see [Inspect Top Integration Flows by Database Connection Usage](#).

- **Show Integration Content**

Inspect the selected artifact (see [Manage Integration Content](#)).

Inspect Top Data Stores by Message Size

Inspect top local and global data stores by their message sizes for a selected time period.

You can choose to display data about global or local data stores by using the **Data Store Visibility** menu. A local data store is accessible only by a single integration flow, while a global data store is accessible by all integration flows deployed on a tenant.

You can also configure the heat map to show either the average message size of the data stores or the maximum size reached at a particular time using the **Message Size** menu.

The heat map displays only data stores identified as having the highest message sizes, sorted top-down by database volume usage. The color shade of a cell reflects the file size level.

i Note

You can use the **Top Data Stores** filter to adjust the number of data stores shown on the chart. The default is six, and the maximum is 15. The total number of data stores displayed is the data stores for which there's available data.

i Note

When you navigate from the usage overview page, the time period you selected is automatically highlighted in the heat map.

Select a cell to get more context information and to get access to the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar (see [Monitor Message Processing](#)).

- [Show Data Stores](#)

Navigate to the [Data Stores](#) tile, [Manage Stores](#) screen, to show all data stores of the tenant (see [Managing Data Stores](#)).

- [Inspect Data Stores](#)

Inspect the usage of the selected data store in more detail.

- [Zoom Out](#) and [Zoom In](#) to extend/reduce the selected time period.
- [Copy Name To Clipboard](#)

You can also access the functions below in the heat maps that show **local** data stores:

- [Show Database Connections](#)

Inspect database connection usage (it can correlate with database consumption by data stores). For more information, see [Inspect Top Integration Flows by Database Connection Usage](#).

- [Show Integration Content](#)

Inspect the selected artifact (see [Manage Integration Content](#)).

Inspect Database Connection Usage

Inspect database connection usage for a given time period and analyze critical situations in which insufficient database connections are available.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Each SAP Cloud Integration tenant is associated with a physical database that has a limited size. The data store is a logical storage that consumes tenant database volume. It can be accessed to by certain integration flows steps (read and write access) to read and write the message payload or variables during runtime. There are two kinds of data stores:

Every integration flow step that implies access to the tenant database requires a database connection. This number of available database connections is a limited integration resource.

Database connection usage depends on the integration flow design and on the volume of the processed data. There can also be situations where not enough connections are available. Furthermore, the configured transaction handling settings have an effect on database connection usage.

Screen Components

The bar chart shows the database connection usage of the tenant database for a defined time period, as shown in the image below.



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval.

i Note

You can use the **Database Connection Usage** filter to choose between displaying the **Average** or **Maximum** usage of database connections. The average is calculated over hourly or daily intervals, depending on the time period you select in the **Time** filter. While the average usage of database connections might seem uncritical, usage could reach warning or critical levels for short periods. To inspect this situation, select **Maximum** in the **Database Connection Usage** filter.

The database connection usage is plotted in a bar graph against time. The horizontal axis shows the time window, and the vertical axis shows the percentage of database connection usage in relation to the maximum usage. The maximum usage corresponds to the situation where all database connections are used.

The usage level is represented by the following elements:

Graphical Element	Database Connection Usage	Meaning
Red Bar	More than 90%	Critical: There's a risk of overloading the database storage
Orange Bar	Between 70% and 90%	Warning: Database connection usage exceeds the given limit.
Green Bar	Less than 70%	OK: Database connection usage is within the given limit.
Red Bullet	Failed attempts	The Cloud Integration runtime can't get any database connections because no free database connection is available.
Green Bullet	Successful attempts	The number of successful attempts to connect to the database. This reflects how actively the database is being used.

i Note

The system reads the resource consumption every hour. This means there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Functions

Choose a bar or a bullet to get more context information and access the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen and inspect the message processing log for the selected time interval. For more information, see [Monitor Message Processing](#).

- **Inspect Top Consumers**

Inspect database connection usage per integration flow.

Navigate to the **Database Connections** screen that allows you to inspect database connection resources in more detail.

The target screen shows:

- The database connection usage for those integration flows that are identified as top consumers of the database connection pool for a given time period (as selected with the **Time** parameter).
- Integration flows that cause critical database connection load (unavailable connections) for a given time period (as selected with the **Time** parameter).

For more information, see:

[Inspect Top Integration Flows by Database Connection Usage](#)

[Inspect Top Integration Flows with Failed Connection Attempts](#)

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

Troubleshooting

If there are not enough database connections, check your top consuming integration flows. In particular, check the **Transaction Handling** parameter (when the Integration Process or Local Integration Process shape is selected in an integration flow model). Check if **Required for JDBC** is selected and, if it is, check if you can choose another option.

More information:

- [Define Proper Transaction Handling](#)
- [Transaction Handling Guidelines](#)
- [Define Transaction Handling](#)

Inspect Top Integration Flows by Database Connection Usage

Inspect database connection usage per integration flow for a given time period (as selected with the **Time** parameter).

The chart shows those integration flows identified as the top consumers of database connections. They're sorted top-down based on either average or maximum usage, as selected with the **Database Connection Usage** filter.

→ Tip

You can use the **Top Integration Flows** filter to adjust the number of integration flows shown on the chart. The default is six, and the maximum is 15. The total number of integration flows displayed are the flows for which there's available data.

The color shade of a cell reflects the level of database connection usage. The darkest shade reflects the highest usage observed during the selected time period.

Choose a cell to display the percent value of connection usage and to access one of the following functions for the selected integration flow (results are filtered according to the setting of the **Time** parameter):

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar (see [Monitor Message Processing](#)).

- **Show Integration Content**

Navigate to the **Manage Integration Content** screen for the related integration flow (see [Manage Integration Content](#)).

- **Inspect Integration Flow**

Inspect the database connection usage of the selected integration flow (see: [Inspect Resource Consumption for Individual Integration Flow](#)).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

- **Copy Name to Clipboard**

Inspect Top Integration Flows with Failed Connection Attempts

Inspect the integration flows that requested a database connection but for which no free database connection was available in the selected time period.

i Note

This section only shows content if the time filter is set so that in the associated time period critical load situations are associated with at least one integration flow.

Integration flows are sorted from top to bottom according to the number of unsuccessful attempts to request a database connection (divided by the absolute number of attempts).

→ Tip

You can use the **Top Integration Flows** filter to adjust the number of integration flows shown on the chart. The default is six, and the maximum is 15. The total number of integration flows displayed are the flows for which there's available data.

The number of unsuccessful attempts to request a database connection during the selected time period is indicated by the cell color (from green for a low number, up to red for a critical number of unsuccessful attempts).

Choose a cell to display the absolute number of unsuccessful attempts to get a database connection for an integration flow (**Unavailable Count** number). You also get access to the following functions for the selected integration flow (results are filtered according to the setting of the **Time** parameter):

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar (see [Monitor Message Processing](#)).

- **Show Integration Content**

Navigate to the **Manage Integration Content** screen for the related integration flow (see [Manage Integration Content](#)).

- **Inspect Integration Flow**

Inspect the database connection usage of the selected integration flow (see: [Inspect Resource Consumption for Individual Integration Flow](#)).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.
- **Copy Name to Clipboard**

Inspect Database Transaction Usage

Inspect database transaction duration for a given time period and analyze critical situations in which long-running transactions occur.

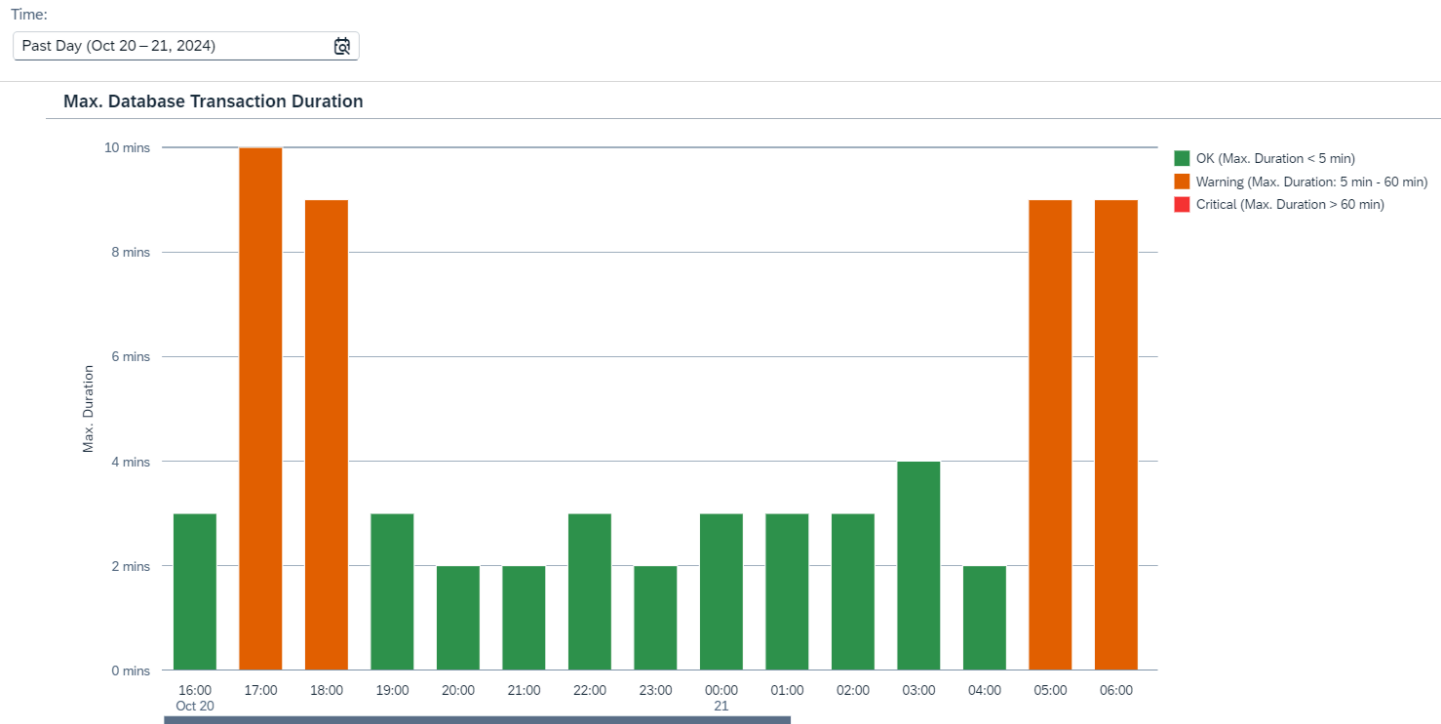
i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

There are no hard limits for the number and duration of database transactions. However, ideally, design your integration scenarios so that transactions are short lived. Long-living transactions can impose a significant load on database resources. A high load on database resources can cause problems in the database management system and impact message processing.

Screen Components

The bar chart shows the maximum transaction duration of the tenant database for a defined time period, as shown in the image below.



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval.

i Note

The maximum duration of database transactions is plotted in a bar graph against time. The horizontal axis shows the time window, and the vertical axis shows the duration of database transactions in minutes.

The duration level is represented by the following elements:

Graphical Element	Database Transaction Usage	Meaning
Red Bar	Maximum duration: more than 60 minutes	Critical: long running transactions can cause issues with the system database.
Orange Bar	Maximum duration: between 5 minutes and 60 minutes	Warning: transactions are running for longer than expected.
Green Bar	Maximum duration: less than 5 minutes	OK

i Note

The system reads the resource consumption every hour. This means there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Functions

Choose a bar to get more context information and access the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen and inspect the message processing log for the selected time interval. For more information, see [Monitor Message Processing](#).

- **Inspect Top Consumers**

Navigate to the **Top Integration Flows** screen and inspect those integration flows that are identified as using the longest running transactions of the tenant database for a specific time period (as selected by the **Time** parameter).

For more information, see [Inspect Top Integration Flows by Maximum Transaction Duration](#).

- **Show Message With Max. Duration**

Navigate to the **Monitor Message Processing** screen and inspect the message processing log with the maximum duration for the selected time interval. For more information, see [Monitor Message Processing](#).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

Troubleshooting

If there are integration flows exhibiting transaction durations at warning or even critical level, check your top consuming integration flows. In particular, check the **Transaction Handling** parameter (when the **Integration Process** or **Local Integration Process** shape is selected in the integration flow model). Check if **Required for JDBC** is selected and, if it's, check if you can choose another option.

More information:

- [Define Proper Transaction Handling](#)
- [Transaction Handling Guidelines](#)

- [Define Transaction Handling](#)

Inspect Top Integration Flows by Maximum Transaction Duration

Inspect maximum duration of database transactions per integration flow for a given time period (as selected with the **Time** parameter).

Only those integration flows that are identified as top consumers of database connections are shown (sorted top-down by average usage).

→ Tip

You can use the **Top Integration Flows** filter to adjust the number of integration flows shown on the chart. The default is six, and the maximum is 15. The total number of integration flows displayed are the flows for which there's available data.

The color shade of a cell reflects the maximum duration of database transactions. The darkest shade reflects the longest duration observed during the selected time period.

Click a cell to display the percent value of connection usage and to access one of the following functions for the selected integration flow (results are filtered according to the setting of the **Time** parameter):

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar. For more information, see [Monitor Message Processing](#).

- **Show Integration Content**

Navigate to the **Manage Integration Content** screen for the related integration flow. For more information, see [Manage Integration Content](#).

- **Show Message With Max. Duration**

Navigate to the **Monitor Message Processing** screen and inspect the message processing log with the maximum duration for the selected integration flow and time interval. For more information, see [Monitor Message Processing](#).

- **Inspect Integration Flow**

Inspect the database transaction duration of the selected integration flow. For more information, see [Inspect Resource Consumption for Individual Integration Flow](#).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

- **Copy Name to Clipboard**

Inspect Monitoring Storage Usage

Inspect usage of the monitoring storage database for a given time period (as selected with the **Time** parameter).

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

At runtime, monitoring data is written to a database. For each integration flow processed at runtime, a message processing log is written that contains information about the processing steps (see [Message Processing Log](#)).

The default value of the assigned database capacity is 35 GB, as documented at [What Is SAP Cloud Integration?](#).

When the assigned database capacity for monitoring storage is exceeded, you're advised to use the **Inspect** feature to examine and minimize storage usage. Exceeding the assigned database limit for monitoring storage doesn't imply that message processing logs can't be persisted anymore. However, in this case SAP can't ensure optimal performance for message monitoring.

Screen Components

The bar chart shows the consumption of the database storage for monitoring data for a defined time period, as shown in the image below:



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval.

i Note

However, the value of a bar at the time period 11:00-12:00 doesn't represent the amount of monitoring data that was added during 11:00-12:00. But rather the total amount of monitoring data found in the database during 11:00-12:00.

The database storage usage is plotted in a bar graph against time. The horizontal axis shows the time window, and the vertical axis indicates the total data store space used in MB.

The thresholds are based on the assigned database capacity for monitoring storage consumption in the monitoring database.

The usage level is represented by the following elements:

Graphical Element	Monitoring Storage Usage	Level of Usage
Orange Bar	Entitlement exceeded	Warning: Monitoring storage usage exceeds the given limit. The bar height shows the data volume (in MB) used by monitoring data. The thresholds are based on the entitlement for storage consumption in the tenant database.
Green Bar	Less than 100%	OK: Monitoring storage usage is within the given limit. The bar height shows the data volume (in MB) used by monitoring data.
Blue Bullet	Message Processing Log Count	The count of message processing logs. You can filter the count of message processing logs by status with the Message Processing Status filter. The default value is All, which means that the total count of message processing logs is displayed.

i Note

The system reads the resource consumption every hour. This means there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Functions

Choose a bar or a bullet to get more context information and access the following functions:

- [Show Messages](#)

Navigate to the [Monitor Message Processing](#) screen for the selected time period. For more information, see [Monitor Message Processing](#).

- [Inspect Top Consumers](#)

Navigate to the [Top Integration Flows](#) screen that allows you to inspect those integration flows that write the most message processing logs for a specific time period. For more information, see: [Inspect Top Integration Flows by Message Processing Log Count](#).

- [Zoom Out](#) and [Zoom In](#) to extend/reduce the selected time period.

Troubleshooting

If the assigned capacity for the monitoring database volume is approached or even exceeded, you can inspect which integration flows are the top contributors (see [Inspect Top Integration Flows by Message Processing Log Count](#)). For the top consuming integration flows you can ensure that the message processing **Log Level** parameter is set to **Info**

To do that, in the **Monitor** section, click a tile in the **Manage Integration Content** section select the integration flow, and configure the **Log Configuration** parameter accordingly.

See: [Setting Log Levels](#)

You can also check if there are integration flows showing many erroneous message processing logs. To filter for such integration flows, set the **Message Processing Status** filter to **Failed**. Check the respective integration flow and fix the root cause for the failures.

You might see a critical status, even though there were not too many message processing logs in the past 24 hours. In this situation you have to extend the time selection to **Past Week** or even **Past Month**, to reveal the point of time when a large amount of message processing logs further in the past could have filled the monitoring storage to a critical level. Note also that monitoring data is removed only through an automated cleanup job, which removes monitoring data for which the retention period has expired (30 days by default).

More information:

[Guidelines and Best Practices for Message Monitoring](#)

Inspect Top Integration Flows by Message Processing Log Count

Inspect the count of message processing logs by integration flows for a given time period (as selected with the **Time** parameter). Only integration flows that have the most message processing logs are displayed.

Integration flows are sorted top-down according to the count of message processing logs.

→ Tip

You can use the **Top Integration Flows** filter to adjust the number of integration flows shown on the chart. The default is six, and the maximum is 15. The total number of integration flows displayed are the flows for which there's available data.

The color shade of a cell reflects the count of message processing logs.

Using the **Message Processing Status** combo box, you can filter the count of message processing logs by status. The default value is **All**. That means, the total count of message processing logs is displayed.

Choose a cell to get more context information and to get access to the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar (see [Monitor Message Processing](#)).

- **Show Integration Content**

Navigate to the **Manage Integration Content** screen for the selected integration flow (see [Manage Integration Content](#)).

- **Inspect integration Flow**

Inspect the message processing log count for the selected integration flow (see [Inspect Resource Consumption for Individual Integration Flow](#))

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

- **Copy Name to Clipboard**

Inspect System Memory Usage

Inspect the usage of the system heap memory for a given time period (as selected with the **Time** parameter).

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

Messages processed by integration flows at runtime consume system heap memory. Consumption of heap memory depends on several factors:

- Message throughput

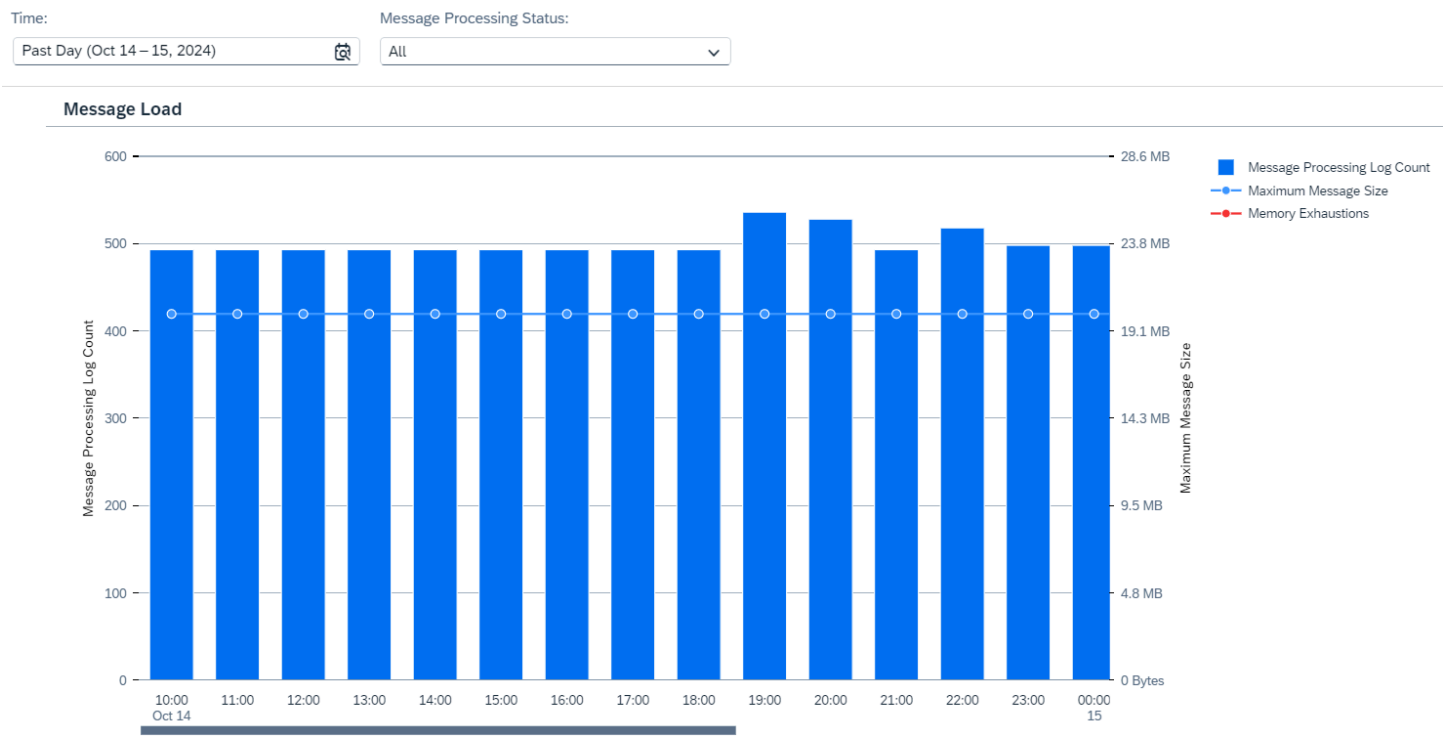
The higher the number of messages that are processed within a given time frame, the higher the consumption of memory. In addition, you can expect that consumption increases with message sizes.
- Integration flow design

An integration flow can contain steps with a particularly high demand for memory, such as (but not limited to) Groovy scripts and message mappings.

Screen Components

Message Load

The blue bar chart shows the message load for a defined time period:



The message load is plotted in two charts against time:

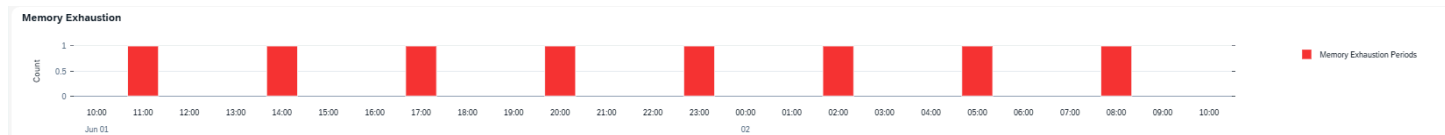
- The total count of message processing logs (MPLs) for a given time period (as selected with the **Time** parameter) is plotted in a bar chart.

You can filter the count of message processing logs by status with the **Message Processing Status** filter. The default value is **All**, which means that the total count of message processing logs is displayed.

The number of MPLs is identical to the number of messages processed during the selected time period.
- The maximum message size within this time period is plotted in a line chart with a blue bullet.

Memory Exhaustion

The red bar chart shows the time periods when heap memory was exhausted:



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval. You can also use the **Previous** and **Next** buttons on the time filter to switch the time interval.

i Note

When you select the option **Custom** for the **Time** filter, you can adjust the date and time intervals with the calendar and watch elements. You can select dates up to 30 days in the past. However, selection of dates in the future is disabled.

i Note

The system reads the resource consumption every hour. This means there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Functions

Choose a bar or bullet to get more context information including the precise timestamp, message process log count, and the count of memory exhaustion events (shown in error state when the count is greater than 0).

You can access to the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the selected time period For more information, see [Monitor Message Processing](#).

- **Inspect Top Consumers**

Navigate to the **Top Integration Flows** screen that allows you to inspect memory allocation by integration flows in more detail. For more information, see: [Inspect Integration Flows By System Memory Usage](#).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

Troubleshooting

If memory exhaustion have been detected, you can identify the corresponding integration flow and flow step on the **Top Flows by Memory Allocation** screen.

Consider the following recommendations for specific flow steps to mitigate memory exhaustion situations.

- General integration flow design measures

You can design integration flows in such a way that memory consumption is minimized.

More information: [Optimize Memory Footprint](#)

- **Script** step

When using XMLSlurper or JSONSlurper in a script, check out the following SAP Community blog: [Stream the XMLSlurper input in Groovy Scripts](#) 🙌.

In general, avoid getting the message body as String wherever possible. If you wish to implement a mapping, consider using a [Message Mapping](#) for XML or JSON payloads.

- JDBC adapter

When using the JDBC receiver adapter, make sure that not too many records are loaded per request. Otherwise, the memory can be overloaded.

- [Splitter](#) step

When using Splitter steps, an incoming message can spawn many messages by placing a high load on memory. If you use an [Iterating Splitter](#) with parallel processing enabled, consider reducing the number of parallel processes. In other cases, consider reducing the bulk size and/or the rate of incoming messages.

More information: [Reduce the Memory Consumption for Splitter Scenarios](#)

- Mapping

Mappings can require a high degree of memory allocation. Therefore, consider reducing the rate of incoming messages in case of memory shortages.

Inspect Integration Flows By System Memory Usage

Inspect relative memory allocation per integration flow for a given time period (as selected with the [Time](#) parameter).

The chart shows memory allocation and allows you to do a relative comparison of integration flows with regard to memory consumption.

i Note

Heap memory is continuously allocated for processing messages, and it's automatically reclaimed by garbage collection when it's no longer used. If the allocated memory is high and can't be reclaimed by the garbage collector for a longer time, you can expect memory issues.

However, it's usual for an integration flow to have high throughput and allocate a lot of memory, as long as the memory is reclaimed fast enough. This is the normal mode of operation for high-throughput scenarios.

The top chart displays only those integration flows that are identified as the top consumers of memory in terms of allocation. The bottom chart shows the integration flows with the largest message sizes.

→ Tip

- Use the [Top Integration Flows](#) filter to adjust the number of integration flows shown on the chart. The default is six, and the maximum is 15. The total number of integration flows displayed are the flows for which there's available data.
- Use the [Aggregation](#) filter to display the top integration flows with the maximum detected memory consumption.

The color shade of a cell reflects the level of memory allocation. The darkest shade reflects the highest usage observed during the selected time period.

i Note

When you navigate from the usage overview page, the time period you selected is automatically highlighted in the heat map.

Choose a cell to display the degree of memory allocation and to access one of the following functions for the selected integration flow (results are filtered according to the setting of the **Time** parameter):

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the time period covered by the bar. For more information, see [Monitor Message Processing](#).

- **Show Integration Content**

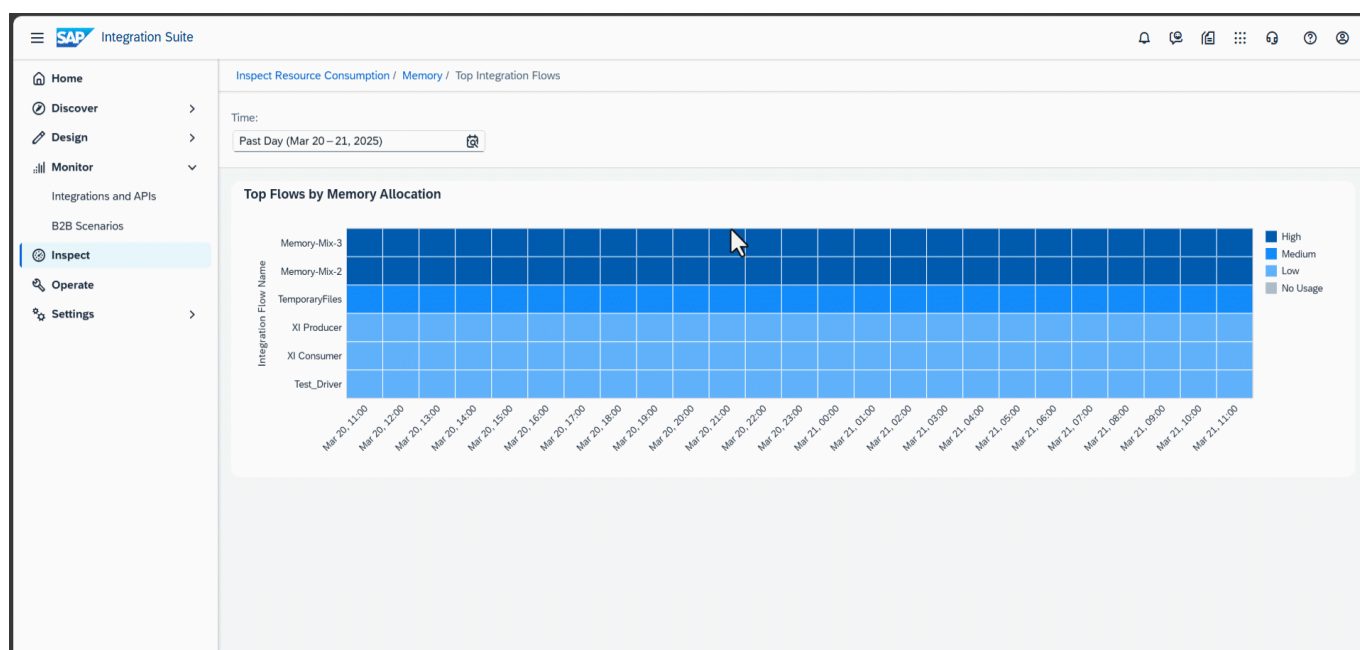
Navigate to the **Manage Integration Content** screen for the related integration flow. For more information, see [Manage Integration Content](#).

- **Inspect Integration Flow**

Inspect the system memory usage of the selected integration flow. For more information, see [Inspect Resource Consumption for Individual Integration Flow](#).

- **Show Flow Step**

Choose this function to navigate to the **Integration Designer**. The specific flow step identified as the main memory consumer is highlighted in the integration flow.



- **Copy Flow Step to Clipboard**

Copy the identifier of the integration flow step that was detected as the main consumer of memory to the clipboard. After navigating to the integration flow designer via the function **Show Integration Content**, you can use this identifier for finding the integration flow step within the integration flow.

→ Tip

On the **Manage Integration Content** screen, you can select **View Deployed Artifact** to navigate to the integration flow editor for the deployed artifact. Alternatively, you can select **Navigate to Artifact Editor** to navigate to the integration flow editor in the **Design** section.

- **Copy Flow ID to Clipboard**

Copy the integration flow identifier to the clipboard. This is useful on screens where you can filter by integration flow using the artifact ID.

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

Inspect Content Size

You can inspect the number of integration artifacts deployed on your SAP Integration Suite tenant (account) and the file size they allocate for a given time period.

Integration flows and other content artifacts get deployed to the file system rather than the database. The **Content Size** feature calculates the total file size of all deployed content artifacts, including integration flows, value mappings, integration adapters, and so on.

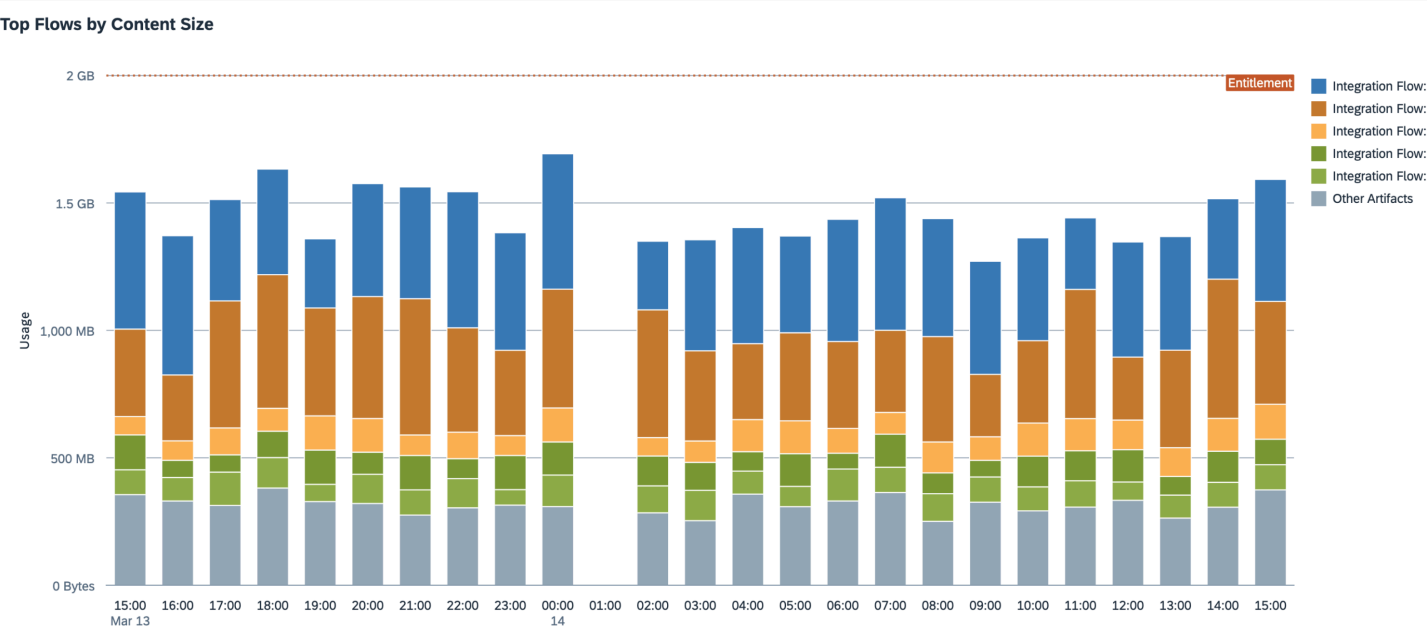
i Note

Content Size displays the compressed size of the content artifacts (the size of the deployed .jar file). In order to see the uncompressed references of your integration flows, navigate to **Manage Integration Content** as described in the **Functions** section of this page. For more information, see [Manage Integration Content](#).

The default value of the entitlement is 2GB, as documented at [What Is SAP Cloud Integration?](#)

Screen Components

The bar chart shows the artifacts that put the highest load on the file system for a given time period, each represented by a different color as shown below:



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval. You can also use the **Previous** and **Next** buttons on the time filter to switch the time interval.

i Note

When you select the option **Custom** for the **Time** filter, you can adjust the date and time intervals with the calendar and watch elements. You can select dates up to 30 days in the past. However, selection of dates in the future is disabled.

The total content size of the deployed artifacts is plotted in a bar graph against time. The horizontal axis shows the time window, and the vertical axis shows the file size used by the artifacts in MB.

The legend lists the top consuming integration artifacts and associates a color to each artifact.

i Note

The system reads the resource consumption every hour. This means that there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Functions

Choose a bar to get more context information and access the following functions:

- **Inspect Content**

Navigate to the more detailed **Inspect Artifact** screen to inspect the selected artifact further. For more information, see [Inspect Resource Consumption for Individual Integration Flow](#).

- **Show Integration Content**

Navigate to **Manage Integration Content** to monitor the specific artifact you've selected. For more information, see [Manage Integration Content](#).

i Note

You can use this function to inspect an artifact and its uncompressed references. In the **Manage Integration Content** screen, go to **Artifact Details > View Deployed Artifact > Integration Flow > References**. You can find all the uncompressed archives, schemas, and scripts of your selected artifact there.

- **Zoom In/Zoom Out**. Extend/reduce the selected time period.

Troubleshooting

If the total file size of the artifacts exceeds the system's storage, limit the size of the deployed integration contents. For more information, see [Limit Size of Deployed Integration Content](#).

Inspect Temporary Storage

You can inspect the total storage usage that temporary files allocate in the file system for a given time period.

At runtime, the system can cache messages in temporary files. The **Temporary Storage** feature calculates the storage volume these temporary files use and shows the maximum file size detected over the selected time period.

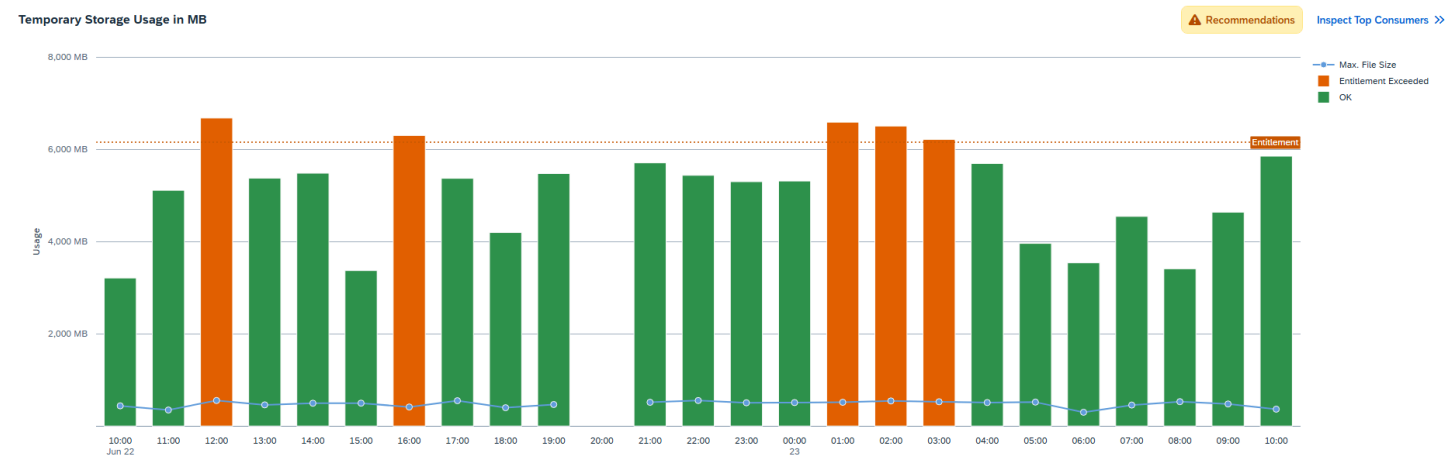
Temporary files become especially relevant when integration flows process large messages.

i Note

The system deletes messages that are temporarily stored from storage when message processing completes.

Screen Components

The bar chart shows the total storage used by temporary files for a defined time period, as shown in the image below:



You can change the displayed time period by selecting a different option in the dropdown box under **Time**. You can select **Past Day**, **Past Week**, **Past Month**, or **Custom** for a custom time interval. You can also use the **Previous** and **Next** buttons on the time filter to switch the time interval.

i Note

When you select the option **Custom** for the **Time** filter, you can adjust the date and time intervals with the calendar and watch elements. You can select dates up to 30 days in the past. However, selection of dates in the future is disabled.

The total storage usage is plotted in a bar graph against time. The horizontal axis shows the time window, and the vertical axis shows the storage usage level in MB.

The usage level is represented by the following elements:

Graphical Element	Temporary Storage Usage	Meaning
Green Bar	OK	Temporary storage usage is within the recommended limit of 6 GB.
Orange Bar	Entitlement Exceeded	Temporary storage usage exceeds the recommended limit of 6 GB.
Blue Bullet	Max. File Size	The maximum size that temporary files reach at the defined time period.

i Note

The system reads the resource consumption every hour. This means that there can be a maximum lag of 1 hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the Inspect feature.

Functions

Choose a bar or a bullet to get more context information and access the following functions:

- **Show Messages**
Navigate to the **Monitor Message Processing** screen for the selected time period. For more information, see [Monitor Message Processing](#).
- **Inspect Top Consumers**

Navigate to the [Top Integration Flows](#) screen that allows you to inspect those integration flows that occupy the most storage in more detail. For more information, see [Inspect Top Integration Flows by Storage Usage](#).

- [Zoom Out](#) and [Zoom In](#) to extend/reduce the selected time period.

Troubleshooting

If temporary files show storage usage at a warning or critical level, consider the following:

- Check if the maximum file size is very large, such as hundreds of MB or even GB. If it is, consider processing smaller messages. For more information, see Disk Space in [System Scope in the Cloud Foundry Environment](#).
- Check for delays in message processing when temporary storage reached a critical level. Message processing delays can cause temporary files to accumulate in storage.

If these recommendations don't resolve the issue, consider adjusting the integration flow design and message payload so that the system processes smaller messages with less latency.

Inspect Top Integration Flows by Storage Usage

Inspect temporary storage usage and maximum file size per integration flow for a given time period.

Inspect Top Integration Flows by Storage Usage

Inspect the volume of temporary storage used by different integration flows during a specific time period (as selected with the [Time](#) parameter). Only those integration flows that are identified as top consumers are shown, sorted top-down by average storage usage.

→ Tip

You can use the [Top Integration Flows](#) filter to adjust the number of integration flows shown on the chart. The default is six, and the maximum is 15. The total number of integration flows displayed are the flows for which there's available data.

The color shade of the cells reflects the level of storage usage. The darkest shade reflects the highest usage observed during the selected time period.

Inspect Top Integration Flows by Maximum File Size

Inspect the maximum file size per integration flow for a given time period (as selected with the [Time](#) parameter). Only those integration flows that are identified as having the maximum file size are shown, sorted top-down by average usage.

The darkest shade of the cells reflects the highest usage observed during the selected time period.

Functions

Choose a cell to display the level of storage usage and file size, and to access one of the following functions for the selected integration flow:

- [Show Messages](#)

Navigate to the [Monitor Message Processing](#) screen for the selected time period (see [Monitor Message Processing](#)).

- [Show Integration Content](#)

Navigate to the [Manage Integration Content](#) screen for the related integration flow (see [Manage Integration Content](#)).

- [Inspect Integration Flow](#)

Inspect the selected integration flow in more detail (see [Inspect Resource Consumption for Individual Integration Flow](#)).

- [Zoom Out](#) and [Zoom In](#) to extend/reduce the selected time period.
- [Copy Name to Clipboard](#)

Inspect Top Consuming Integration Flows

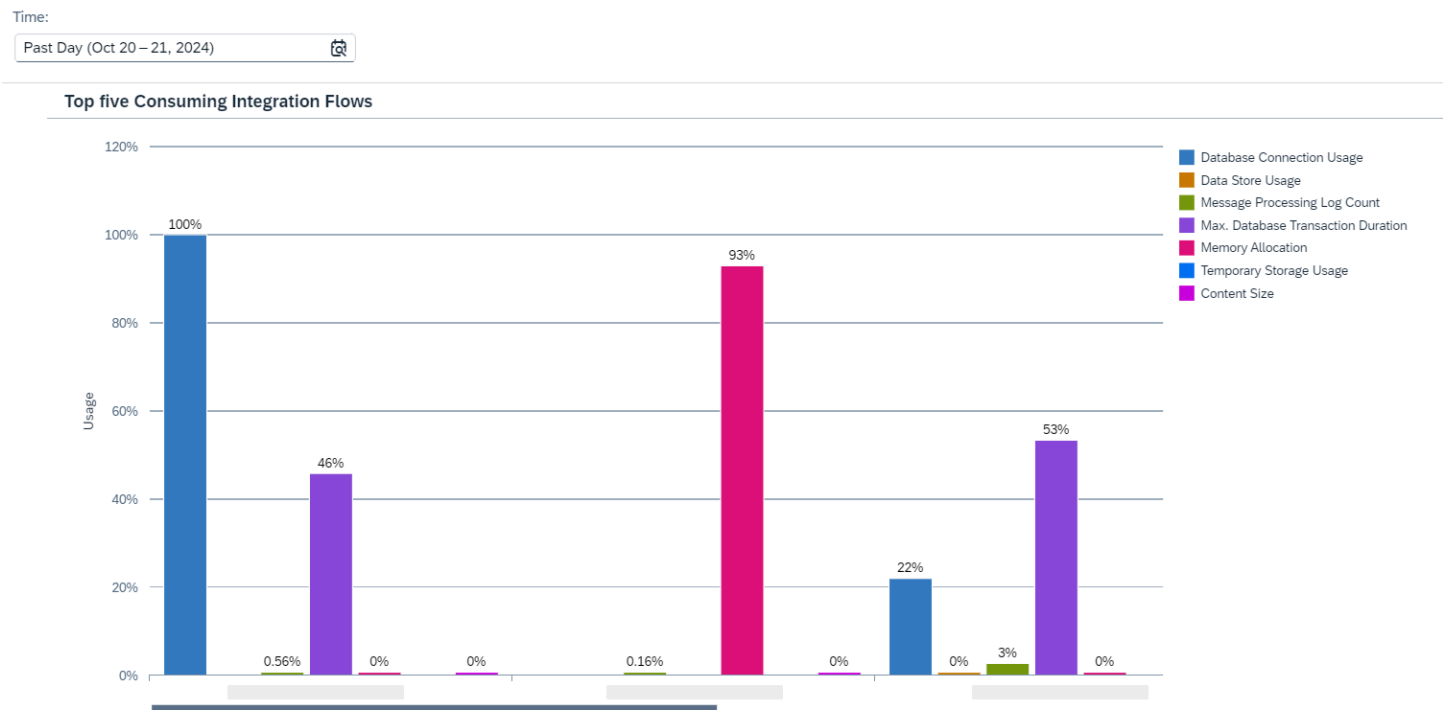
You can inspect the top five integration flows that consume the most resources.

The [Integration Flows](#) feature allows you to analyze and compare the integration flows that allocate the most resources among all your deployed integration flows. The bar graph also helps you to identify which specific resources are being consumed the most within each integration flow

Screen Components

The bar chart shows the five integration flows with the highest resource consumption for the defined time period.

Each flow displays different colored bars representing the levels of consumption of the different resources within the individual integration flows, as shown in the image below.



You can change the displayed time period by selecting a different option in the dropdown box under [Time](#). You can select [Past Day](#), [Past Week](#), [Past Month](#), or [Custom](#) for a custom time interval.

i Note

When you select the option [Custom](#) for the [Time](#) filter, you can adjust the date and time intervals with the calendar and watch elements. You can select dates up to 30 days in the past. However, selection of dates in the future is disabled.

The total resource consumption is presented in a bar graph. The horizontal axis displays the names of the integration flows, while the vertical axis shows the percentage of total resource consumption over the selected time period.

You can choose one of the bars to learn more about the consumption level of a specific resource. Depending on the resource type, you can see:

- **Usage relative to the entitlement.** This shows the maximum resource usage based on the resource's own entitlement.
- **Usage relative to other integration flows.** This indicates the usage level based on the total consumption of all integration flows.
- Maximum detected usage, duration, or time of the selected resource within the integration flow. The color of the value indicates whether the resource's entitlement was exceeded. If the consumption is at a critical level, it appears in orange or red. If the usage is within the given limit, it appears in green. If the value is in black, despite being high, it means there's no critical risk.

i Note

The system reads the resource consumption every hour. This means that there can be a maximum lag of one hour between processing an integration flow with a certain transaction setting and displaying the latest integration flow usage in the **Inspect** feature.

Functions

Choose a bar to get more context information and access the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen for the selected time period. For more information, see [Monitor Message Processing](#).

- **Show Integration Content**

Navigate to **Manage Integration Content** to monitor the specific integration flow you've selected. For more information, see [Manage Integration Content](#).

- **Inspect Integration Flow**

Navigate to the more detailed **Inspect Integration Flow** screen to further inspect the selected resource's consumption. For more information, see [Inspect Resource Consumption for Individual Integration Flow](#).

Inspect Resource Consumption for Individual Integration Flow

Inspect the resource consumption for a specific integration flow for a given time period.

i Note

This information is relevant only when you use SAP Cloud Integration in the Cloud Foundry environment.

You can navigate to this dashboard from the following screens for a selected integration flow:

- **Database Connection Usage > Top Integration Flows** (see [Inspect Top Integration Flows by Database Connection Usage](#))
- **Database Transactions > Top Integration Flows** (see [Inspect Top Integration Flows by Maximum Transaction Duration](#))
- **Monitoring Storage > Top Integration Flows** (see [Inspect Top Integration Flows by Message Processing Log Count](#))
- **Memory Usage > Top Integration Flows** (see [Inspect Integration Flows By System Memory Usage](#))

- **Content Size** (see [Inspect Content Size](#))
- **Temporary Storage** **Top Integration Flows** (see [Inspect Top Integration Flows by Storage Usage](#))

Select the time interval for which you want to display resource consumption. If you've selected the resource type **Monitoring Storage** or **Memory Usage**, you can further filter the display by monitoring status.

You can select the resource type for which you want to display the consumption in a dropdown list.

The resource consumption for an individual integration flow is shown for the given time interval in a diagram. Depending on the chosen resource type, the diagram shows the following data:

Selected Resource Type	Diagram Shows ...
Database Connection Usage	The database connection usage (in percent) is plotted in a bar graph against time. The bar height shows usage in percentage of the maximum usage. The duration covered by a bar is: Maximum usage corresponds to the situation where all database connections are used. The level of usage is indicated by the bar height and color (from green for low usage, up to red for critical usage). • Critical (red) indicates more than 90% • Warning (orange) indicates between 70% and 90% • OK (green) indicates less than 70%
Database Transactions	The maximum duration of database transactions (in minutes) is plotted in a bar graph against time. The level of usage is indicated by the bar height and color (from green for low duration, up to red for critical duration). • Critical (red) indicates maximum duration: more than 60 minutes • Warning (orange) indicates maximum duration: between 5 minutes and 60 minutes • OK (green) indicates maximum duration: less than 5 minutes
Monitoring Storage	The number of message processing logs stored in a dedicated time interval is plotted in a trajectory against time. A blue bullet represents the total count of message processing logs.
Memory Usage	The message throughput is plotted in two charts against time. • The total count of message processing logs (MPLs) for a given time period is plotted in a bar chart. The number of MPLs is identical to the number of messages processed during the selected time period. • The maximum message size within this time period is plotted in a line chart with a green bullet.
Temporary Storage	The total storage usage that temporary files allocate in the database against time. • The total storage usage (in MB) is plotted in a bar graph against time.

Selected Resource Type	Diagram Shows ...
	Each bar also has a bullet that indicates the size of the file that uses the maximum storage.
Content Size	The file size allocated to the selected artifact (in MB) is plotted in a bar graph against time. The bar height shows the total content size that the artifact occupies in the defined time period.

You can choose a bar or a bullet to get more context information and access the following functions:

- **Show Messages**

Navigate to the **Monitor Message Processing** screen. For more information, see [Monitor Message Processing](#).

- **Show Integration Content**

Navigate to the **Manage Integration Content** screen for the related integration flow. For more information, see [Manage Integration Content](#).

- **Zoom Out** and **Zoom In** to extend/reduce the selected time period.

Support Tasks

When the customer-managed operating model is applied, the tenant cluster is managed by the customer. If support activities have to be performed on the tenant cluster by an SAP expert, the following steps are required.

As tenant administrator (at customer side) perform the following steps:

1. Ask your SAP contact to entitle an SAP expert as system development expert.

SAP provides you with the corresponding user ID of the dedicated SAP expert.

2. **Temporarily** (for the time the support activities are going on) assign the authorization group **AuthGroup.SystemDeveloper** to this user.

Provide the tenant administrator (at customer side) with the user ID of the SAP expert and ask him or her to **temporarily** (for the time the support activities are going on) assign the authorization group **AuthGroup.SystemDeveloper** to this user.

Related Information

[Personas](#)

Additional Features

Related Information

[Health checks and recommended actions for SAP Integration Advisor Node](#)

[Enabling SAP Solution Manager to Act as Additional Alert Consumer](#)

Health checks and recommended actions for SAP Integration Advisor Node

Currently, the following health checks are covered for Integration Advisor node:

Integration Advisor State	The MBean checks the health of Integration Advisor. The threshold check is String 'Available'. In case Integration Advisor's health is good the check returns ' Available' and 'Unavailable' in case Integration Advisor's health is poor.
Availability Check	It checks the availability of the application via HTTP ping. The check is executed every second. Default values of 50 seconds for warning and 60 seconds for critical are currently assigned to this check. This means that if the ping response time is more than 50 or 60 seconds a warning or critical alert will be generated respectively.

Enabling SAP Solution Manager to Act as Additional Alert Consumer

You can enable SAP Solution Manager to display alerts.

Context

Currently, alerts are supported that are raised if no receiver can be determined during routing.

To set up this scenario, the SaaS administrator and the administrators of the involved SAP Solution Manager systems have to perform the following steps.

i Note

For more information on the tasks of the administrator of the SAP Solution Manager systems, see the documentation for SAP Solution Manager.

Procedure

1. Provide the administrator of the SAP Solution Manager system with the tenant management node URL.

The tenant management node URL is contained in the email that is sent out to the customer after the tenant provisioning process.

2. Create a user that enables the SAP Solution Manager system to connect to the integration platform as a client.
3. Assign the following role to this user: `IntegrationOperationServer.consumealerts`.
4. Provide the administrator of the SAP Solution Manager system with this user.
5. Make sure that the SAP Solution Manager system is registered as an alert consumer.

To request this feature, create a ticket on component LOD - HCI - PI - ET - OP.

Results

Once the connected SAP Solution Manager systems and tenants have been configured accordingly, alerts of the above-mentioned type are displayed in the Exception Management inbox of the involved SAP Solution Manager systems.